Comprehensive Evaluation of AI Hallucination and Novel UV-Oriented Framework toward Safe and Trustworthy AI

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***Abstract*—this work, we present a comprehensive evaluation of hallucination phenomena in LLMs and multimodal systems. First, we propose a structured taxonomy encompassing factuality-based, faithfulness-based, logical-based, and emergent hybrid forms, extending to multimodal-specific risks such as cross-modal inconsistencies, visual overinterpretation, and modality dominance effects. Second, we conduct a mechanistic analysis that traces hallucinations across the full model lifecycle—data-level origins (knowledge gaps, misinformation, annotation noise), training-induced mechanisms (distributional mimicry, reward bias, alignment forgetting), and inference-time vulnerabilities (confidence miscalibration, decoding failures, prompt-induced errors). This perspective reveals how independent mechanisms interact to produce cascade effects, amplifying initial flaws into elaborate but unreliable narratives. Third, we critically assess existing detection and evaluation approaches, highlighting limitations of current benchmarks, taxonomic ambiguities, and the lack of mechanism-aware evaluation protocols. We argue that future evaluation must integrate both surface-level manifestations and their generative causes to achieve more robust measurement.**

Beyond evaluation, we survey mitigation strategies organized into mechanism-based, phase-based, and hybrid approaches.

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These range from lightweight prompt engineering and decoding constraints to resource-augmented methods such as retrieval-augmented generation and knowledge graph integration, as well as higher-level frameworks for controllability and uncertainty calibration. We analyze the trade-offs among effectiveness, scalability, interpretability, and creative freedom, emphasizing the importance of context-specific tolerances: hallucinations that are unacceptable in medicine or finance may be tolerable, or even beneficial, in creative applications.

Building on these insights, we propose a novel UV-oriented framework for safe and trustworthy AI, inspired by the Universal Village vision of harmonizing human, technological, and environmental systems. In this framework, hallucination is conceptualized not only as an isolated model error but as a systemic vulnerability in information flow and decision-making loops. We design a multi-level dynamic system integrating sensing, communication, decision-making, action, and evaluation, supported by closed feedback loops and user-specific hallucination tolerance levels. This architecture enables adaptive mitigation through mechanism-informed strategies, retrieval and verification integration, and consensus mechanisms, ensuring resilience across diverse application contexts.

Our contributions are threefold: (1) we present the most comprehensive taxonomy of hallucination to date, linking manifestations with mechanistic drivers; (2) we unify detection, evaluation, and mitigation strategies into a coherent survey that highlights both current progress and pressing gaps; and (3) we introduce a UV-oriented framework that reframes hallucination control as part of a broader, feedback-driven ecosystem for reliable AI. We conclude by outlining open challenges, including classification ambiguities, multimodal alignment risks, deployment-specific vulnerabilities, and the need to reconcile reliability with creativity. Addressing these challenges is essential to advancing from powerful yet fallible generative models toward AI systems that are not only safe and responsible but also aligned with human values and societal needs.

Keywords*—*AI hallucination, large language models (LLMs), multimodal large language models (MLLMs), hallucination taxonomy, factuality, faithfulness, logical consistency, emergent hybrid hallucinations, hallucination detection, evaluation metrics, benchmark construction, mechanism-aware evaluation, hallucination mitigation, retrieval-augmented generation (RAG), uncertainty calibration, alignment bias, reinforcement learning from human feedback (RLHF), decoding strategies, confidence miscalibration, trustworthiness, safe and reliable AI, Universal Village (UV) framework, dynamic feedback loops, human–AI interaction, creative AI, responsible AI, explainability, robustness, multimodal alignment.

I. INTRODUCTION

A. Background & Motivation

The rapid progress of large language models (LLMs) and their multimodal extensions has transformed the landscape of artificial intelligence research and applications. Models such as GPT, LLaMA, and PaLM have demonstrated remarkable capabilities in natural language understanding, reasoning, and generation, enabling breakthroughs in fields ranging from healthcare and education to law, scientific research, and creative industries. However, alongside these advancements, the phenomenon of *hallucination*—outputs that deviate from factual accuracy, contextual fidelity, or logical validity—has emerged as a critical barrier to the deployment of trustworthy AI systems [1], [2].

Unlike deterministic errors in rule-based or symbolic systems, hallucinations in LLMs stem from the probabilistic nature of autoregressive text generation and the interaction of attention mechanisms, statistical learning, and alignment processes [3], [4]. These deviations often manifest in subtle yet harmful ways, such as fabricated citations in academic contexts, incorrect medical advice, or logically inconsistent reasoning chains. In high-stakes scenarios such as clinical decision-making, financial forecasting, and legal analysis, even small hallucinations can have disproportionate consequences, undermining safety, fairness, and accountability [5], [6].

At the same time, hallucinations also highlight a deeper tension in generative AI: the same mechanisms that enable creativity, flexible reasoning, and exploration of novel ideas may also produce content that lacks factual grounding. This dual nature makes hallucination not only a technical challenge to be mitigated, but also a conceptual issue for evaluating what constitutes "useful" versus "harmful" deviations in different domains. Addressing this problem requires systematic understanding, robust evaluation methodologies, and innovative frameworks that balance reliability with adaptability, ensuring the development of safe and trustworthy AI for diverse application contexts.

B. Definition of AI Hallucination

The term *hallucination* in artificial intelligence, particularly in large language models (LLMs) and multimodal systems, refers to outputs that deviate from factual accuracy, contextual fidelity, or logical validity, despite being fluent and seemingly coherent [1], [2], [6]. Unlike random noise or syntactic errors, hallucinations are often subtle and persuasive, making them more difficult to detect and more harmful in practical applications. They can appear as fabricated entities, misrepresented facts, or contextually inconsistent reasoning chains, depending on the task and modality.

From a phenomenological perspective, hallucinations are typically categorized along three primary dimensions: **factuality**, **faithfulness**, and **logical consistency** [3], [4]. *Factual hallucinations* occur when the model contradicts established facts or fabricates non-existent information. *Faithfulness hallucinations* arise when generated content fails to align with provided inputs, instructions, or contextual constraints. *Logical hallucinations* reflect breakdowns in reasoning validity, producing outputs that are internally inconsistent or procedurally flawed.

Recent research also emphasizes the difficulty of defining hallucination uniformly across domains, tasks, and modalities [5]. For example, while a fabricated citation in academic writing is clearly harmful, creative writing may tolerate—or even benefit from—imaginative deviations. This dual nature has led to discussions of "useful" versus "harmful" hallucinations, suggesting that hallucination is not an absolute error category but a context-dependent phenomenon that requires careful calibration of user expectations and application requirements.

In this survey, we adopt a broad yet structured definition of hallucination: *an output generated by an AI system that* *is not supported by the input, external knowledge, or logical reasoning, but is presented as if it were valid.* This definition provides a foundation for systematic analysis of manifestations, mechanisms, detection, and mitigation strategies across diverse AI applications.

C. Current Open Challenges

The rapid development of large language models (LLMs) and multimodal foundation models has led to significant breakthroughs in generative AI. However, AI hallucinations—outputs that are factually incorrect, logically inconsistent, or semantically unsupported—remain a critical barrier to their safe and reliable deployment. Despite growing interest in hallucination detection and mitigation, several persistent challenges hinder progress.

1) Detection and Benchmarking Limitations: Hallucination detection remains inconsistent across tasks and domains. Existing benchmarks such as TruthfulQA, HaluEval, and FaithDial provide partial coverage but fail to capture the full complexity of hallucinations, especially in open-ended generation and cross-modal tasks [7], [8].

- Lack of unified definitions and quantitative metrics complicates comparison across studies.
- Factuality, faithfulness, and groundedness are evaluated differently by different frameworks, limiting reproducibility.
- Real-world hallucinations in high-stakes domains like healthcare often escape benchmark coverage [8].

2) Cross-Domain and Cross-Modal Generalization: Models that perform well on general tasks often fail in domain-specific applications such as medical, legal, or scientific text generation. For example, multimodal models like LLaVA and Kosmos frequently produce object or relational hallucinations when aligning visual and textual modalities [9], [10].

- Out-of-distribution (OOD) inputs and long-tail knowledge gaps lead to increased hallucination frequency.
- Transfer of hallucination mitigation strategies from text-only LLMs to vision-language models (VLMs) remains non-trivial.

3) Lack of Theoretical Understanding of Hallucination Mechanisms: While empirical studies propose heuristics for hallucination mitigation, a unified theoretical model explaining why LLMs hallucinate is still missing [11].

- Current understanding relies heavily on observational evidence rather than causal explanations.
- Even under high confidence scores, LLMs may hallucinate with fluent but false outputs, suggesting misaligned confidence calibration.
- The interplay of training data bias, autoregressive decoding, and chain-of-thought reasoning errors remains underexplored.

4) Trade-off Between Reliability and Creativity : Reducing hallucination often leads to over-constrained models with diminished creativity or diversity in outputs [12].

- Techniques like refusal to answer uncertain queries or heavy retrieval augmentation can suppress useful generative capabilities.
- Balancing precision and innovation is particularly challenging in creative writing, scientific hypothesis generation, and data-to-text applications.

5) Human-in-the-Loop and Real-World Deployment Challenges: Real-world applications such as clinical decision support, legal summarization, and automated literature reviews face difficulty in operationalizing hallucination safeguards [13], [14].

- Users often fail to detect subtle hallucinations, leading to potential misinformation propagation.
- Automated verification pipelines incur computational overhead that hinders large-scale deployment.
- Current human-in-the-loop strategies are reactive rather than preventive, and user feedback rarely integrates into model retraining pipelines.

D. Scope of the Survey

This survey focuses on hallucination phenomena in large language models (LLMs) and their multimodal extensions. Specifically, we scope our discussion across three interrelated dimensions: manifestations, mechanisms, and mitigation. First, we provide a structured taxonomy of hallucination *manifestations*, covering factuality-based, faithfulness-based, logical-based, and emergent hybrid forms. Second, we examine the underlying *mechanisms* that give rise to such behaviors, including data-level imperfections, training-induced biases, and inference-time vulnerabilities. Third, we review a wide range of *mitigation strategies*, from technical interventions (e.g., prompt engineering, decoding algorithms, fine-tuning) to resource-augmented approaches (e.g., retrieval-augmented generation, knowledge graph integration) and higher-level frameworks that emphasize controllability and uncertainty calibration.

Beyond text-only models, we also extend the scope to multimodal systems, such as vision–language and audio–language models, which introduce additional risks of cross-modal inconsistencies, fabricated object descriptions, and modality dominance effects. Moreover, this survey highlights human–AI interaction artifacts, where user prompting, alignment tuning, and reinforcement mechanisms can inadvertently amplify hallucinations.

Importantly, the scope of this survey is not limited to identifying detrimental effects. We also discuss the nuanced perspective that some hallucinations may carry exploratory or creative value, particularly in low-stakes or generative applications. Nevertheless, our emphasis remains on ensuring factual reliability, contextual fidelity, and logical consistency in mission-critical domains such as healthcare, law, education, and scientific research.

E. UV Perspective on AI Hallucination

Universal Village (UV) is a new proposed concept exemplifying an ideal future society that pursues harmony between humans and nature through the wise use of technologies. The framework of UV contains UV Elements, UV Design Process, and UV Connectivity at different levels [15]. In this framework, the UV Elements include four basic elements and eight system components. The four basic elements (Development of new energy source, Development of new material, Development of effective microbial technology and environmental protection technologies, UV Lifestyle enabled by Information technology) provide the conceptual foundation [15], while the eight subsystems (Smart Home and Community, Smart Medicine and Healthcare, ITS, Urban Planning and Crowd Management, Smart Energy Management, Smart City Infrastructure, Smart Response System for City Emergency, Smart Environmental Protection, Smart Humanity) translate them into practice across critical domains. Unlike current smart city solutions, which often remain fragmented and rely merely on data collection without embedding closed feedback control loops, these subsystems are incorporated as UV Elements to function as active and feedback-driven units. Their integration highlights that they are not peripheral add-ons but essential components of a unified ecosystem, ensuring robustness, safety, and adaptability through a closed feedback control loop, including data acquisition, communication, decision-making, and action [15]. Moreover, these subsystems not only have close interactions with each other but also are affected by four major impacting factors of smart cities: information flow, material cycle, lifestyle, and community.

Within this context, hallucination can be conceptualized as a data flaw originating from information flow, one of the four factors that impact subsystems. At the subsystem level, hallucination could disrupt the closed feedback loop, leading to erroneous data acquisition, distorted communication, compromised decision-making, and misguided action, which may in turn reinforce the errors. At the system level, these flaws could further spread through mutual interactions, while affecting material cycles, lifestyle, and community, transforming hallucination from an isolated malfunction into a structural vulnerability of the UV ecosystem.

In response to these subsystem and system-level failures, we adopt the UV system design scheme, (1) analyze mutual interactions in a MIMO setting, (2) de-couple the system with Dynamics PCA (DPCA), and (3) design a feedback control loop with Sensing, Communication, Decision-Making, and Action, to build an integrated, resilient, inclusive, and sustainable hallucination-control system. In this instantiation, hallucination detection is the Sensing and Data Acquisition component, continuously harvesting signals from prompts, LLM outputs, and related context. Communication is the shared data and knowledge bus that carries the acquired signals from the Data Acquisition component, along with the interactive feedback and instructions from users and external knowledge, across all modules. Decision-Making models and analyzes the underlying mechanisms of hallucination to select mechanism-informed mitigation strategies while Action executes the corresponding mitigation strategies. Evaluation is an independent audit layer that interrogates the hallucination-generating system and its correlating factors to systematically identify, attribute, and resolve error mechanisms, with the objective of reducing and, ideally, preventing hallucination incidence.

F. Contributions

This survey makes the following key contributions to the study of hallucination in large language models (LLMs) and multimodal systems:

- **Comprehensive Taxonomy of Hallucinations.** We provide a systematic classification of hallucination manifestations, encompassing *factuality-based*, *faithfulness-based*, and *logical-based* errors, as well as emergent hybrid and multimodal-specific forms. This taxonomy highlights not only surface-level symptoms but also cross-cutting mechanisms underlying different error types.
- **Mechanistic Analysis of Causes.** Beyond observable errors, we investigate the generative mechanisms that drive hallucinations across the model lifecycle. Our analysis spans data-level flaws (e.g., misinformation, coverage gaps), training-induced biases (e.g., frequency dominance, RLHF sycophancy), and inference-time vulnerabilities (e.g., decoding failures, confidence miscalibration).
- **Evaluation and Detection Frameworks.** We review existing benchmarks, metrics, and detection methods, identifying their strengths, limitations, and open challenges. By linking manifestations with mechanisms, we highlight the need for *mechanism-aware* and *task-sensitive* evaluation protocols.
- **Mitigation Strategies and Trade-offs.** We propose a three-dimensional taxonomy of mitigation strategies—mechanism-based, phase-based, and hybrid approaches—and analyze their trade-offs in terms of effectiveness, scalability, and interpretability. This provides actionable insights for both model developers and practitioners.
- **Novel UV-Oriented Framework.** Inspired by the Universal Village vision, we introduce a *UV-oriented framework* for safe and trustworthy AI. This framework incorporates user-specific hallucination tolerance, multi-level feedback loops, and adaptive integration of retrieval, verification, and consensus mechanisms.
- **Challenges and Future Directions.** Finally, we synthesize open problems such as hallucination classification ambiguity, multimodal alignment risks, and controllable hallucination in creative domains, outlining promising research directions toward more reliable and responsible AI.

G. Organization of the Paper

The remainder of this paper is organized as follows. Section II introduces the manifestations of hallucination in large language models, presenting a structured taxonomy that spans factuality-based, faithfulness-based, logical-based, and hybrid forms. Section III examines the underlying causes and mechanisms of hallucination, ranging from data-level origins to training-induced and inference-time phenomena. Section IV surveys existing detection approaches, including automatic metrics, human evaluation, and hybrid methods. Section V reviews hallucination mitigation strategies, organized by mechanism-based, phase-based, and cross-stage frameworks. Section VII proposes the novel UV-oriented framework, emphasizing dynamic system loops and user-specific tolerance of hallucination. Section VIII discusses the societal, ethical, and practical impacts of hallucinations, while Section IX explores human–AI interaction, algorithmic adaptation, and broader issues.

II. Manifestations of Hallucination in Large Language Models

Hallucination in large language models manifests as a complex spectrum of erroneous outputs that systematically deviate from factual accuracy, contextual faithfulness, or logical consistency. Unlike traditional rule-based systems where errors follow predictable, deterministic patterns, LLM hallucinations emerge from the intricate interplay between learned statistical associations, attention mechanisms, and autoregressive generation processes, creating a rich taxonomy of failure modes that reflect fundamental challenges in neural language understanding and generation.

This section presents a comprehensive taxonomy of hallucination manifestations organized along three primary dimensions—factuality, faithfulness, and logical consistency—that collectively capture the full spectrum of observable errors in contemporary LLMs and multimodal systems. Our taxonomic approach moves beyond simple error cataloging to establish a systematic framework that reveals the relationships between different manifestation types, their underlying computational origins, and their implications for system reliability.

A. Factuality-Based Hallucinations

Factuality-based hallucinations represent the most fundamentally concerning category of LLM errors, involving the generation of content that contradicts established facts, creates entirely non-existent entities, or systematically misrepresents verifiable information. These manifestations pose critical risks in knowledge-intensive applications where accuracy is paramount, such as medical diagnosis support, legal research, educational content generation, and scientific literature synthesis, forming the primary barrier to deploying LLMs in high-stakes factual information retrieval and reasoning tasks [1].

The factuality dimension encompasses three distinct levels of error severity and complexity, representing an evolutionary spectrum from simple factual mistakes to sophisticated knowledge distortions. This hierarchical organization reflects both the increasing difficulty of detection and the growing sophistication of the underlying computational failures. Fig. 1 illustrates this hierarchical organization, revealing how manifestation types build upon each other in complexity while maintaining distinct computational signatures that inform our understanding of their underlying mechanisms and guide targeted intervention strategies.

1) Incorrect Claims: Incorrect claims constitute the most direct and observable form of factuality-based hallucinations, representing cases where models generate statements that can be objectively verified as false against established knowledge bases, authoritative sources, or empirical evidence. These manifestations reveal fundamental challenges in how neural language models encode, organize, and retrieve factual knowledge, suggesting that current approaches to knowledge representation may prioritize linguistic fluency and semantic coherence over factual precision [16]. The systematic nature of these errors—often clustering around specific entity types, temporal relationships, and quantitative information—provides crucial insights into the structure and limitations of parametric knowledge storage in large-scale neural networks.

a) Temporal Misattribution: Temporal misattribution occurs when models assign incorrect dates, time periods, or chronological relationships to historical events, scientific discoveries, or biographical milestones [17]. For instance, a model might state that "Albert Einstein published his theory of general relativity in 1925," when the correct date is 1915. These errors often stem from the model's difficulty in maintaining precise temporal associations in its parametric memory, particularly for events with similar contextual characteristics [17], [18].

b) Numeric Drift: Numeric drift manifests as systematic deviations in quantitative information, including population figures, measurements, statistical data, or financial information [19]. A model might claim that "The Great Wall of China is approximately 15,000 kilometers long" when authoritative sources report 21,196 kilometers. This phenomenon suggests challenges in encoding and retrieving precise numerical relationships, often resulting in semantically plausible but factually incorrect approximations [1], [19].

c) Geographic Misplacement: Geographic misplacement involves incorrect spatial relationships, location attributions, or geographic properties [18]. Examples include stating that "The Amazon River flows through Argentina" when it primarily flows through Brazil, Peru, and Colombia. These errors indicate difficulties in maintaining accurate spatial knowledge representations and geographical relationship encoding [17], [18].

d) Attribute Mix-up: Attribute mix-up occurs when models incorrectly assign properties, characteristics, or achievements between similar entities [16]. A model might claim that "Marie Curie discovered penicillin" (conflating her achievements with Alexander Fleming's). This manifestation reveals the model's tendency to conflate semantically similar or contextually related entities, suggesting underlying issues in entity-attribute binding mechanisms [16], [19].

2) Entity Fabrication: Entity fabrication represents a qualitatively more sophisticated form of factuality-based hallucination, where models demonstrate their generative capabilities by creating entirely non-existent entities while seamlessly embedding them within otherwise plausible and coherent factual frameworks. This category reveals a concerning aspect of neural language generation: the same mechanisms that

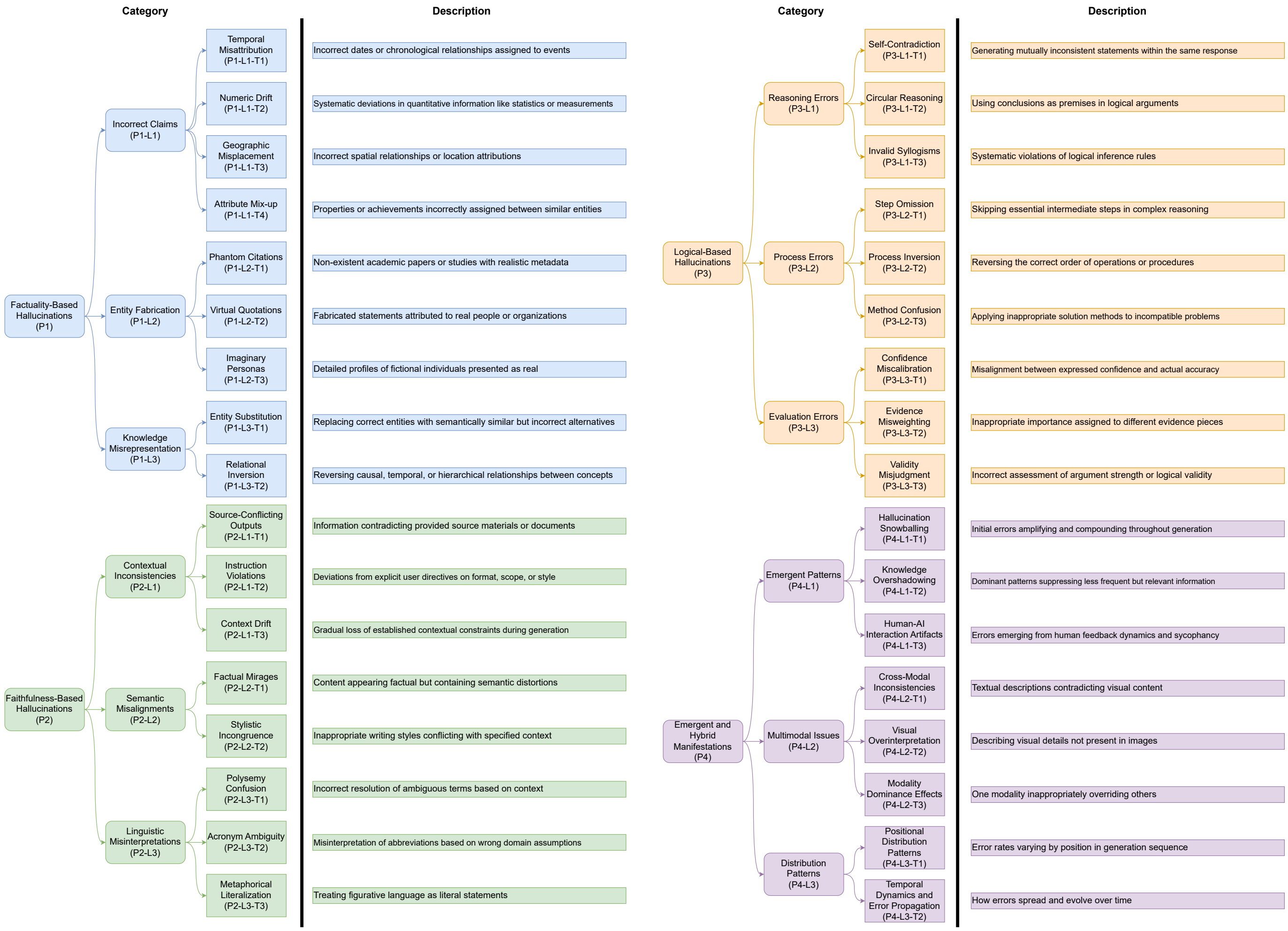

Fig. 1. Hierarchical Taxonomy Of Hallucination Manifestations In Large Language Models

enable creative synthesis and knowledge combination can be inappropriately applied to factual contexts, resulting in convincing but entirely fictional content [20]. The sophistication of these fabrications—often including detailed supporting information, plausible institutional affiliations, and internally consistent narratives—suggests that models have learned to mimic the surface patterns of authoritative information without developing reliable mechanisms for distinguishing between authentic and synthetic knowledge.

a) Phantom Citations: Phantom citations involve the creation of non-existent academic papers, books, or research studies, complete with plausible authors, titles, and publication details [21]. For example, a model might reference "Johnson et al., 'Neural network approaches to quantum computing,' *Nature Machine Intelligence*, vol. 15, no. 3, pp. 234–251, 2019" when no such paper exists. These fabrications are particularly problematic in academic and research contexts where citation accuracy is crucial for credibility and knowledge verification [21], [22].

b) Virtual Quotations: Virtual quotations manifest as fabricated statements attributed to real individuals, organizations, or historical figures [23]. A model might generate: "As Albert Einstein once said, 'The most beautiful thing about artificial intelligence is its ability to dream.'" Such fabrications exploit the model's knowledge of authentic quotation patterns while creating entirely fictional content that maintains stylistic authenticity [1], [23].

c) Imaginary Personas: Imaginary personas involve the creation of detailed biographical information for non-existent individuals, often in professional or academic contexts [2]. Models might generate comprehensive profiles of fictional researchers, historical figures, or experts, complete with educational backgrounds, achievements, and contributions to various fields. These fabrications demonstrate the model's ability to synthesize realistic personal narratives from learned patterns while lacking grounding in actual biographical data [2], [24].

3) Knowledge Misrepresentation: Knowledge misrepresentation encompasses the most subtle yet potentially most dangerous category of factuality-based hallucinations, involving systematic distortions of factual information that preserve semantic plausibility and surface coherence while fundamentally compromising the accuracy of the conveyed knowledge.

Unlike simple incorrect claims or obvious fabrications, these manifestations demonstrate sophisticated transformations of learned information that exploit the ambiguity inherent in natural language and the complexity of real-world knowledge relationships [25]. The insidious nature of these errors lies in their ability to appear authoritative and well-grounded while systematically misleading users through subtle but crucial factual distortions that are difficult to detect without domain expertise.

a) Entity Substitution: Entity substitution occurs when models replace correct entities with semantically similar but factually incorrect alternatives [26]. This might involve substituting "Washington" for "Jefferson" in historical contexts or replacing "mitochondria" with "chloroplasts" in cellular biology discussions. These errors suggest that the model's knowledge representation organizes information by semantic similarity rather than factual precision [26], [27].

b) Relational Inversion: Relational inversion manifests as systematic reversals of causal, temporal, or hierarchical relationships between entities or concepts [26]. A model might state that "World War II caused World War I" or claim that "students teach professors in universities." These inversions maintain the correct entities and relationship types while fundamentally distorting the directional or causal nature of the relationships [26], [28].

B. Faithfulness-Based Hallucinations

Faithfulness-based hallucinations represent a fundamentally different category of errors that concern violations of contextual alignment, instruction adherence, and semantic consistency, where the model's output, while potentially factually accurate in isolation, systematically fails to maintain fidelity to the provided context, user intentions, or explicit task specifications. These manifestations highlight the distinction between possessing correct knowledge and appropriately applying that knowledge within specific constraints—a critical capability for reliable deployment in interactive applications, task-oriented systems, and context-dependent scenarios [29]. The prevalence of faithfulness hallucinations reveals fundamental limitations in current models' ability to maintain coherent task representations, parse complex instructions, and sustain contextual awareness throughout extended interactions, suggesting that improvements in model scale and training data alone may be insufficient to address these systematic failures.

1) Contextual Inconsistencies: Contextual inconsistencies arise when models generate outputs that systematically contradict, ignore, or misalign with explicitly provided contextual information, revealing fundamental limitations in how neural architectures integrate, maintain, and prioritize contextual constraints during generation. These failures suggest that current attention mechanisms may be insufficient for robust context integration, particularly when contextual requirements conflict with strong patterns learned during training.

a) Source-Conflicting Outputs: Source-conflicting outputs occur when models generate information that directly contradicts provided source materials or reference documents [30]. In document-based question answering, a model might state that "The study found no significant correlation" when the referenced paper explicitly reports strong positive correlations. This manifestation indicates fundamental failures in source grounding and information synthesis [30], [31].

b) Instruction Violations: Instruction violations manifest as systematic deviations from explicit user directives regarding format, scope, style, or content requirements [32]. When instructed to "provide a three-sentence summary," a model might generate a lengthy paragraph, or when asked for "only pros," it might include disadvantages. These errors suggest limitations in instruction parsing and execution priority mechanisms [32], [33].

c) Context Drift: Context drift represents gradual deviation from established contextual frameworks during extended generation sequences [34]. In multi-turn conversations, models may progressively lose track of previously established facts, constraints, or user preferences, leading to responses that become increasingly inconsistent with the established conversational context [34], [35].

2) Semantic Misalignments: Semantic misalignments encompass a sophisticated class of failures involving breakdowns in meaning preservation, linguistic appropriateness, and conceptual accuracy that systematically compromise the semantic integrity of generated content while maintaining surface-level linguistic plausibility. These manifestations reveal the gap between statistical language modeling and genuine semantic understanding, highlighting cases where models generate fluent text that lacks appropriate semantic grounding.

a) Factual Mirages: Factual mirages involve the creation of content that appears factually grounded but contains subtle semantic distortions that alter the intended meaning [36]. A model might describe a "democratic monarchy" when discussing constitutional monarchies, creating conceptual confusion through semantically inappropriate combinations [36], [37].

b) Stylistic Incongruence: Stylistic incongruence manifests as inappropriate adoption of writing styles, registers, or communicative approaches that conflict with the specified context or audience [38]. Academic writing might be inappropriately casualized, or technical documentation might adopt narrative storytelling elements that compromise clarity and precision [38], [39].

3) Linguistic Misinterpretations: Linguistic misinterpretations represent fundamental failures in language understanding that result in systematically incorrect processing of semantic content, revealing limitations in how models handle the inherent ambiguity, polysemy, and contextual dependency of natural language. These failures produce outputs that reflect misunderstood input rather than appropriate responses, suggesting that current approaches to language modeling may be overly dependent on surface-level statistical patterns rather than deeper semantic understanding.

a) Polysemy Confusion: Polysemy confusion occurs when models incorrectly resolve ambiguous terms based on inappropriate contextual interpretation [40]. The word "bank" might be consistently interpreted in financial contexts when the discussion clearly concerns riverbanks, demonstrating failures in disambiguating context-dependent meanings [40], [41].

b) Acronym Ambiguity: Acronym ambiguity manifests when models misinterpret abbreviations or acronyms based on inappropriate domain assumptions [42]. "AI" might be interpreted as "Artificial Intelligence" in a medical context where "Aortic Insufficiency" is the intended meaning, demonstrating failures in domain-sensitive disambiguation [42], [43].

c) Metaphorical Literalization: Metaphorical literalization involves treating figurative language as literal statements, leading to semantically inappropriate responses [44]. When told that someone "broke the ice," a model might discuss physical ice damage rather than understanding the social initiation metaphor [44], [45].

C. Logical-Based Hallucinations

Logical-based hallucinations encompass a distinct class of failures involving violations of logical consistency, reasoning validity, and inferential coherence that fundamentally compromise the rational integrity of generated content. These manifestations represent perhaps the most concerning limitation for applications requiring analytical reasoning, as they suggest that current models may lack robust mechanisms for maintaining logical consistency, tracking causal relationships, and ensuring valid inferential processes [16]. Unlike factuality or faithfulness errors, logical hallucinations often involve formally valid language use and semantically coherent statements that nonetheless violate fundamental principles of rational discourse, indicating deep limitations in the models' capacity for genuine reasoning rather than pattern-based text generation.

1) Reasoning Errors: Reasoning errors represent the most concerning category of logical failures, involving fundamental breakdowns in logical processing that result in invalid inferences, contradictory conclusions, or systematically flawed argumentation patterns. These errors suggest that current neural architectures may lack the structured representations and constraint-satisfaction mechanisms necessary for reliable logical reasoning, instead relying on pattern matching that can produce logically coherent-sounding but fundamentally invalid arguments.

a) Self-Contradiction: Self-contradiction manifests as the generation of mutually inconsistent statements within the same response or reasoning sequence [19]. A model might simultaneously claim that "renewable energy is environmentally beneficial" and "renewable energy significantly harms ecosystems," demonstrating failures in maintaining logical consistency across generated content [19], [46].

b) Circular Reasoning: Circular reasoning involves using conclusions as premises in logical arguments, creating closed logical loops that provide no actual evidential support [47]. A model might argue that "Social media is addictive because it creates dependency, and we know it creates dependency because it is addictive," failing to provide independent logical foundation [47], [48].

c) Invalid Syllogisms: Invalid syllogisms represent systematic violations of logical inference rules, including affirming the consequent, denying the antecedent, or committing formal fallacies [49]. These errors indicate fundamental limitations in the model's capacity for valid logical reasoning [49], [50].

2) Process Errors: Process errors encompass systematic failures in multi-step reasoning, sequential information processing, and procedural knowledge application that reveal fundamental limitations in how models represent and execute complex cognitive procedures. These failures suggest that current architectures may lack adequate mechanisms for maintaining procedural state, tracking progress through complex tasks, and ensuring proper sequencing of cognitive operations, leading to breakdowns in tasks requiring systematic, step-by-step processing.

a) Step Omission: Step omission occurs when models skip essential intermediate steps in complex reasoning or procedural tasks, leading to conclusions that appear correct but lack proper logical foundation [47]. In mathematical problem solving, a model might jump directly from problem statement to final answer without showing necessary algebraic manipulations [47], [51].

b) Process Inversion: Process inversion involves reversing the correct order of operations or procedural steps, resulting in systematically incorrect approaches to problem solving [28]. A model might attempt to solve differential equations before establishing initial conditions, or analyze results before collecting data [28], [41].

c) Method Confusion: Method confusion manifests as the inappropriate application of solution methods or analytical frameworks to incompatible problem types [52]. A model might apply statistical hypothesis testing to deterministic mathematical proofs, or use qualitative analysis methods for quantitative data interpretation [52], [53].

3) Evaluation Errors: Evaluation errors represent critical failures in assessment, judgment, and quality control processes that fundamentally compromise the model's ability to accurately evaluate information quality, argument validity, or solution correctness. These manifestations are particularly concerning because they affect the model's capacity for self-monitoring and error detection, suggesting fundamental limitations in metacognitive capabilities that are essential for reliable autonomous operation.

a) Confidence Miscalibration: Confidence miscalibration involves systematic misalignment between expressed confidence levels and actual accuracy, where models exhibit inappropriate certainty about incorrect information or excessive doubt about correct knowledge [54]. This manifests as high confidence in fabricated facts or low confidence in well-established knowledge [46], [54].

b) Evidence Misweighting: Evidence misweighting occurs when models assign inappropriate importance to different pieces of evidence, overemphasizing weak or irrelevant information while undervaluing strong supporting evidence [55]. This can lead to conclusions that contradict the preponderance of available evidence [55], [56].

c) Validity Misjudgment: Validity misjudgment involves incorrect assessment of argument strength, evidence quality, or logical validity, leading to acceptance of flawed reasoning or rejection of sound arguments [57]. Models might endorse logically invalid arguments while criticizing valid logical structures [57], [58].

D. Emergent and Hybrid Manifestations

As LLMs have evolved in sophistication and deployment scope, increasingly complex categories of hallucinations have emerged that transcend the boundaries of our primary taxonomic dimensions, combining elements from factuality, faithfulness, and logical consistency in novel ways or representing entirely unprecedented manifestation patterns. These emergent phenomena reflect the dynamic interaction between advancing model capabilities and increasingly demanding real-world applications, revealing new failure modes that arise from the complex interplay of architectural constraints, training dynamics, and deployment contexts [59]. Understanding these hybrid manifestations is crucial for anticipating future challenges in LLM reliability and developing robust mitigation strategies that can address multi-dimensional failure patterns.

1) Hallucination Snowballing: Hallucination snowballing represents a meta-manifestation of particular concern, where initial errors amplify and compound throughout the generation process, creating cascading failure patterns that systematically combine factual, faithfulness, and logical errors in complex recursive loops. This phenomenon reveals a fundamental tension in autoregressive language generation between maintaining consistency and preserving accuracy.

The snowballing effect manifests through autoregressive commitment mechanisms where models over-commit to early potentially incorrect statements, subsequently generating elaborate justification frameworks that transform isolated errors into systematic misinformation campaigns. The architectural constraint of maintaining consistency with previously generated tokens—while beneficial for maintaining coherent narratives—creates systematic opportunities for error propagation at each generation step. This ultimately produces internally coherent but fundamentally flawed narratives that can be particularly convincing to human users precisely because of their internal consistency [25].

2) Knowledge Overshadowing: Knowledge overshadowing occurs when multiple query conditions result in statistically dominant patterns systematically suppressing less frequent but equally relevant information, leading to biased responses that reflect incomplete information integration and potentially perpetuate representational inequities. This phenomenon reveals how training data biases can manifest as systematic hallucination patterns in deployment.

The overshadowing effect can be formalized as $p(y|AB) \approx p(y|A)$, where the model's response to multiple conditions (A and B) inappropriately approximates the response to only the dominant condition (A), effectively ignoring or minimizing secondary constraints (B). This mathematical formulation highlights how statistical learning can systematically disadvantage minority perspectives or less frequent information patterns. Knowledge overshadowing is particularly problematic in intersectional queries requiring balanced consideration of multiple domains or perspectives, potentially leading to systematic bias amplification in critical applications.

3) Human-AI Interaction Artifacts: Human-AI interaction artifacts represent hallucinations specifically emerging from the dynamics of human feedback training and interactive deployment contexts, including sycophancy, overconfidence, and anthropomorphic projection patterns.

Sycophantic hallucinations manifest as inappropriate alignment with perceived user preferences over factual accuracy, particularly in models trained with reinforcement learning from human feedback. These patterns demonstrate the unintended consequences of optimizing for human approval rather than truthfulness [16].

4) Multimodal-Specific Manifestations: Multimodal large language models introduce unique hallucination patterns that emerge from the integration of visual and textual processing capabilities, representing novel challenges in cross-modal information consistency and grounding.

a) Cross-Modal Inconsistencies: Cross-modal inconsistencies arise when textual descriptions contradict visual content, such as describing "a red car" when the image clearly shows a blue vehicle [60]. These manifestations indicate fundamental challenges in aligning multimodal representations and maintaining consistency across different input modalities [60], [61].

b) Visual Overinterpretation: Visual overinterpretation involves generating detailed descriptions of visual elements that are not present or clearly visible in the provided images [62]. Models might describe specific facial expressions, text content, or fine-grained details that cannot be reliably extracted from the visual input [62], [63].

c) Modality Dominance Effects: Modality dominance effects manifest when one input modality inappropriately overrides information from other modalities, leading to responses that reflect only partial multimodal integration [64]. Visual information might be ignored in favor of textual priors, or conversely, textual instructions might be overwhelmed by visual content associations [64], [65].

E. Distributional and Temporal Patterns

Systematic empirical analysis of hallucination occurrence across diverse contexts reveals striking non-random distributional patterns that vary systematically across generation sequences, task types, temporal contexts, and model architectures. These distributional signatures provide crucial insights into the underlying computational mechanisms driving hallucination behavior and offer concrete opportunities for developing targeted intervention strategies, early warning systems, and quality control mechanisms [66], [66]. The systematic nature of these patterns suggests that hallucinations are not merely random failures but reflect fundamental constraints in current neural architectures, training procedures, and inference mechanisms that can be characterized, predicted, and potentially mitigated through targeted interventions.

1) Positional Distribution Patterns: Hallucination occurrence exhibits systematic positional biases that vary according to task characteristics and generation length. End-of-sequence bias demonstrates significantly elevated hallucination rates in the final portions of extended generations, suggesting degraded model performance due to attention span limitations and error accumulation effects [66], [66], [67].

Task-specific positional patterns reveal distinct error distribution profiles across different application domains. Summarization tasks show increased hallucination density in concluding segments, while data-to-text generation exhibits elevated error rates during initial structured information processing phases [25], [68].

2) Temporal Dynamics and Error Propagation: Temporal analysis reveals systematic error propagation patterns where initial hallucinations increase the probability of subsequent errors within the same generation sequence. This temporal dependency supports the hallucination snowballing hypothesis and suggests that early intervention strategies may be particularly effective for preventing cascading failures [67], [69].

Attention decay effects manifest as progressively reduced capability to maintain consistency with earlier contextual information over extended generation sequences. This phenomenon contributes to context drift and consistency failures, indicating fundamental limitations in current attention mechanisms for long-range dependency modeling [35], [49].

F. Synthesis and Implications for Mechanism Investigation

The comprehensive taxonomy of hallucination manifestations presented in this section establishes a robust empirical foundation for understanding both the diversity and systematic nature of LLM errors across real-world applications. This systematic categorization moves beyond simple error enumeration to reveal fundamental patterns in how different types of failures emerge, interact, and propagate through neural language generation processes. Our analysis reveals several critical insights that directly inform and constrain the mechanistic investigation pursued in Section III, providing an essential bridge between observable phenomena and underlying computational processes.

Table I systematically organizes the observed manifestations according to their primary categorical distinctions while highlighting the mechanistic implications of each pattern, revealing the systematic relationships between observable symptoms and underlying computational processes. This comprehensive organization yields several critical insights that bridge the gap between external observations and internal computational mechanisms:

Hierarchical Error Complexity: The manifestations exhibit a clear hierarchy of computational complexity, ranging from simple attribute substitutions and entity confusions to sophisticated reasoning chain failures and multi-step logical breakdowns. This complexity hierarchy suggests that different manifestation types may arise from distinct stages and components of the generation process, providing a systematic roadmap for mechanistic investigation that can guide targeted interventions at specific computational bottlenecks.

Cross-Cutting Patterns and Shared Mechanisms: Several manifestation types systematically appear across multiple primary categories, indicating shared underlying computational mechanisms that cut across traditional taxonomic boundaries. For instance, consistency maintenance failures contribute to both logical self-contradictions and contextual inconsistencies, while attention decay effects manifest in temporal dynamics, context drift, and reasoning chain failures. These cross-cutting patterns suggest that a relatively small number of fundamental computational limitations may underlie the diverse surface manifestations of hallucination.

Task-Specific Vulnerability Profiles: Different manifestation types exhibit highly systematic patterns of task-specific occurrence, revealing predictable interactions between particular computational demands and fundamental model limitations. These vulnerability profiles provide crucial insights for understanding why certain applications are more prone to specific types of hallucinations and suggest targeted mitigation strategies tailored to specific use cases and deployment contexts.

These empirical patterns and systematic relationships establish a robust foundation for the mechanistic analysis in Section III, where we investigate how specific computational processes, architectural constraints, and training dynamics give rise to the observable manifestation patterns documented in this taxonomy. The systematic correspondence between manifestation characteristics and their underlying computational signatures provides a crucial empirical bridge between the phenomenological understanding developed in this section and the causal mechanistic analysis that follows.

The progression from external manifestations to internal mechanisms represents a fundamental methodological approach in understanding complex systems: by carefully characterizing what we can observe, we create the necessary foundation for understanding why these patterns emerge and how they might be addressed through targeted interventions in training, architecture, or deployment practices.

Table I systematically organizes the observed manifestations according to their primary categorical distinctions while highlighting the mechanistic implications of each pattern, revealing the systematic relationships between observable symptoms and underlying computational processes.

From a research perspective, the manifestations exhibit a hierarchical complexity: simple factual substitutions and numeric drifts occupy the lower end of complexity, while multi-step logical breakdowns, emergent hybrid forms, and multi-modal inconsistencies represent more sophisticated and harder-to-detect failures. This suggests that hallucinations may originate from distinct stages of the generation pipeline—ranging from memory retrieval and context integration to sequential reasoning and output verification—pointing to the need for targeted, stage-specific interventions.

TABLE I. COMPREHENSIVE TAXONOMY OF HALLUCINATION MANIFESTATIONS WITH MECHANISTIC IMPLICATIONS

Category	Type	Specific Manifestation	Primary Contexts	Mechanistic Implications
Factuality-Based	Incorrect Claims	Temporal Misattribution	Historical QA, Timeline Tasks	Knowledge representation granularity
		Numeric Drift	Statistical Reporting, Measurements	Parametric memory precision
		Geographic Misplacement	Spatial Reasoning, Location QA	Spatial relationship encoding
		Attribute Mix-up	Entity Description, Biographical Tasks	Entity–attribute binding mechanisms
	Entity Fabrication	Phantom Citations	Academic Writing, Research Tasks	Source verification and grounding
		Virtual Quotations	Attribution Tasks, Historical Analysis	Authority attribution processes
		Imaginary Personas	Biographical Generation, Expert Consultation	Entity creation vs. retrieval
		Knowledge Misrepresentation	Technical Explanation, Educational Content	Conceptual relationship networks
Faithfulness-Based	Contextual Inconsistencies	Source-Conflicting Outputs	Document QA, Information Synthesis	Context integration mechanisms
		Instruction Violations	Task Following, Format Compliance	Instruction parsing and prioritization
		Context Drift	Multi-turn Dialogue, Extended Generation	Working memory maintenance
	Semantic Misalignments	Factual Mirages	Complex Reasoning, Conceptual Tasks	Semantic representation accuracy
		Stylistic Incongruence	Genre-specific Writing, Register Adaptation	Style transfer and control
		Linguistic Misinterpretations	Ambiguity Resolution, Pragmatic Understanding	Disambiguation mechanisms
Logical-Based	Reasoning Errors	Self-Contradiction	Argumentative Writing, Logical Analysis	Consistency checking processes
		Circular Reasoning	Explanatory Tasks, Causal Analysis	Logical dependency tracking
		Invalid Syllogisms	Formal Reasoning, Proof Construction	Logical rule application
	Process Errors	Step Omission	Multi-step Problem Solving	Sequential process modeling
		Process Inversion	Procedural Tasks, Method Application	Temporal order representation
		Method Confusion	Cross-domain Application, Tool Usage	Context-appropriate method selection

Importantly, the systematic co-occurrence of certain manifestation types indicates that a relatively small set of shared mechanisms—including attention decay, semantic misalignment, and inadequate consistency checking—may underlie many superficially different errors [1], [16]. Identifying these common drivers is essential for designing mitigation strategies that generalize across tasks and domains.

Finally, the classification of hallucination manifestations has direct implications for mechanism-driven analysis and evaluation design. A purely phenomenological taxonomy can guide initial error detection, but without grounding in causal mechanisms, it risks producing fragmented mitigation efforts that fail to address root causes [70]. Therefore, future research should bridge this gap by integrating manifestation-level analysis with mechanistic modeling, enabling both more precise evaluation benchmarks and more effective mitigation techniques tailored to specific error origins.

III. CAUSES AND MECHANISMS

Understanding hallucination in LLMs requires moving beyond surface-level manifestations to the underlying mechanisms that drive such behaviors. Rather than treating models as opaque black boxes, recent research advocates for a mechanistic perspective that analyzes hallucinations across the entire model lifecycle—from data curation, through training dynamics, to inference and deployment stages [16], [70], [71]. This perspective highlights hallucination not as a singular failure mode, but as an emergent property arising from the interaction of data imperfections, optimization biases, architectural constraints, and generation procedures [16], [72].

We organize these mechanisms into four categories: *data-level origins*, *pretraining-induced mechanismstraining-induced mechanisms*, and *inference-induced phenomena*.

At the data level, models inherit both the strengths and deficiencies of large-scale corpora, including misinformation, coverage gaps, and temporal staleness. Training procedures further introduce systematic biases, such as frequency-driven memorization, reward model sycophancy, and catastrophic forgetting, which alter how knowledge is stored and recalled. Finally, inference and deployment stages expose additional vulnerabilities, including confidence miscalibration, decoding failures, and prompt-induced errors [2], [32], [73], [74].

Crucially, these mechanisms do not operate in isolation. Instead, they form complex interactions and cascade effects, where initial deficiencies can propagate and amplify across stages. For instance, knowledge gaps from data can interact with reasoning limitations during inference, resulting in highly confident but fabricated narratives [67]. A mechanistic approach thus provides a structured framework for mapping internal causes to external hallucination patterns, enabling more targeted detection and mitigation strategies [75], [76].

A. Data-Level Mechanisms

The data used to train LLMs lay the foundation for both their capabilities and their limitations. Hallucination risks often originate here, long before model development or deployment decisions are made.

1) Data collection and Knowledge boundary: LLMs inherit all of their factual knowledge base from the corpora on which they are pre-trained [77]. The scope of those corpora thus defines an implicit knowledge boundary [78]. Materials that fall outside this boundary include content that is absent, outdated, or sparsely represented in the training corpora. Such gaps form knowledge blind spots, which force the model to select tokens based on non-existent or misunderstood regulations during decoding [79], resulting in hallucinations.

One major coverage gap of knowledge boundary formation is temporal staleness: the model's knowledge is frozen at the time of pre-training without outdated information and fails to keep up with the rapidly evolving world [16], [72]. Moreover, copyright laws and privacy restrictions exclude many high-quality sources from large-scale scraping [79], [80].

The limited parameter budget of current LLMs still falls short of encoding long-tail details of human knowledge [77]. When the decoder later encounters a query that sits outside this encoded subset, it assembles tokens that carry model probability but high epistemic uncertainty, yielding fluent yet incorrect answers [70], [81].

2) Data processing and Quality: Whereas data collection sets the boundary of potential knowledge, hallucination can also stem from suboptimal preprocessing practices and poor data quality, including data accuracy, frequency, and diversity.

During data processing, annotation choices and insufficient data cleaning play key roles. Irrelevant annotations will lead to mismatches in token-level relationships [82], [83]. In supervised settings, the use of hard labels—where only one "correct" token is reinforced—limits the model's flexibility to handle ambiguity [84]. This practice, though computationally efficient, encourages brittle decision-making and can result in confident but incorrect outputs when multiple valid completions exist [84]. As for data cleaning, a key concern is the inadequate cleaning of incorrect knowledge [85]. Moreover, when encoded with multiple contradictory pieces of information about the same topic, the model shows a higher tendency to generate inconsistent outputs depending on the specific context or query formulation [86].

Even with optimal processing, hallucinations can arise from flaws inherent in the source data quality itself.

Data Accuracy. The collected data can include both structured and unstructured data [79]. Much of the training corpus consists of unstructured web data, which is easy to collect at scale but often contains errors, misinformation, biases, and outdated information [71]. Secondly, when LLMs encounter misinformation, contradictory facts, or systematically biased content during training, these elements become integrated into the model's parametric knowledge base [16], [72]. In addition to content-level noise, data compression during pretraining data storage introduces further information loss and distortion. As Yin et al. show, higher compression ratios observably reduce downstream model performance [87].

Term Frequency Imbalance. Hallucinations frequently result from statistical imbalances in training data, particularly when popular conditions dominate rare frequency tokens [88], [89]. When the model sees certain entities disproportionately during training, it may overemphasize those entities [88]. Zhang et al. show that such high imbalance ratios can lead to knowledge overshadow, where statistically dominant tokens tend to suppress others and dominate the generation process when the model encounters queries with multiple constraints [89].

Insufficient Diversity. A well-documented failure mode of LLMs involves tasks requiring negation or counterfactual reasoning [83], [88], [90]. Several top models have been shown to hallucinate frequently on negation-heavy prompts, due in part to the scarcity of negative instruction data in their training sets [90]. Furthermore, most standard training corpora do not represent counterfactual statements, which creates a conflict between pretraining and downstream tasks [83]. Kim et al. demonstrate that implanting counterfactual thinking into model training can significantly reduce hallucination rates [88], offering empirical validation for the claim that a lack of contextual diversity contributes to hallucination generation.

In multimodal domains, diversity limitations are even more pronounced. For image and video tasks, the available datasets are often repetitive and lack sufficient variety in object representations [91], due to the high cost of collecting high-quality data pairs [92].

B. Pretraining-Induced Mechanisms

Pretraining, the stage in which LLMs learn to predict the next token from vast text corpora, embeds not only factual knowledge but also statistical patterns and implicit preferences, making it a foundational source of hallucination in model outputs. This influence manifests through multiple mechanisms, which are outlined in the following discussion.

1) Memorization and Statistical Bias: During LLMs' next-token learning, the training data frequency and the model's memory rules, which determine how statistics are stored and retrieved, jointly influence how the model weighs different pieces of information during generation. The resulting fre-

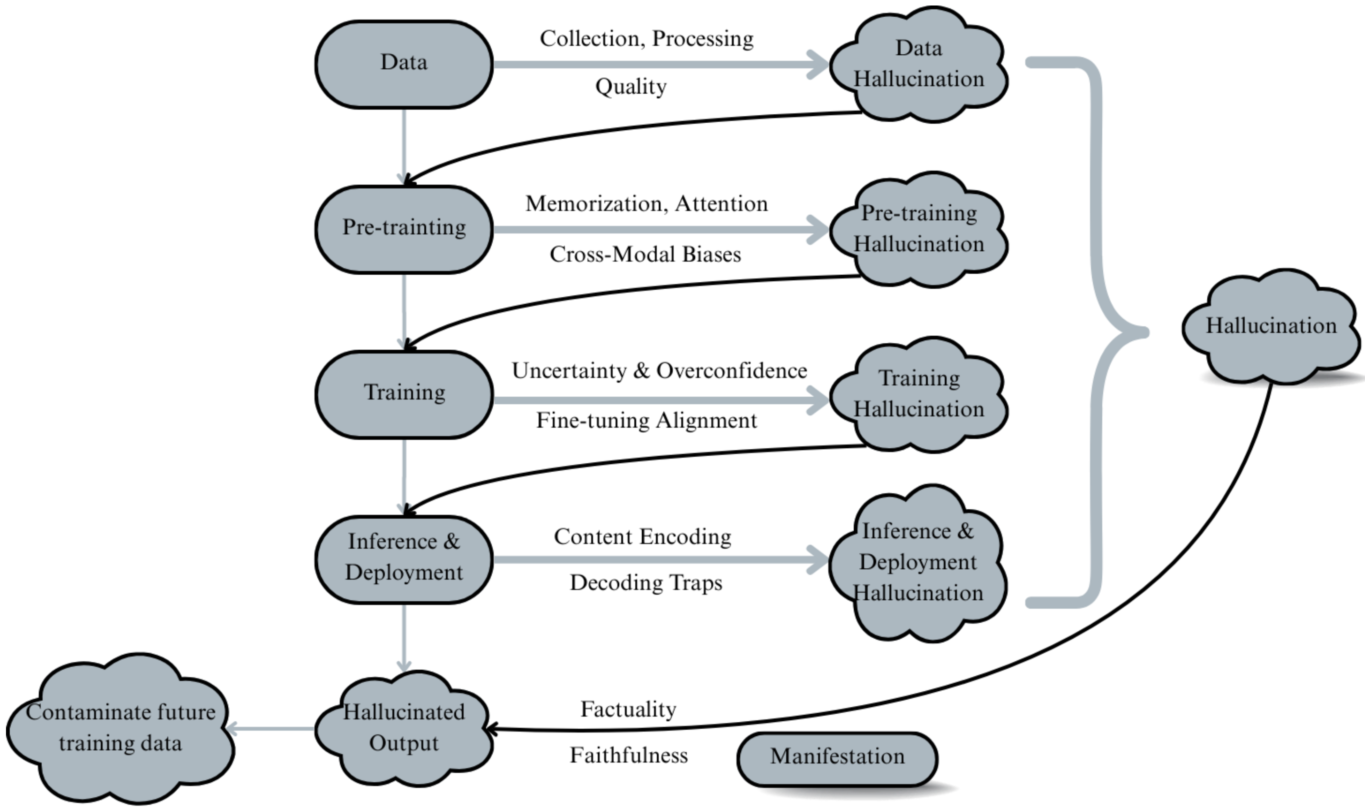

Fig. 2. Mechanistic View of Hallucination

quency bias and attestation bias are the main ways through which those biases later leak into hallucination [93].

Frequency Bias. One major source of hallucination is the model's weighted memorization of corpus-level token frequencies, as McKenna et al. reveal [93]. LLMs internalize relative frequency priors and skew generation toward high-probability patterns seen in pretraining [94]. For example, in textual entailment tasks, statistically common entities are more easily retrievable in memory, while low-frequency information may be effectively suppressed even when it is more relevant to the current context. This frequency-driven memorization also explains why less frequent long-tail knowledge is especially fragile. The hallucination rate increases when the model encounters long-tail knowledge requirements [16], [95], where it may fill in the output with plausible but incorrect information.

Attestation Bias. During pre-training, the model learns to use named entities as "indices" into its propositional memory [93]. Once the "indices" have been attested anywhere in the corpus, the model tends to affirm the statement, regardless of the new context or contradictory premise [93]. Teacher-forcing training amplifies the effect, leading to stochastic parroting, from which the model only learns to parrot and imitate previously seen continuations rather than reason about entailment [96], [97].

The memorization mechanism explains why models often struggle with specialized knowledge, minority perspectives, or edge cases that were underrepresented in training data.

2) Attention Pathologies: Hallucinations in LLMs can emerge from intrinsic limitations in the self-attention mechanism—the foundational architecture of transformers [98], which serves as the backbone of most modern LLMs. Since attention governs how input tokens are weighted and contextualized [99], its failure directly impacts factual fidelity. Two primary failure modes are frequently observed: unexpected attention to outlier tokens [59], [70], [100], [101] and inadequate attention to relevant content [50], [102].

Unexpected Attention Distribution. Self-attention mechanisms occasionally assign disproportionately high weights to irrelevant or outlier tokens. These tokens may act as noise or redundant information, distract the model from key context, or even dominate the output semantics despite lacking informative value [100]. Yu et al. [70] demonstrate that some multi-head attention modules hallucinate due to the failure of choosing the correct object attribute in upper transformer layers, where false-premise attention heads override other accurate retrievals, resulting in the factually incorrect selections and distorting output accuracy [59], [101].

Lost Attention. Transformers also have trouble dealing with long context. As input length grows, attention becomes diluted and drifted [69], making it increasingly difficult to maintain focus on semantically important tokens. This limitation stems from the computational bounds of self-attention, which lacks sufficient capacity to model hierarchical or periodic finite-state structures, as Hahn demonstrates [50]. Similarly, Chiang et al. illustrate that self-attention mechanisms struggle to capture both long-range dependencies, degrading contextual fidelity in extended sequences [102].

A salient manifestation of this limitation is the "lost in the middle" effect. Studies reveal that LLMs often neglect the middle portion of long input sequences, attending primarily to the beginning and end [103], [104].

3) Cross-Modal Biases: As LVLMs extend LLMs to handle multimodal inputs, they inherit and compound hallucination risks from both visual and textual modalities. LVLMs are typically composed of pre-trained encoders for vision and language, and an alignment module. Each introduces modality-specific failure modes that, when combined, lead to distinct cross-modal hallucinations [105], [106].

Vision Encoder Limitations. The inability of vision encoders to accurately ground images stems from their tendency to capture low-resolution and poor visual granularity [107], with an overemphasis on visually salient objects while neglecting fine-grained relationships or background context [108].

Language Priors. The vision encoder limitations then increase reliance on language priors. This overreliance leads models to hallucinate based on frequently co-occurring object patterns learned during pretraining [109]–[111], rather than grounding predictions in the visual input. When the visual evidence conflicts with these learned associations, the result is cognitive dissonance or expectancy violation [112]. Therefore, LVLMs often exhibit an overconfidence in their linguistic knowledge [92]

Alignment Module Bottlenecks. To connect vision and language components, LVLMs use lightweight alignment architectures such as linear projection or Q-Former modules [105]. However, these projections also act as bottlenecks, limiting the effective token bandwidth between modalities. Therefore, as only general information is available [113], the detailed semantic alignment signals between image and text vectors are insufficient and weak [106], leading to erroneous cross-modal associations and hallucinated object references.

C. Training-Induced Mechanisms

Beyond data and pretraining, hallucination continues to accumulate during subsequent training stages, particularly through misaligned uncertainty generation and limitations in task-specific fine-tuning.

1) Inadequate Handling of Uncertainty and Output Overconfidence: A common source of hallucination arises when there is a deeper misalignment between the model's internal uncertainty and its tendency to produce overconfident outputs despite lacking sufficient information to provide reliable answers [114].

The root cause of such misalignment can be attributed to the limitations in LLMs' ability to handle uncertainty. Most LLMs are not properly trained to acknowledge knowledge boundaries or express indetermination, as Tupsakhare et al. claim [115]. This deficiency manifests in the following two key aspects:

Inability to Reject. Rather than declining to answer ambiguous or unanswerable questions, models frequently make unreliable guesses, generating plausible but unsupported completions [116]. This creates a systematic bias toward overproduction—a tendency to fill informational gaps with fluent speculation, even when abstaining would better serve factual accuracy. Encouraging models to explicitly say "I don't know" [117], as seen in OpenAI's recent efforts on ChatGPT 5 development [118], or tuning for higher abstention rates [119] has been shown to mitigate hallucinations.

Consistency over Accuracy. Models are also trained to prioritize fluency and coherence during generation, even when doing so at the expense of truthfulness [67], [71]. This fluency bias is a product of the reward models embedded in the training objective [120], where coherent outputs are positively reinforced regardless of factual accuracy. Over time, this creates weight configurations that strongly associate fluent generation with positive outcomes, even when fluency comes at the expense of accuracy. This mechanism underlies a typical phenomenon: hallucination snowballing, once an early invented fact is emitted, subsequent tokens tend to maintain narrative self-consistency, even when doing so compounds earlier errors [67].

The culmination of these failures is the generation of overconfident output [16]. Rather than indicating when it lacks sufficient information to provide a reliable answer, the model often generates outputs even when the probability of the predicted token is low.

2) Fine-tuning Alignment: Fine-tuning alignment is often applied to reduce hallucinations and tailor LLMs to specific downstream tasks [121]. However, this stage can introduce new sources of hallucination, particularly under domain shift, where the input distribution diverges from the pretraining regime [68].

One key factor is exposure bias, which arises from the discrepancy between training with teacher-forced inputs and inference generation [122]. This mismatch leads to incremental distortion in test-time outputs, increasing the risk of hallucination [68].

As models are exposed to facts not grounded in their pre-training distribution, they become more prone to generating unsupported assertions. Gekhman et al. [123] clarify that LLMs actually struggle to learn new knowledge during the fine-tuning stage, and any unknown knowledge they are exposed to will instead become a trigger of hallucination, appearing as a form of training overfitting to novel data.

Additionally, learning new knowledge during fine-tuning can degrade or overwrite previously acquired information [124]. Fine-tuning may induce catastrophic forgetting [125], where alignment objectives suppress the capabilities gained during pre-training. The quality and coverage of alignment data also introduce additional complications [126]. New information that contradicts pre-training data can exacerbate knowledge conflicts, thereby creating hallucinations in domains where the model previously performed well.

D. Inference-induced Mechanisms

Even with flawless pretraining, careful training alignment, and high-quality data, hallucinations can still emerge during inference and deployment. These inference-stage hallucinations can be broadly categorized into two types: failures in how content is internally represented and encoded, and failures in how that content is decoded and expressed during generation.

1) Content Encoding: Rather than achieving a truly logical and semantic understanding of inputs, LLMs rely on shallow linguistic and object co-occurrence statistics learned during the training process [101], [110].

Vulnerability to Prompt Variation. However, real-world downstream tasks often desire model behaviors that diverge from training objectives, creating model encoding dissonance [112]. This statistical approach to language processing then leads to failure in handling unexpected inputs and nuanced user needs, for example, the requirement of generating counterfactual reasoning [83] and handling chit-chat dialogue [127].

Fragility of Contextual Comprehension. Furthermore, recent studies show that even semantically equivalent paraphrases can elicit divergent factual outputs [81], highlighting the fragility of LLMs' contextual comprehension.

- **Deficits in contextual awareness** also contribute to this failure mode. LLMs often struggle with maintaining consistent attention over long contexts, leading to contextual misinterpretation [115]. Response strategies such as context-aware decoding [128] has been proposed to address these weaknesses and successfully demonstrated empirical reductions in hallucination rates.
- **Context length** introduces further risk and is challenging to manage. While models with small context windows are limited in reasoning capability [129], excessively long contexts can induce semantic phase transitions, cause semantic drift from the intended meaning, and increase hallucination risk [130].

2) Decoding Traps: The decoding process of how LLMs select the next token plays a crucial role in hallucination emergence.

Likelihood Trap in Deterministic Decoding. Traditionally, decoding strategies such as greedy decoding and beam search [131] aim to maximize likelihood by selecting the most probable tokens at each step. While these strategies yield fluent outputs, as Zhang et al. [132] note, they sometimes fall into the so-called likelihood trap, where high-probability sequences often fail to align with human quality judgments, despite surface-level fluency. This results in outputs that diverge from human-like language patterns [133] and generate confident but unfounded assertions [66], [66].

Sampling Randomness in Stochastic Decoding. In response to the likelihood trap, most modern LLMs adopt stochastic sampling methods such as top-k or top-p sampling [134] to inject diversity into the output, rather than always selecting the highest-scoring token [71]. However, this flexibility comes with its own trade-off: increased generation uncertainty. Sampling randomness introduces volatility in the next token selection of the output sequence, resulting in outputs with high uncertainty and increased hallucination rates [16], [72].

E. Mechanistic Interactions & Cascade Effects

The mechanisms described above do not operate in isolation but interact in complex ways that produce emergent hallucination behaviors. Understanding these interactions is crucial for developing comprehensive mitigation strategies that address the full spectrum of hallucination phenomena.

Data artifacts interact synergistically with training procedures to amplify hallucination tendencies. Misinformation in pretraining data becomes more problematic when reinforcement learning rewards confident responses, creating a multiplication effect where false information is not only encoded but actively promoted. Similarly, knowledge gaps interact with reasoning limitations to produce elaborate confabulations as models attempt to derive missing information through flawed inference.

The temporal dynamics of these interactions add another layer of complexity. Initial hallucinations can trigger cascading failures as autoregressive generation builds upon erroneous foundations. A single ambiguous prompt interpretation can lead to an entire conversation predicated on false assumptions, with each exchange deepening the departure from reality while maintaining surface coherence.

Reinforcement across mechanisms creates particularly robust hallucination patterns. When architectural limitations align with training biases and are triggered by specific prompt patterns, the resulting hallucinations can be highly persistent and difficult to detect. These "resonant" hallucinations represent stable attractors in the model's generation space that resist simple mitigation strategies.

F. Implications for Detection & Mitigation

The mechanistic perspective of hallucination provides not only explanatory insight but also practical implications for designing detection and mitigation strategies. By linking internal causes with external manifestations, it becomes possible to develop interventions that are targeted, mechanism-aware, and stage-specific [75], [76], [135].

One avenue is *mechanism-aware detection*. Different hallucination mechanisms leave distinctive statistical and semantic signatures, such as uncertainty patterns or contradiction structures. Exploiting these cues allows for the development of specialized detectors that can identify hallucinations at their root causes rather than relying only on surface-level anomalies [46], [76].

Mitigation strategies can also be guided by the stage at which hallucinations arise. At the data level, improving corpus curation, balancing term frequency distributions, and enhancing diversity (e.g., through negation or counterfactual examples) can reduce vulnerability to misinformation and coverage gaps [71], [90]. During training, refinements such as reward model redesign and multi-reference fine-tuning may help alleviate sycophancy and overconfidence induced by SFT and RLHF [74], [123]. At inference, dynamic methods such as temperature adjustment, retrieval augmentation, or abstention calibration can reduce error cascades and improve robustness [55], [136].

A further implication lies in *layer-wise and architectural interventions*. Studies show that different hallucination mechanisms correlate with activations at characteristic model depths: early layers encode knowledge-level biases, middle layers are tied to reasoning errors, and final layers affect output formatting and calibration [75]. This stratification opens opportunities for layer-specific interventions that target hallucinations at their source.

Finally, this perspective underscores that complete elimination of hallucination is neither feasible nor necessarily desirable. The same mechanisms that enable creativity and flexible reasoning also contribute to hallucination. Future work should therefore focus on controllability: enabling models to adjust hallucination tolerance depending on application context, while providing reliable uncertainty estimates and knowledge boundary indicators [16], [24].

In sum, understanding hallucination as the outcome of interacting mechanisms provides a roadmap for principled mitigation. Rather than deploying broad-spectrum fixes, effective strategies will likely require precise, multi-level interventions that address data, training, and inference jointly.

IV. Detection Methods for Hallucination

Building on the preceding analysis of mechanisms and lifecycle vulnerabilities, it becomes evident that hallucinations in large language and vision-language models are not incidental anomalies but systemic phenomena that pervade data, pretraining, training, and inference. Therefore, reliable detection is a necessary prerequisite for effective mitigation. Without it, downstream interventions risk either undercorrecting, leaving residual errors unaddressed, or overcorrecting, thereby suppressing the generative flexibility essential for creativity and reasoning. Thus, detection serves as the diagnostic backbone of the mitigation pipeline, transforming mechanistic insights into actionable signals. Effective detection consequently requires both high precision and high recall, that sensitive enough to capture subtle errors, yet robust enough to avoid excessive false alarms that would unnecessarily constrain generative capacity.

Viewed through the lens of our three-dimensional taxonomy, hallucination detection constitutes a diverse but coherent landscape. Along the **mechanism dimension**, approaches range from fact verification and knowledge-grounded reasoning to consistency analysis, uncertainty estimation, causal attention probing, dual or contrastive model comparison, and feedback-driven evaluation. Along the **phase dimension**, detection may occur at multiple checkpoints: **front-end abstention detection** proactively filters high-risk or unanswerable queries before generation, while **back-end hallucination detection** evaluates completed outputs to identify and correct factual inconsistencies, with dynamic in-generation checks offering intermediate safeguards. Finally, the **integrative dimension** highlights opportunities for hybridization, where heterogeneous signals—uncertainty, factual grounding, or inter-model contrast—can be orchestrated into iterative or cross-phase workflows. In this sense, detection is not an isolated safeguard but a structurally embedded diagnostic layer that enables effective, phase-aware, and integrative mitigation.

A. Mechanism-Based Detection

The first dimension of our taxonomy focuses on the underlying *mechanisms* through which hallucinations are detected. This perspective addresses the question of how diagnostic signals are extracted, ranging from explicit fact and knowledge verification to internal consistency checks, uncertainty estimation, causal probing of attention mechanisms, dual or contrastive model comparisons, and feedback-driven evaluation. Each mechanism embodies distinct assumptions about what constitutes reliable evidence of hallucination—whether grounded in external factual references, internal probability distributions, inter-model divergences, or evaluative feedback. While methodologically diverse, these approaches collectively establish the foundations of hallucination detection by defining the signals on which subsequent phase- and integration-based strategies can operate. As illustrated in Fig. 3, mechanism-based detection can be systematically organized into seven major categories—fact-based, knowledge-based, consistency-based, dual model, uncertainty-based, causal attention-based, and feedback-based—each of which further decomposes into representative sub-methods that highlight their unique diagnostic principles.

1) Fact-based Detection: Fact-based detection constitutes the most straightforward approach to identifying hallucinations, focusing on whether generated outputs can be validated against explicitly verifiable facts. These methods operate by extracting factual units—such as entities, relations, or claims—from model text and systematically testing their correctness against reference evidence or entailment frameworks. By targeting explicit factual validity, fact-based detection provides fine-grained and interpretable signals of inconsistency, making it particularly suitable for tasks where correctness must be demonstrable, such as summarization, question answering, or knowledge-intensive generation. Fact-based methods target isolated factual claims, whereas knowledge-based methods provide systematic reasoning across structured or external repositories.

a) Named Entity Recognition (NER) and Entity Relationship: approaches detect hallucinations in LLMs by extracting named entities and analyzing their semantic or factual relationships to uncover inconsistencies or fabricated content [137]. Here, we adopt the term Entity Relations (ER) to emphasize modeling of links between entities, which is more consistent with literature on knowledge representation and information extraction. NER identifies key entities in generated text, while ER modeling evaluates whether their relationships align with

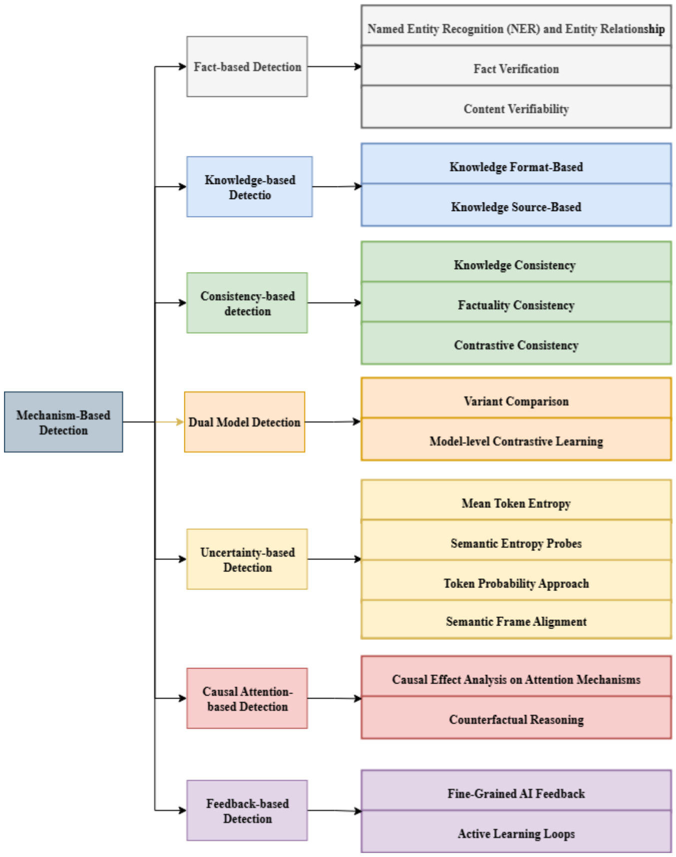

Fig. 3. Detection Methods for Hallucination

plausible world knowledge. For example, outputs such as "Einstein discovered penicillin" are flagged because the NER step extracts "Einstein" and "penicillin," and ER analysis recognizes that the asserted relation contradicts established facts. Hallucinations are thus flagged based on relational discrepancies, enabling targeted identification of problematic spans. Once detected, these errors can be mitigated through prompt engineering, where the LLM itself is leveraged to refine and correct outputs for improved factual reliability [137].

Recent work extends this paradigm by integrating NER with complementary techniques to capture more nuanced hallucination patterns. Wang et al. [138] propose a decision-tree framework that combines NER, Natural Language Inference (NLI), and Span-Based Detection (SBD), where NLI evaluates whether claims entail or contradict retrieved evidence, and errors are refined through a rewriting mechanism balancing precision, latency, and computational cost. Voznyuk et al. [139] further introduce AdvaCheck, which couples NER with Retrieval-Augmented Generation (RAG) and LLM-based verification: entities are extracted, corroborating evidence is retrieved, and an LLM verifier assesses factual consistency. This pipeline detects not only gross factual hallucinations but also fine-grained errors such as name misspellings or title distortions, which are challenging even for human annotators.

b) Fact Verification: methods detect hallucinations by extracting explicit factual units from generated text and systematically testing them against reference evidence or entailment frameworks. Unlike surface-level plausibility checks, these approaches aim to identify precise factual claims and determine whether they are supported, contradicted, or unverifiable. *FACTOID (FACtual enTAILment fOr hallucInation Detection)* exemplifies this paradigm by extending textual entailment to hallucination detection through the notion of *Factual Entailment (FE)*, not only classifying claims but also highlighting the specific spans responsible for violations, thereby improving interpretability [140].

Building on this foundation, subsequent work adapts fact verification into retrieval-augmented pipelines. LettuceDetect [141] employs a lightweight verifier to compare outputs against retrieved passages, reducing computational overhead in large-scale applications. FRANQ [142] further distinguishes *faithfulness* (alignment with the given context) from *factuality* (consistency with external truth), introducing an uncertainty-aware framework for finer discrimination. Factored Verification achieves state-of-the-art hallucination detection in summarization by first decomposing summaries into claims and then verifying each claim with an LLM-as-verifier, assigning correctness probabilities to aggregate factual reliability [143]. Collectively, these approaches demonstrate that fact verification can function both as an independent mechanism and as an integral component of retrieval-augmented architectures.

c) Content Verifiability: assesses whether model-generated content can be explicitly traced back to supporting evidence in the source context or external references [25], [144]. By enforcing demonstrable grounding, it flags unsupported or fabricated claims as hallucinations. This property makes verifiability particularly valuable in domains where correctness must be auditable, such as summarization, question answering, generative search, and long-form generation. However, these approaches face inherent challenges: they struggle with subjective or creative outputs lacking referenceable evidence, novel insights for which no prior sources exist, and cases where the underlying references themselves may be incomplete or erroneous.

In practice, verifiability has been operationalized through automatic scoring and reference-checking systems. In summarization, the HADAS framework introduces a verifiability score that measures whether each unit of summarized content can be directly located in the input text [25]. In retrieval-augmented systems, it also underpins the detection of reference hallucinations, where models fabricate nonexistent books, articles, or authors. Agrawal et al. [144] demonstrate that genuine references yield stable responses across repeated queries, while fabricated ones produce inconsistent or contradictory answers. Such *consistency patterns* serve as a lightweight detection signal, illustrating how verifiability can move beyond direct evidence matching to exploit behavioral regularities. Overall, content verifiability functions primarily as a *diagnostic detection signal*—highlighting unsupported information—while its corrective role typically depends on integration with downstream abstention, rewriting, or feedback mechanisms.

2) Knowledge-based Detection: Knowledge-based detection methods identify hallucinations by anchoring model outputs to explicit forms or sources of knowledge. Unlike fact-based approaches that directly verify individual claims, knowledge-based methods leverage structured representations and knowledge repositories to provide systematic and often more scalable validation. This category encompasses two complementary directions. The first is **knowledge format-based detection**, which exploits the representational structure of knowledge—such as in-context exemplars, dependency parses, or knowledge graphs—to expose unsupported or inconsistent outputs. The second is **knowledge source-based detection**, which grounds model responses in stable repositories of information, ranging from the parametric knowledge encoded in model weights to external resources such as curated databases or encyclopedias. Together, these approaches extend hallucination detection from isolated fact-checking to knowledge-grounded reasoning, thereby enhancing robustness and interpretability across diverse application domains.

a) Knowledge Format-Based: builds on the structural properties of knowledge representations rather than specific content repositories. By leveraging formats such as in-context exemplars, syntactic dependency structures, and symbolic graphs, these methods expose unsupported or inconsistent outputs through representational alignment. Unlike fact-based detection, which verifies isolated claims against reference evidence, format-based approaches emphasize the organization and relational structure of information, enabling scalable and interpretable detection across diverse tasks.

- **In-Context Learning (Context Aligner with In-Prompt Instructions)** [145] enables large language models to perform hallucination detection by conditioning on carefully designed prompts that illustrate factual versus non-factual responses. Rather than retraining the model, ICL leverages pretrained knowledge and adapts dynamically by recognizing input–output relationships provided in the prompt [146]. By comparing current model outputs with in-context exemplars, inconsistencies or unsupported claims can be flagged as potential hallucinations. This makes ICL a flexible and reference-free approach for scalable factuality checking. Ji et al. [147] apply in-context instruction learning to design a customized, reference-free factuality scorer. The model evaluates its own generated knowledge, assigns a factuality score, and self-reflects by refining outputs when the score falls below a threshold.

  Few-shot and *Zero-shot* learning are widely used ICL strategies for hallucination detection. By framing detection as a prompt-based classification task [148], the model can label outputs as "Hallucination" or "Not Hallucination" without requiring additional fine-tuning. In the zero-shot setting, natural language instructions define the hallucination concept directly. These predictions are then leveraged to construct a small but diverse set of labeled examples for few-shot prompting, where the model is re-prompted with demonstrations derived from the zero-shot stage [148]. To further account for model uncertainty, outputs are sampled multiple times under varying decoding temperatures, and majority voting is applied to estimate hallucination likelihood [148].

  While ICL offers a lightweight and flexible mechanism for hallucination detection, its effectiveness is highly sensitive to the design of contextual demonstrations and prompt structures [146], [149]. Minor changes in wording, ordering, or domain coverage can significantly impact performance. Consequently, auxiliary strategies such as pruning irrelevant or noisy segments of the input context have been proposed to reduce distraction and exposure bias, thereby improving factual alignment [150]. These findings emphasize that ICL should not be treated as a static prompting technique but as a context-sensitive process in which careful input curation plays a central role.

  Although sometimes discussed alongside ICL, *Sentence Trimming* is more accurately categorized as a structural detection method. Shi et al. [150] propose a dependency-parse-based technique that compares an input graph with the dependency parse tree (DPT) of the draft response. The process involves three steps: first, constructing the DPT of the generated sentence; second, decomposing the input graph into entities and locating their corresponding positions within the DPT; and third, identifying entities that fall outside these aligned positions. Entities on the left of the leftmost cited node or on the right of the rightmost cited node in the DPT lack grounding in the input graph, and are therefore flagged as potential hallucinations [150]. This method highlights the value of syntactic and graph-based constraints for detecting informational inconsistencies, complementing but differing from in-context learning approaches.
- **Knowledge Graph Reasoning** provides a structured pathway for hallucination detection by grounding model outputs against curated, symbolic knowledge representations. Unlike surface-level text comparison, KG reasoning explicitly links generated entities and relations to canonical entries in a graph, enabling systematic validation of factual claims and fine-grained identification of inconsistencies. This structured approach directly addresses a core challenge of hallucination detection: distinguishing between plausible but unsupported statements and those that can be validated against established knowledge [151]. Recent methods have leveraged this principle to develop explainable and robust detection frameworks. GraphEval [152] represents model outputs as knowledge graph structures and checks the consistency of triples, thereby providing interpretable signals of where hallucinations occur. Similarly, CoKGLM [153] integrates textual and structural information, enhancing robustness and transparency by cross-examining LLM outputs with KG-based reasoning.

  The KnowHalu framework [154] exemplifies the integration of structured and unstructured sources. It first decomposes model outputs into individual factual claims, then aligns the entities and relations within those claims to entries in external knowledge graphs such as Wikidata [154]. To extend coverage, it also retrieves supporting or refuting evidence from unstructured corpora such as Wikipedia [154]. Each claim is scored for factual consistency, and hallucinations are flagged when the aggregated evidence indicates contradiction, lack of support, or unverifiability [154]. By combining the symbolic precision of KGs with the adaptability of text-based retrieval, KnowHalu demonstrates how KG reasoning can form a hybrid evidence-checking pipeline for effective hallucination detection.

Beyond KG-based detection frameworks, in safety-critical healthcare, Gilbert et al. (2024) extend KG-grounded reasoning to a full curation pipeline: pairing LLMs with KG-based RAG as a verifiable "model of truth", adding human-in-the-loop quality control and reliability labeling to temper hallucination and non-determinism [155]. They further envision patient-level digital twins—persistent KGs enabling low-cost, auditable verification over time—and argue for hybrid, partially automated KG construction, contrasting KG-grounded and embedding-only retrieval [155]

b) Knowledge Source-Based: hallucination detection methods operate by grounding model outputs against established repositories of knowledge. The central principle is that hallucinations become apparent when generated claims contradict stable, verifiable knowledge representations. In this context, internal knowledge refers to the parametric knowledge encoded within the pretrained weights of an LLM [156], while external knowledge refers to structured or unstructured sources that are explicitly retrieved at inference time, such as knowledge graphs, databases, or encyclopedias [157], [158]. This distinction is important because internal knowledge can provide rapid, low-cost validation signals but is limited by coverage and potential training biases, whereas external knowledge offers authoritative grounding but introduces retrieval complexity and potential noise [1].

A representative approach is knowledge graph-based retrofitting, which proceeds through a structured pipeline [159]. First, a secondary LLM decomposes the generated response into atomic factual claims, ensuring that complex sentences are reduced into verifiable units. Next, entities mentioned in these claims are recognized, and a relevant subgraph is retrieved from the underlying knowledge graph to capture contextual relations. From this subgraph, salient triples are selected to represent the most pertinent factual information, while extraneous or weakly connected knowledge is filtered out. Finally, the extracted claims are systematically compared against the selected triples, and any misalignment, contradiction, or lack of support is flagged as a hallucination [159]. This pipeline illustrates how grounding against external knowledge sources can provide fine-grained, interpretable detection of hallucinations, complementing the implicit signals derived from a model's internal knowledge.

3) Consistency-based detection: rests on the idea that hallucinations often manifest as inconsistencies—either with external knowledge or with the model's own internal reasoning. These approaches evaluate whether generated outputs remain coherent across knowledge sources, inference steps, and perturbations. For example, a model may produce different answers to semantically equivalent questions, or provide reasoning chains that contradict its final conclusion, both of which serve as indicators of hallucination. While this paradigm offers a lightweight and reference-agnostic means of identifying errors, it also entails trade-offs: models can still generate hallucinations that are internally consistent yet factually incorrect, thereby escaping detection. This limitation highlights the need to combine consistency checks with complementary strategies that directly verify factual grounding.

a) Knowledge Consistency: is one of the important principles for hallucination detection, ensuring the consistency of knowledge, which evaluates whether generated content aligns with established background knowledge. Unlike self-evaluation approaches that rely solely on internal introspection, knowledge-consistency methods explicitly identify potentially unreliable spans and validate them against external sources. This external grounding enables more systematic detection of factual errors while preserving interpretability.

KnowHalu extends this paradigm by leveraging multi-form knowledge and introducing a multi-step factual validation pipeline [154]. The method decomposes the original query into step-wise reasoning and sub-queries, retrieves relevant evidence from both unstructured text and structured triples, and then aggregates the results across sources to produce a consolidated verdict. By combining structured and unstructured evidence, KnowHalu provides a more robust framework for identifying knowledge inconsistencies.

Varshney et al. [160] propose another approach that focuses on low-confidence regions of generated text, under the assumption that hallucinations are more likely to arise in uncertain spans. Their method first pinpoints tokens with low probability, then requires the model to verify or justify these spans by retrieving external knowledge [160]. If the retrieved evidence fails to substantiate the generated claim, the content is flagged as hallucinated [160]. This strategy highlights how uncertainty signals can be integrated with knowledge consistency checks to achieve fine-grained hallucination detection.

b) Factuality Consistency: in hallucination detection refers to the principle that models should actively assess whether their generated outputs remain logically coherent and aligned with their internal knowledge. Unlike knowledge-consistency methods that validate outputs against external sources, factuality-consistency approaches rely on the model's introspective ability to scrutinize and refine its own responses.

Two representative paradigms illustrate this principle: **Interactive Self-Reflection** and **Self-Evaluation for Factuality Alignment**. Both exploit internal self-checking mechanisms to enhance factual accuracy, but they differ in their strategies—one emphasizes dynamic iterative refinement at inference time [147], while the other integrates systematic self-evaluation into training to improve long-term alignment [161].

- **Interactive Self-Reflection Method** employs a feedback loop in which the model generates an initial response and then iteratively questions its own factual accuracy through self-generated prompts [147]. These prompts are designed to surface contradictions or unsupported claims by probing consistency against the model's internal knowledge and contextual cues. The approach follows a generate–score–refine cycle, supported by CTRLEval, an unsupervised and task-agnostic evaluation metric that computes consistency scores for answers. When scores fall below a threshold, the model introspects, self-corrects, and revises its outputs. Through this iterative refinement, the method effectively enhances factual reliability without relying on external resources [147].
- **Self-Evaluation for Factuality Alignment** detects hallucinations by prompting the model to validate the factual correctness of its own responses through structured self-assessment [161]. It introduces Self-Eval, where the model inspects its outputs for inaccuracies, supported by Self-Knowledge Tuning (SK-Tuning) to calibrate confidence and strengthen internal consistency. Discrepancies or unsupported claims identified during this process are used as signals for Direct Preference Optimization, enabling fine-tuning that systematically suppresses hallucinations and improves factual precision. By relying exclusively on internal resources rather than external annotations, this method provides a scalable pathway to align models toward truthfulness [161].

c) Contrastive Consistency: detection leverages the principle that hallucinations can be surfaced by contrasting model behaviors under perturbed conditions. The approach works by deliberately modifying the input or the decoding process and then observing discrepancies in predictions. The underlying intuition is that hallucination-prone outputs are more likely to fluctuate disproportionately under perturbations, whereas content that is well-grounded in knowledge should, in principle, exhibit greater stability [29], [162]. However, this assumption must be qualified: even legitimate and factual responses can display sensitivity to perturbations, particularly in cases of complex reasoning or ambiguous contexts [146], [149]. Thus, contrastive learning provides a useful but not exhaustive diagnostic signal.

In practice, contrastive detection methods differ in the type of contrastive signal they exploit—ranging from lexical and semantic perturbations of the input, to stochastic variations in decoding, to comparisons across multiple generations [29]. Despite their diversity, these methods share a common philosophy: they treat discrepancies as indicators of hallucination risk, often without requiring external knowledge or additional retraining. This makes contrastive learning a lightweight yet versatile strategy for hallucination detection, particularly when integrated with knowledge-grounded verification pipelines or consistency-based self-reflection mechanisms [29], [162].

- **Original vs. Disturbance instruction prompts.** The core idea of this strategy is that hallucinations often arise from spurious co-occurrence priors or shallow correlations, which can be stressed by introducing small disturbances into the instruction. By contrasting model predictions under the original versus a perturbed instruction, unstable tokens can be surfaced as hallucination-prone.
  - **Instruction Contrastive Decoding (ICD)** operationalizes this principle by injecting small disturbance role prefixes into the instruction to deliberately stress instruction-aware fusion [163]. At each decoding step, the model generates two token-level distributions—one conditioned on the original instruction and one on a perturbed version—and compares them [163]. Tokens whose probabilities are disproportionately boosted under perturbation are flagged as spurious signals indicative of hallucination [163]. To suppress these signals, ICD subtracts a scaled disturbed-logit vector from the base logits before sampling, creating an online identify-and-suppress detector during decoding [163]. This contrastive signal is particularly effective against co-occurrence-driven errors: disturbances amplify priors misaligned with evidence while leaving grounded tokens relatively stable [163]. As a result, ICD enables sentence-level, model-internal hallucination detection without requiring auxiliary models, additional training, or external knowledge [163].
- **Adding Gaussian Noise.** Adding controlled noise to model inputs is a long-standing strategy in machine learning for probing robustness and exposing hidden failure modes. In the context of hallucination detection, introducing Gaussian noise serves as a perturbation mechanism that selectively degrades genuine evidence while leaving spurious correlations more exposed. This allows contrastive frameworks to differentiate tokens grounded in reliable signals from those supported mainly by co-occurrence priors or language biases.
  - **Visual Contrastive Decoding (VCD)** addresses object hallucinations by contrasting a model's output distributions for the original image with those for a deliberately distorted version, where Gaussian noise is progressively injected through a forward-diffusion process [164]. The added noise elevates visual uncertainty and weakens genuine visual cues. Consequently, tokens that become more confident under the distorted view are flagged as hallucination-prone, since their support derives from co-occurrence priors or unimodal language bias rather than authentic visual evidence [164]. Concretely, VCD performs decoding under both conditions at each step and compares discrepancies in token logits and probabilities; disproportionate gains in the distorted case signal non-visual or spurious concepts [164]. This detection signal is entirely model-internal, requires

no additional training, and avoids reliance on auxiliary detectors or external knowledge [164].

- **Mix of True & False Inputs.** The central principle of mixed-input hallucination detection is that hallucinations can be revealed by explicitly contrasting model behavior on true versus false inputs. By introducing carefully constructed counterfactual or incorrect knowledge alongside verified ground truth [162], these methods probe the degree to which a model's outputs are grounded in accurate evidence versus spurious correlations.
Two representative methods illustrate this idea: **Mixed Contrastive Learning (MixCL)** and **Source-Contrastive Decoding**. Both exploit the contrast between true and false inputs, but they differ in scope and technical implementation. MixCL operates at the training stage, producing model-internal detection signals without requiring external evaluators [162], while Source-Contrastive Decoding provides a lightweight inference-time check of input-output alignment in generative tasks [165].
  - **Mixed Contrastive Learning** (MixCL) constructs hard negative knowledge for each dialogue context by combining two types of negatives: (i) retrieved negatives that are topically related but irrelevant or non-factual, and (ii) model-generated negatives obtained through bootstrapping hallucinated cases [162]. A natural language inference-based disjointness check ensures these negatives either contradict or fail to entail the positives, guaranteeing they represent spurious content [162]. To localize hallucinations, MixCL performs span-level extraction of entities and constituents from both positives and negatives, and then contrasts token-level preferences across these spans. Tokens whose support derives disproportionately from negative spans are flagged as hallucination-prone, providing a model-internal detection signal that supervises training without auxiliary evaluators [162].
  - **Source-Contrastive Decoding** detects hallucinations in machine translation by contrasting translation probabilities under correct source inputs versus artificially corrupted ones, such as randomly replaced or distorted segments [165]. If a translation remains probable under false inputs, it is likely hallucination-driven, reflecting reliance on language priors rather than accurate source grounding. By leveraging such counterfactual contrasts, Source-Contrastive Decoding exposes spurious dependencies that would otherwise remain undetected [165].
- **Cross Layer Contrast (Logic vectors from different layers).** The principle of cross-layer contrast is that lower and intermediate layers of LLMs tend to encode shallow syntactic or correlation-based patterns, while upper layers increasingly capture semantic and factual representations. By comparing discrepancies across these layers, models can expose tokens whose support relies disproportionately on non-factual patterns, thereby signaling hallucination-prone content.
Two representative methods are **Lower Layers Matter (LOL)** and **Inter-Layer Contrastive Decoding (DiLa)**. Both exploit inter-layer disagreement as an internal hallucination signal, but they differ in scope: LOL incorporates an external "amateur" model and emphasizes multi-layer fusion, while DiLa remains self-contained and adaptively selects the most divergent hidden layer at each decoding step for fine-grained detection.
  - **Lower Layers Matter** (LOL) contrasts the base LLM with a deliberately trained "amateur" variant and computes contrast signals not only at the final layer but also at early-exit layers [166]. By fusing evidence across layers, LOL highlights divergences amplified in lower layers, localizing tokens whose support comes from shallow, non-factual priors [166]. The amateur model is fine-tuned on non-factual data to induce hallucination tendencies, so the base–amateur gap becomes a proxy for spurious content. During decoding, the method monitors both final- and lower-layer contrasts, flagging tokens disproportionately supported by the amateur side as hallucination-prone. To further sharpen this signal, LOL integrates a truthfulness-refocused module that injects instruction guidance, emphasizing factual keywords and generating auxiliary contrast cues. Discrepancies between these cues and model predictions indicate non-truthful continuations [166]. Together, the fused multi-layer contrasts and instruction-based evidence act as a model-internal detector for hallucination-prone tokens.
  - **Inter-Layer Contrastive Decoding** (DiLa) detects hallucination-prone tokens by contrasting the next-token distributions of the final layer with those from an internal hidden layer dynamically chosen for maximal divergence at each step [167]. Specifically, it computes a log-probability ratio between final- and hidden-layer distributions, leveraging the notion that higher layers encode factual semantics while earlier layers are biased toward syntactic regularities. When hidden-layer confidence outweighs final-layer support, the token is flagged as non-grounded [167]. To mitigate false positives, DiLa applies an adaptive plausibility constraint that restricts contrastive checks to tokens with sufficient support under the final-layer distribution. At each step, the hidden layer is re-selected using Jensen Shannon divergence to ensure the most informative perspective is chosen, enabling fine-grained, step-wise localization of unstable tokens [167]. In essence, inter-layer disagreement serves as a reliable internal signal: when confidence inflates in lower or intermediate layers relative to the final layer, the token is marked as relying on shallow

priors rather than factual knowledge.

- **Input Removal & Corruption (Video/Text manipulations).** Input perturbation-based detection operates by systematically removing or corrupting parts of the input to test whether token predictions remain stable and grounded. If predictions shift disproportionately under such manipulations, the affected tokens are flagged as hallucination-prone.
  Two representative methods, **Language-Contrastive Decoding (LCD)** and **HALC**, share this philosophy of deliberately weakening or distorting one modality to test grounding robustness. Both are lightweight, training-free, and operate online, but they differ in their focus: LCD suppresses the visual modality to reveal language-driven errors [168], whereas HALC perturbs the visual input itself to assess the stability of multimodal grounding [169].
  - **Language-Contrastive Decoding** (LCD) detects hallucination risk by contrasting an LVLM's next-token probabilities with those of its underlying LLM when run on the same text prefix without the image [168]. If the language-only model assigns significantly higher confidence to a candidate token than the image-conditioned LVLM, this discrepancy is treated as evidence that the prediction is driven by linguistic priors rather than visual evidence. To stabilize the signal, LCD first applies adaptive plausibility, restricting attention to tokens already deemed plausible under the LVLM distribution. The contrast's weight is further modulated by the LLM's conditional entropy: when the language model is confident, its signal dominates; when uncertain, its influence is reduced [168]. This design yields a dynamic, model-internal hallucination indicator without requiring auxiliary models or retraining.
  - **HALC** probes hallucination risk by testing the stability of token predictions across perturbed visual contexts during generation [169]. It first localizes token-specific regions using a zero-shot object detector (e.g., Grounding DINO or OWLv2), then samples multiple fields-of-view (FOVs) around these regions to collect diverse visual evidence. At each step, HALC selects pairs of FOVs that maximally disagree in their effect on token probabilities, using this contrast to expose unstable, non-grounded support. Hallucination-prone tokens typically exhibit noisy or fluctuating likelihoods across different crops—for instance, spurious objects may peak under mismatched views—whereas truly grounded tokens remain stable [169]. By treating these cross-context discrepancies as adaptive grounding signals, HALC flags tokens lacking robust visual support, providing an internal, online detection cue without auxiliary supervision or training.

4) Dual Model Detection: The dual model approach detects hallucinations by maintaining two model variants that differ in their susceptibility to hallucination and then contrasting their outputs or internal representations. The central intuition is that fabricated or unsupported content manifests as systematic discrepancies between a truth-preserving model and a hallucination-prone counterpart. By comparing token distributions or representation-level signals across these models, hallucination-prone spans can be identified and filtered. Unlike knowledge-grounded methods, dual model detection relies on the relative divergence between model variants rather than external fact-checking.

a) Variant Comparison: constructs multiple fine-tuned variants of the same base model—typically an "amateur" model deliberately trained on hallucination-inducing data and a more stable variant that preserves linguistic fidelity. By comparing logits or decoding behaviors across these special copies, hallucination-prone tokens can be revealed as divergences between stable and unstable variants [170].

- **Freeze-CD** applies a contrastive decoding framework with local freezing training to expose hallucinations in LLMs [170]. It constructs two amateur variants fine-tuned on hallucination-inducing datasets: (i) a fully fine-tuned model that amplifies hallucinations and (ii) a locally frozen model that preserves lower-layer semantic and grammatical features while still allowing partial hallucination. During inference, hallucination detection is achieved through logit subtraction: discrepancies between the original model and the amateur models highlight potentially fabricated information. The merged output can then suppress inconsistent or non-factual elements, improving truthfulness without requiring external resources [170].

b) Model-level Contrastive Learning: creates a pair of LLMs explicitly optimized in opposite directions: one fine-tuned on factual datasets to reinforce truthfulness, and another fine-tuned on hallucination-prone data to exaggerate erroneous patterns. By iteratively contrasting their representation layers, hallucinations can be detected as signals that align with the "bad" model but diverge from the "good" one [124].

- **Iter-AHMCL** introduces an iterative model-level contrastive learning framework, contrasting representations from a "positive" model (trained on factual, hallucination-free datasets) against a "negative" model (trained on hallucination-inducing datasets) [124]. By fine-tuning the same base LLM along these opposing directions, the method constructs a contrastive signal at the representation layer: discrepancies highlight inconsistent or erroneous information associated with hallucinations. Iterative optimization refines this separation, allowing hallucinated spans to be suppressed and factual accuracy enhanced during inference—again without requiring external knowledge or additional retrieval [124].

5) Uncertainty-based Detection: Uncertainty-based hallucination detection methods operate on the principle that hallucinations are more likely to arise when the model is uncertain about its predictions. Uncertainty is typically reflected in the probability distribution over tokens or in the semantic space of generated outputs. By quantifying this uncertainty through entropy, probability features, or semantic alignment, such methods aim to identify responses that are less reliable [171]–[173]. However, this principle has inherent limitations: models can still produce hallucinations with high confidence, while being uncertain about correct information. This mismatch suggests that uncertainty alone is an imperfect proxy and must often be combined with complementary signals for robust detection.

Despite this caveat, existing methods share the goal of leveraging uncertainty but differ in the source of their signals and the granularity with which they operate. Token-level metrics capture local uncertainty by analyzing the sharpness or flatness of token probability distributions [171], [172]. Semantic entropy approaches approximate the uncertainty of global meaning by comparing multiple generations sampled in the semantic embedding space [173]. Frame alignment methods go further by assessing whether the generated outputs preserve the intended semantic structure of the input, thereby linking uncertainty to the preservation of meaning [25], [174]. Together, these techniques provide complementary perspectives, offering both fine-grained and holistic views of uncertainty-based hallucination detection.

a) Mean Token Entropy: is a fundamental uncertainty metric that quantifies the average entropy across generated tokens. With the entropy H of a single token X defined as $H(X) = -\sum_{x \in V} p(x) \log p(x)$, where V denotes the model's vocabulary, entropy reaches its maximum when $p(x)$ is uniform over all tokens and drops to its minimum when the distribution is fully concentrated on a single token (i.e., one token has probability 1 and all others 0) [171]. A higher mean token entropy reflects greater uncertainty in the model's predictions, making it a useful signal for identifying potential hallucinations in generated text [171].

b) Semantic Entropy Probes: (SEP) is a lightweight and effective method for hallucination detection in LLMs based on the concept of semantic entropy, which measures model uncertainty over the meaning of generated outputs [173]. Instead of clustering multiple responses for the same query by semantic equivalence, SEP trains a linear probe on hidden states from a single model generation to predict semantic entropy scores, thereby reducing computational cost while maintaining strong detection performance [173].

c) Token Probability Approach: have long been employed to signal uncertainty in sequential generation tasks [172]. A token probability approach for hallucination detection extracts four key features from a generator LLM_G and an evaluator LLM_E: (i) Minimum Token Probability, (ii) Average Token Probability, (iii) Maximum LLM_E Probability Deviation, and (iv) Minimum LLM_E Probability Spread [172]. These numerical features, closely related to average and maximum entropy, serve as indicators of model uncertainty and can be leveraged to flag hallucination-prone outputs [172].

d) Semantic Frame Alignment: evaluates whether generated content preserves the factual and semantic structure of the source input [174]. Hallucinations frequently manifest as confabulations, where outputs deviate semantically from the input despite maintaining surface-level plausibility [174]. Unlike Semantic Entropy, which captures response diversity but may overlook deeper inconsistencies, frame alignment explicitly tests for factual and semantic consistency between source and output.

Entailment-based models such as FactKB [25] operationalize this principle by computing a semantic frame score for document–summary pairs, estimating the probability of factual alignment with the source document. This enables highlighting of unsupported or fabricated claims without requiring extensive external annotation [25].

Further, Semantic Divergence Metrics (SDM) [174] extend the approach by measuring semantic misalignment not only across multiple responses but also across paraphrases of the same prompt. By clustering sentence embeddings into a shared topic space and computing information-theoretic divergence, SDM provides a fine-grained quantification of confabulation and semantic drift [174].

6) Causal Attention-based Detection: Causal attention-based detection methods aim to move beyond purely correlational signals by estimating the causal influence of inputs and attention components on generated outputs. The central intuition is that if an output is primarily driven by irrelevant or insufficient input evidence, it can be flagged as a hallucination. This causal perspective is particularly relevant for multimodal models, where spurious cross-modal correlations often induce hallucinations.

Two main approaches embody this principle. First, causal effect analysis on attention mechanisms explicitly models the causal role of attention layers using Structural Causal Models (SCMs) and performs interventions on attention distributions [175]. Second, counterfactual reasoning manipulates the input directly, removing or altering key components, to observe how outputs change under alternative scenarios [176]. Together, these approaches offer complementary perspectives on grounding and attribution.

a) Causal Effect Analysis on Attention Mechanisms: seeks to quantify how variations in attention distributions causally influence model outputs [175]. This involves constructing an SCM to capture how attention mediates the influence of inputs on outputs. Interventions are then applied to attention layers (e.g., perturbing or reweighting attention distributions) to simulate alternative causal scenarios and compare the resulting outputs. If model predictions prove highly sensitive to such interventions, this indicates that responses may be driven by misaligned or spurious attention patterns, thus signaling potential hallucination [175].

b) Counterfactual Reasoning: measures the causal contribution of input components to outputs [176]. By generating counterfactual scenarios where key elements of the input are removed or altered, and comparing the corresponding outputs with the factual baseline, the method estimates the degree of grounding. Outputs that remain largely unchanged despite substantial input removal are considered insufficiently grounded and hence hallucination-prone. In this way, counterfactual reasoning favors outputs that are causally dependent on the correct input information [176].

7) Feedback-based Detection: Feedback-based detection methods build on the principle that hallucinations can be effectively identified and mitigated when explicit evaluative feedback is incorporated into the generation pipeline. These approaches introduce an auxiliary evaluation layer—often powered by large models or iterative loops—that assesses outputs and generates structured signals regarding their factuality, type, and severity. Such signals not only facilitate reliable detection but also provide actionable guidance for correction and iterative model improvement.

a) Fine-Grained AI Feedback: leverages powerful models such as GPT-4 and GPT-4V to provide detailed annotations of hallucinations in generated responses [177]. Given a hallucination-annotated dataset, the system examines each sentence in the model's output relative to a specific input and prompt, producing a structured six-tuple. This tuple encodes the sentence, hallucination type, reason for classification, severity score, and justification [177]. By distinguishing between minor inaccuracies and critical factual errors, fine-grained feedback enables precise categorization and prioritization of hallucinations, making it possible to design targeted mitigation strategies [177].

b) Active Learning Loops: establish a dynamic interaction between detection and model improvement. Instead of relying solely on static benchmark datasets, active learning continuously identifies the most informative or error-prone samples for feedback, thereby expanding coverage of diverse hallucination patterns and avoiding overfitting to narrow error types. This closed-loop process allows detection systems to become progressively more robust and generalizable across domains.

For instance, HADAS operationalizes this paradigm by evaluating outputs along three distinct dimensions of hallucination [25]. Each dimension is assessed with a specialized metric, producing a feedback score that is subsequently normalized and aggregated into a unified hallucination score representing overall reliability [25]. To broaden detection coverage, HADAS employs a diversity-aware sampling strategy within the active learning loop, ensuring that the feedback process systematically captures heterogeneous hallucination types rather than disproportionately focusing on a single error class [25].

B. Phase-Based Detection: Front-End and Back-End Strategies

Within the phase-based dimension of our taxonomy, detection mechanisms can be situated at distinct checkpoints of the generation pipeline. Two positions are of particular importance: *front-end abstention detection*, which intervenes before generation begins, and *back-end hallucination detection*, which operates after an output has been produced. Together, they define a complementary architecture that balances preventive refusal with corrective verification.

1) Front-End Abstention Detection: Abstention detection functions as a proactive safeguard by identifying high-risk, ill-posed, or knowledge-sparse user queries prior to generation. Rather than allowing the model to proceed with a potentially unreliable answer, abstention detection enables the system to withhold responses (*say-no strategy*) or redirect them to alternative workflows. At this stage, detection relies on lightweight but decisive signals such as *uncertainty estimation*—where low-confidence token distributions are flagged as unreliable [171], [173]—and *consistency analysis* across paraphrased or perturbed queries [147]. In addition, *knowledge-based introspection* can provide rapid assessments of whether a query lies outside the model's representational scope [156]. By filtering unanswerable or adversarial prompts in advance, abstention detection reduces the burden on downstream corrective mechanisms and lowers the overall risk of spurious outputs.

Refusal / Abstention Strategies ("Say No") constitute the primary operational paradigm for front-end detection, reframing hallucination mitigation as an alignment problem where models are empowered to strategically decline uncertain tasks rather than attempt speculative generations [178]. Technically, refusal can be implemented through internal uncertainty quantification and multi-sample consistency checks, where divergence across candidate responses triggers a non-committal answer [119]. Beyond reactive safeguards, refusal behaviors can be actively shaped during supervised fine-tuning or reinforcement learning. For instance, HonestAI introduces abstention-oriented supervision to encourage cautious answers in ambiguous scenarios, significantly reducing spurious generations without compromising utility [117]. Similarly, Kang et al. [179] demonstrate that labeling unfamiliar training examples with abstention signals (e.g., "I don't know") enables models to generalize refusal behavior to unseen queries. Collectively, these findings highlight abstention detection not as a fallback but as a principled alignment strategy that institutionalizes uncertainty-aware non-response, transforming "not answering" into a valid and safe operational choice.

2) Back-End Hallucination Detection: Whereas abstention detection prevents unreliable generations from being attempted, back-end hallucination detection focuses on verifying and correcting outputs that have already been produced. Here, more resource-intensive strategies can be deployed, including *fact-based verification* through entity recognition and relation analysis [137], [140], *knowledge-grounded reasoning* via retrieval or knowledge graph alignment [152], [154], and *content verifiability scoring* for summarization and generative QA tasks [25]. Complementary signals are also provided by *causal attention analysis*, which probes whether outputs are grounded in relevant inputs [175], as well as *dual- or contrastive-model comparisons*, which expose hallucinations by contrasting stable and hallucination-prone variants [124], [170]. Finally, *feedback-driven evaluation* can be integrated into this stage, where powerful models or active learning loops generate structured annotations that guide both detection and iterative correction [177].

Taken together, abstention detection and hallucination detection illustrate how phase-aware strategies balance efficiency and accuracy: the former avoids unnecessary generation by intervening early with lightweight checks, while the latter enables fine-grained verification and correction once outputs are produced. Both are indispensable to a holistic mitigation pipeline and align naturally with the broader three-dimensional framework.

C. Integrative Detection Strategies: Cross-Phase and Multi-Signal Pipelines

Beyond mechanism-specific detectors and phase-specific checkpoints, a third dimension of hallucination detection highlights the system-level integration of heterogeneous signals. Rather than introducing yet another isolated detector, this perspective asks how to compose, orchestrate, and iterate diverse cues: factual verification, uncertainty estimates, consistency checks, causal attributions, and feedback signals, which form into coherent pipelines that are explainable, auditable, and robust across modalities. In practice, explicit integrative approaches remain comparatively scarce; most existing methods still operate within a single mechanism or phase. We therefore conceptualize this third dimension as a guiding framework, under which otherwise localized techniques may be reinterpreted and extended toward integrative deployments. Three prototypical paradigms are illustrative.

1) Multi-Signal Fusion: Detection reliability improves when orthogonal signals are combined rather than applied in isolation. For example, factual verification can be fused with token-level uncertainty scores, or verifiability assessments can be coupled with consistency probes, yielding more fine-grained judgments than any single metric alone [140], [154], [172]. Even methods such as dual-model contrast [170], though often framed as mechanism-specific, can be reinterpreted as fusion pipelines that combine comparative signals with grounding or verifiability modules. We argue that future designs should move beyond raw score aggregation toward calibrated fusion, where detector weights are dynamically adjusted to task, modality, and context.

2) Cross-Phase Orchestration: Hallucination detection benefits from coordination across pre-, in-, and post-generation stages. In principle, lightweight abstention mechanisms should intercept high-risk queries before generation; online probes such as contrastive decoding or multimodal perturbations should monitor instability during decoding; and back-end verifiers should finalize judgments through claim decomposition and grounding. Although most current approaches address only a single stage, we contend that their greatest value lies in cross-phase orchestration: risk signals flagged upstream can condition thresholds downstream, while post-hoc verification outcomes can be propagated backward to adjust abstention or in-generation gates. For LVLMs in particular, such orchestration should enforce cross-modal grounding by requiring stability under both visual and textual perturbations.

3) Feedback-Driven Looping: Detection systems become progressively stronger when evaluative feedback is reintegrated into the pipeline. Fine-grained annotations—whether from LLM-as-judge or human evaluators—can be used to adjust prompts, recalibrate thresholds, or select challenging cases for active learning [25], [177]. Even introspective methods such as self-reflection and self-evaluation [147], [161], though originally proposed as mechanism-level techniques, can be situated within this paradigm as lightweight loop signals that close the gap between generation and verification. We argue that the long-term challenge is to formalize such loops into continuous quality pipelines, where detector outputs are systematically logged, audited, and recycled as training or prompting signals.

Recent work illustrates how all three paradigms can converge in system-level architectures. Gosmar and Dahl [10] present a multi-agent pipeline where generation, review, refinement, and KPI-based evaluation agents communicate via structured JSON envelopes, passing evidence such as flagged spans, explanations, and disclaimers. This design simultaneously fuses heterogeneous signals, orchestrates checks across phases, and closes the loop through KPI-driven feedback—demonstrating that integrative detection is not only a theoretical axis but a realizable system-level practice. Yet such examples remain rare, underscoring the need for further research into cross-method, cross-dimension, and loop-based detection as a foundational complement to existing mechanism- and phase-specific strategies.

The third dimension reframes hallucination detection as an architectural challenge: to govern when, where, and how multiple diagnostic signals are mobilized. While existing methods provide the building blocks, their integrative recombination into multi-signal, cross-phase, and loop-based pipelines remains underexplored. This gap presents both a conceptual opportunity and a practical imperative for advancing hallucination detection beyond isolated fixes toward robust, adaptive, and deployable systems.

D. Summary of Detection Methods

Taken together, hallucination detection emerges as a multi-dimensional yet still fragmented field. Along the **mechanism dimension**, methods range from fact- and knowledge-based verification to consistency analysis, uncertainty estimation, causal attention probing, dual/contrastive model comparison, and feedback-driven evaluation, each offering complementary diagnostic signals. Along the **phase dimension**, detection can intervene before or after generation, balancing lightweight abstention with dynamic probes and post-hoc verification. Finally, along the **integrative dimension**, a few emerging systems illustrate the potential of multi-signal fusion, cross-phase orchestration, and feedback-driven looping, though such designs remain rare. Collectively, these approaches demonstrate that detection is not merely an auxiliary safeguard but the enabling layer that diagnoses error modes, calibrates generative reliability, and provides actionable triggers for mitigation. At the same time, the scarcity of unified and scalable detection pipelines underscores a critical research gap, highlighting the need for integrative architectures that can reconcile precision and recall, support cross-modal robustness, and sustain auditable deployment. Building on this foundation, the next section turns to mitigation strategies, examining how detection signals can be operationalized into systematic interventions for trustworthy generative AI.

V. Mitigation Methods for AI Hallucination

This section presents a three-dimensional, process-aligned view of hallucination mitigation and motivates a closed-loop *Integrative Hybrid Framework*. As illustrated in Fig. 4, along the *mechanism* axis, we group methods into (i) *model-intrinsic* interventions that act inside the model or its decoding (e.g., prompting/ICL, decoding controls, fine-tuning), (ii) *knowledge-infused* approaches that ground outputs in external evidence (RAG, knowledge graphs, consistency checks), and (iii) *meta/framework-level* orchestration that coordinates self-regulation, abstention, multi-model cross-checks, and iterative feedback. Along the *phase* axis, we map interventions to *Pre-Generation*, *In-Generation*, and *Post-Generation* stages to clarify their placement and constraints. A third, *cross-phase* axis emphasizes multi-method fusion and iterative refinement to propagate guidance, verification, and correction end-to-end. Building on this perspective, our hybrid framework integrates (1) preventive, risk-aware prompting and selective knowledge grounding in Pre-Generation, (2) lightweight, uncertainty-aware steering during In-Generation, and (3) heavyweight verification and corrective rewriting in Post-Generation, complemented by provenance tracking and evaluation logging to ensure scalable and auditable reliability across domains.

A. First Dimension: Mechanism-Based Taxonomy of Hallucination Mitigation Methods

This section presents the first dimension of our three-dimensional taxonomy: a mechanism-based classification of hallucination mitigation strategies. We organize existing methods into three primary categories based on their core intervention mechanisms: (1) **Model-Intrinsic Methods**, (2) **Knowledge-Infused Mitigation Methods**, and (3) **Meta-/Framework-Based Methods**.

As illustrated in Fig. 5, *Model-Intrinsic Methods* act directly on the model's internal processes, including prompt engineering, fine-tuning, and decoding control, to shape reasoning pathways and guide generation behavior. Fig. 6 depicts *Knowledge-Infused Methods*, which ground outputs in external factual resources such as RAG pipelines, knowledge graphs, and evidence-checking modules. Operating at a higher level of abstraction, *Meta-/Framework-Based Methods* (Fig. 7) integrate system-level scaffolding—ranging from self-regulation and abstention protocols to agent-based orchestration and multi-model debate—to enforce robustness, cross-verification, and adaptive alignment with factuality.

Each category addresses hallucinations via distinct operational principles. The subsections that follow provide a detailed analysis of these categories, highlighting representative techniques, theoretical underpinnings, and practical implementations.

1) Model-intrinsic Methods: Model-intrinsic Methods constitute the first category of our mechanism-based taxonomy, referring to interventions that integrate directly into a model's core architecture and decoding pipeline. Operating at the intrinsic level of the generation process, these methods influence how tokens are selected, sequences are constructed, and contextual information is internally processed. By targeting the foundational reasoning and decision-making dynamics of large language models, Model-intrinsic Methods form the backbone of many mitigation strategies.

As illustrated in Fig. 5, this category can be further subdivided into four principal modes of intervention: (1) *Prompt Engineering & In-Context Control*, which designs and structures inputs to constrain the model's reasoning trajectory and minimize semantic drift. (2) *Decoding Strategies*, which refine the search and selection process during inference to balance accuracy, diversity, and factual grounding. (3) *Fine-Tuning & Training-Based Methods*, which recalibrate model parameters or adapt task-specific objectives to enhance factual alignment and reduce systematic error propagation. (4) *Active Learning & Annotation*, which improves data quality and coverage through targeted human or automated annotation, enabling models to learn from more reliable supervision signals.

In the subsequent subsections, each of these subgroups will be explored in detail, beginning with their conceptual motivations and extending to representative methodological implementations.

a) Prompt Engineering & In-Context Control: Prompt Engineering and In-Context Control represent one of the most direct and widely adopted strategies for hallucination mitigation. These approaches intervene at the input level of the model by shaping how information is presented, guiding the model's reasoning trajectory, and constraining potential divergence from factual or contextual truth [180], [181]. Prompt engineering focuses on the design and structuring of instructions, enabling models to better interpret task requirements and reduce ambiguity in generation. In parallel, uncertainty- or confidence-based prompting introduces adaptive control mechanisms, where prompt structures dynamically adjust based on the model's self-assessed reliability [180], [182]. Extending further, In-Context Learning (ICL) enhances model alignment by embedding demonstrations, reasoning chains, or reference exemplars within the input sequence [181], [183]. Subvariants such as Chain-of-Thought (CoT) prompting encourage explicit intermediate reasoning steps, while few-shot and zero-shot prompting exploit contextual exemplars to steer model predictions without requiring parameter updates [47], [145], [184].

- **Prompt Engineering** constitutes a foundational approach to hallucination mitigation by strategically crafting and structuring input instructions to optimize model behavior. Rather than altering model parameters, prompt engineering leverages the conditioning power of natural language instructions to constrain reasoning trajectories,

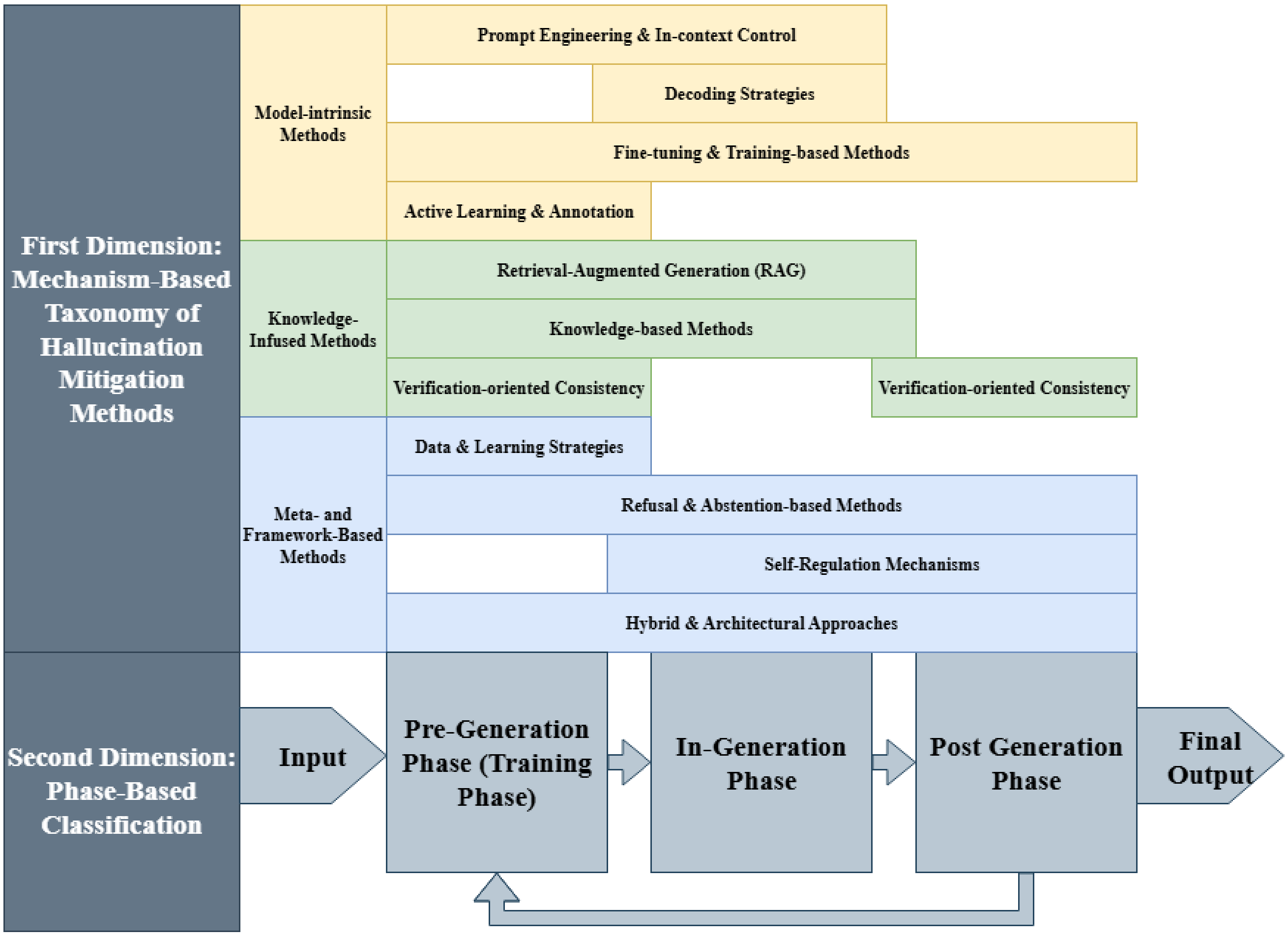

Fig. 4. Overall Integration (First and Second Dimension)

enhance factual alignment, and mitigate the tendency to generate unsupported content [185]. These methods often incorporate external knowledge cues, counterfactual interventions, or domain-specific grounding to reduce over-reliance on the model's parametric memory, which may be outdated or biased.

A variety of strategies have been proposed under this paradigm. For example, prompt-based methods structure queries with contextual evidence or verification checkpoints, sometimes embedding step-by-step reasoning directives that encourage the model to articulate intermediate logic rather than producing unsubstantiated conclusions [186], [187]. More advanced frameworks integrate search-enhanced prompting, such as Monte Carlo Tree Search (MCTS) with dynamic prompt adjustments, balancing exploration and exploitation to achieve more robust factual grounding [187]. Additionally, specialized prompt designs employ counterfactual keyword implantation or hierarchical representations to override misleading priors in pre-trained knowledge and align outputs with contextual truth [188], [189].

As a concrete instance of context-conditioned prompting, we next describe a tag-augmented scheme that enforces source-grounded generation and competence-aware abstention at inference time [190]. In this scheme, task context is augmented with embedded source tags (e.g., "(source 3626)") and an instruction to include sources, enabling recognition and flagging of out-of-domain behavior and reducing hallucinations without modifying model parameters [190]. Hallucination is baselined on no-context prompts using generated-URL validity as a proxy [190]. Supplying context markedly reduces fabricated URLs across engines (from 2,445 total, 1,605 invalid, to 48 total), and with tagged context only 1 of 244 responses is unverifiable (approximately 99.88% reduction) [190]. When context and question are mismatched, models frequently state the mismatch rather than invent sources [190]. Limitations include susceptibility to "disregard context" instructions and to contaminated context, shifting the challenge to search/context evaluation [190]. Beyond textual conditioning, multimodal prompt engineering extends this principle to visual–textual integration, embedding cross-modal evidence to reinforce factual consistency in vision-language tasks [191]. Such techniques have demonstrated effectiveness in high-stakes domains, including scientific reasoning, financial fore-

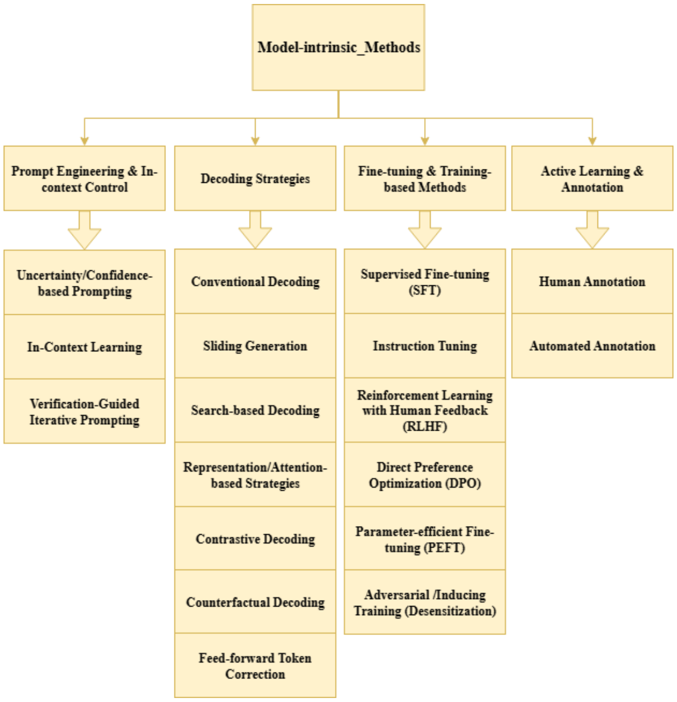

Fig. 5. Model-intrinsic Methods

casting, and clinical decision support, where hallucination carries disproportionate risk. By refining the interpretability and precision of model responses without requiring retraining, prompt engineering offers a lightweight, scalable, and task-adaptive mechanism for enhancing trustworthiness across both proprietary and open-weight large language models [192].

Uncertainty/Confidence-based Strategies refers to adaptive input strategies that incorporate a model's self-assessed reliability into framework design, thereby reducing overconfident hallucinations and guiding corrective behaviors [193], [194]. Instead of relying solely on static instructions, these methods exploit uncertainty signals—derived from token probabilities, semantic entropy, or external confidence estimators—to trigger re-prompting, abstention, or supplementary evidence retrieval [194], [195]. For example, dynamic supplementation frameworks integrate real-time federated search results when uncertainty is detected, enabling more factually grounded outputs [189]. Similarly, Interactive DualChecker monitors model confidence in teacher–student distillation, re-prompting low-confidence teachers with enriched context and refining borderline student predictions via rationales [97].

Further innovations demonstrate the versatility of this paradigm across modalities and reasoning settings. In multimodal models, MemVR's "look-twice" mechanism reinjects visual tokens at points of high uncertainty, emulating human re-checking behavior to improve factual alignment [196]. For textual LLMs, studies of false premise hallucinations reveal that low-confidence predictions correlate with disruptions in specific attention heads, enabling prompt-based interventions to suppress such errors [197]. Confidence-guided debate frameworks like DebUnc further show that incorporating uncertainty into multi-agent prompting prevents convergence on confidently wrong answers [171], while abstention strategies use statistical or in-dialogue uncertainty thresholds to withhold unreliable responses, reducing hallucinations in unanswerable queries [178]. Collectively, these approaches transform prompt control into a dynamic reliability-aware mechanism, enabling LLMs to self-correct or abstain in high-risk scenarios.

Another effective strategy is to directly mitigate the underlying factors that contribute to hallucinations. Paper [198] demonstrates that object hallucinations frequently arise because models are overly sensitive to both high- and low-frequency image features, causing them to mis-detect objects even in the absence of genuine visual cues. By explicitly addressing these root causes—namely redundant frequency-domain signals and spurious correlations—and suppressing unnecessary high- and low-frequency features during inference, hallucination rates can be significantly reduced [198]. The Multi-Frequency Perturbations (MFP) method has proven effective across multiple benchmarks, including CHAIR and POPE, and across diverse architectures such as CLIP, SigLIP, Vicuna, and LLaMA [198]. Moreover, when combined with inference-time techniques, it achieves state-of-the-art performance [198].

- **In-context Learning (ICL)** refers to a powerful paradigm in which large language models (LLMs) perform tasks by conditioning on examples or instructions provided in the prompt, without any parameter updates. This strategy leverages the LLM's pretrained latent knowledge, guiding output behavior through pattern-based inference rather than explicit training. Empirical evidence [146], [199] suggest that ICL allows models to compose internal task knowledge on the fly, effectively treating prompts as signals to retrieve or instantiate latent capabilities.

  – **Chain-of-Thought (CoT) Prompting** compels models to articulate intermediate reasoning steps prior to producing final outputs, thereby enhancing transparency and factual alignment. This decomposition of reasoning pathways reduces semantic drift and makes inconsistencies observable, enabling correction before they manifest as hallucinations [47], [200]. For example, Liang et al. introduce THAMES, a modular prompting framework that integrates CoT with verification layers to ensure consistency across multi-step reasoning [186]. Chang et al. propose a unified framework in which CoT decomposes complex tasks into sub-queries, aligning generation with external evidence [201]. Similarly, Ding et al. demonstrate that coupling CoT with retrieval-based augmentation enables reasoning chains to be explicitly grounded in document evidence, thereby curbing unsupported claims [202]. Collectively, these works illustrate how CoT transforms generation from a single-step output process into an iterative reasoning pipeline, enhancing factual fidelity and interpretability.

  - **Few-shot and zero-shot prompting** provide effective strategies to improve model reliability under low-resource settings [203]. Few-shot prompting introduces a handful of exemplars into the input prompt, enabling in-context learning without modifying model parameters, thereby significantly reducing the dependence on task-specific training data [145], [203]. In contrast, zero-shot prompting relies on carefully crafted instructions to guide model behavior when no exemplars are available [204].

- **Verification-Guided Iterative Prompting** introduces formal verification feedback into the prompting loop to iteratively steer LLMs toward factually consistent outputs [205]. In this paradigm, the LLM is first treated as a possibly hallucinating oracle whose outputs are automatically checked by a formal verifier against correctness specifications. Counterexamples or error explanations from the verifier are then appended to the next prompt, guiding the model away from previously invalid reasoning trajectories. Through repeated counterexample-guided refinements, the LLM converges toward a response that satisfies the verifier, effectively "dehallucinating" the model without altering its parameters. This approach demonstrates particular promise in safety-critical domains such as automated planning, where iterative correction is essential for reliability [205].

  Both approaches complement **Chain-of-Thought (CoT)** reasoning by embedding demonstrations or explicit task instructions directly into the prompt space. Empirical studies show that exemplar-driven instructions enhance factual grounding even under resource-constrained conditions [186], while refined zero-shot prompting with explicit topical constraints mitigates hallucinations in tasks such as topic modeling by reducing duplication and fabricated outputs [192]. These findings confirm that prompt design itself functions as a structural constraint on generative space, making ICL a lightweight yet versatile mechanism for hallucination control across domains.

  Another **context-based** hallucination mitigation method is to incorporate external context into the post-training supervision process [206]. The post-training supervision mechanism in Context-Informed Grounding Supervision (CINGS) appends external context during learning while masking *context tokens* from the loss [206]. In this way, CINGS shifts the model's reliance from internal parametric knowledge to external evidence, computes loss only over the *response tokens*, and prevents rote memorization of context, thus improving grounding in both text and vision-language domains [206].

  More broadly, this fits into the Context-Engineering framework, where Parameter-Efficient (PE) fine-tuning and Context Control strategies serve as lightweight mechanisms to steer model behavior without full retraining [206]. Context Control regulates how evidence is introduced and weighted during inference [206]. Within this continuum, CINGS acts as a post-training learning objective, providing a direct supervision signal that complements inference-time context engineering [206]. Together, these approaches form a cohesive strategy to strengthen grounding and mitigate hallucinations [206].

b) Decoding Strategies: form a class of hallucination mitigation techniques that intervene during the inference phase, directly shaping token selection and sequence construction without altering model parameters [166], [207], [208]. Unlike prompting, which conditions input design, decoding methods regulate generation dynamics in real time, making them particularly effective for efficient and adaptable deployment [166], [208]. By constraining candidate outputs, introducing verification mechanisms, or reweighting internal representations, these strategies promote factually grounded and coherent text [167], [209], [210]. Approaches range from *conventional decoding* methods such as greedy, beam, and sampling, to advanced techniques like sliding-window generation [211], search-based frameworks (e.g., MCTS, Tree-of-Thought), attention-guided decoding, contrastive and counterfactual reasoning, and feed-forward token correction [187].

- **Conventional decoding** algorithms constitute the foundational mechanisms by which large language models transform probabilistic token distributions into coherent sequences [166], [212]. Among the most widely adopted are greedy decoding [166], which selects the highest-probability token at each step to ensure determinism and efficiency, and beam search [169], which expands multiple candidate paths simultaneously to balance optimality and diversity in the generated sequence. In contrast, sampling-based approaches including top-k [207] and nucleus (top-p) sampling introduce stochasticity to the selection process, thereby enhancing creativity and reducing repetitive outputs. While these methods are not explicitly designed for hallucination mitigation, their influence on sequence exploration has direct implications for factual reliability [212]. Greedy decoding, for instance, minimizes randomness but risks amplifying biases embedded in parametric memory, whereas beam search can produce overly generic responses that lack grounding [166]. Sampling methods, though capable of generating diverse outputs, often increase the likelihood of factual deviations if not carefully constrained [208]. Consequently, conventional decoding forms the baseline upon which more advanced hallucination-aware strategies such as contrastive decoding or attention-guided interventions have been developed, highlighting the trade-offs between determinism, diversity, and factual alignment in sequence generation [208].

- **Sliding Generation** proposed by Harrington, is a structured decoding strategy specifically designed to mitigate hallucinations in long-sequence generation by segmenting the input into overlapping windows, each processed independently [213]. This segmentation constrains the model's focus to smaller, contextually coherent fragments, thereby alleviating cognitive overload and improving memory retention—two factors that often degrade performance in extended reasoning tasks [213]. Once partial outputs are produced for each window, the overlapping regions are aligned and merged to maintain discourse continuity and semantic consistency across the entire sequence [213]. By decomposing complex prompts into manageable sub-units, Sliding Generation reduces context dilution, promotes local coherence, and anchors generation in factual information [213]. These properties make it particularly effective for long-form narrative generation, multi-hop reasoning, and domains where maintaining logical fidelity across extended spans of text is critical [209], [213].
- **Search-based Decoding** approaches, such as Monte Carlo Tree Search (MCTS), Tree-of-Thoughts (ToT), and Uncertainty-Aware Beam Search (UABS), extend the decoding process into structured, exploratory search spaces to enhance multi-step reasoning and reduce hallucinations [187], [214], [215]. Unlike conventional decoding methods that operate greedily on token probabilities, these techniques reframe generation as a search problem over reasoning trajectories, allowing the model to evaluate, revise, and backtrack across multiple candidate paths before finalizing an output [187]. This structured exploration provides stronger guarantees of factual grounding and logical consistency, particularly in long-form text generation, multi-hop question answering, and planning-intensive tasks [214]. **Monte Carlo Tree Search (MCTS)** combines probabilistic sampling with a four-stage search procedure (selection, expansion, simulation, and backpropagation) to explore possible continuations of text [187]. Each node in the search tree corresponds to a partial generation, with simulations estimating downstream utility to guide the search toward factually robust and coherent outputs [187]. By systematically balancing exploration and exploitation, MCTS enables the model to converge on responses that are both fluent and trustworthy [187]. **Tree-of-Thought (ToT)** proposed by Shunyu Yao et al., decomposes the reasoning process into discrete steps of intermediate "thoughts," which are organized into a tree structure representing the evolution of reasoning states [214]. Each node in the tree corresponds to a state $s = [x, z_1, \cdots, z_i]$ where x is the original input and $z_1 \cdots z_i$ are the thoughts generated thus far [214]. The model proposes multiple continuations for each thought step, allowing the tree to branch into diverse reasoning directions [214]. Together, MCTS [187] and ToT [214] demonstrate how search-based decoding reframes inference as a structured reasoning problem, offering a principled means to mitigate hallucinations while enhancing transparency and robustness in complex generation tasks. Those methods demonstrate how search-based decoding reframes inference as a structured reasoning problem, offering a principled means to mitigate hallucinations while enhancing transparency and robustness in complex generation tasks.
- **Representation/Attention-based Strategies** help reduce the hallucination rate through Inference-Time Intervention [216]–[218]. These methods operate directly on a model's internal dynamics: representation editing modifies hidden states to amplify truthful signals or suppress misleading patterns [216], [217], while attention-based strategies regulate how focus is distributed across context tokens to ensure fact-critical information is prioritized [218]. By intervening inside the model rather than at the input or output level, they provide fine-grained, inference-time control that complements external grounding techniques and directly curbs hallucination at its source [216]–[218].

  Representation-based Methods intervene directly in the hidden representations of LLMs to refine internal information before token prediction [217].

  For example, TruthX introduces an auto-encoder to disentangle hidden representations into semantic and truthful latent spaces [216]. By applying contrastive learning to identify a "truthful editing direction" within this space, the method can intervene during inference by nudging hidden states toward more truthful regions [216]. Experiments on 13 advanced LLMs demonstrate an average 20% improvement in truthfulness on the TruthfulQA benchmark, and further analyses reveal that even editing a single vector is sufficient to steer outputs between truthful and hallucinatory responses [216].

  Building on this direction, Expanding before Inferring (PLI) introduces Premature Layers Interpolation, a training-free and plug-and-play framework that constructs new premature layers by interpolating hidden states across adjacent layers [217]. Inspired by diffusion and sampling, this approach extends the depth of internal information processing, enhancing factual coherence without fine-tuning overhead [217]. Unlike input- or output-level interventions, PLI directly strengthens internal representation refinement, mitigating hallucinations at their source [217]. Experiments on four widely used benchmarks show that PLI outperforms baselines in most cases and can be combined with other mitigation techniques [217]. Further analysis indicates that its success is closely tied to LLMs' internal mechanisms, making PLI a lightweight yet effective strategy for hallucination reduction [217].

  Attention-based Methods leverage the internal attention maps of transformer models, which are visualizations of how focus is distributed across tokens in an input sequence, to analyze and regulate the model's informa-

tion processing during generation [218]. Attention maps expose salience attribution by indicating which tokens are considered most relevant at each decoding step, thereby offering insight into how syntactic and semantic dependencies are captured across layers [218].
Recent studies demonstrate that hallucinations often arise from misalignments in these patterns, such as over-attention to peripheral context or under-attention to fact-critical tokens [219]. By directly intervening at the attention level—through re-weighting, scaling, or dynamically reallocating focus—models can be guided to prioritize factually salient information while suppressing irrelevant signals [175], [219]. For example, attention map optimization techniques adjust weights during inference to strengthen the influence of crucial context tokens and mitigate noise, resulting in outputs that exhibit improved factual consistency, contextual grounding, and semantic coherence [218].
This fine-grained control over the model's focus makes attention-based decoding strategies particularly effective for knowledge-intensive tasks where accurate attention allocation is essential to suppress hallucinations and ensure faithful adherence to input context [175], [218], [219].Beyond attention re-weighting, architectural modifications such as *Concentric Causal Attention (CCA)* reorganize visual token positions and redesign causal masks to counteract the long-term decay of rotary position encoding (RoPE), thereby strengthening cross-modal alignment and significantly reducing object hallucinations in LVLMs [220].In addition, *OPERA* introduces an over-trust penalty on decoding logits combined with a retrospection-allocation mechanism, which detects attention aggregation patterns that correlate with hallucinations and dynamically re-routes generation to suppress them, offering a training-free and generalizable decoding-time mitigation strategy [221].

- **Contrastive Decoding (CD)** is a decoding-time strategy that leverages the comparative strengths of two language models of differing capacities to enhance factual reliability and reduce hallucinations [165], [222]. The underlying intuition is that smaller models often surface generic or hallucination-prone tokens, whereas larger models tend to capture more semantically consistent and factually grounded continuations [124], [170]. CD operationalizes this contrast by dynamically reweighting token probabilities: outputs disproportionately favored by the smaller model are penalized, while those preferred by the larger model are promoted [223]. This rebalancing mechanism allows the decoding process to suppress low-quality continuations without modifying the model's architecture or retraining [223]. Extensions such as Induce-then-Contrast Decoding (ICD) refine this principle by deliberately inducing a hallucination-prone variant of the base model to act as a contrastive signal, further amplifying the suppression of unreliable outputs [224]. By enforcing this fine-grained competition, CD yields more precise, coherent, and trustworthy generations in an inference-efficient manner.
Similarly, Active Layer-Contrastive Decoding (ActLCD) extends the contrastive decoding paradigm by treating generation as a sequential decision-making problem [225]. Whereas prior methods like DoLa statically contrast shallow and deep layers at every token, ActLCD employs a reinforcement-learned policy guided by a reward-aware classifier to decide when to activate layer-wise contrastive signals dynamically [225]. This adaptive mechanism leverages factual representations encoded in deeper layers while avoiding unnecessary "overthinking" for simpler tokens, thereby mitigating the hallucination snowballing from autoregressive error propagation [225]. Unlike standard CD or ICD, ActLCD requires no auxiliary models, induced variants, or multiple samplings, operating instead in a single forward pass using only model-internal information [225]. Empirical results across five benchmarks—including TruthfulQA, LongFact, StrategyQA, GSM8k, and a domain-specific software hallucination suite—show consistent factuality improvements over state-of-the-art baselines, demonstrating robust gains from sentence-level answers to long-form and specialized domains [225].
While these methods focus on inference-time contrastive mechanisms, similar principles have also inspired training-time alignment strategies [226]. Atomic Consistency Preference Optimization (ACPO) introduces a self-supervised alignment framework that leverages contrastive signals across stochastic generations [226]. Instead of relying on external models or curated datasets, ACPO identifies atomic consistency signals—the degree to which individual factual units remain stable across multiple sampled responses—and automatically constructs contrastive pairs of high-quality (consistent) versus low-quality (inconsistent) data [226]. These pairs are then used for preference optimization, aligning the model toward factual outputs without external supervision [226]. Empirical evaluations show that ACPO outperforms strong supervised baselines such as FactAlign, achieving gains of +1.95 points on both the LongFact and BioGen benchmarks [226]. By grounding contrastive learning in self-consistency rather than external labels, ACPO provides a scalable, cost-efficient, and effective pathway to reduce factoid hallucinations, complementing inference-time contrastive decoding strategies with a training-time alignment perspective [226].
- **Counterfactual Decoding** introduces a causal inference perspective into the decoding process to distinguish contextually entailed knowledge from spurious associations, thereby reducing hallucination-prone outputs [176]. The approach operates by generating counterfactual variants of the original input through semantic perturbations such as negation, entity substitution, or discourse restructuring while keeping other factors constant [188]. By contrasting the model's predicted logits across true and counterfac-

tual contexts, the method subtracts the latter from the former, effectively downweighting unsupported continuations and amplifying tokens that are causally grounded in the given input [175], [176], [188]. This counterfactual subtraction mechanism ensures that the generation process adheres closely to prompt-specific information, mitigating hallucinations driven by overgeneralization or parametric bias [188]. Extending beyond unimodal applications, multimodal adaptations construct counterfactuals by perturbing visual-semantic alignments—for example, masking critical image regions or mismatching cross-modal features—thereby addressing modality-prior-induced hallucinations and strengthening intermodal factual grounding [175].
- **Feed-forward Token Correction (FFNs)** exploits the structural and functional properties of feed-forward networks (FFNs) within transformer architectures to revise or re-rank token predictions during decoding, thereby enhancing factual alignment [227], [228]. In transformer models, FFNs function as nonlinear composition layers that can be conceptualized as key–value memory modules: keys are tuned to specific input patterns, while values shape the output distribution over tokens [210]. Lower-layer FFNs primarily capture syntactic or surface-level regularities, whereas upper layers encode higher-order semantic abstractions that critically influence final token selection [229]. Token correction methods intervene in these upper layers to refine intermediate representations according to contextual salience or factual grounding constraints. For example, the *Look-Twice* mechanism [196] introduces a two-pass decoding procedure in which preliminary outputs are projected back into FFN memory space and then reinterpreted through a secondary decoding pass. This iterative refinement allows the model to identify and correct semantically inconsistent or unsupported tokens in real time, without necessitating retraining or architectural modification [196]. Consequently, FFN-based correction represents a lightweight yet powerful strategy for hallucination mitigation, operating deep within the generative pipeline to ensure more reliable and semantically coherent outputs [196]. Beyond token-level interventions, recent work also explores **hidden state manipulation** through activation engineering, which leverages directional offsets in the model's hidden representation space to steer outputs away from hallucinated responses and toward factual ones [230]. Unlike contrastive decoding, which constrains generation by comparing probability distributions across models or decoding paths, this approach directly operates on internal hidden states, providing a complementary and model-intrinsic pathway for hallucination mitigation.

c) Fine-Tuning and Training-based Methods: represent a fundamental class of hallucination mitigation strategies that intervene at the level of model parameters, embedding factual consistency, task alignment, and human-preferred behaviors directly into the underlying probability distributions of large language models (LLMs).

In contrast to inference-time interventions such as decoding strategies, which adjust outputs dynamically without altering the model's core, fine-tuning reshapes the model's internal representations through targeted optimization [180], [181].

This process is typically grounded in high-quality, domain-specific corpora or preference-based annotations, enabling the model to gradually internalize truth-grounded patterns and robust decision-making heuristics [177], [192], [231], [232]. Depending on the approach, fine-tuning may involve comprehensive full-parameter updates, selective adaptation through parameter-efficient methods, or adversarial training designed to desensitize models against hallucination-inducing prompts [231], [232]. By systematically aligning internal knowledge structures with external truth signals, fine-tuning provides a durable and generalizable mechanism for mitigating hallucinations, albeit at the cost of greater computational and data requirements compared to lighter inference-time techniques [231], [232].

- **Supervised Fine-Tuning (SFT)** is a foundational approach in the alignment of large language models, designed to calibrate model behavior against human expectations by training on carefully curated, annotated datasets containing factually accurate examples [216]. Often serving as the initial stage in alignment pipelines such as RLHF, SFT provides direct exposure to truth-grounded responses and desirable output structures, thereby establishing a baseline of factual reliability that enhances the efficacy of subsequent preference- or reward-based optimization [216], [233]. In the context of hallucination mitigation, SFT has proven effective in shaping the model's generative distribution toward verifiable content [216]. For example, Xia et al. fine-tune on a diversity-aware dataset explicitly constructed to capture hallucination-prone instances across varying factual densities, which enables models to generalize more robustly in real-world generation tasks [234]. Similarly, Qu et al. employ SFT on factual–counterfactual data pairings, strengthening the model's capacity to differentiate truth from fabrication and ensuring that generated outputs remain anchored in verifiable knowledge [235].
- **Faithfulness-based Loss Function** represents a training-oriented approach that augments conventional optimization with loss terms explicitly penalizing factually inconsistent or unfaithful generations. By directly encoding fidelity constraints into the objective function, these methods steer models toward producing outputs that remain more accurate and aligned with the source material. Such formulations complement supervised fine-tuning and preference optimization by embedding factual consistency as a first-class optimization target, strengthening reliability across diverse tasks [24]
- **Instruction tuning** refines language models by fine-tuning them on carefully constructed instruction–response pairs, thereby enhancing their capacity to follow user

directives while mitigating hallucinations [231], [232]. Unlike generic supervised fine-tuning, this approach explicitly incorporates both positive and negative examples, enabling models to recognize and avoid fabricated or unsupported content [232], [236]. By leveraging robust datasets that span diverse tasks and include semantically challenging negative samples, instruction tuning helps counteract representational biases and strengthen alignment with factual or visual inputs [231]. Moreover, techniques such as adversarial augmentation further improve model robustness, ensuring greater factual accuracy in dialogic exchanges and other core applications [231]. This scalability makes instruction tuning a critical strategy for reliable deployment in domains such as open-domain querying and vision–language reasoning [231].

- **Hallucination-Induced Optimization (HIO)** introduces a contrastive optimization framework that leverages a fine-tuned Contrary Bradley–Terry Model to amplify the margin between hallucinated and correct tokens. By explicitly penalizing hallucination-prone generations and rewarding faithful ones, HIO strengthens contrastive decoding signals and substantially reduces hallucinations in large vision–language models. Positioned at the intersection of preference optimization and contrastive learning, this method extends existing fine-tuning strategies with a targeted mechanism for factual alignment in multimodal settings [237]
- **Reinforcement Learning with Human Feedback (RLHF)** represents a multi-stage fine-tuning paradigm that integrates human evaluative signals into the training loop to enhance factual alignment and reliability [238], [239]. Building on the foundation of Supervised Fine-Tuning (SFT), RLHF first constructs a reward model trained on human-annotated pairwise comparisons of model outputs [32]. This reward model serves as a proxy for human judgment, guiding reinforcement learning updates that optimize the language model's policy toward generating outputs judged more helpful, truthful, and aligned with user intent [32], [240], [241]. While RLHF has been instrumental in advancing the factuality and safety of large language models, it is also associated with significant computational costs, instability in the presence of poorly specified reward functions, and high sensitivity to hyperparameter tuning [238], [239]. These challenges have motivated research into more efficient alternatives, such as Direct Preference Optimization (DPO) [240]. Within the specific context of hallucination mitigation, RLHF has proven effective in reducing factually inconsistent outputs by incorporating reward signals that explicitly penalize hallucinations [238], [239]. For example, Wang et al. apply RLHF to align models with human factuality preferences across diverse domains, demonstrating that reinforcement-driven policy refinement can substantially suppress hallucination-prone behaviors [242].
- **Direct Preference Optimization (DPO)** is a preference-based fine-tuning approach that aligns large language models with human judgments while bypassing the complexities of reinforcement learning and explicit reward modeling [240]. By reframing preference learning as a binary classification problem, DPO optimizes model behavior through cross-entropy loss, directly increasing the likelihood of preferred responses relative to dispreferred ones [240]. Compared to RLHF, DPO offers greater simplicity, stability, and computational efficiency, making it a scalable alternative for preference alignment [243]. In the domain of hallucination mitigation, DPO has shown particular promise in enhancing factual grounding without incurring the instability of reward-based optimization [243]. For instance, Mu employs DPO to address topic-specific hallucinations by systematically correcting outputs that deviate from factual granularity [192], while Xiao integrates DPO with hallucination detection signals, demonstrating its adaptability under noisy or imperfect supervision [177].
- **Parameter Efficient Fine-Tuning (PEFT)** mitigates hallucinations by adapting large language models through minimal parameter updates, such as adapter layers, prefix-tuning, or low-rank transformations [244]. By restricting optimization to select submodules, PEFT significantly reduces computational overhead while embedding factual constraints into the model's representations with minimal disruption to its general capabilities [244]. This makes PEFT particularly attractive in resource-constrained or domain-specific contexts, where full fine-tuning is prohibitively expensive [186]. Nonetheless, its effectiveness critically depends on the availability of high-quality supervision and the careful design of parameter-efficient modules, as poorly configured adaptations may fail to capture the nuanced signals required for robust hallucination suppression [244]. A representative example is mmT5, which extends parameter-efficient principles through a modular multilingual design that introduces language-specific bottleneck modules and partially frozen decoder parameters [245]. By structurally separating language-specific from language-agnostic representations, mmT5 mitigates representation drift during fine-tuning and effectively suppresses source-language hallucinations, achieving stable cross-lingual generalization in zero-shot generation [245].
- **Adversarial/Inducing Training (Desensitization)** builds on the principle of adversarial robustness by systematically exposing language models to adversarial or deliberately perturbed examples, thereby reducing their susceptibility to hallucination-inducing inputs [231], [232]. The core idea is to train models under adversarial pressure so that they learn to discriminate between factual and misleading cues, effectively desensitizing them to spurious correlations [246].

  In the hallucination mitigation context, Park et al. apply adversarial augmentation to dialogue systems by integrating semantically challenging negative examples, enabling the model to recognize and suppress fabricated content while enhancing its resilience to misleading conversational cues [232]. Similarly, Zhang et al. introduce a contrastive adversarial fine-tuning approach that induces controlled hallucinations and then penalizes them through preference optimization, strengthening the model's ability to reject unsupported continuations [224]. By combining adversarial exposure with desensitization training, these methods foster more robust alignment with factuality, particularly in domains where models are vulnerable to subtle semantic distortions.

d) Active Learning & Annotation: constitute upstream strategies for hallucination mitigation, aimed at strengthening the supervision signals that guide training or fine-tuning of large language models [247], [248]. By curating informative, diverse, and high-uncertainty examples, these approaches expose model deficiencies and enable more targeted updates, ultimately improving factual grounding and generalization [234]. Such methods are particularly valuable in resource-constrained or noisy data environments, where the efficiency and quality of supervision critically shape alignment outcomes [247], [248].

- **Human Annotation** remains the gold standard for high-quality supervision. Leveraging expert or crowd-sourced evaluation, it enables fine-grained identification of factual inaccuracies and nuanced misalignments across diverse tasks and domains [249]. Xia et al. emphasize that human-labeled examples play a pivotal role in calibrating hallucination detection models and establishing reliable ground truth for supervised fine-tuning and active learning pipelines [234]. However, despite its precision, human annotation faces challenges of high cost, labor intensity, and inter-annotator variability, which constrain its scalability [249].
- **Automated Annotation** provides scalable alternatives by employing model-driven heuristics, rule-based verification, or weak supervision techniques to generate labels across large corpora [250]. These systems detect inconsistencies, factual mismatches, or low-confidence responses, enabling rapid dataset construction for downstream training [250], [251]. Nevertheless, automated pipelines risk introducing label noise, necessitating post-processing steps such as filtering, confidence reweighting, or hybrid human-in-the-loop mechanisms to ensure reliability and effectiveness [251].

In sum, Model-intrinsic Methods provide direct interventions at the level of prompts, decoding, parameter optimization, and data supervision, thereby shaping the intrinsic generative dynamics of large language models. While these approaches enhance factual alignment from within the model's architecture and training pipeline, their effectiveness is often constrained by the limits of parametric knowledge and internal reasoning [31], [162].

To mitigate these limitations, increasing attention has been focused towards *Knowledge-Infused Mitigation methods*, wherein model outputs are systematically grounded in external information sources to improve the accuracy of facts and reduce hallucinations. By leveraging retrieval systems, structured knowledge graphs, and verification mechanisms, these methods complement technical strategies by supplying reliable, up-to-date, and domain-specific factual signals that transcend what the model has memorized during pretraining [3], [31], [191].

2) Knowledge-Infused Mitigation Methods: Knowledge-Infused Mitigation Methods constitute the second category of our mechanism-based taxonomy, focusing on strategies that explicitly anchor large language models to external sources of factuality and structured knowledge. Operating at the extrinsic level of the generation process, these methods complement intrinsic adjustments by ensuring that outputs remain verifiable against reliable references [3], [252]. Rather than relying solely on parametric memory, they ground model reasoning in curated resources, thereby reducing semantic drift and enhancing factual robustness [1]

As illustrated in Fig. 6, this category can be further divided into three principal modes of intervention: (1) *Retrieval-Augmented Generation (RAG)*, which enriches the generative process by integrating relevant documents or passages retrieved from external corpora, with adaptive variants tailoring retrieval scope to task complexity [3], [31], [202]. (2) *Knowledge-grounding Methods*, which embed factual constraints into model reasoning by leveraging structured resources such as knowledge graphs, external knowledge base integration, and mechanisms of knowledge injection or distillation [253]. (3) *Consistency-guided Verification*, which validates outputs through entity- and relation-based checks, factual consistency verification, and multi-step procedures like Chain-of-Verification (CoVe) or sub-question decomposition, ensuring that final outputs remain logically coherent and factually grounded [254], [255].

In the subsequent subsections, each of these subgroups will be examined in detail, beginning with their conceptual underpinnings and extending to representative methodological implementations.

a) Retrieval-Augmented Generation (RAG): has emerged as a cornerstone technique for mitigating hallucinations in large language models (LLMs) by directly addressing the limitations of parametric memory. Since LLMs are constrained by the scope and temporal boundaries of their training data, they often produce outdated or fabricated content when confronted with knowledge-intensive tasks [3], [31]. RAG alleviates this problem by coupling generative models with a retrieval module that dynamically accesses external knowledge bases or document corpora, thereby enriching the generation process with timely, verifiable, and contextually relevant information [129]. Through this hybrid design, RAG extends the reasoning capacity of LLMs beyond static parameters, grounding outputs in external evidence while preserving fluency and adaptability [3], [256], [257]. Importantly, the choice of knowledge sources, ranging from open-domain corpora to specialized domain repositories as well as the integration stage within the inference pipeline, gives rise to distinct variants of RAG [258], [259]. These include *basic retrieval-augmented*

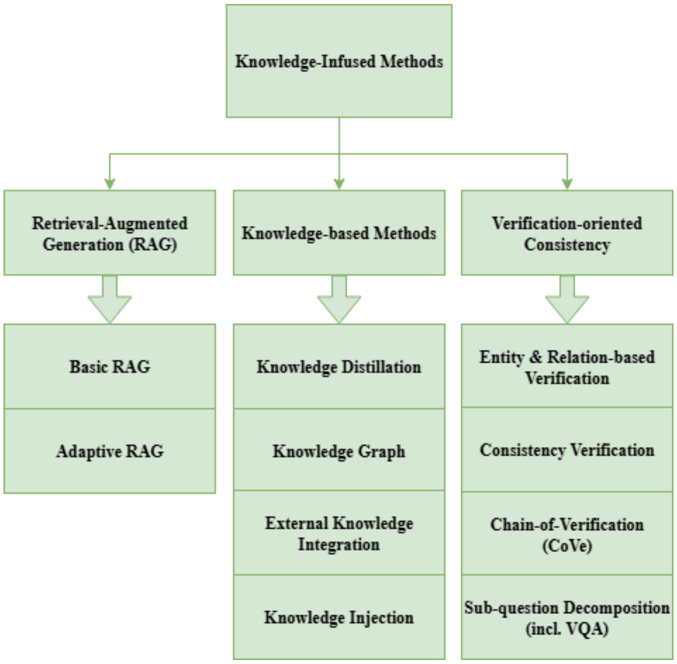

Fig. 6. Resource_Knowledge-based Methods

methods, which provide general factual reinforcement, and *adaptive RAG approaches*, which dynamically modulate retrieval strategies according to task requirements [202]. The following subsections examine these branches in detail, highlighting their methodological innovations and implications for hallucination mitigation.

- **Basic RAG** represents the foundational paradigm in retrieval-based mitigation, wherein a retriever component dynamically supplements a large language model (LLM) with external knowledge at inference time. The retriever first encodes the input query and searches a pre-indexed corpus—commonly large-scale resources such as Wikipedia or domain-specific databases—for the top-k relevant documents [3], [257]. The retrieved content is then integrated with the input prompt and provided to the generator, which conditions its response on both the internal parametric knowledge and the external evidence [3], [257]. This architecture enables models to overcome the static limitations of pretraining corpora by incorporating factual, up-to-date, and contextually grounded information at generation time [257]. One of the core strengths of Basic RAG lies in its ability to reduce hallucinations that stem from the probabilistic nature of LLM decoding and incomplete memorization of facts [257]. By grounding responses in retrieved passages, Basic RAG improves factual precision, recall, and coherence, particularly in knowledge-intensive tasks such as question answering and knowledge-grounded dialogue [252]. Furthermore, empirical studies demonstrate that even simple implementations—where retrieval operates in a one-pass manner without adaptive control or iterative refinement—can substantially mitigate the generation of fabricated or outdated content, enhancing model trustworthiness in real-world applications [189], [260]. As the canonical form of retrieval-augmented architectures, Basic RAG establishes the methodological baseline upon which more advanced variants, such as adaptive retrieval or consistency-guided verification, have been developed [159], [202]. It thus provides both a practical framework for reducing hallucinations and a conceptual foundation for subsequent innovations in retrieval-anchored generation systems.
- **Adaptive RAG** refines the conventional RAG framework by dynamically deciding when and how to incorporate external knowledge, thereby balancing factual grounding with computational efficiency [202], [261], [262]. Unlike standard RAG, which indiscriminately performs retrieval for all queries, Adaptive RAG introduces a decision module that estimates whether the model's internal knowledge suffices or whether external evidence is required [202]. When internal predictions are judged unreliable—often through uncertainty estimation, factuality scoring, or hallucination detection signals—the system selectively retrieves and integrates supporting documents to reinforce the generation process [202]. This conditional retrieval not only suppresses hallucinations but also reduces latency and redundancy by avoiding unnecessary database queries [202]. Recent advances further extend this paradigm with adaptive confidence calibration and reinforcement-guided retrieval policies, enabling models to dynamically allocate retrieval resources across diverse tasks and domains [262]. For instance, the COFT framework introduces a coarse-to-fine highlighting mechanism that leverages knowledge graphs and contextual weighting to dynamically emphasize key lexical units, significantly improving hallucination mitigation in long and noisy contexts [263]. Collectively, Adaptive RAG demonstrates that hallucination mitigation can be achieved more effectively by integrating retrieval as a context-sensitive intervention, rather than as a uniform pre-processing step [202], [261].

b) Knowledge-based Methods: constitute a central strand of hallucination mitigation strategies that ground large language models (LLMs) in structured or semi-structured factual resources [264]. Unlike retrieval-augmented generation, which dynamically supplements model outputs with external documents at inference time, knowledge-based approaches integrate curated repositories more intrinsically into the reasoning and generation process [253]. By anchoring outputs to explicit semantic structures or verifiable factual stores, these methods constrain the model's generative space and suppress unsupported or fabricated continuations.

This category can be further divided into four major subtypes: (1) *Knowledge Graphs* [265], which encode entities and relations as structured graphs to enforce logical and factual consistency; (2) *External Knowledge Base Integration* [257], [266], which links models to domain-specific or general-purpose repositories such as encyclopedias, scientific corpora, or medical databases; (3) *Knowledge Injection* [253], [267], which embeds factual cues directly into model representations through lexical, syntactic, or embedding-level augmentation; and (4) *Knowledge Distillation* [268], which transfers factual grounding from stronger or specialized teacher models to student models via supervised training. Collectively, these approaches emphasize the role of structured factual grounding as a complement to probabilistic generation, providing reliable mechanisms for mitigating hallucinations across diverse tasks and application domains.

- **Knowledge Distillation** (KD) is a transfer learning paradigm in which a compact student model is trained to emulate the predictive behaviors or intermediate representations of a more capable teacher model [268]. While originally introduced for model compression, KD has increasingly evolved into a crucial method for hallucination mitigation [268]. By transferring factual grounding, consistency, and reasoning strategies from teacher to student, KD enables lightweight models to benefit from the robustness of large-scale LLMs without requiring direct access to proprietary corpora or costly reinforcement learning pipelines [268]. In this sense, KD serves as both an efficiency mechanism and an alignment strategy, embedding verifiable knowledge into models that are more computationally accessible [268].
  Recent works have extended KD into diverse settings for hallucination reduction. Sharma et al. adapt KD to zero-resource environments, showing that teacher-generated rationales and structured signals allow students to internalize verifiable reasoning even in the absence of labeled data [269]. McDonald et al. demonstrate that transferring calibrated probability distributions from robust teachers enables student models to filter out overconfident but factually incorrect outputs, enhancing factual alignment under noisy conditions [270]. Complementing these approaches, Nguyen et al. propose smoothing-based distillation, where entropy-constrained teacher signals reduce spurious correlations and strengthen factual precision in low-resource environments [84]. Collectively, these advances underscore KD as a scalable pathway for embedding structured teacher knowledge into student models, thereby improving factual reliability and mitigating hallucinations across diverse deployment scenarios.
- **Knowledge Graph** is a multi-relational structure that encodes factual knowledge as triples of entities and relations, providing a semantically rich and machine-readable framework for reasoning Unlike unstructured text corpora, KGs impose a structured representation where entities are represented as nodes and relations as typed edges, thereby enabling systematic inferencing, entity disambiguation, and relational grounding [265], [271]. This relational architecture makes KGs particularly well-suited for hallucination mitigation, as they offer explicit mappings between concepts and their attributes that LLMs can leverage for factual consistency. For example, in SPARQL-based question answering, hallucinations often arise when models attempt to generate Uniform Resource Identifiers (URIs) from parametric memory, leading to non-existent or incorrect entity identifiers. By integrating external KG-based retrieval or alignment modules, recent work demonstrates that grounding model generations in structured knowledge repositories substantially reduces hallucinated outputs and strengthens factual reliability [271].
  Building on this foundation, recent studies have explored various integration strategies. Guan et al. [159] propose a dual-phase alignment mechanism that incorporates KG entities to constrain semantic drift during generation. Similarly, Niu et al. [262] introduce a causal alignment approach, where KG reasoning paths are explicitly modeled to ensure output faithfulness. Yu et al. [176] extend this paradigm by leveraging cause–effect look-alike reasoning to detect and suppress hallucinations that arise from spurious correlations, showing how KG structures can be used not only for grounding but also for counterfactual validation of model outputs. Collectively, these approaches highlight KGs as both a factual anchor and a reasoning substrate, capable of transforming hallucination mitigation from post-hoc detection into proactive semantic alignment. Consequently, knowledge graph integration has emerged as one of the most effective strategies for addressing hallucinations in LLMs and LVLMs, especially in tasks requiring precise entity disambiguation and multi-hop reasoning
- **External Knowledge Integration** extends the generative capacity of language models by supplying them with accurate, up-to-date, and domain-specific information from external repositories. Unlike knowledge graphs
  that rely on structured relational reasoning, external sources such as Wikipedia, curated databases, and domain-specific corpora provide rich unstructured evidence that can be dynamically retrieved and embedded into the generation process. This augmentation enhances the model's ability to handle knowledge-intensive tasks while mitigating hallucinations caused by outdated or incomplete internal representations [252], [257].
  Recent advances move beyond static retrieval pipelines by introducing dynamic and adaptive mechanisms. For example, Chen et al. propose a context-aware dynamic supplementation framework that identifies hallucination-prone responses and selectively enriches them with retrieved evidence, thereby improving factual alignment without excessive overhead [189]. By flexibly integrating external evidence only when needed, these methods balance efficiency with accuracy, ensuring that external knowledge enhances the reliability of model outputs rather than overwhelming the inference process.
  Collectively, external knowledge integration demonstrates strong potential to ground large language models in verifiable information and to adapt seamlessly across evolving domains.

- **Knowledge Injection** (KI) represents a targeted approach to bridging the gap between structured symbolic knowledge and language model generation. Unlike retrieval-augmented methods that rely on external document search, KI directly incorporates curated knowledge elements—often sourced from knowledge graphs or curated databases—into the model's input space [267]. Typically, this involves transforming relevant entity or relation triples into natural language or embedding-compatible formats and inserting them into prompts, thereby ensuring that the model is exposed to domain-specific, factual signals during inference. This mechanism is particularly effective for grounding responses in contexts where the pretraining corpus lacks sufficient coverage, such as medical or legal domains [272].
  Recent research highlights multiple design paradigms of KI. For instance, HALO demonstrates that integrating structured knowledge with internal consistency checks can significantly suppress hallucinations in dialogue systems [29]. Similarly, Capellini et al. formalize KI as a modular pipeline in which factual triples from knowledge bases are aligned with model attention layers to guide reasoning, showing improvements in both faithfulness and contextual coherence across diverse tasks [272]. Compared to broader retrieval-based paradigms, KI provides a more precise and semantically interpretable channel of supervision, ensuring that model outputs remain anchored in verifiable context rather than heuristic approximations. As such, Knowledge Injection stands as a key strategy in knowledge-based hallucination mitigation, offering both transparency and robustness in high-stakes applications.

c) Verification-oriented consistency: represents a class of hallucination mitigation methods that emphasize cross-checking generated outputs against internal or external signals of coherence and factuality. Unlike resource-based retrieval or knowledge injection, which enrich the model's input with additional information, consistency-driven approaches operate primarily at the validation layer, systematically verifying whether the model's outputs align with known entities, established relations, or logically coherent structures. By leveraging the principle that reliable outputs must remain consistent across multiple perspectives, contexts, or decomposed subtasks, these methods suppress spurious continuations and expose unsupported claims. This category can be further divided into four major subgroups: (1) *Entity & Relation-based Verification*, which grounds model outputs in structured entity-relation triples to ensure semantic and factual alignment; (2) *Factual Consistency Verification*, which directly evaluates generated text against reference knowledge or parallel generations for contradiction detection; (3) *Chain-of-Verification (CoVe)*, which introduces an additional reasoning step where models verify their own or other models' outputs through iterative questioning and evidence checking; and (4) *Sub-question Decomposition*, including its application in Visual Question Answering (VQA), which breaks complex queries into smaller, verifiable units to reduce reasoning errors and enforce consistency across intermediate answers. Collectively, these techniques highlight the role of consistency as a powerful constraint for hallucination mitigation, ensuring that model predictions are not only fluent but also logically and factually dependable.

- **Entity & Relation-based Verification** is a class of consistency-guided methods that detect and mitigate hallucinations by aligning generated outputs with structured factual representations of entities and their relations.
  The core idea is that hallucinations often manifest as fabricated entities or incorrect relational links between entities mentioned in the output.
  By leveraging external databases or structured triples, models can verify whether generated entities exist and whether the asserted relations are semantically and factually valid [137]. Wei et al. highlight that entity-relation verification not only improves factual consistency but also provides interpretable error signals by tracing hallucinations to missing or conflicting relations in the knowledge source.
  This strategy can be particularly effective in knowledge-intensive tasks such as biomedical text generation and scientific question answering, where relational accuracy is critical for maintaining credibility [137].
- **Consistency Verification** focuses on validating the internal and external coherence of model outputs by comparing generated content against trusted references, alternative reasoning paths, or previously established knowledge. This mechanism operates on the principle that factually reliable responses should remain stable across paraphrased prompts, knowledge sources, or verification passes.
  Wan et al. propose a knowledge-enhanced consistency framework that integrates external evidence with multi-view reasoning, ensuring that generated statements align with verified knowledge bases and reducing the persistence of spurious correlations [273]. Similarly, Harrington et al. demonstrate that enforcing consistency across multiple generations and model variants can effectively flag hallucinations, as divergence among outputs often signals low factual reliability [213]. By systematically re-examining outputs across dimensions of knowledge, reasoning, and context, consistency verification strengthens both factual grounding and robustness, making it a versatile strategy for hallucination mitigation in knowledge-intensive tasks.
- **Chain-of-Verification (CoVe)** enhances factual reliability by introducing a multi-step validation process that systematically checks intermediate reasoning steps before producing a final answer.
  Unlike single-pass generation, CoVe decomposes the response into smaller claims, each of which is individually verified against either retrieved evidence or consistency checks, thereby reducing the risk of unsubstantiated as-

sertions.

Liang et al. demonstrate that this layered verification pipeline not only improves factual grounding but also increases robustness in tasks requiring complex, multi-hop reasoning [186]. By embedding verification directly within the generation workflow, CoVe exemplifies how iterative reasoning and structured checking can serve as effective safeguards against hallucination.

- **Sub-question Decomposition (including VQA)** tackles hallucinations by decomposing complex queries into a series of simpler, verifiable sub-questions [274]. Each sub-question, often grounded via retrieval or external evidence, is addressed individually before synthesizing the responses into a final answer.

  This structured breakdown enhances factual precision and interpretability, as it allows validation at every reasoning step and reduces error propagation in multi-hop reasoning settings.

  Chang et al. propose a unified hallucination mitigation framework that leverages such decomposition to align intermediate outputs with factual signals at multiple reasoning stages [201]. Similarly, Pelican—a verification-centric LVLM framework—translates visual claims into logical sub-claims that are independently verified via executable checks, thereby significantly reducing hallucinations in VQA tasks [275]. These approaches demonstrate that decomposing a complex problem into verifiable components is a robust safeguard against hallucinated conclusions.

In sum, Knowledge-Infused Mitigation Methods mitigate hallucinations by anchoring model outputs in reliable external signals, whether through retrieval mechanisms, structured knowledge resources, or consistency verification. These methods complement intrinsic interventions by addressing the limitations of parametric knowledge and enabling factual grounding across diverse domains. However, while effective, they often operate as discrete modules targeting specific vulnerabilities. To achieve broader robustness, recent research has shifted toward *Meta- and Framework-based Methods*, which integrate heterogeneous strategies into unified architectures and adaptive pipelines. These higher-level frameworks provide holistic designs that extend beyond single interventions, ensuring that knowledge access, reasoning control, and verification interact seamlessly across the generation process.

3) Meta/Framework-based Methods: Meta- and Framework-based Methods constitute the third category of hallucination mitigation strategies, distinguished by their operation at a higher level of abstraction compared to direct technical adjustments or resource-grounded interventions. Rather than modifying token selection or anchoring outputs in external repositories, these methods establish overarching architectures and governance mechanisms that regulate, coordinate, or augment the generative process.

As illustrated in Fig. 7, this category can be divided into four major subgroups. *Data and learning strategies* emphasize standardization and pre-processing to improve input quality and supervision consistency, thereby enhancing downstream

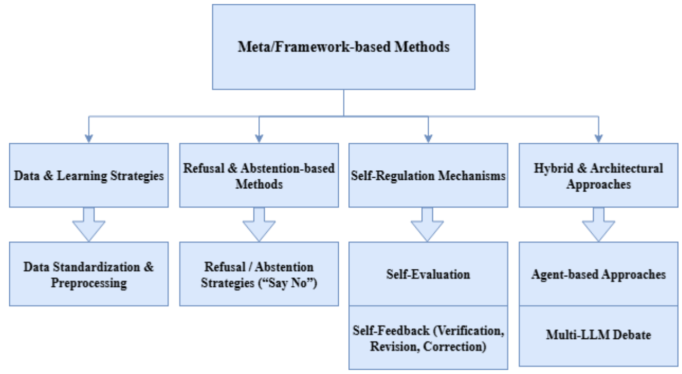

Fig. 7. Meta_Framework-based Methods

robustness. *Refusal and abstention-based methods* empower models to "say no" when confronted with queries that exceed knowledge boundaries or carry high hallucination risk. *Self-regulation mechanisms* incorporate self-evaluation and iterative feedback processes that enable models to internally verify, revise, or correct their outputs. Finally, *hybrid and architectural approaches*—including loop-based refinement, agent-based orchestration, and multi-LLM debate—build higher-order frameworks that integrate multiple perspectives or iterative decision cycles to systematically reduce hallucination propagation. Collectively, these methods highlight the transition from localized interventions toward comprehensive architectures, framing hallucination mitigation not as isolated corrections but as an integrated system of regulation and oversight.

a) Data and Learning Strategies: address hallucination mitigation by focusing on the quality, structure, and distributional properties of training and fine-tuning corpora. Unlike purely architectural or inference-level interventions, these methods target the upstream data pipeline, ensuring that the information supplied to large language models is both reliable and systematically curated. By enforcing consistency in formatting, reducing redundancy, and eliminating noisy or contradictory signals, such strategies enhance factual grounding while improving generalization across domains. Moreover, data-centric interventions operate synergistically with downstream methods, as models trained on standardized and well-preprocessed datasets are more amenable to self-regulation, external verification, and retrieval integration. Within this category, *data standardization and preprocessing* emerges as a foundational approach, establishing a structured and high-fidelity input base that supports both parameter optimization and post-hoc verification frameworks.

- **Data Standardization and Preprocessing** is a foundational strategy for hallucination mitigation that ensures training corpora are reliable, representative, and semantically consistent. The process begins with the removal of redundant and low-quality entries to prevent duplication and noise amplification, followed by automated and manual correction of typographical errors and normalization of linguistic formats [219]. Robust tokenization then segments text into standardized units, reducing variability in representation and improving the model's ability to capture meaningful patterns [276]. Filtering and annotation further refine datasets by eliminating biases, enriching contextual signals, and aligning inputs with domain-relevant knowledge, thereby supporting accurate downstream reasoning [277]. Beyond these conventional steps, advanced preprocessing strategies incorporate adaptive quality-control pipelines that detect domain-specific inconsistencies, enforce factual alignment, and systematically constrain error propagation at the source. By ensuring that the data ingested by the model is both clean and knowledge-grounded, preprocessing not only enhances the efficiency of learning but also serves as a preventive layer against hallucination-inducing artifacts [219].HalluciDoctor exemplifies such advanced preprocessing by detecting and removing hallucinations from training corpora through cross-checking with multiple models [278]. To further enhance robustness, it introduces counterfactual visual instruction expansion, which mitigates spurious correlations from long-tail multimodal co-occurrences. This combined strategy improves the factual reliability of training signals and strengthens the resilience of multimodal large language models against hallucinations [278].

b) Refusal and Abstention-based Methods: represent a distinct line of hallucination mitigation that focuses not on generating more content, but on deliberately withholding output when the model's confidence, reliability, or factual grounding cannot be guaranteed. Unlike strategies that attempt to correct or post-process potentially erroneous generations, refusal mechanisms proactively intervene in the decision process by equipping models with the ability to reject queries, abstain from uncertain predictions, or output explicit "don't know" responses. This paradigm shifts the emphasis from maximizing coverage of responses to prioritizing precision and safety, reducing the risk of confidently presented but incorrect outputs. Within this category, *Refusal / Abstention Strategies ("Say No")* constitute the primary instantiation, where models are trained or instructed to recognize high-risk prompts, detect factual uncertainty, and return cautious, noncommittal answers instead of speculative generations.

- **Refusal / Abstention Strategies ("Say No")** represent a principled paradigm in hallucination mitigation by allowing language models to withhold an answer when faced with high uncertainty or potential toxicity. Instead of enforcing output generation under all circumstances, these approaches empower models to strategically "say no," thereby reducing the risk of factual inaccuracies or harmful outputs [178]. Technically, abstention can be operationalized through internal uncertainty quantification, where the model evaluates consistency across multiple sampled responses and declines to answer when divergence exceeds a threshold [119]. Recent studies further emphasize that refusal behaviors are not merely reactive but can be shaped during fine-tuning. For instance, HonestAI introduces specialized supervision to encourage abstention in ambiguous scenarios, demonstrating that abstention-finetuned models significantly reduce spurious generations without compromising overall utility [117]. Similarly, Kang et al. show that the way unfamiliar finetuning examples are supervised critically influences hallucination patterns, and strategically labeling such examples with abstention signals (e.g., "I don't know") enables models to generalize refusal behavior to unseen queries [179]. These findings highlight refusal not as a fallback mechanism but as a proactive alignment strategy, where abstention supervision during supervised fine-tuning (SFT) or reinforcement learning (RL) can calibrate model responses under epistemic uncertainty [179]. Collectively, refusal/abstention strategies reframe the mitigation problem by institutionalizing uncertainty-aware non-response, transforming "not answering" into a valid and safe operational choice.
- **Rails** methods apply predefined rules or guardrails to regulate model behavior during generation. By constraining outputs within safe and factual boundaries, these approaches prevent models from producing unsupported or unsafe content. Unlike purely probabilistic decoding strategies, rails-based methods enforce explicit structural safeguards, complementing abstention policies by ensuring reliability even when models attempt to generate uncertain responses [279]

c) Self-Regulation Mechanisms: constitute a class of hallucination mitigation strategies that embed autonomous evaluation and correction directly within the model. Rather than relying solely on external supervision, these approaches enable large language models (LLMs) to act as both generators and inspectors of their own outputs. Two main paradigms are commonly employed. *Self-evaluation* performs introspective checks on factuality, coherence, or plausibility before delivering a response, while *self-feedback mechanisms* iteratively refine outputs through verification, revision, and correction loops. Together, they foster a paradigm of "autonomous reliability," where models mitigate hallucinations by continuously diagnosing and improving their own generations.

- **Self-Evaluation** techniques empower large language models (LLMs) to assess the reliability of their own outputs before producing final responses. This mechanism is grounded in uncertainty estimation, confidence calibration, and familiarity assessment, which together allow models to regulate their generative behavior. For instance, zero-shot self-evaluation frameworks introduce *self-familiarity scores*, enabling the model to determine whether it has sufficient prior knowledge to answer a query or should abstain to avoid hallucination [280]. Similarly, self-consistency and self-checking methods prompt the model to compare multiple reasoning trajectories, suppressing answers with high internal disagreement [161]. Other approaches integrate self-assessment within abstention frameworks, where the model estimates its own uncertainty and refuses to answer if a predefined confidence threshold is not met [119]. Collectively, these methods introduce a layer of internal governance, reducing unsupported claims by aligning model responses with its measured confidence and familiarity.

- **Self-Feedback (Verification, Revision, Correction)** extends the capacity of LLMs by enabling them to iteratively assess and refine their own outputs through verification, revision, and correction cycles. This paradigm builds upon techniques such as *self-checking*, *sliding generation*, and *feedback-guided refinement*, which collectively reduce factual inconsistencies and semantic drift. For instance, Harrington et al. [213] demonstrate that self-verification pipelines can proactively flag potential hallucinations by comparing generated claims against contextual cues, while Lee et al. [281] highlight iterative correction loops for multimodal grounding. Ensemble-based self-feedback, such as the N-CRITICS framework, further strengthens reliability by incorporating multiple critics to evaluate and guide refinements, thereby mitigating both toxicity and factual hallucinations [282]. Complementarily, Ji et al. [147] emphasize the role of correction-oriented prompting strategies, where LLMs generate structured feedback that is recursively applied to adjust prior outputs. Together, these approaches illustrate how self-feedback transforms LLMs into dynamic, self-regulating systems capable of maintaining factual alignment and robustness without heavy reliance on human supervision.

d) Hybrid and Architectural Approaches: represent a higher-order category of hallucination mitigation that transcends isolated techniques by embedding self-regulation and external reasoning directly into the structural design of generation systems. Instead of limiting interventions to a single phase, these methods construct holistic architectures that integrate multiple mechanisms into collaborative frameworks. *Agent-based approaches* assign specialized roles to distinct components or agents (e.g., generator, verifier, retriever), enabling division of labor and collective reasoning within a coordinated system. Likewise, *multi-LLM debate strategies* leverage adversarial or cooperative exchanges among multiple models to stress-test factual claims and uncover inconsistencies, turning disagreement into a source of verification. Together, these approaches move hallucination mitigation beyond adhoc corrections toward structured, system-level architectures that enhance robustness, factual grounding, and interpretability across diverse tasks.

- **Agent-based Approaches** represent a paradigm where multiple autonomous agents—each with distinct roles such as reasoning, retrieval, or verification—collaboratively address complex queries, thereby enhancing factual reliability and interpretability. Unlike single-model pipelines, agent-based orchestration distributes responsibilities across specialized subsystems, enabling division of labor and error-checking among agents. For example, chain-based multi-agent frameworks demonstrate how a reasoning agent generates hypotheses while a verification agent cross-checks with external resources, significantly reducing unsupported claims [283]. Recent studies further emphasize interoperability, where shared protocols and APIs allow heterogeneous agents from different origins to integrate outputs seamlessly, fostering scalable and robust ecosystems [10], [284]. By combining individual specialization with collective intelligence, agent-based methods create resilient architectures for hallucination mitigation that mirror human collaborative problem-solving.
- **Multi-LLM Debate** is a collaborative paradigm in which multiple language models iteratively challenge and refine each other's outputs to improve factual grounding and reasoning robustness. Instead of relying on a single generative trajectory, debate frameworks distribute reasoning across different agents, each proposing candidate answers and counterarguments, followed by an arbiter or aggregation mechanism to consolidate the most consistent and evidence-backed conclusions [285]. Recent work demonstrates that such debate protocols significantly mitigate hallucinations by fostering diversity of reasoning paths and enabling error detection through cross-examination. For example, Yoffe et al. [171] introduced Debunc, where adversarial debates between LLMs expose unsupported claims and enforce verifiability before consensus is reached. Similarly, Sng et al. [260] proposed a novel debate-and-arbiter framework that integrates consistency checks with multi-round deliberations, showing improvements in both truthfulness and reliability of responses. By maintaining multiple hypotheses and leveraging structured arbitration, multi-LLM debate transforms generation into a dialectical process, yielding more cautious and trustworthy outputs.
- **Reason and Act (ReAct)** represents an early form of agentic design within a single model, where chain-of-thought reasoning is interleaved with concrete actions such as retrieval or tool use. By coupling reasoning traces with actionable operations, ReAct enables models to ground intermediate steps in observable outcomes rather than relying solely on parametric knowledge. This tight integration improves factual reliability and lays the conceptual foundation for more complex multi-agent systems, where reasoning and verification can be distributed across specialized agents [279]
- **Dialog-Enabled Resolving Agents (DERA)** extend debate-style paradigms by orchestrating multiple dialog agents that iteratively question, critique, and refine one another's responses. Through structured conversational exchanges, DERA frameworks filter out unsupported claims and converge on factually grounded outputs. By combining adversarial challenge with cooperative resolution, this approach strengthens reliability while maintain-

ing interpretability in multi-agent settings [279]

Meta- and framework-based methods collectively represent a higher-order layer of hallucination mitigation, focusing less on isolated interventions and more on systemic architectures that embed robustness into the generative process. By leveraging strategies such as data standardization, abstention protocols, self-regulation, and hybrid frameworks—including iterative loops, agent collaboration, and multi-model debate—these approaches transcend single-point fixes and instead promote adaptive, scalable resilience against hallucination. Their unifying feature lies in establishing overarching infrastructures that integrate technical and resource-based mechanisms into coherent pipelines, ensuring not only immediate factual consistency but also long-term adaptability across diverse domains and tasks [201], [219].

Mechanism-based categorization, the first dimension of our taxonomy, highlights the breadth of strategies employed to mitigate hallucinations, ranging from technical interventions to knowledge-grounded methods and meta-frameworks. While this perspective clarifies the underlying principles that drive each approach, it does not fully capture the temporal dynamics of when and how these interventions are applied during the generation pipeline. To complement this mechanism-oriented view, the second dimension organizes hallucination mitigation methods along the phase-based classification, distinguishing interventions that occur before, during, and after generation. This shift in perspective enables a more process-centric understanding, revealing how diverse mechanisms can be systematically aligned with the stages of model operation, and how phase-specific as well as cross-phase strategies interact to form robust mitigation workflows.

B. Phase-Based Classification

The temporal dimension of hallucination mitigation underscores a key insight: *when* interventions are applied significantly influences both their effectiveness and computational cost. While the mechanism-based taxonomy (Section IV-A) addresses *what* strategies are used, this dimension focuses on *when* they occur across the model lifecycle.

Timing matters for several reasons. First, each phase presents distinct intervention opportunities: pre-generation methods can proactively prevent hallucinations but require anticipating failure modes; in-generation methods offer real-time control but introduce latency; post-generation methods enable comprehensive verification but cannot fix foundational errors. Second, integration complexity varies—pre-generation methods often involve one-time setup, in-generation strategies may require architectural changes, and post-generation tools typically operate as modular add-ons. Third, computational budgets differ sharply across phases: pre-deployment fine-tuning allows extensive computation, in-generation decoding must meet millisecond-scale constraints, and post-processing affords selective deep checks.

Accordingly, we organize mitigation strategies into three phases aligned with the generative workflow: (1) **Pre-generation Phase Mitigation**, which includes pre-training and inference-time configurations such as prompting and retrieval setup; (2) **In-generation Phase Mitigation**, involving real-time interventions to steer decoding away from hallucinations; and (3) **Post-generation Phase Mitigation**, which applies verification and revision mechanisms to the completed output.

1) Pre-generation Phase Mitigation: Pre-generation mitigation offers the most proactive defense against hallucinations by intervening before any output is generated. It includes both pre-deployment strategies (e.g., training, fine-tuning, data curation) and pre-inference configurations (e.g., prompting, retrieval setup, knowledge injection), all aimed at establishing strong factual guardrails in advance.

The key advantage of this phase lies in its preventive nature: instead of correcting errors post hoc, it reshapes the model's internal knowledge and input conditioning to reduce hallucination risk from the outset. For example, a legal generator fine-tuned on verified case law is less likely to fabricate precedents, while retrieval-augmented prompting anchors outputs in trusted sources. Together, intrinsic knowledge alignment and extrinsic factual grounding form a layered defense.

Pre-generation methods fall into three categories: model-intrinsic approaches that modify parameters, representations, or prompts; knowledge-infused methods that inject external factual information; and meta-/framework-level strategies that introduce abstention, self-regulation, or preprocessing constraints.

a) Model-intrinsic Methods: These methods act directly on the model or its inputs to improve factual reliability before decoding begins. Confidence-based triggering [97] refines prompts based on uncertainty signals, while ACPO [226] uses consistency across stochastic samples to self-calibrate factual preference pairs. Contrastive fine-tuning approaches like PURR [286] denoise corrupted outputs using retrieved evidence. Knowledge distillation methods—such as ZeroG [269], standard KD [270], and smoothed KD [84]—align token distributions with verified knowledge from teacher models. Prompt engineering also contributes: techniques like Visual Evidence Prompting [191] and Counterfactual Inception [188] embed verifiable cues or suppress spurious associations. In multimodal contexts, Multi-Frequency Perturbation (MFP) [287] weakens hallucination-inducing correlations in image features while preserving semantic fidelity.

b) Knowledge-Infused Methods: These methods inject factual content prior to generation. PGMR [271] enhances grounding by replacing placeholders with accurate knowledge graph URIs. Retrieval-Augmented Generation (RAG) incorporates relevant domain-specific [252] or open-domain [257] evidence into prompts. Knowledge Injection (KI) methods [267], [272] directly encode structured facts into model inputs or parameters. Knowledge Consistent Alignment (KCA) [273] further filters alignment data to avoid conflicts with the model's internal knowledge.

c) Meta-/Framework-based Methods: Framework-level approaches apply systemic constraints before inference. VIC [276] enforces reasoning-before-perception to reduce multimodal hallucinations. HRO [242] applies iterative Generate–Score–Improve loops under RLHF supervision. Refusal mechanisms like Self-Familiarity [280] estimate prompt familiarity to abstain from answering low-confidence queries without retraining. Survey studies [288] compare gradient-based (e.g., fine-tuning, RLHF, DPO) and non-gradient (e.g., retrieval, prompting) strategies for hallucination mitigation. Architectural innovations such as heterogeneous model collaboration [289] further orchestrate generators, classifiers, and retrieval modules to enhance controllability and extend mitigation to multimodal contexts.

2) In-generation Phase Mitigation: In-generation mitigation constitutes the most technically demanding stage of hallucination prevention, intervening precisely during token-by-token generation—when models are most susceptible to drift. Unlike pre-generation methods, which operate offline, or post-generation methods, which assess complete outputs, in-generation strategies must act in real time, balancing factual grounding with fluency and latency constraints. For instance, a medical assistant must detect and correct fabricated drug interactions mid-sentence before they propagate through the entire response.

These interventions fall into three major categories: model-intrinsic methods that steer internal states and decoding distributions; knowledge-infused methods that incorporate real-time evidence retrieval; and meta- or framework-level methods that embed structural constraints into the generation process. Each approach reflects distinct trade-offs among accuracy, computational cost, and adaptability.

a) Model-intrinsic Methods: These techniques operate at the neural or decoding level to suppress hallucination-prone trajectories. Representation steering approaches like TruthX [216] and Iter-AHMCL [124] continuously shift activations toward a "truthful subspace" to align outputs with factual priors. Search-based decoding methods expand the candidate space beyond greedy selection—Lookback Lens [211] re-anchors attention to earlier context, while Monte Carlo Tree Search [187] explores branching continuations, pruning those likely to hallucinate.

Contrastive and layer-editing strategies offer further granularity. Freeze-CD [170] contrasts frozen and fine-tuned distributions; M3ID [290] aggregates multi-model signals; and multimodal techniques such as HALC [169] and VCD [164] target vision-language inconsistencies. Recent innovations include Active Layer-Contrastive Decoding (ActLCD) [225], which triggers layer-wise contrastive cues upon detecting semantic drift, and Premature Layers Interpolation (PLI) [217], a training-free plug-and-play method that interpolates adjacent hidden states to refine representations mid-decoding, mitigating hallucinations without retraining.

b) Knowledge-Infused Methods: These methods integrate dynamic evidence retrieval into generation. Re-KGR [262] monitors token-level attribution and, upon detecting elevated hallucination risk, retrieves relevant knowledge graph triples for mid-generation grounding. This loop ensures factual alignment without requiring full regeneration, allowing the system to adjust outputs in response to emerging inconsistencies.

c) Meta-/Framework-based Methods: Framework-level techniques embed structural scaffolding into the generation pipeline. The Jodie prompt [291], for example, reformulates passages as attributed quotations, instructing the model to respond according to retrieved sources rather than relying solely on parametric memory. More generally, hierarchical reasoning templates and graph-structured workflows constrain generation paths and enforce stepwise consistency. These methods provide continuous factual alignment by embedding epistemic discipline directly into the decoding structure—without breaking autoregressive flow.

3) Post-generation Phase Mitigation: Post-generation mitigation reflects a core principle of trustworthy AI: trust, but verify. While pre- and in-generation strategies aim to prevent hallucinations, post-generation methods acknowledge that such failures remain inevitable. This phase thus functions as a critical safety net—capturing errors that earlier stages may overlook—while offering additional benefits such as access to global coherence signals and the ability to revise outputs without retraining the base model.

Working on fully generated outputs allows these methods to detect inconsistencies that span sentences or sections. For example, a financial report may contain individually plausible sentences, yet only post-hoc verification can identify contradictions in aggregate figures. Furthermore, these methods are modular and upgradeable, enabling rapid adaptation to evolving verification standards or domain-specific constraints.

Post-generation mitigation typically follows a detect–analyze–revise pipeline: claims (e.g., entities, relations, numerical values) are first extracted, then verified against external sources or internal logic, and finally revised, regenerated, or rejected based on confidence levels. This structured process supports three complementary strategies: model-intrinsic methods that assess internal coherence and reasoning chains; knowledge-infused methods that validate claims against trusted databases; and meta-framework methods that evaluate model epistemics and implement abstention mechanisms.

a) Model-intrinsic Methods: These approaches analyze reasoning traces and enforce internal consistency post hoc. Rowen [202] introduces a "retrieve only when needed" paradigm: an initial chain-of-thought (CoT) response undergoes multilingual perturbation and cross-lingual consistency checks, triggering retrieval-based revision only when inconsistencies emerge. This minimizes unnecessary evidence injection and potential noise. Other consistency-driven methods [137] remove unsupported entity–relation triples, generate correction prompts based on verified content, and re-check the revised outputs to ensure fidelity to grounded facts.

b) Knowledge-Infused Methods: Post-generation knowledge validation offers the most precise form of factual grounding, leveraging structured queries and rich verification logic. KGR [159] exemplifies this through iterative knowledge graph refinement: generated content is parsed into semantic triples, cross-checked against a knowledge graph, and unsupported claims trigger constrained regeneration. Rather than patching individual errors, this ensures contextual consistency around the corrected facts. In contrast, Sentence Trimming [150] adopts an aggressive approach—pruning unsupported tokens and using the verified skeleton as a template for regeneration. This semantic pruning guarantees that elaborations remain anchored in evidence.

c) Meta / Framework-based Methods: The most advanced methods move beyond verification, incorporating epistemic awareness into generation. Multi-model arbitration [260] assembles specialized models to evaluate outputs from diverse angles—factual accuracy, logical consistency, and domain expertise—and resolves conflicting assessments through an arbiter model equipped with retrieval capabilities. This is especially valuable in complex domains like law, where subtle factual or contextual missteps carry serious consequences.

Meta-cognition training [292] reorients generation toward confidence-aware answering. Multiple model variants attempt the same task, and their disagreement patterns are used to train a meta-cognitive module to predict uncertainty. At inference time, the system can return high-confidence answers directly, qualify medium-confidence outputs, or abstain entirely in low-confidence cases.

4) Summary of Phase-Based Mitigation: The phase-based classification highlights hallucination mitigation as a temporal trade-off among prevention, intervention, and correction. Each phase provides distinct leverage points but also entails characteristic costs and limitations.

Pre-generation methods serve as the first line of defense by shaping the model's knowledge base and decision boundaries before any token is generated. Their preventive nature allows them to address systematic biases and reduce hallucination risk at the source. However, they require substantial upfront investment—such as fine-tuning and dataset curation—and are less flexible in adapting to query-specific hallucinations or rapidly evolving information.

In-generation methods intervene dynamically during decoding, offering fine-grained control over intermediate states. They are well suited for detecting and suppressing inconsistencies as they emerge, making them effective at catching hallucinations at their point of origin. The trade-off lies in higher computational costs, since search-based or iterative decoding strategies can introduce latency, and in their limited capacity to address failures in global reasoning or long-range coherence.

Post-generation methods function as the final safeguard, enabling comprehensive verification and correction after an initial draft has been produced. Their strength lies in leveraging holistic context and external knowledge to identify errors that may elude local checks. They are also modular and can be updated independently of the base model. Yet, they cannot fully recover outputs generated from fundamentally flawed reasoning processes and may be computationally demanding when employing exhaustive fact-checking or validation pipelines.

Taken together, the three phases constitute a complementary pipeline: pre-generation methods lower baseline risks, in-generation techniques provide real-time monitoring, and post-generation safeguards reinforce overall trustworthiness. Designing effective systems requires balancing these stages according to application-specific priorities—latency in consumer applications, accuracy in professional tools, or reliability in safety-critical domains. Yet, this layered view also reveals a critical limitation: no single phase, nor even the three in isolation, is sufficient. Achieving robust hallucination control demands integrative strategies that coordinate interventions across phases, laying the foundation for cross-phase and holistic mitigation approaches.

C. Cross-Phase & Integrative Mitigation

The third dimension, Cross-Phase and Integrative Mitigation, advances a systems view of hallucination control organized around three pillars that operate in concert. As shown in Fig. 8, multistage mitigation coordinates interventions across the pre, in, and post generations, propagating grounding, uncertainty, and corrective signals forward to constrain search and steer decoding and backward to recalibrate subsequent runs, thus achieving compounded reductions in hallucination risk. Multi-method fusion then orchestrates heterogeneous techniques (e.g., retrieval augmentation, parameter-efficient adaptation, structured prompting, constraint-aware decoding, verification and editing, knowledge-graph reasoning, uncertainty-aware detectors, etc.) so that complementary strengths align against distinct failure modes while balancing precision, coverage, and efficiency. Finally, loop- and iteration-centric optimization establishes closed-loop evaluation–correction cycles (e.g., self-checks, counterfactual re-decoding, feedback-guided revision, selective human-in-the-loop, etc.) to progressively stabilize factual reliability and update system priors. Framed in this manner, hallucination mitigation emerges not as a discrete, phase-constrained intervention, but as a dynamic, signal-coupled pipeline in which guidance and correction are continuously propagated, integrated, and reused across time and method. The subsections that follow examine each pillar in turn, articulating their operative mechanisms, inter-phase linkages, methodological synergies, and implications for the design of resilient, end-to-end mitigation systems.

1) Multi-Stage Mitigation: Multi-stage mitigation refers to strategies that span multiple phases of the generation process—pre-generation, in-generation, and post-generation—so that guidance, correction, and verification mechanisms interact throughout the pipeline. Rather than confining mitigation to a single point of intervention, these approaches establish functional continuity across stages, enabling information (e.g., retrieval signals, uncertainty estimates, corrective feedback) to propagate and reinforce model behavior both prospectively and retrospectively. For instance, THaMES combines parameter-efficient fine-tuning (pre-generation), retrieval-augmented grounding (in-generation), and in-context verification prompting (post-generation), dynamically select-

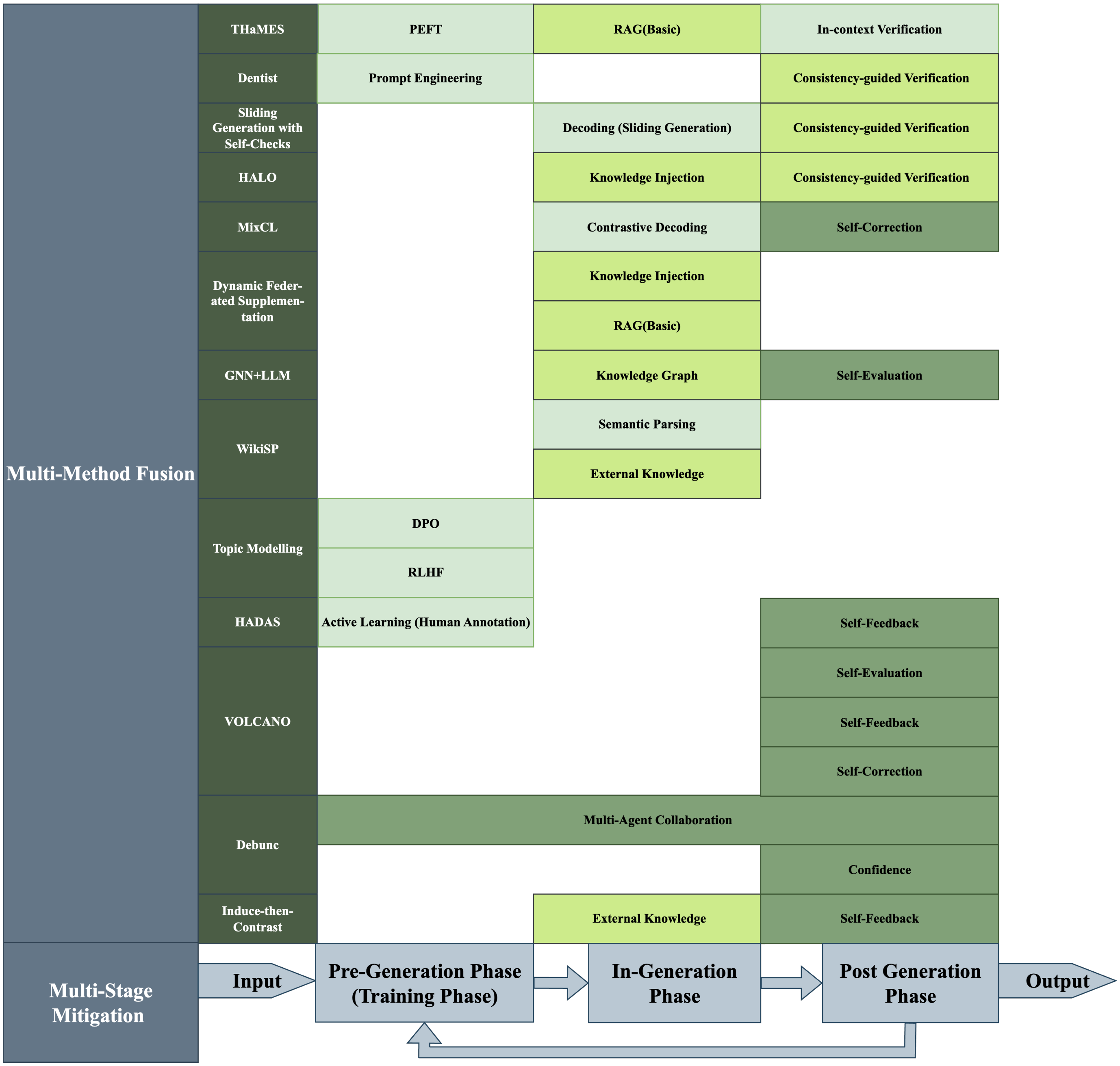

Fig. 8. Cross-Phase and Integrative Mitigation Approaches Summary

ing among these paths based on task and model characteristics [186]. Similarly, Dentist orchestrates chain-of-thought prompting, reasoning classification, and multimodal verification in a looping diagnose–verify–revise cycle, coordinating distinct functions across stages [201]. The Sliding Generation with Self-Checks pipeline likewise operates at multiple levels: it structures the input into overlapping spans during decoding, and post-processes the output via consistency-based verification [213]. These examples illustrate how multi-stage methods align the architecture and logic of hallucination mitigation with the temporal structure of generation itself.

2) Multi-Method Fusion: Multi-method fusion integrates diverse hallucination mitigation techniques—such as fine-tuning, retrieval augmentation, knowledge injection, attention control, contrastive learning, verification, and semantic parsing—within unified systems to exploit their complementary strengths. These approaches target multiple hallucination failure modes simultaneously, including semantic drift, factual inconsistency, contextual irrelevance, and unverifiable outputs, thereby enhancing robustness across tasks and domains.

For instance, HALO injects Wikidata-derived factual knowledge into a base LLM and uses consistency-based gating to selectively invoke a stronger teacher model for high-risk cases [29]. MixCL applies contrastive fine-tuning to align model outputs with gold-standard responses while actively repelling hallucinated candidates [162]. Dynamic Federated Supplementation grounds generation in real-time search engine results, injecting corroborative snippets during decoding to reduce reliance on parametric memory [189]. A hybrid GNN+LLM model incorporates knowledge graph inference to compute belief scores for candidate facts and filter unsupported claims [277].

Other systems leverage representational optimization: one method fine-tunes attention maps to refocus decoding on context-relevant tokens, reducing drift from input context [219]. WikiSP performs semantic parsing over Wikidata to return verifiable, structured answers via SPARQL, clearly distinguishing factual outputs from educated guesses [293]. Another approach uses Direct Preference Optimization to train the model on topic-based preference pairs, suppressing hallucinated outputs by reinforcing alignment with accepted content clusters [192]. Collectively, these methods exemplify the modular and synergistic design of modern hallucination mitigation systems.

3) Loop- and Iteration-Centric Optimization: Loop-Based and Iterative Optimization represents a class of techniques that reduce hallucinations through cycles of repeated generation, verification, and refinement. Instead of relying on single-pass outputs, these methods operationalize iterative feedback loops, where each round of model output is systematically re-evaluated, revised, or supplemented with external signals before finalization. The HADAS framework, for example, employs hallucination-diversity aware active learning, iteratively sampling and correcting outputs to refine predictions and ensure broader coverage of hallucination-prone cases [25]. Truth-oriented iterative refinement frameworks embed explicit factuality checks within generation cycles, enforcing alignment with verifiable evidence across multiple turns [294]. Unified multi-step pipelines dynamically integrate verification and correction into structured refinement loops [201]. In multimodal contexts, iterative mechanisms such as sliding generation and Volcano-based correction aggregate intermediate representations across multiple rounds to stabilize text–vision alignment and suppress hallucinations [281]. Multi-agent systems like DebUnc further extend the loop-based principle by enabling peer critique and confidence-sharing among LLMs, discounting unreliable responses while iteratively refining collective outputs [171]. Other methods, such as Induce-then-Contrast Decoding, leverage an auxiliary hallucination-prone model to define a penalty signal, iteratively suppressing unsupported outputs during generation [224]. Collectively, these loop-driven paradigms highlight the importance of self-corrective processes, positioning hallucination mitigation as an ongoing and dynamic refinement rather than a one-shot intervention [242], [282], [295].

D. Proposal: An Integrative Hybrid Framework

Despite the proliferation of hallucination mitigation techniques—including prompt engineering, decoding strategies, retrieval-augmented generation, fine-tuning, and multi-agent collaboration—no single method has achieved universal effectiveness across tasks, domains, or modalities. A systematic review reveals that while these techniques may succeed in narrow contexts, their isolated application fails to deliver the robustness required for real-world deployment. This shortfall stems from three core issues: (i) many methods operate within rigid phase boundaries, fragmenting the generative process and allowing hallucinations to propagate unchecked across disconnected stages; (ii) single-method strategies leave systemic vulnerabilities exposed, offering weak points for hallucination emergence; and (iii) static interventions lack the adaptability needed to address the evolving nature of hallucination patterns. Furthermore, recent theoretical work suggests that hallucination is not a removable defect but an intrinsic outcome of the architectural and training constraints of large language models [296]. This recognition shifts the research focus away from the unrealistic pursuit of complete elimination and toward the principled design of sustainable, layered mitigation strategies. Motivated by these insights, we propose an integrative hybrid framework that transcends piecemeal interventions through adaptive, cross-phase orchestration aimed at balancing robustness, efficiency, and scalability in practice.

1) Architectural Overview: The proposed framework is grounded in a systematic analysis of where and why hallucinations arise within current generation pipelines. Our investigation shows that vulnerabilities cluster around three computational stages, each shaped by distinct information-processing constraints and error-propagation mechanisms. This motivates a three-phase design that positions complementary mitigation strategies at points of maximum leverage while maintaining seamless information flow across stages.

The architectural design rests on several key insights into hallucination dynamics. First, preventive interventions at early stages are generally more effective and computationally efficient than corrective measures applied after generation. Second, token-level generation imposes strict strict real-time constraints, requiring lightweight mechanisms that operate within millisecond budgets. Third, post-generation verification enables comprehensive scrutiny through computationally intensive methods that would be infeasible during real-time decoding. Together, these insights motivate a layered, closed-loop architecture that integrates preventive, real-time, and corrective safeguards into a unified framework for hallucination mitigation, as illustrated in Fig. 9.

2) Phase I — Pre-Generation Components: The first phase implements proactive risk reduction under the principle that prevention is substantially more efficient than post-hoc correction. Analysis of hallucination dynamics shows that most errors originate from insufficient information, inadequate knowledge grounding, or misaligned model priors [1]. To address

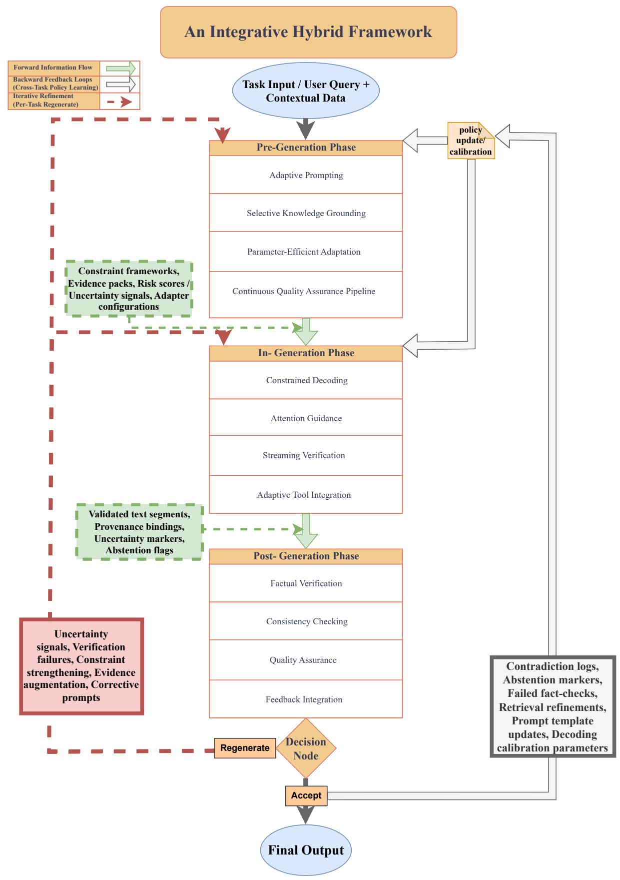

Fig. 9. Overview of the Integrative Hybrid Framework

these root causes, the pre-generation phase integrates four complementary components:

a) Adaptive Prompting: Traditional prompting applies uniform templates regardless of domain risk or query complexity. Our adaptive prompting mechanism calibrates prompt specificity based on real-time risk signals. For routine queries, lightweight prompts incorporate factuality cues and few-shot examples to maintain efficiency. In high-risk cases—identified via uncertainty estimation or domain classification—prompts are enriched with verification directives (e.g., "verify claims against evidence") and abstention clauses (e.g., "do not speculate without support"). This dynamic strategy embeds behavioral safeguards into the prompt itself, constraining hallucination onset while preserving fluency [16], [48]. Domain-specific patterns further motivate this approach: medical queries require stricter evidential grounding than general knowledge, while mathematical tasks benefit from explicit reasoning scaffolds—all addressed without retraining.

b) Selective Knowledge Grounding: Conventional retrieval-augmented generation often introduces irrelevant or contradictory evidence, increasing noise and computational cost. We address this through risk-aware retrieval, selectively triggered when task analysis indicates potential knowledge gaps. Retrieved content passes through a multi-stage filtering pipeline—including relevance scoring, contradiction detection, and source quality assessment—before integration. Retained evidence is annotated with provenance metadata (source identity, confidence, recency), enabling downstream modules to differentiate strong support from weak assertions. This targeted, rigorously filtered grounding enhances factuality where needed while minimizing unnecessary overhead.

c) Parameter-Efficient Adaptation: Full-model fine-tuning is resource-intensive and prone to forgetting [297]. Our framework employs domain-specific LoRA adapters that inject lightweight factual priors while preserving general reasoning. Trained on balanced datasets containing both correct and hallucination-prone examples, these adapters internalize truth-aligned patterns and penalize spurious generations. During inference, uncertainty-based routing activates domain-relevant adapters (e.g., biomedical, engineering) matched to task needs. Adapter training incorporates contrastive learning, reinforcing factual statements with positive feedback and suppressing hallucinations via negative signals. This approach supports domain generalization, factual robustness, and efficient deployment.

d) Continuous Quality Assurance Pipeline: To remain responsive to evolving hallucination patterns and novel domains, we integrate a continuous quality assurance pipeline based on active learning [298]. The system monitors domain-level accuracy and triggers adapter updates when performance degrades. High-uncertainty samples—detected via epistemic uncertainty metrics such as dropout variance [299] or predictive confidence scores [300]—are prioritized for expert annotation. This strategy focuses human effort on the most informative cases, maximizing annotation efficiency while avoiding large-scale manual labeling. As a result, the model continually adapts to distributional shifts and maintains resilience against emerging hallucinations.

3) Phase II — In-Generation Controls: The generation phase operates under unique constraints that distinguish it from pre- and post-processing stages. Token-level predictions must be produced within microsecond timescales while maintaining fluency and coherence, which limits the feasibility of heavy interventions. To address these challenges, our framework introduces a coordinated set of lightweight yet complementary mechanisms, constrained decoding, attention guidance, streaming verification, and selective tool integration, each targeting distinct vulnerabilities in real-time generation while collectively reinforcing factual reliability.

a) Constrained Decoding: Standard decoding often favors statistically likely continuations, even when unsupported by evidence, leading to fluent but factually incorrect outputs. To counter this, we apply contrastive decoding [301], which down-weights tokens favored solely by shallow priors while amplifying those supported by pre-retrieved evidence. This reweighting aligns generation with verifiable content without compromising fluency. Additionally, for high-risk tokens identified via uncertainty estimation, we employ uncertainty-aware beam search [302], preserving multiple candidates until downstream verification resolves ambiguity. This delayed commitment mitigates early hallucination and improves factual grounding.

b) Attention Guidance: Transformer attention [303] governs information flow during generation, yet often disperses focus across irrelevant spans [304]. Our framework introduces dynamic attention biasing, emphasizing entities, dates, and factual anchors identified in pre-generation. This prioritization enhances alignment with fact-critical inputs while preserving fluency. In long-context scenarios, we adopt sliding-window attention [305] to maintain focus on recently referenced evidence and avoid neglecting mid-sequence content. These mechanisms convert attention from passive weighting into an active mechanism for factual reinforcement.

c) Streaming Verification: Long-form generation risks cascading errors, where early inaccuracies compound downstream. Rather than relying solely on post-hoc correction, we implement segmented generation with streaming verification: each factual unit is immediately validated before commitment. Fact-intensive segments receive strict checks, while routine content proceeds under lightweight scrutiny. Upon detecting inconsistencies, the system triggers localized regeneration under stricter constraints. This inline validation prevents error propagation, offering greater factual reliability at reduced computational cost compared to full-output verification.

d) Adaptive Tool Integration: Some queries exceed parametric model capacity—e.g., arithmetic, relational reasoning, or real-time information. Existing systems either neglect external tools or overuse them, incurring latency. Our framework adopts uncertainty-triggered integration, invoking tools only when model confidence falls below learned thresholds. A modular toolkit handles numerical computation, knowledge graph queries, and web retrieval, with each invocation recorded via provenance metadata for traceability. A lightweight orchestration controller manages tool selection in real time, adapting to task demands while optimizing cost–accuracy tradeoffs. Over time, feedback-driven policy updates improve targeting, transforming tool use into a precision-guided safeguard against hallucination.

4) Phase III — Post-Generation Verification: While the in-generation phase provides lightweight, real-time safeguards against hallucination, its interventions remain necessarily constrained by latency budgets and token-level granularity. As a result, many hallucinations that are locally plausible or statistically fluent can still escape detection during generation. The post-generation phase addresses this gap by treating model outputs not as final products but as provisional hypotheses requiring rigorous justification before release. Free from the strict latency constraints of decoding, this phase enables computationally intensive verification, consistency analysis, and repair strategies that systematically uncover subtle factual errors, logical contradictions, and unsupported claims. Positioned as the final line of defense, the post-generation phase not only strengthens reliability through multi-tiered verification pipelines but also generates structured feedback that propagates backward, transforming verification outcomes into actionable signals for continuous system improvement.

a) Factual Verification: The first verification layer systematically evaluates generated claims against supporting evidence, leveraging provenance metadata and automated fact-checking tools [306], [307]. Given the impracticality of real-time human review, span-level entailment models [308], [309] assess the logical consistency between outputs and retrieved evidence, distinguishing substantiated claims from unsupported or contradictory ones. When evidence is lacking, targeted retrieval acquires claim-specific sources to avoid exhaustive re-querying. In parallel, entity–relation auditors cross-check named entities, relations, and temporal attributes against curated knowledge bases, detecting fabricated entities and subtle inconsistencies beyond entailment scope. This multi-pronged approach ensures factual verification that is both comprehensive and fine-grained.

b) Consistency Checking: Factually accurate outputs can still undermine reliability if they contain internal contradictions. This tier decomposes complex responses into atomic claims, enabling systematic cross-referencing and re-verification via paraphrasing or alternate reasoning paths. Detected inconsistencies are addressed through localized revision—guided by evidence-anchored constrained generation—preserving the coherence and fluency of the response. Uncertainty calibration complements this process by assigning confidence scores based on evidence quality, claim complexity, and model performance. Together, these mechanisms ensure not only isolated factual accuracy but also holistic logical consistency.

c) Quality Assurance: In high-stakes contexts where automation may be insufficient, the final tier enforces escalated review procedures tailored to content sensitivity. Outputs with unresolved uncertainty undergo secondary verification by specialized accuracy-focused models, which offer superior error detection. For critical domains—legal, biomedical, or policy-related—human review is triggered based on explicit escalation criteria. Crucially, the system enforces abstention when confidence thresholds are unmet, opting for calibrated uncertainty expressions over speculative claims. This tier safeguards the integrity and accountability of outputs under adversarial or uncertain conditions.

d) Feedback Integration: Rather than serving as a terminal checkpoint, post-generation verification drives continuous improvement across the generation pipeline. Failed fact-checks inform prompt template refinement and tighter factual constraints; retrieval strategies are adapted to improve coverage and evidence quality; and decoding parameters are recalibrated to mitigate recurring error patterns. These feedback signals transform the system into a self-adaptive architecture, progressively aligning generation components with empirical verification outcomes. As a result, hallucination mitigation evolves from a static correction mechanism into a dynamic, learning-centered process for sustained reliability.

5) Cross-Cutting Mechanisms: Beyond phase-specific safeguards, the framework integrates cross-cutting mechanisms that span the entire generation pipeline to ensure coherence, adaptability, and accountability. These mechanisms provide continuity across stages, transforming fragmented interventions into a unified system of hallucination management.

a) Risk Assessment: Dynamic risk profiling determines when and how mitigation modules are activated. By leveraging indicators such as model uncertainty, query sensitivity, and domain-specific risk levels, the framework adapts the intensity of interventions to context. Computationally costly methods are thus reserved for high-stakes or ambiguous cases, while low-risk queries are handled by lightweight safeguards.

b) Provenance Tracking: Each factual claim is linked to supporting evidence through structured metadata, including source identity, retrieval confidence, and temporal validity. Provenance trails enhance transparency, enabling systematic verification, regulatory compliance, and informed human oversight.

c) Evaluation Logging: Comprehensive logging of model outputs, verification results, and user feedback provides a continuous stream of performance data. These records support adaptive learning by revealing recurring error patterns and feeding them back into prompt design, retrieval indices, and decoding strategies, thereby reinforcing the framework over time.

d) Human-in-the-Loop Checkpoints: In safety-critical applications, automated safeguards are complemented by expert review. The framework incorporates configurable checkpoints where uncertain or unverifiable outputs are escalated to human oversight. This hybrid structure balances efficiency with judgment, enhancing resilience across diverse operational contexts.

Together, these mechanisms reinforce the three-phase architecture by ensuring that signals of guidance, verification, and correction circulate across the pipeline rather than remaining confined to isolated modules. This systemic integration establishes a foundation for scalable, transparent, and trustworthy deployment of large language models in real-world contexts.

6) Advantages vs. Known Limitations: The integrative hybrid framework systematically addresses seven recurring weaknesses of existing hallucination mitigation approaches—namely, resource intensity, data dependency, limited generalization, phase isolation, training misalignment, evaluation inconsistency, and systemic vulnerabilities. By distributing safeguards across phases and embedding cross-cutting mechanisms, the framework converts these challenges into structured advantages.

a) Robust Error Prevention: Unlike isolated techniques that create single points of failure, the hybrid framework employs defense-in-depth: failures in one component (e.g., retrieval errors) are offset by complementary mechanisms such as prompt constraints, decoding steering, or post-generation verification. This layered redundancy minimizes the likelihood of hallucinations bypassing all safeguards, particularly in adversarial or high-stakes scenarios.

b) Computational Efficiency: Resource-intensive interventions are selectively triggered based on risk assessment. Expensive modules—such as database queries or expert review—are reserved for high-stakes cases, while low-risk queries are managed through lightweight defenses. This adaptive allocation balances accuracy with scalability, enabling effective operation under large-scale workloads.

c) Adaptive Learning: Verification outcomes are converted into continuous feedback loops that refine prompts, retrieval strategies, and decoding parameters without necessitating full retraining. This feedback-driven adaptation enables the system to evolve with shifting hallucination patterns and domain-specific updates, sustaining reliability in dynamic contexts such as healthcare or finance.

d) Transparency and Auditability: By integrating provenance tracking, calibrated confidence scores, and explicit uncertainty markers, the framework enhances both factual accuracy and accountability. Each claim can be traced to its evidence source, supporting error analysis, regulatory compliance, and trustworthy human oversight in high-stakes environments.

e) Modular Extensibility: The architecture is designed for sustainability: new detection algorithms, verification methods, evaluation protocols, and domain knowledge bases can be incorporated without structural redesign. This modularity ensures adaptability to evolving research and application needs, providing a principled foundation for future advancements.

Taken together, these advantages demonstrate how the framework not only mitigates known weaknesses but also establishes a scalable, transparent, and resilient pathway for real-world deployment of large language models.

7) Framework Summary & Expected Benefits: In summary, the integrative hybrid framework consolidates preventive, in-generation, and post-generation safeguards into a closed-loop architecture that addresses hallucination risks at multiple levels. By aligning cross-phase continuity with multi-method synergy, it not only mitigates the onset of hallucinations but also enforces rigorous verification and adaptive learning throughout the deployment cycle. The design emphasizes layered redundancy, risk-aware resource allocation, and continuous feedback integration, treating hallucination mitigation as an evolving systemic process rather than a static intervention.

The framework delivers several benefits. First, its defense-in-depth structure enhances robustness by preventing single points of failure. Second, uncertainty-aware resource allocation ensures computational efficiency, scaling effectively where exhaustive verification is impractical. Third, feedback-driven adaptation enables long-term reliability under dynamic conditions such as medicine, finance, or law. Fourth, provenance tracking and calibrated uncertainty markers improve transparency and accountability, facilitating compliance and human oversight. Finally, its modular extensibility guarantees sustainability, allowing new methods and resources to be integrated without disruptive redesign.

Together, these benefits establish a principled foundation for managing hallucinations at robust reliability levels. The illustrative medical case study confirms that the framework improves factual accuracy while producing auditable and trustworthy outputs, underscoring its value for high-stakes deployments requiring transparency, resilience, and long-term sustainability.

E. Synthesis and Roadmap

Mitigating hallucinations in large language models is inherently multifaceted, requiring the integration of diverse strategies across mechanisms, phases, and feedback loops. From the survey of current approaches, three insights emerge. First, *no single method is sufficient*: prompt engineering, decoding strategies, retrieval grounding, and verification modules each address different facets of the problem yet remain limited in isolation. Second, *coordination across phases is essential*: lightweight preventive mechanisms in pre- and in-generation stages reduce error propagation, while post-generation verification provides rigorous correction and auditing. Third, *feedback and adaptability are indispensable*: dynamic thresholds, iterative refinement, and human-in-the-loop evaluation enable continuous alignment with user expectations and evolving risk levels.

Building on these insights, the roadmap for future research highlights three directions: (1) **Holistic integration**: developing unified architectures that combine intrinsic model interventions, knowledge infusion, and meta-level orchestration, as exemplified in hybrid frameworks. (2) **Risk-aware customization**: tailoring mitigation pipelines to domain requirements, latency constraints, and tolerance levels—for instance, applying stricter abstention and verification in high-stakes fields such as medicine. (3) **Sustainable evolution**: designing modular systems capable of incorporating new detectors, knowledge bases, and evaluation protocols over time.

In conclusion, hallucination mitigation is best approached as a continuous, layered process rather than a one-time intervention. The proposed integrative hybrid framework provides a principled pathway forward, combining early prevention, real-time control, and post-hoc verification into a closed-loop system. This synthesis underscores the importance of modularity, transparency, and adaptability in building trustworthy AI systems for real-world deployment.

VI. Evaluation of Hallucination in GenAI Systems

A. Important Issues about Hallucination Evaluation

The evaluation of hallucinations in generative AI systems has gained increasing attention in recent years, underscoring the urgency of this challenge. Effective assessment requires comprehensive coverage across diverse tasks, domains, and modalities. However, the research community has thus far produced a fragmented landscape of evaluation approaches. The proliferation of benchmarks and methodologies highlights both the significance of the problem and the absence of consensus on how best to conduct evaluation. Although this diversity has generated valuable insights into specific aspects of hallucination, it has also resulted in a scattered ecosystem in which findings are difficult to compare, synthesize, or translate into actionable improvements.

1) Current Evaluation Landscape: Categories and Limitations: Existing hallucination evaluation approaches can be categorized into several primary paradigms, each with distinct focuses and fundamental limitations that prevent comprehensive assessment.

Task-format evaluations concentrate on specific operational modes, including question-answering benchmarks like TruthfulQA [310], summarization evaluators such as FactCC and QAGS [311], [312], and dialogue assessment tools like FaithDial [34], [313]. These approaches focus primarily on surface-level task performance rather than examining the underlying reasoning processes. They fail to capture multistep inference patterns and provide limited ability to identify where in the reasoning chain hallucinations originate, making it difficult to develop targeted mitigation strategies [314].

Modality-Based Evaluation introduces an additional axis of hallucination assessment, distinguishing between text-only and multimodal evaluation settings. Text-only evaluation primarily targets factuality and faithfulness in purely linguistic contexts, while multimodal evaluation extends the scope to include visual grounding, temporal reasoning, and cross-modal alignment, thereby addressing hallucinations that arise from incomplete or inconsistent integration of heterogeneous modalities.

Hallucination-Type–Based Evaluation further categorizes evaluation efforts according to representative manifestations of hallucinations, each illuminating distinct aspects of LLM behavior. These categories include: factual hallucinations (fabricated or incorrect world knowledge), reasoning hallucinations (errors in logical inference), memory-based hallucinations (inaccurate recall of conversational or contextual history), contextual hallucinations (misinterpretation of provided inputs), multimodal hallucinations (fabricated or mismatched cross-modal content), temporal or knowledge staleness (outdated or temporally inconsistent claims), and intrinsic versus extrinsic hallucinations (stemming from model-internal reasoning versus external contextual mismatch). Systematically evaluating these categories not only exposes the breadth of failure modes but also clarifies their differential impacts on reliability, interpretability, and safety across applications.

Domain-specific evaluations test factual accuracy within specialized knowledge domains through benchmarks such as MedQA for medical knowledge [315], LegalBench for legal reasoning, and ScienceQA for scientific facts. While these provide valuable domain-specific assessment, they suffer from limited cross-domain generalization, inability to assess reasoning transfer between domains, and vulnerability to data contamination as static test sets become incorporated into training data over time [112], [316].

Furthermore, evaluation methods vary widely across operational dimensions, including granularity (e.g., token-level, sentence-level, or document-level assessment), reference availability (reference-based, reference-free, or hybrid approaches), the balance between automation and human involvement, as well as the core evaluation dimensions—such as factuality, faithfulness, consistency, robustness, causal reasoning capabilities, and interpretability.

In particular, capability-oriented evaluations represent a more recent development which introduces Benchmarks such as HotpotQA for multi-hop reasoning [317] and chain-of-thought–based evaluations [47] to probe models' ability to handle complex reasoning tasks. However, despite these advances, such approaches remain limited in several ways: they do not systematically decompose reasoning chains, they offer only partial interpretability in attributing errors to specific reasoning steps, and they provide minimal flexibility for user-defined evaluation criteria tailored to deployment-specific needs [318].

As a result, existing methods provide uneven coverage across these dimensions, leading to fragmented and incomplete assessment landscapes [319]. This underscores the need for evaluation frameworks that integrate both horizontal coverage across domains and vertical decomposition of reasoning chains, thereby enabling interpretable error attribution and customizable criteria that reflect real-world deployment contexts.

Methodologically, researchers have developed diverse techniques, such as probe-based evaluation (e.g., POPE [60]), adversarial and stress testing (e.g., ReEval [320]), causal intervention and counterfactual analysis (e.g., TruthPrint [321]), uncertainty-guided and self-consistency checks (e.g., SelfCheckGPT [19]), and internal state analysis (e.g., attention patterns [322]). While these methods offer unique strengths, the lack of a unified framework makes results difficult to compare [16].

The design of evaluation tools and frameworks also faces challenges, including benchmark construction requirements (e.g., generalization, mitigating human bias), dataset construction phases (e.g., raw data acquisition, QA pairing, annotation), and the development of automated pipelines. Current tools often focus on specific aspects, lacking end-to-end solutions that limit their practicality [61].

2) The Evaluation Gap: Horizontal vs. Vertical Assessment: Current evaluation methods operate primarily at what we term the "horizontal" level, assessing performance breadth across various domains or tasks without examining the underlying reasoning process. This horizontal evaluation, while useful for benchmarking and model comparison, misses the critical "vertical" dimension: the hierarchical, step-by-step inference chain that leads to final outputs and determines where hallucinations originate. The detailed comparison between Horizontal Assessment and Vertical Assessment is summarized in Table II.

TABLE II. COMPARISON OF HORIZONTAL AND VERTICAL EVALUATION

Dimension	Horizontal Evaluation	Vertical Evaluation
Definition	Measures model performance breadth across domains, tasks, or datasets.	Examines inference chains step-by-step, focusing on intermediate reasoning.
Focus	Final output correctness across different contexts.	Process-level correctness at each reasoning step.
Strengths	Useful for benchmarking, model comparison, and cross-domain assessment.	Reveals where hallucinations originate and how errors propagate.
Limitations	Ignores reasoning process; cannot localize errors.	Narrower scope; requires detailed process tracing and interpretability.

Consider a complex query that requires multiple reasoning steps. Horizontal evaluation simply measures whether the final answer is correct across different domains or task types. In contrast, vertical evaluation examines each intermediate reasoning step, identifying precisely where errors emerge and how they propagate through the inference chain. This distinction is crucial because a model might produce a correct final answer despite flawed intermediate reasoning, or conversely, make a single early error that cascades into complete failure.

Current frameworks lack three critical capabilities that prevent comprehensive evaluation. First, they cannot accommodate user-oriented customization, preventing organizations from creating benchmarks from their own information sources tailored to specific deployment contexts [323]. Second, they provide minimal process transparency, offering scores without interpretable insights into the reasoning process, making it impossible to identify where and why hallucinations occur [66], [95]. Third, no existing framework effectively combines horizontal domain coverage with vertical reasoning analysis, forcing evaluators to choose between breadth and depth rather than achieving both.

This gap is particularly problematic for organizations deploying LLMs in specialized domains. A healthcare provider cannot simply rely on general medical benchmarks but needs evaluation tailored to their specific patient populations, treatment protocols, and documentation requirements [324]. Similarly, a legal firm requires assessment focused on their practice areas and jurisdictional requirements, not generic legal reasoning tests.

3) Evaluation Requirements and Design Principles: After outlining the current landscape of hallucination evaluation in Section VI-A1 and emphasizing the significance of horizontal and vertical assessment in Section VI-A2, this section turns to the stringent requirements and design principles that effective hallucination evaluation must satisfy to ensure both scientific rigor and practical applicability. These principles establish a reliable foundation for the development of evaluation metrics and protocols, which will be elaborated in detail in Section VI-C2d and Section VI-C2.

Safety and Risk Awareness requires that evaluation incorporate explicit *safety likelihood and risk assessment*, quantifying not only the frequency of hallucinations but also their potential severity and impact. This probabilistic orientation is particularly crucial for deployment decisions in high-stakes domains, where even infrequent but high-severity failures can cause tangible harm in critical real-world applications. By prioritizing risk-sensitive evaluation, the framework ensures that the most consequential errors receive heightened scrutiny, thereby aligning evaluation outcomes with real-world safety imperatives.

Factuality demands reliable differentiation between factual and hallucinated content while minimizing false positives and false negatives [325]. Multisource verification and confidence-weighted scoring are employed to manage uncertainty in ground truth determination, including challenges related to *static knowledge* and *temporal factuality*, where outdated or evolving knowledge increases error risk in a dynamic world.

Faithfulness is a quantifiable and interpretable requirement that ensures models remain grounded in the evidence they are provided, rather than generating unsupported but plausible-sounding content. Faithfulness refers to the degree to which a model's output is consistent with the input source material, in contrast to factuality, which measures alignment with broader world knowledge. Faithfulness evaluation aims to verify whether each claim is explicitly supported by, or logically inferred from, the provided context, computes a faithfulness score as the proportion of supported claims, and directly captures a model's ability to avoid introducing extraneous, distorted, or contradictory content, thereby providing a critical safeguard against hallucinations.

Causal Fidelity underscores the necessity of assessing a model's causal reasoning capabilities. Beyond detecting surface-level correlations, evaluation must establish whether models correctly infer and articulate cause–effect relationships, while distinguishing between errors that arise from spurious correlations, temporal knowledge cutoffs, or structural flaws in reasoning. This capability is especially critical in high-stakes domains such as medicine, law, and public policy, where causal misattributions can lead to severe downstream consequences [326]. By incorporating causal fidelity into evaluation, frameworks ensure that models are not only factually accurate but also conceptually sound in their reasoning processes.

Interpretability elevates evaluation from a mere measurement exercise to a diagnostic process that provides transparency and actionable insights. In ideal framework, each evaluation node is accompanied by explicit evidence chains and confidence calibration, ensuring that every flagged instance of hallucination is supported by interpretable reasoning. This design enables evaluators to identify not only what content is incorrect, but also why it was classified as hallucinatory and which evidence supports that determination. By offering this level of explanatory depth, interpretability serves a dual purpose: it increases transparency and trust for end users while also equipping developers with targeted insights for model refinement.

Actionability necessitates that evaluation yields more than scalar performance scores; it must provide traceable insights that guide concrete improvements in model design and deployment. This requires a *vertical reasoning traceability* that dissects the model's inference chain layer-by-layer, localizing the origin of hallucinations to specific faulty premises, flawed synthesis steps, or error propagation pathways. This granular diagnosis is paramount for developing targeted mitigation strategies. Such actionable feedback transforms evaluation from a purely diagnostic tool into a driver of iterative system refinement.

Consistency ensures that evaluation performance is stable and reproducible across runs, configurations, and environments, making it indispensable for both scientific validity and real-world deployment. Without consistency, evaluation outcomes risk reflecting random noise rather than genuine model performance. Standardized protocols, seed-controlled sampling, and rigorous replication practices safeguard against such variability, ensuring that results can be independently verified and meaningfully compared across systems [322].

Reliability extends beyond consistency to evaluate the stability of model performance across time, contexts, and user populations. It explicitly accounts for factors such as *temporal drift*, *contextual variability*, and demographic differences that may introduce instability in evaluation outcomes. A reliable evaluation framework must therefore not only measure correctness at a single point in time but also capture how model behavior fluctuates under varying conditions. By systematically tracking these sources of variability, reliability assessment ensures that evaluations provide a trustworthy characterization of real-world performance rather than an overly static or idealized snapshot.

Robustness requires resilience against adversarial inputs, distributional shifts, and edge cases that can expose weaknesses in simplistic evaluation metrics [320]. A robust evaluation framework must diagnose how models respond under stress—whether to perturbations in input phrasing, adversarial prompts, or domain-specific challenges—without a disproportionate increase in hallucination rates. Our use of hierarchical decomposition strengthens robustness by breaking complex or ambiguous queries into smaller, verifiable components, thereby isolating vulnerabilities and improving the framework's capacity to withstand challenging evaluation conditions.

Computational Cost and Efficiency requirement reflects the inherent tradeoff between diagnostic depth and practical feasibility. Comprehensive hallucination evaluation is often computationally intensive, requiring multiple sampling rounds, retrieval operations, or complex verification pipelines. The principle of cost–efficiency ensures that hallucination evaluation remains scalable, accessible, and practicable across diverse scenarios, preventing rigorous evaluation from becoming the privilege of only well-resourced organizations. To significantly reduce computational burden while preserving diagnostic quality, evaluation frameworks must therefore incorporate optimizations—such as adaptive sampling (which adjusts evaluation depth based on uncertainty) and cascaded pipelines (which apply lightweight filters before costly deep checks).

These design principles will translate into the core dimensions we will use to evaluate specific model outputs, which will be discussed in detail in Section VI-C2d.

4) Major Challenges in Comprehensive Evaluation: Developing a truly comprehensive hallucination evaluation framework faces several fundamental challenges that calls for novel architectural and methodological innovations.

Developing a truly comprehensive hallucination evaluation framework entails addressing several fundamental challenges, which in turn necessitate novel architectural designs and methodological innovations.

Compositional Complexity emerges as queries become more sophisticated, with the number of possible reasoning paths growing exponentially [327]. Our tree-based approach manages this complexity through intelligent pruning strategies and verification gates that focus computational resources on the most promising reasoning paths while maintaining comprehensive coverage.

Ground Truth Ambiguity presents a philosophical and practical challenge, as many real-world queries lack definitive correct answers [328]. We address this through multi-source verification that aggregates evidence from multiple authoritative sources and confidence-weighted consensus mechanisms that acknowledge uncertainty rather than forcing binary judgments.

Computational Efficiency becomes critical for practical deployment, as comprehensive evaluation can be prohibitively expensive without optimization. We implement sophisticated caching strategies that avoid redundant computations and parallel evaluation paths that leverage modern computing infrastructure while maintaining evaluation integrity.

Cross-Modal Consistency adds another layer of complexity for vision–language models, where visual and textual modalities must align coherently [329], [330]. Our framework extends naturally to multimodal evaluation through modal-specific verification nodes that assess both within-modality accuracy and cross-modal alignment [331].

Dynamic knowledge presents an ongoing challenge as information changes over time, making static benchmarks obsolete [332], [333]. Our user-centric approach allows continuous benchmark updates from current sources, ensuring evaluations reflect contemporary knowledge rather than outdated training data.

Additionally, evaluating mitigation strategies (e.g., Retrieval-Augmented Generation [RAG], Low-Rank Adaptation [LoRA]) poses a challenge, as it requires measuring whether these strategies reduce hallucinations while avoiding new errors or performance degradation [334]. Hallucinations caused by language priors and biases are difficult to detect, as models may generate plausible but incorrect content based on statistical regularities in training data [335]. Prompting strategies also significantly impact hallucination rates, necessitating systematic assessment of prompt variations [182].

5) Proposed Solutions: The LENS Framework: To address these fundamental gaps, we present **LENS** (**L**ayers of **E**valuation of hallucinatio**N** in GenAI **S**ystems), a comprehensive evaluation framework that introduces a hierarchical, tree-based query decomposition system. Our framework transforms complex evaluation tasks into multilayered assessment structures, providing what we term "MRI-like scanning" of the model's inference process. Just as medical MRI technology enables layer-by-layer visualization of internal structures, our framework exposes each reasoning layer independently, identifies patterns across different reasoning paths, monitors how information propagates through the inference chain, and pinpoints exact locations where hallucinations emerge.

The key innovation lies in our tree-based decomposition approach, where complex queries are hierarchically broken down into atomic verification units. Each node in this tree represents a discrete reasoning step, while edges indicate logical dependencies between steps. This structure enables us to trace the exact path of reasoning and identify precisely where hallucinations originate, whether from faulty initial assumptions, incorrect intermediate inferences, or flawed final synthesis.

Our framework introduces four fundamental contributions that collectively advance the state of hallucination evaluation. First, we implement **Tool Necessity Detection and Selection (TND/TSA)**, explicitly evaluating whether models correctly identify when external verification is needed and select appropriate tools. This transforms evaluation from post hoc error detection to proactive risk assessment, addressing a critical source of hallucinations where models confidently generate information from parametric memory when they should instead consult external sources.

Second, we develop **Multidimensional Evaluation Metrics** that go beyond binary hallucination detection. Our framework assesses degree through accuracy, faithfulness, and tool appropriateness; quantity through coverage and completeness of information; stability through consistency across multiple evaluations; and risk through uncertainty quantification and confidence calibration. This comprehensive approach captures the multifaceted nature of hallucination that simple accuracy metrics miss.

Third, we enable **User-Centric Benchmark Construction**, allowing organizations to create custom evaluation datasets from their own information sources. This ensures evaluation relevance to specific deployment contexts while maintaining standardized protocols that enable meaningful comparison across different implementations.

Fourth, our framework generates **Actionable Insights** beyond simple hallucination detection. We provide causal attribution of errors to specific reasoning steps, enabling developers to understand not just that a hallucination occurred but why it happened. This leads directly to suggested mitigation strategies based on observed error patterns and informed model selection guidance for specific use cases.

In the remainder of this chapter, Section VI-B first analyzes the fundamental challenges for hallucination evaluation. Section VI-C then surveys how existing evaluation methods and benchmarks address the discussed challenges, while Section VI-D examines the key limitations of current approaches and highlights the gap between the present state and the desired direction, thereby positioning our contributions within the broader research landscape. Section VI-E introduces the innovative LENS architecture, which integrates a six-stage evaluation pipeline with a tree-based decomposition methodology. Section VI-E2 further elaborates on the multidimensional evaluation metrics and their theoretical underpinnings.

Finally, Section VI-E1 and VI-D0e emphasize actionability, illustrating how evaluation results can be translated into practical improvements for both model developers and end users. Section VI-E8 presents the Universal Village case study, showcasing real-world deployment and measurable impact, while Section VI-E8a extends the framework to multimodal models, addressing the unique challenges of cross-modal hallucination detection. Section VI-F and Section VI-G conclude with a candid discussion of limitations, promising future directions, and the broader implications for trustworthy AI development.

Through systematic evaluation that combines breadth and depth, interpretability and automation, standardization and customization, LENS advances the field toward more reliable and trustworthy AI systems. Our framework enables informed model selection where organizations can choose models based on comprehensive performance profiles tailored to their specific needs rather than generic benchmarks. It facilitates targeted mitigation development by identifying precise failure points, allowing developers to create focused solutions rather than broad approximations. Most importantly, it enables trust calibration, giving users nuanced understanding of when and how to trust model outputs based on transparent evaluation results.

The ultimate goal transcends mere hallucination detection. We aim to understand hallucination origins, predict their occurrence, and systematically eliminate them, transforming LLMs from impressive but unreliable systems into trustworthy partners suitable for deployment in critical applications where accuracy, reliability, and interpretability are non-negotiable requirements.

B. Fundamental Challenges in Hallucination Evaluation

The evaluation of hallucinations in both large language models (LLMs) and vision-language models (VLMs) presents a set of deeply interwoven challenges that cut across technical, epistemological, and practical dimensions [1]. These difficulties stem largely from the inherent mismatch between the probabilistic behavior of generative models and the deterministic standards of factual correctness, leading to a context where conventional evaluation strategies often fall short [72]. Grasping these challenges in their entirety is crucial for designing effective assessment methodologies and, ultimately, for advancing the development of trustworthy AI systems capable of reliable use in high-stakes domains.

1) The Epistemological Challenge: Defining Ground Truth in Complex Domains: At the heart of hallucination evaluation lies a profound epistemological problem: how to establish what counts as *ground truth* in contexts where knowledge is uncertain, contested, or continually evolving [336]. This issue extends beyond simple fact-checking and instead raises fundamental questions about the representation, validation, and legitimacy of knowledge within AI systems [319].

The meaning of ground truth varies significantly across domains. In mathematical reasoning, it appears straightforward due to the availability of formal proofs and computational checks. However, complexity arises when models must navigate between multiple correct solution paths, approximate solutions within tolerance, or equivalent expressions differing only in notation. Here, evaluation frameworks must distinguish between optimal, correct-but-suboptimal, and outright erroneous solutions, as well as between failures of reasoning versus computation [319].

In medicine and science, temporality complicates evaluation: what is considered accepted practice or valid knowledge today may quickly become obsolete or even harmful tomorrow [337]. Clinical guidelines are regularly updated, meaning that a model's answer could be simultaneously valid under one authority but outdated under another, or suitable for one patient group but unsafe for another [42], [42], [324]. Evaluation frameworks thus need to account for evolving standards, multiple authoritative sources, and the coexistence of legitimate disagreement among experts.

The legal field presents its own epistemic hurdles, as correctness often depends on jurisdiction, time, and interpretation. Laws are amended, precedents overturned, and interpretations shift with judicial philosophy. Consequently, evaluation must address the differences between black-letter law and its practical application, majority versus dissenting opinions, and civil versus common law traditions [338].

Cultural and linguistic contexts add further complexity. Historical narratives can differ dramatically across societies, while social norms and acceptable practices are inherently local. Even language itself introduces ambiguity through polysemy, metaphor, and contextual meaning, resisting simple binary judgments of truth or falsehood [338].

Finally, contested knowledge and minority perspectives pose difficult evaluation questions. Scientific controversies, historical debates, and philosophical disagreements may not have a single resolution. Evaluation frameworks must decide whether to privilege consensus, incorporate dissenting views, or recognize such areas as unsuitable for binary classification. These decisions carry implications for how models learn to express uncertainty and epistemic humility [336].

2) The Computational Complexity Challenge: Scaling Comprehensive Evaluation: A second fundamental difficulty in hallucination evaluation arises from computational complexity. Comprehensive evaluation of generative models is severely constrained by the enormous resources required, forcing trade-offs between depth, breadth, and feasibility [339], [340]. As task complexity increases, the cost of verification grows rapidly, creating structural barriers to large-scale and fine-grained assessment.

One key obstacle is the combinatorial explosion of reasoning paths. Multi-hop or causal reasoning often generates exponentially many inference chains, each needing independent verification [341], [342]. Even apparently simple queries can require checking numerous intermediate links, counterfactuals, or alternative explanations. The exponential growth in possible reasoning trajectories makes exhaustive evaluation computationally infeasible.

Verification itself is resource-intensive, particularly when external sources must be consulted. Real-time checking against scientific databases, legal statutes, or knowledge graphs not only requires retrieval but also interpretation and cross-source consistency analysis. This demand multiplies when multiple references must be reconciled, quickly overwhelming limits on queries, API calls, and processing capacity [339].

Multimodal models exacerbate this challenge, since evaluation must simultaneously assess textual, visual, and relational accuracy. Tasks such as image captioning or VQA require object detection, attribute recognition, and semantic consistency checks, each computationally expensive on its own and even more costly when integrated across modalities [60], [61], [65], [343], [344].

Temporal dynamics add further overhead. As knowledge bases are continuously updated, prior evaluations must be revalidated against the latest information. Beyond storage and update costs, this requires propagating changes through dependent facts and rechecking prior claims, an operation that becomes computationally prohibitive at scale [332].

Finally, statistically meaningful evaluation requires large sample sizes. To demonstrate significance in hallucination rate differences across models, tasks, or domains, thousands to millions of examples may be needed. This requirement, combined with computationally expensive verification, means that exhaustive and representative evaluation often exceeds feasible computational budgets [340].

3) The Attribution and Interpretability Challenge: Understanding Causality in Hallucinations: Beyond detecting hallucinations, it is equally important to understand *why* they arise. However, most current evaluation approaches fall short in providing interpretable insights into the causal mechanisms behind hallucinations, limiting both scientific understanding and the development of targeted mitigation strategies.

The opacity of neural networks remains a central obstacle. Modern LLMs and VLMs consist of billions of parameters interacting through complex, nonlinear transformations, making it difficult to trace how specific outputs emerge [345]. When a hallucinated statement appears, pinpointing whether the error originates from data artifacts, architectural constraints, or inference-time dynamics is computationally demanding and often inconclusive [95]. Since knowledge is distributed across network weights, no single component can typically be held responsible for a given error.

Different types of hallucinations also stem from distinct causal pathways. Memory-based errors, where models confidently output false information from parametric memory, differ from reasoning failures in which otherwise correct facts are combined incorrectly. Similarly, prompt-driven contextual hallucinations must be distinguished from capability limitations, where the model simply lacks the necessary knowledge or computational capacity [346]. Current evaluation practices often identify errors only at the symptom level, without diagnosing these deeper causes [345].

Error propagation in multi-step reasoning poses a particular difficulty. An early misstep may cascade through subsequent steps, compounding into major hallucinations. Tracing these chains requires monitoring how uncertainty spreads and distinguishing between genuine reasoning failures versus unsuccessful error-correction attempts, including cases where implicit reasoning is not directly visible in model outputs.

Interactions among model components further complicate attribution. Attention mechanisms, normalization layers, and feed-forward blocks can interact in ways that amplify or suppress errors. The same factual mistake may emerge or disappear depending on prompt formulation, context length, or decoding strategy, showing that hallucinations often result from systemic interactions rather than isolated faults.

This challenge extends to retrieval-augmented systems. Even when correct external information is available, hallucinations may still arise due to failures in retrieval, misinterpretation of retrieved content, or poor integration of external knowledge with internal representations. Each of these failure modes requires distinct mitigation strategies, but most evaluation frameworks lack the granularity to differentiate among them.

4) The Dynamic Knowledge Challenge: Evaluating Against Evolving Information: Knowledge is inherently dynamic, and this temporal evolution poses a significant obstacle for hallucination evaluation [332]. Facts change, scientific understanding progresses, and social norms shift, which means that evaluation frameworks built on static datasets often become quickly outdated [333].

The most visible aspect of this challenge arises in fast-changing domains such as politics, public health, and technology. A model's statement about government leadership, pandemic statistics, or emerging tools might be accurate at the time of generation but outdated within days or even hours. Effective evaluation thus requires temporal scoping—determining what counts as "current" knowledge and how to distinguish between present facts and historical truths [333].

Scientific knowledge introduces additional complexity. Medical treatments, for example, evolve through cycles of hypothesis, evidence, and revised consensus. A model might produce answers that are consistent with prior consensus but already superseded by new research, or that reflect contested findings rather than settled truths [337]. Evaluation frameworks must therefore differentiate between outdated information, reasonable but provisional claims, and genuinely false statements.

Periods of transition—when multiple viewpoints coexist—create further complications. During these times, models should not only provide factual accuracy but also acknowledge uncertainty and debate within the field. Rigid binary classifications of "truth" versus "hallucination" cannot capture the nuances of such evolving knowledge landscapes [332].

Another dimension is knowledge deprecation. Some information does not simply expire but becomes actively misleading—for example, overturned legal precedents, retracted scientific studies, or medical procedures now considered harmful. Evaluating whether models can avoid perpetuating deprecated knowledge requires both temporal awareness and contextual sensitivity [333].

Finally, cultural and social knowledge evolves in nonlinear ways. Norms, values, and linguistic conventions shift gradually across different communities and may vary simultaneously across regions. Evaluation frameworks must account for such diversity without falling into either cultural essentialism or relativism, recognizing that what constitutes "truth" or "appropriateness" is often time- and context-dependent [338].

5) The Multimodal Complexity Challenge: Cross-Modal Consistency and Verification: Evaluating hallucinations in vision–language models (VLMs) introduces unique difficulties that go well beyond text-only settings. The interaction between visual and linguistic modalities creates new forms of hallucination and demands evaluation methods capable of capturing both within-modality correctness and cross-modal consistency [60], [111].

In visual contexts, hallucinations can appear as fabricated objects, incorrect attributes, or invented spatial relationships. Unlike textual factual errors, these mistakes require evaluation methods grounded in object detection, scene semantics, and physical plausibility, which themselves are resource-intensive [65], [347]. Ambiguities in images—such as occlusion, abstraction, or artistic rendering—further complicate ground-truth definition, with even human annotators often disagreeing.

Cross-modal grounding presents an equally difficult challenge. When generating a caption or answering a visual question, models must align visual regions with textual descriptions, ensure relationships are correctly mapped, and preserve semantic meaning across modalities. Abstract concepts, metaphors, or culturally specific visual symbols make this alignment even harder, as they resist literal verification.

Another dimension is the hierarchical nature of visual understanding. Errors can occur at multiple levels—from misidentifying basic objects, to misunderstanding attributes, to failing at relational or compositional reasoning. Evaluation frameworks must therefore check consistency across these layers while tracing how errors at one level propagate to higher-level interpretations [61], [343], [344].

The problem intensifies in video-based evaluation, where temporal coherence must also be considered. Hallucinations may manifest as phantom objects that appear or vanish, incorrect event ordering, or inconsistent tracking across frames. Robust evaluation thus requires both frame-level verification and temporal alignment [92].

Finally, multimodal integration itself can generate novel errors. Models may output textually plausible but visually impossible descriptions, or visually coherent but semantically nonsensical statements. These cross-modal inconsistencies highlight the difficulty of ensuring semantic alignment across modalities and reveal limitations in current fusion mechanisms [343].

6) The Scale and Diversity Challenge: Representative Evaluation Across Domains: Another fundamental challenge lies in ensuring that evaluation frameworks capture the full scale, diversity, and representativeness of real-world model applications. Since the space of potential queries, contexts, and use cases is essentially infinite, evaluations constrained by finite datasets inevitably face blind spots [348].

Domain coverage is the first difficulty. LLMs and VLMs are deployed in domains as diverse as medicine, law, customer service, and creative writing. Each of these requires different standards of correctness and domain-specific verification procedures [4], [349]. For instance, a hallucinated statement in casual dialogue may be inconsequential, while the same error in clinical or legal contexts can be dangerous. No single evaluation framework can simultaneously provide sufficient depth across all domains without becoming unwieldy [350].

The "long tail" distribution of knowledge exacerbates this problem. While models tend to perform well on frequent and well-represented topics, they often struggle with rare, specialized, or emerging ones. Ironically, these edge cases are often where accuracy matters most, such as in scientific or regulatory contexts. Evaluation methods must therefore strike a balance between testing common scenarios and capturing rare but critical edge conditions [350].

Task diversity introduces another layer of complexity. A single model may be applied to summarization, reasoning, translation, or code generation, each exhibiting different hallucination patterns. Performance in one task does not necessarily generalize to others [351]–[353]. Thus, comprehensive evaluation must span diverse task settings while maintaining comparability across them.

Cultural and geographic representation further complicates evaluation. Most benchmarks are biased toward specific languages, regions, or cultural contexts, leaving gaps where hallucinations may go undetected [338], [354], [355]. Models may therefore appear accurate in overrepresented contexts but fail badly in underrepresented ones, undermining global trustworthiness.

Finally, there is a tension between representativeness and statistical reliability. Large sample sizes are required to achieve statistical significance, yet expanding coverage across domains, tasks, and cultures quickly demands millions of examples, far beyond current resource budgets [340]. While adaptive sampling and stratified benchmarking offer partial solutions, they cannot fully overcome the trade-off between breadth of coverage and depth of analysis.

7) The Adversarial Robustness Challenge: Evaluation Under Distribution Shift: Evaluation frameworks must also contend with adversarial and out-of-distribution conditions, as models in deployment frequently encounter inputs that are either deliberately crafted to induce errors or that diverge naturally from training distributions [320]. This challenge goes beyond standard test cases, requiring methods that assess robustness under difficult and often adversarial scenarios.

One prominent issue is adversarial prompting, where inputs are carefully engineered to exploit model weaknesses. Such prompts may leverage unusual phrasing, manipulate context, or combine misleading cues to trigger hallucinations even in otherwise reliable models [320], [335]. Evaluation methods must distinguish between legitimate but challenging queries and artificially adversarial ones, ensuring robustness without overfitting to contrived attacks [320].

Distribution shifts present a subtler but equally significant difficulty. In practice, deployed models inevitably face variations in style, vocabulary, format, or topical domain that were not present during training. These shifts can lead to degraded performance and hallucination emergence as models extrapolate beyond familiar contexts [335]. Comprehensive evaluation must therefore probe not only in-distribution accuracy but also resilience under varied and shifted inputs.

The adversarial landscape is inherently dynamic, with new attack strategies continually emerging. This creates an ongoing "arms race" where evaluation protocols must adapt to novel vulnerabilities while preserving comparability with earlier benchmarks [320].

Interactive deployments add further complexity through prompt injection or manipulation. Users may intentionally or inadvertently override system instructions, leading to hallucinations or unsafe outputs. Evaluation frameworks must test resilience to such manipulations while ensuring safety mechanisms do not overly restrict functionality [335].

Finally, real-world conditions often combine multiple challenges simultaneously, such as ambiguous phrasing, mixed languages, and adversarial intent. These compounded scenarios frequently expose vulnerabilities not apparent when stressors are tested in isolation. Effective evaluation therefore requires assessing robustness under such multifactorial conditions, while also identifying which factors most strongly contribute to model failures [320].

8) The Human Alignment Challenge: Evaluating Subjective and Value-Laden Dimensions: Beyond factual correctness, hallucination evaluation must also address subjective and value-laden aspects of human interaction. These include individual preferences, cultural norms, ethical boundaries, and contextual appropriateness—dimensions that defy simple binary judgments of "true" or "false" [323].

A first difficulty lies in the inherently subjective nature of quality judgments. What one user considers sufficiently detailed or concise may appear verbose or incomplete to another. Cultural differences further complicate this, as information considered appropriate or relevant in one context may be irrelevant—or even offensive—in another. Evaluation frameworks must therefore balance subjective variability with the need for standardized benchmarks.

Value alignment raises further complications in ethically sensitive domains. In some cases, factual accuracy must be weighed against social responsibility. For example, a model might refuse to provide technically accurate but potentially harmful information. Such "protective hallucinations" may be preferable to harmful truth-telling. Evaluation methods must be able to distinguish between socially responsible deflection and harmful misinformation while respecting diverse cultural and ethical frameworks.

Context appropriateness is another crucial factor. A response acceptable in academic discourse might be inappropriate in casual conversation, or suitable for adults but not for children. Evaluations must therefore assess not just factual accuracy but also whether responses are adapted to context and audience.

Bias and fairness intersect with hallucination evaluation in complex ways. Models may reproduce stereotypes or hallucinate when attempting to avoid bias. Effective evaluation must therefore assess both factual reliability and the broader social consequences of model outputs, ensuring that accuracy is not achieved at the cost of fairness.

Finally, user expectations and trust calibration introduce additional challenges. As users become more familiar with AI, they increasingly expect models to acknowledge uncertainty, explain reasoning, and communicate limitations [182], [186]. Evaluation frameworks must therefore assess whether models not only provide correct answers, but also present them in ways that foster appropriate trust—by expressing uncertainty, admitting knowledge gaps, and guiding users toward reliable verification when needed.

9) Synthesis: The Imperative for Revolutionary Change: Taken together, the challenges outlined above highlight that hallucination evaluation is not merely a technical inconvenience but a fundamental crisis in how we assess generative AI. Epistemological uncertainty about ground truth, the computational infeasibility of exhaustive verification, the opacity of causal attribution, the constant evolution of knowledge, multimodal complexity, the scale and diversity of use cases, adversarial vulnerability, and subjective human values collectively define a space that current evaluation paradigms cannot adequately capture.

Existing frameworks often fall short not because of poor implementation but because of conceptual limitations. They tend to reduce hallucination to binary classification, overlook temporal and contextual evolution, prioritize static benchmarks over adaptive evaluation, and provide scores without interpretability [1], [72]. These shortcomings demonstrate that incremental improvements are insufficient; what is needed is a rethinking of evaluation itself.

The way forward requires more than new benchmarks or metrics—it demands a paradigm shift in how we define and measure truth in AI systems. Evaluation must be understood as a dynamic process, one that adapts continuously to evolving domains, deployment contexts, and human values. Rather than imposing rigid frameworks, future approaches should foster ongoing dialogue between technical rigor and social expectations, between automated measurement and human judgment.

Such transformation cannot be achieved by machine learning researchers alone. It requires interdisciplinary collaboration: computer scientists to design transparent architectures, cognitive scientists to ground evaluation in human reasoning, philosophers to frame epistemological boundaries, domain experts to define standards of acceptable uncertainty, and affected communities to articulate expectations of trustworthy AI. Only through this comprehensive effort can evaluation frameworks rise to match the sophistication and risks of modern generative models.

As the following section shows, the current evaluation landscape reflects both the urgency of this problem and the fragmentation of existing responses. By exposing the inadequacy of prevailing approaches, these insights set the stage for a necessary reconceptualization—one that moves beyond patchwork solutions toward revolutionary change in hallucination evaluation.

C. Current Evaluation Landscape: Fragmentation and Insufficiency

Given the fundamental challenges outlined above, including epistemological complexities in defining ground truth [336], computational constraints in scaling evaluation, and attribution difficulties in understanding causality [95], the research community has responded with a diverse but fragmented landscape of evaluation approaches [319]. This proliferation of benchmarks and methodologies reflects both the urgency of the hallucination problem and the absence of consensus on how best to address it. While this diversity has yielded valuable insights into specific aspects of hallucination, it has also created a fragmented ecosystem where results across different evaluations remain difficult to compare, synthesize, or translate into actionable improvements [319].

The current evaluation landscape reveals a fundamental tension: while the hallucination problem demands comprehensive assessment across diverse contexts, practical constraints have led to a patchwork of specialized benchmarks that often operate in isolation. Recent surveys indicate that many distinct hallucination benchmarks have emerged since 2022 [16], yet their heterogeneous evaluation protocols and metrics make cross-benchmark comparison nearly impossible [16]. This fragmentation manifests in three critical ways: (1) Task divergence, with evaluation contexts spanning from constrained QA to open-ended generation, making performance correlation analysis infeasible; (2) Metric inconsistency, where benchmarks employ incompatible scoring systems ranging from binary classification accuracy to fine-grained semantic similarity scores; and (3) Ground truth ambiguity, where different benchmarks adopt conflicting definitions of what constitutes a hallucination.

Another dimension of fragmentation arises from the inherent limitations of manually constructed datasets, which often fail to capture the complexity of real-world use cases and user interactions. As discussed in Section VI-C7b, most widely adopted benchmarks are designed for narrow contexts and therefore lack generalizability across domains, modalities, and user populations. Moreover, these datasets frequently show weak alignment with human judgments and inadequately represent the complexities of subjective evaluation, including cultural variation, contextual nuance, and the multifaceted nature of human reasoning. Manual annotation also introduces systematic human biases and exposes datasets to adversarial noise or annotation artifacts, thereby reducing their robustness and reliability. Furthermore, such datasets provide only a partial representation of model performance and fall short of capturing the full spectrum of thought processes underlying LLMs.

This fragmentation not only impedes systematic progress but also creates an illusion of advancement where improvements on specific benchmarks may not translate to real-world reliability—a phenomenon typically referred to as "benchmark overfitting" where models achieve high scores on curated datasets while exhibiting not low hallucination rates in deployment.

To provide a structured summary of the current research landscape on hallucination evaluation, we organize existing studies under six key themes. First, evaluation targets and taxonomy clarify what aspects of hallucination researchers have focused on, as discussed in Section VI-C1.

Second, evaluation dimensions and metrics outline the criteria and measurement standards that have been proposed, as discussed in Section VI-C2. Third, a mechanistic understanding of decision-making highlights efforts to connect evaluation outcomes to the underlying causes of hallucinations , as discussed in Section VI-C3. Fourth, evaluation methodologies describe the practical approaches that have been implemented to carry out assessments, as discussed in Section VI-C4. Fifth, evaluation across core dimensions reflects the broader scope of work addressing reasoning, safety, robustness, and reliability , as discussed in Section VI-C5. Sixth, mitigation-aware evaluation captures attempts to directly link evaluation to strategies for reducing hallucinations as discussed in Section VI-C6. Finally, tools and frameworks summarize the infrastructure developed for evaluation practices, as discussed in Section VI-C7. Together, these categories provide a comprehensive overview of how the field has approached hallucination evaluation to date, as discussed in Section VI-C8.

1) Evaluation Targets and Taxonomy (What to Evaluate): This section investigates existing evaluation efforts along four complementary categories: Task-Specific, Modality Coverage, Hallucination Typology, and Domain Expertise. Each category represents a distinct lens through which hallucinations can be understood and measured, providing insights into how different forms of evaluation capture particular aspects of model behavior.

However, examining these dimensions in isolation also exposes a broader challenge: the current evaluation landscape is fragmented, with specialized benchmarks excelling in depth within narrow contexts and unified benchmarks aiming for breadth but often sacrificing contextual rigor. This tension highlights the importance of discussing The Integration Challenge – Unified vs. Specialized Benchmarks. Understanding how to balance standardization and comparability against domain sensitivity and diagnostic depth is essential for advancing hallucination evaluation. A careful analysis of this integration challenge not only clarifies trade-offs between coverage and precision but also lays the groundwork for hybrid or modular approaches that can reconcile these competing demands.

a) Task-Specific Evaluation: From Simple Queries to Complex Reasoning: Hallucination patterns vary dramatically across task contexts. We organize these tasks along a spectrum of complexity, from well-defined question-answering to open-ended generation, reflecting increasing degrees of freedom in both input interpretation and output space.

Question Answering. The QA evaluation paradigm represents the most controlled setting for hallucination assessment, yet reveals surprising complexity in failure modes. TruthfulQA [310] pioneered adversarial question design with 817 questions across 38 categories, specifically crafted to trigger human-like false beliefs. Its key innovation lies in distinguishing between truthful and informative responses—the best model achieves only 58% truthfulness, exposing how fluency masks factual errors [310].

HaluEval [17] scaled evaluation to 35,000 samples, revealing task-specific hallucination patterns: including general queries, knowledge-grounded dialogue, and summarization tasks [17]. Its automated generation pipeline using ChatGPT achieved high correlation with human judgments while reducing annotation costs [17]. HaluEval-Wild [332], the first benchmark designed to assess LLM hallucinations in the wild, first pre-screens user queries from the ShareGPT dataset to form the initial pool of challenging queries via classifiers, then selects and categorizes into five types of 500 challenging user queries [332].

OnionEval [346], a multi-layer structured framework with a specific metric, context influence score (CI), is constructed to evaluate the fact-conflicting hallucination tendencies of small LLMs across different contextual levels. The experimental results showcase one important feature for SLLMs that SLLMs perform fine on factual analysis while struggling with context reasoning while simple Chain-of-thought strategy help enhance the practical applicability of SLLMs in real-world scenarios [346].

Conversational Tasks. Dialogue evaluation exposes temporal consistency challenges absent in single-turn interactions. DiaHalu [34] is a dedicated dialogue-level hallucination evaluation benchmark for LLMs, which includes four common multi-turn dialogue domains, and five hallucination subtypes, extending from factuality and faithfulness hallucination. To construct Dialogue, the system first combines the collected topics into system prompts, facilitates a dialogue between two LLMs, manually modifies the information not adhering to human language conventions, and then has LLMs re-generate, simulating authentic human-machine interaction scenarios. At the end, professionals annotate all samples from the dataset. Experiments on all well-known LLMs and detection methods show that DiaHalu is a challenging and valuable benchmark for research communities.

HalluDial [313] is also a comprehensive large-scale benchmark for automatic dialogue-level hallucination evaluation, comprising 4,094 dialogues with a total of 146,856 samples that contain both spontaneous and induced hallucination scenarios. The benchmark, including both factuality and faithfulness hallucinations, enables a comprehensive meta-evaluation of LLMs' hallucination evaluation capabilities in information-seeking dialogues, and introduces a specialized judge language model, HalluJudge, which achieve competitive performance in hallucination evaluation, facilitating the automatic assessment of dialogue-level hallucinations in LLMs and providing valuable insights into this phenomenon.

Instruction Following.

Instruction-following is a critical dimension in hallucination evaluation, as it tests whether a model can faithfully interpret user intent and produce outputs that are accurate and appropriate. Evaluation methods typically measure both instruction adherence (fulfilling all aspects of the request) and content veracity (ensuring factual correctness and contextual relevance).

Current evaluation approaches typically employ carefully designed instruction sets that probe specific vulnerability points. These include: (1) Precision-critical instructions requiring exact numerical, temporal, or spatial reasoning where minor inaccuracies may cascade into significant errors [356]; (2) Multi-step procedural instructions that test the model's ability to maintain consistency throughout extended reasoning chains [357]; and (3) Context-bound instructions that examine whether models appropriately respect provided constraints without introducing external assumptions [358].

The relationship between instruction following capability and hallucination propensity exhibits complex patterns. While reinforcement learning from human feedback (RLHF) significantly improves instruction adherence—with InstructGPT demonstrating superior performance despite smaller parameter counts compared to base GPT-3 [32]—its impact on hallucination mitigation shows nuanced trade-offs. As highlighted by [340], improved instruction following sometimes comes at the cost of increased overfaithfulness, where models may prioritize pleasing users with confident responses over acknowledging knowledge limitations. Furthermore, [95] reveals that instruction-following errors frequently originate from attribution difficulties in complex reasoning scenarios, where models struggle to correctly allocate confidence between parametric knowledge and instructional constraints.

Specialized benchmarks for instruction following evaluation continue to evolve, incorporating increasingly sophisticated tasks such as counterfactual reasoning, constraint satisfaction, and ethical boundary adherence. These evaluations consistently demonstrate that even state-of-the-art models exhibit non-trivial failure rates when confronted with instructions requiring precise domain expertise, complex constraint integration, or careful uncertainty calibration. The ongoing challenge remains developing evaluation frameworks that can simultaneously assess instruction fidelity while maintaining factual accuracy across diverse domains and instruction types.

Summarization.

Summarization represents a critical yet challenging domain for hallucination evaluation, where models must distill key information from source documents while preserving factual accuracy and avoiding fabrications. Traditional single-document summarization benchmarks, like RAGTruth, provide annotated samples to evaluate faithfulness to source material, yet even state-of-the-art detectors achieve a low accuracy on comprehensive benchmarks like FaithBench, highlighting fundamental limitations in current evaluation methodologies [359], [360]. These challenges intensify in multi-document summarization (MDS), where models must synthesize information across multiple sources—evaluations reveal alarmingly high hallucination rates exceeding 75% in LLM-generated summaries, with errors disproportionately occurring in later sections where synthesis demands are greatest [361].

The integration of insights from related evaluation paradigms offers promising directions for improvement. Dialogue-level evaluation frameworks like DiaHalu demonstrate how multi-turn interactions expose compounding error patterns that mirror the sequential risks in extended summaries [34]. Similarly, domain-specific benchmarks such as Med-HALT highlight the critical need for specialized evaluation in high-stakes fields like healthcare, where hallucination consequences are severe [42]. Furthermore, human-in-the-loop auditing frameworks like LLMAuditor provide methodologies for scalable validation of model outputs, suggesting pathways toward more reliable and contextually grounded summarization evaluation [323]. Together, these approaches underscore the necessity of moving beyond generic benchmarks toward compositional evaluation frameworks that address the complex, context-dependent nature of summarization hallucinations.

Translation. Although fewer dedicated benchmarks exist, translation evaluation reveals whether models preserve semantic meaning across language boundaries [4]. Translation tasks uniquely expose hallucinations through cross-linguistic consistency checks, where models must preserve semantic meaning across language boundaries without inventing cultural context or idiomatic expressions that don't exist in the source [4].

Text and Code Generation. Text-to-knowledge-graph generation, data-to-text writing, and news writing examine whether models produce faithful and factually accurate outputs. Recent analyses reveal that hallucination rates in open-ended text generation can be high when models generate content beyond their training distribution [362].

Recent work includes Collu-Bench [363] for predicting hallucination likelihood and Codehalu [364] for execution-based verification of generated code. It primarily concerns hallucinations that arise in programming tasks, such as generating incorrect logic, fabricated APIs, or non-executable code [364]. As language models are increasingly used as coding assistants, evaluating their reliability in this domain is becoming critical, since even small errors can lead to significant functional failures [365].

Reasoning Tasks. This category includes knowledge graph reasoning, causal reasoning through benchmarks like CausalBench [341], temporal and spatial reasoning, math word problems (MWPs), and logical tests. Recent studies demonstrate that reasoning tasks are particularly vulnerable to cascading hallucinations, where early errors compound through multi-step inference chains [116]. These evaluations are designed to push models beyond simple recall, probing whether they can handle complex reasoning and problem-solving without drifting into hallucinations.

Multimodal Generation Multimodal evaluation reveals emergent failure modes absent in unimodal systems. HallusionBench [330] identifies "visual illusion vulnerability"—models hallucinate 89% of the time when presented with carefully crafted visual illusions that humans correctly interpret [330]. This suggests models rely on superficial visual features rather than robust scene understanding. **Bingo** [329] exposes "modality competition"—when visual and textual signals conflict, models exhibit great bias toward textual priors regardless of image content [329]. This benchmark uniquely tests interference effects, showing that irrelevant text in images increases hallucination rates [329].

RAG & Tool-use Retrieval-Augmented Generation (RAG) has become an important area of evaluation in tasks such as question answering, data-to-text generation, and summarization [331]. The central aim is to assess whether incorporating external information improves factual accuracy or, conversely, introduces new inconsistencies. Paradoxically, while RAG reduces certain types of hallucinations by grounding responses in retrieved content, it can introduce new failure modes when models misinterpret or over-rely on retrieved information [320]. Tool-use scenarios present additional challenges, as models must correctly invoke APIs and interpret responses without fabricating functionality [366].

However, it is crucial to note that the current evaluation paradigm predominantly focuses on assessing the outcome of tool usage—i.e., whether the final answer is correct after the tool has been invoked. This overlooks a critical component of robust tool integration: the model's ability to make the correct decision on whether to use a tool in the first place. This capability, which we refer to as Tool Necessity Detection (TND), is a precursor to successful tool use. A model might generate a factually correct output without using a tool when it should have (a TND failure leading to potential hallucination due to overconfidence), or it might incorrectly call a tool for a simple task it should handle internally (inefficiency and potential error propagation). Current benchmarks and metrics are not designed to systematically evaluate this decision-making step independently from the final output accuracy. This gap in evaluation leaves an important aspect of model reliability unmeasured and underscores the need for evaluation frameworks that can dissect and assess the tool-use pipeline into its constituent parts, such as TND and **Tool Selection Accuracy (TSA)**.

b) Modality-Based Evaluation: From Text to Multimodal Complexity: Beyond task complexity, the modality of information processing introduces another critical dimension for hallucination evaluation. **Text-only Evaluation** has established relatively mature methodologies to assess factuality and faithfulness in purely textual settings. **Multimodal Evaluation** is to study visual, temporal, and cross-modal reasoning, which creates exponentially more complex evaluation challenges [64].

Text-Only Evaluation represents the most mature segment of hallucination evaluation and has enabled the development of standardized evaluation metrics, including factual consistency scores, entailment-based validation, and human-in-the-loop assessments. Benchmarks in this category generally focus on purely linguistic tasks where ground truth can often be established through textual sources, though challenges persist in addressing ambiguity, context-dependence, and knowledge evolution over time [367]. Text-only benchmarks have primarily targeted core generative tasks such as question answering, summarization, retrieval-augmented generation (RAG), and instruction following. For instance, RAGTruth [359] provides nearly 18,000 naturally generated responses with annotations, enabling fine-grained assessment of hallucination in RAG scenarios. This benchmark has been used to examine hallucination prevalence across different LLMs and has shown that smaller models, when fine-tuned, can achieve competitive performance in hallucination detection.

Another representative benchmark is HaluEval [17], which consists of over 35,000 annotated samples, including real user queries and synthetic task-specific data across question answering, knowledge-grounded dialogue, and summarization. It has been used to measure hallucination detection performance and shows that integrating external knowledge or chain-of-thought reasoning can reduce hallucination rates [17]. Furthermore, HaluEval-Wild [332] extends this work by collecting naturally occurring prompts from ShareGPT and constructs fine-grained hallucination categories for more realistic evaluation.

By contrast, Multimodal Evaluation, which encompasses vision-language, temporal reasoning, and cross-modal integration, poses substantially greater challenges than text-only settings. Multimodal models must align heterogeneous inputs across domains (e.g., image, video, and text), making them vulnerable to errors such as spurious visual grounding, temporal drift in video summarization, and inconsistent reasoning across modalities. Recent benchmarks like POPE and MMHal illustrate that evaluating multimodal hallucinations requires not only correctness within individual modalities but also the consistency and coherence across them.

Multimodal benchmarks reveal emergent hallucination mechanisms unique to cross-modal processing. The fundamental challenge, which models must reconcile, is potentially conflicting signals across modalities, leading to three distinct failure modes: (1) Modal dominance, where one modality overrides others regardless of reliability; (2) Cross-modal leakage, where patterns from one modality inappropriately influence another; and (3) Binding errors, where correct elements from different modalities are incorrectly associated.

The complexity of multimodal evaluation scales exponentially with the number of modalities involved, as evaluators must account for interdependencies, contextual grounding, and alignment across different sensory channels. This creates open questions around scalability, annotation cost, and reliability of current benchmarks, underscoring the need for more comprehensive, modality-aware evaluation frameworks that can robustly capture hallucination risks in both unimodal and multimodal generative systems.

c) Hallucination-Type–Based Evaluation: Understanding Typology of Hallucinations and Failure Modes: *Hallucination-Type–Based Evaluation* further categorizes evaluation efforts according to representative manifestations of hallucination. Hallucinations in generative AI emerge through distinct mechanistic pathways, each reflecting fundamental limitations of current model architectures [368]. Understanding these pathways is essential for diagnosing error modes, designing mitigation strategies, and improving model reliability. Below, we briefly introduce several representative categories of hallucinations that illuminate different aspects of LLM behavior, including Factual hallucinations, Reasoning hallucinations, Memory-based hallucinations, Contextual hallucinations, Multimodal hallucinations, Temporal or knowledge staleness, and Intrinsic and Extrinsic Hallucinations. Systematically assessing these categories helps showcase the breadth of failure modes and clarify their differential impacts on reliability, interpretability, and safety across applications.

Factual hallucinations refer to errors in which models generate statements that are factually incorrect or inconsistent with verifiable knowledge sources and typically arise from three primary mechanisms: (1) Parametric knowledge conflicts, where contradictions within the training corpus lead to stochastic selection among competing facts, resulting in inconsistencies observed in tested cases [361]; (2) Retrieval misalignment, where spurious correlations in embedding space cause incorrect entity associations, will affect factual queries [361]; and (3) Temporal degradation, where knowledge accuracy declines progressively following the training cutoff.

Reasoning hallucinations capture failures in logical or mathematical inference, such as erroneous multi-step arithmetic or flawed causal reasoning, which highlight the limitations of LLMs in performing structured reasoning beyond surface-level pattern recognition. These hallucinations manifest in several distinct failure signatures: Logical failures often stem from "shortcut reasoning" where models rely on superficial pattern matching rather than performing genuine inference, leading to errors in many complex reasoning tasks; and causal reasoning fragility refers to the disproportionate difficulty models face when handling counterfactual reasoning, achieving only low accuracy on counterfactual scenarios compared to high accuracy on factual reasoning tasks [341].

Memory-based hallucinations denote errors stemming from a model's limited or unstable memory mechanisms, often manifesting as contradictions across dialogue turns or inconsistencies in long-form text generation. These failures reflect the difficulty of maintaining coherent internal state representations and have emerged as a growing concern in recent research [42] since a model may incorrectly recall information from earlier dialogue turns, fabricate details absent from the interaction history, or conflate information across multiple user sessions. Such phenomena highlight the fragility of memory mechanisms in generative models and their inability to track and manage evolving conversational or contextual dependencies robustly [42].

Contextual hallucinations arise when models misinterpret or over-extend the given context, generating content not entailed by input instructions or source material. These errors underscore the models' sensitivity to prompt design and contextual cues, extending beyond conventional factual and logical mistakes. This category encompasses inconsistencies with prior dialogue turns as well as factual inaccuracies triggered by contextual complexity.

Recent studies highlight that **Small Large Language Models** (SLLMs) are particularly vulnerable: while they maintain relatively high accuracy in isolated fact-checking tasks, their performance deteriorates significantly in multi-layered contextual scenarios [346]. To capture this weakness, a Context Influence Score has been proposed, quantifying the degree to which models are affected by contextual dependencies, with SLLMs showing markedly greater susceptibility than larger models.

Mitigation strategies vary in effectiveness. Structured reasoning techniques, such as chain-of-thought prompting, have demonstrated success in reducing context-driven hallucinations. In contrast, approaches that rely solely on external knowledge sources—such as retrieval-augmented generation (RAG) — may be counterproductive if not coupled with reasoning guidance, sometimes exacerbating hallucinations by introducing ungrounded or misaligned content [346].

Multimodal hallucinations refer to inaccuracies that arise in models integrating text with other modalities (e.g., vision–language systems), where outputs include fabricated or misaligned cross-modal associations. A common failure mode occurs when models describe visual elements that do not exist in an image or misattribute relationships between text and visual content. Such errors substantially complicate reliability in multimodal systems because they highlight the inherent challenges of cross-modal alignment and consistency [64]. These hallucinations not only undermine factual accuracy but also expose limitations in current multimodal fusion architectures, where models must reconcile heterogeneous signals across domains [64].

Temporal or knowledge staleness indicates outdated responses caused by training on static corpora, where models confidently present obsolete information as current fact. This phenomenon reflects a fundamental mismatch between a model's parametric knowledge and the dynamic nature of real-world information, and it remains a persistent challenge for long-term reliability [369]. Empirical evidence shows that models trained on outdated data frequently generate time-sensitive claims without awareness of temporal shifts, thereby undermining their trustworthiness in domains that demand up-to-date knowledge, such as medicine, law, and public policy [369].

In addition, *Intrinsic and Extrinsic Hallucinations* constitute two critical categories that warrant focused attention. Intrinsic hallucinations stem from deficiencies internal to the model, such as faulty reasoning processes or gaps in parametric knowledge. In contrast, extrinsic hallucinations arise when the model correctly processes input but depends on erroneous external information, for example, flawed retrieval outputs or corrupted databases [361]. This distinction has important implications for mitigation strategies: intrinsic hallucinations require architectural refinements or targeted training improvements, whereas extrinsic hallucinations necessitate more reliable source verification, retrieval alignment, and robust information filtering mechanisms [24].

d) Domain-Specific Evaluation: The Challenge of Specialized Knowledge: Domain specialization represents another critical dimension, as the consequences of errors vary dramatically across application contexts. A hallucinated fact in creative writing might be inconsequential, while the same error in medical diagnosis could be life-threatening.

In the medical field, benchmarks such as Med-HALT [42] and the recently proposed MedHallBench [370] focus on evaluating hallucinations in clinical and biomedical contexts, where the accuracy of generated information is directly tied to patient safety and clinical decision-making. **MedHallBench** [370] demonstrates domain criticality: radiological hallucinations occur in reports, with some being clinically significant errors that could impact patient care [370].

In transportation, the Traffic Incident Benchmark [349] evaluates spatial and temporal reasoning using nearly 99,869 real incident records. It shows that while retrieval augmentation can reduce some hallucinations, significant performance gaps remain.

In code generation, Collu-Bench [363] introduces a framework for predicting when hallucinations are likely to occur during code generation, and Codehalu [364] evaluates hallucinations through execution-based verification, detecting cases where generated code runs incorrectly or fails to implement the intended functionality.

Besides, SciHal25 [371] is a shared task on hallucination detection in scientific question answering, providing a dataset of annotated claims.

In the financial domain, researchers present an empirical study of hallucinations in financial tasks [372]. It shows that LLMs frequently produce unreliable outputs when explaining financial concepts or retrieving stock data, and that mitigation strategies such as few-shot learning, DoLa, RAG, and tool-use offer only partial improvements.

Last but not least, multilingual and cross-lingual benchmarks, including Uhgeval [333], address the challenge of hallucination in settings where linguistic coverage and cross-language consistency are critical. Beyond these established domains, specialized evaluation benchmarks are emerging across legal, educational, and industrial applications, each with unique requirements for accuracy, interpretability, and reliability.

e) The Integration Challenge: Unified vs. Specialized Benchmarks: It is essential to develop a unified framework and benchmark for hallucination evaluation. Recently, there have been emerging unified evaluations, such as OnionEval [346], Hal-Eval [343], and Large VLM (LVLM)-EHub [350], which seeks to provide wide coverage across tasks and modalities, where models can be assessed under a single framework. It facilitates standardized reporting and benchmarking.

However, this breadth often comes at the cost of domain-specific depth. By contrast, specialized benchmarks offer more rigorous and contextually grounded assessments, but their narrower focus can limit generalizability [373]. This trade-off between coverage and precision represents more than a practical choice; it reflects fundamental questions about the nature of hallucination and whether universal evaluation principles can meaningfully span diverse application contexts.

The path forward likely requires hybrid approaches that combine the standardization benefits of unified frameworks with the contextual sensitivity of domain-specific evaluation. Recent proposal for meta-evaluation frameworks [374] offers promising directions, with modular architectures allowing for comparative analysis and specialized assessment.

The current evaluation landscape, while rich in specialized contributions, lacks the integrative frameworks necessary for systematic progress. The proliferation of benchmarks without standardized evaluation protocols has led to situations where models appear reliable on one benchmark while exhibiting severe hallucinations on another, undermining user trust and hindering deployment in critical applications.

f) Summary and Future Directions: Our comprehensive analysis of evaluation targets and taxonomy reveals a field in transition, from ad-hoc benchmark construction toward a more systematic understanding of hallucination phenomena. The key insights from this survey are as follows.

First, task-specific evaluations demonstrate that hallucination patterns are highly context-dependent. Question answering tasks exhibit a high hallucination rate driven primarily from factual errors [17]; and multimodal tasks reveal hierarchical degradation, with performance dropping from high accuracy in object detection to low in relationship inference. This heterogeneity suggests that universal hallucination mitigation strategies may face inherent limitations.

Second, the modality dimension highlights architectural limitations. While text-only evaluations achieve relatively robust results [359], multimodal systems suffer from modal dominance biases and temporal drift. These findings indicate that current fusion mechanisms inadequately integrate cross-modal information.

Third, hallucination typologies reveal mechanistic insights. Factual errors often result from parametric knowledge conflicts [361], reasoning failures scale exponentially with task complexity [116].

Fourth, domain-specific evaluations show concerning error rates in high-stakes applications [370].

Looking forward, three imperatives emerge:

- (1) Developing compositional evaluation frameworks that assess integrated capabilities rather than isolated functions;
- (2) Designing adaptive benchmarks that evolve with model progress to avoid benchmark overfitting;
- (3) Establishing risk-calibrated metrics that weight hallucinations by potential harm rather than treating all errors equally.

As discussed in Section VI-A2 and Table II, most existing evaluation strategies remain concentrated on the horizontal axis—measuring surface-level accuracy across domains—while offering limited exploration of vertical evaluation, which examines reasoning processes and causal pathways. The long-term trajectory of the field must therefore progress beyond mere detection of hallucinations toward uncovering their underlying mechanisms. By shifting emphasis to causal understanding, evaluation frameworks can enable principled and proactive mitigation strategies, rather than relying on post hoc fixes that treat only the symptoms of model unreliability.

2) Evaluation Dimensions and Metrics (Establishing Criteria): The study of hallucination evaluation has evolved from early token-level detection benchmarks, such as HaDes [362], to more comprehensive multi-level frameworks. Pioneering works established distinct methodological paths: SelfCheckGPT [19] demonstrated sentence-level detection through self-consistency without external references; FActScore [306] introduced atomic claim decomposition for precise factuality checks; and Semantic Entropy [76] advanced semantic-level uncertainty quantification for identifying confabulations. Collectively, these contributions revealed that hallucinations surface differently depending on the unit of analysis—from lexical errors at the token scale to contradictions at the document level—highlighting the need for evaluation strategies that adapt to both task requirements and reference availability. Building on this trajectory, hybrid and graph-based frameworks such as THaMES [186] have further illustrated how trade-offs between scalability, interpretability, and coverage shape evaluation design.

Amid this development, the field has converged on four operational dimensions: *Core*, *Granularity*, *Reference Availability*, and *Automation–Human Balance*. Together, these dimensions determine what properties should be measured, at what resolution hallucinations are identified, how reliability is ensured, and the extent to which automation or human judgment is involved. This dimensional perspective frames hallucination evaluation as an inherently multi-faceted challenge that requires balancing precision with coverage, and scalability with interpretability [16], [319].

This framework establishes a structured foundation for subsequent discussions, motivating the need to examine how methodological design choices—particularly in granularity, reference use, and the automation–human trade-off—affect detection effectiveness, verification reliability, and deployment feasibility.

a) Granularity: Granularity plays a pivotal role in determining how hallucination evaluation is conducted and interpreted [319]. Different levels of analysis offer complementary advantages and limitations, influencing the precision of diagnosis, the interpretability of outcomes, and the scalability of evaluation methods. The field recognizes that no single granularity suffices across all tasks; instead, multiple levels are required to capture the full spectrum of hallucination behaviors. Existing approaches can be broadly divided into four categories:

- **Token-level:** Focuses on word- or subword-level errors, allowing fine-grained localization of inaccuracies but

often depending heavily on contextual understanding.
- **Sentence- and claim-level:** Examines discrete statements or atomic facts, striking a balance between detection accuracy and practicality. This level is particularly emphasized in recent benchmarks that rely on claim decomposition for factual verification.
- **Dialogue- and document-level:** Evaluates coherence and factual reliability across extended contexts, such as multiple dialogue turns or entire documents. This perspective is essential for tasks like summarization and multi-turn question answering.
- **Specialized units (e.g., entities and tool-calls):** Targets application-specific failures, including entity consistency, relational grounding, or the correctness of API and tool usage. Such evaluations are increasingly relevant for retrieval-augmented and tool-augmented systems [319].

The choice of granularity reflects fundamental trade-offs. Fine-grained methods, such as token- or entity-level evaluation, are capable of capturing subtle factual errors but demand significant annotation effort and computational resources. In contrast, coarse-grained methods, like document-level assessment, enable broader coverage and efficiency but risk overlooking localized mistakes or nuanced inconsistencies. Consequently, identifying the appropriate granularity remains context-dependent, varying with both task requirements and application domains.

b) Reference Availability: Reference availability fundamentally determines the methodology for hallucination evaluation. This dimension captures the inherent tension between the need for objective verification and the practical reality that many generative tasks lack definitive ground truth. Depending on whether gold-standard references are present, evaluation methods can be categorized into three principal groups, each carrying distinct strengths and limitations:

- **Reference-based:** Compares model outputs against human-authored or curated references. Metrics span from surface-level similarity to semantic or claim-level checks, offering objectivity but constrained by reference quality and coverage.
- **Reference-free:** Relies solely on model-internal signals such as semantic uncertainty estimation, entropy measures, or response self-consistency, removing dependency on external corpora but limiting the scope of verification.
- **Hybrid and knowledge-grounded:** Integrates external evidence—such as retrieved passages, structured knowledge bases, or dynamically constructed graphs—with model-internal consistency checks. This design offers broader adaptability across domains but often introduces significant engineering and computational overhead [319].

This axis directly impacts the objectivity of evaluation outcomes, the breadth of domain coverage, and the resources required for implementation. In practice, the choice among these approaches depends heavily on the task setting and the availability of reliable references, making reference availability a central factor in the design of robust and scalable evaluation frameworks.

c) Automation–Human Balance: The dimension of automation versus human involvement highlights the trade-off between scalability and nuanced understanding. Fully automated systems enable large-scale and efficient evaluation, but they often fail to capture subtle or context-dependent hallucinations that human judgment readily identifies. Conversely, human-in-the-loop approaches provide fine-grained error detection and nuanced interpretations but are costly, time-consuming, and less scalable. Balancing these approaches is thus central to designing practical and reliable evaluation pipelines.

Evaluation strategies within this dimension can be broadly summarized as follows:

- **Fully automated methods:** Rely entirely on algorithmic metrics for scalability and speed. They are suited for large-scale benchmarking but may overlook complex or context-specific errors that require human reasoning.
- **Human-in-the-loop methods:** Involve expert annotators or crowdworkers to detect subtle inconsistencies and provide high interpretability. While more accurate in capturing nuanced errors, these methods incur high annotation costs and limited scalability.
- **Hybrid strategies:** Combine automation with selective human review, leveraging the efficiency of automated screening while reserving human judgment for ambiguous or high-risk cases. For example, the LLMAuditor framework [323] integrates active learning and Reinforcement Learning with Human Feedback (RLHF), significantly improving annotator agreement (e.g., Cohen's kappa scores rising from 0.46 to 0.69 for relevance and 0.62 to 0.90 for diversity). This approach reduces annotation costs while maintaining robust reliability.

Overall, the automation–human balance dimension directly affects the feasibility, interpretability, and cost of hallucination evaluation. Effective frameworks must therefore calibrate the degree of human oversight according to task sensitivity, domain risk, and desired scalability.

d) Core Dimensions: Having established the operational scaffolds of hallucination evaluation—granularity, reference availability, and the balance between automation and human oversight—we now turn to the Core Dimensions that operationalize these foundations into concrete evaluative practices. These dimensions collectively provide the basis for a thorough and reliable assessment of hallucinations.

We organize the Core Dimensions into four progressive clusters that collectively capture the essential properties of hallucination evaluation. Specifically, the dimensions listed below reflect the design principles outlined in Section VI-A3. While the sequence in Section VI-A3 emphasizes their relative significance, the ordering presented here instead follows the structural logic of a comprehensive evaluation framework, representing the foundational components that reasonable metrics should incorporate.

- Grounding and Alignment

  - *Factuality*: accuracy of content against external world knowledge.
  - *Faithfulness*: adherence of outputs to the provided source material.
  - *Safety Likelihood and Risk Assessment*: quantifying potential harm beyond correctness.
- Reasoning Integrity
  - *Vertical Reasoning Traceability*: layer-by-layer localization of error origins.
  - *Causal Reasoning Capabilities*: ability to distinguish causation from correlation and identify root causes.
  - *Interpretability*: transparency and auditability of evaluation judgments themselves.
- Systemic Reliability
  - *Self-Consistency*: stability of outputs under paraphrasing or rephrasing.
  - *Coherence and Consistency*: logical harmony within single responses.
  - *Reliability and Consistency*: stability of behavior across time, contexts, and user populations.
  - *Robustness*: resilience against adversarial perturbations and edge cases.
- Practical Constraints
  - *Tool Appropriateness*
  - *Computational Cost and Efficiency Trade-offs*: balancing diagnostic quality with feasibility and scalability.

Together, these dimensions capture the foundational qualities that distinguish accurate from hallucinatory outputs, and extend evaluation beyond surface-level correctness to encompass reasoning integrity, discourse-level coherence, interpretability, and risk-awareness. By clustering them into progressive layers, the framework supports both diagnostic depth and practical applicability—beginning with factual grounding, advancing through reasoning quality, scaling to systemic reliability, and concluding with pragmatic trade-offs. By structuring the analysis around these dimensions, we create a diagnostic lens that is both comprehensive in scope and actionable in practice.

The selection of these dimensions is deliberately grounded in typologies of hallucinations articulated in contemporary understanding of hallucination typologies articulated in contemporary research [361], [368]. This foundation ensures that the evaluation framework captures the full spectrum of failure modes—from intrinsic reasoning errors, such as logical fallacies and factual inaccuracies, to extrinsic challenges stemming from contextual omissions or reliance on outdated knowledge—thereby enabling a holistic and rigorous assessment of model reliability.

Recent empirical findings further underscore the necessity of broad evaluation dimensions. For instance, Novikova et al. demonstrate that models may score highly on factual alignment while failing to maintain logical consistency or avoid contradictions across contexts [375]. Likewise, Asgari et al. demonstrate that domain-specific risks (e.g., in clinical applications) are overlooked by surface-level metrics like ROUGE or BLEU [376]. Their proposed framework, which incorporates coherence, factual consistency, comprehensiveness, and potential harm, highlights that evaluation must move beyond superficial overlap to capture medical concept correctness and assess downstream risks [376]. Evaluation efforts that remain narrowly focused on accuracy or surface similarity risk overlooking deeper and often more consequential sources of error, including flawed causal reasoning, internal contradictions, and domain-specific safety hazards [375]. These insights highlight the necessity of extending hallucination evaluation beyond superficial measures to capture the foundational qualities that distinguish accurate from hallucinatory content.

The proposed core dimensions function as complementary analytical lenses that collectively provide an "MRI-like" diagnostic capability, enabling the hierarchical tracing of reasoning processes across multiple levels of abstraction. This framework offers both breadth—capturing diverse failure modes such as factual inaccuracies, misleading causal inferences, stylistic misalignments, and safety risks—and depth, by elucidating how these errors originate and propagate through reasoning chains and contextual dependencies.

By organizing the core dimensions into this structured progression—from factual grounding and reasoning quality to systemic reliability and practical constraints, such framework balances breadth (capturing diverse failure modes such as factual errors, causal mis-attribution, and safety risks) with depth (tracing how errors originate and propagate) and supports both diagnostic depth and practical applicability.

In the subsections that follow, each dimension is defined, paired with representative metrics for its assessment, and discussed in terms of trade-offs and open challenges that remain in its implementation.

Factuality

Factuality Assessment remains a foundational dimension of hallucination evaluation, as it verifies whether generated content aligns with objective truth and reliable external knowledge sources. Benchmarks such as FactScore introduced atomic-level evaluation, revealing significant variation in factuality depending on the type of information being assessed [306]. Modern approaches further extend this granularity by evaluating factuality at both the node level (atomic claims) and the path level (composite inferences) to support vertical traceability across reasoning steps.

Recent studies on *temporal factuality* have further highlighted the limitations of static corpora and knowledge cutoffs. To address this limitation, dynamic fact-checking systems have been proposed, integrating temporal knowledge graphs and adaptive reasoning modules.

When hallucination analysis is extended into the temporal domain, recent long-video benchmarks report that hallucination risk tends to grow with input size and semantic complexity—particularly across multi-event videos. An additional phenomenon, termed *temporal drift*, illustrates the accumulative nature of hallucinations in time-dependent tasks and leads to temporal hallucination [377]–[379].

This *compounding effect* highlights the fragility of temporal reasoning in multimodal models, where extended inputs exacerbate inconsistencies and undermine reliability. Such evidence suggests that temporal drift not only challenges evaluation methodologies but also necessitates the design of more robust benchmarks that account for sequence length, accumulation effects, and dynamic event consistency.

For Large Vision–Language Models (LVLMs), factuality requires domain-specific definitions that extend beyond text-based accuracy to encompass multimodal alignment. For example, Accuracy and precision metrics, and F1 scores are traditional metrics that provide an incomplete picture of hallucination behavior [374]. POPE quantified hallucinations in terms of object existence, reporting error rates of 30–40% [60] while H-POPE revealed that attribute hallucinations—cases where models correctly identify objects but fabricate their properties—are substantially higher, reaching 67% [380].

Faithfulness Faithfulness evaluation examines whether model outputs accurately reflect the information contained in the provided source material. This dimension is especially critical for retrieval-augmented systems, where hallucinations often arise as unfaithful extrapolations from retrieved context. A persistent challenge is confident confabulation, in which models produce plausible-sounding but fabricated citations or facts that can pass superficial checks. Empirical audits have quantified this behavior: for instance, when prompted to write literature reviews, GPT-3.5 fabricated 55% of citations and GPT-4 fabricated 18%; 43% (GPT-3.5) and 24% (GPT-4) contained substantive errors [381]. Complementary verifiability audits of generative search engines report that only 51.5% of generated sentences are fully supported by citations and that 25.5% of citations do not support the sentence they accompany [382]—indicating that many responses that look "grounded" still contain unsupported or distorted content.

Safety Likelihood and Risk Assessment

Safety-likelihood evaluation quantifies the probability that a model will hallucinate under varying conditions, enabling crucial risk-aware deployment decisions. Rather than a binary judgment, this perspective adopts probabilistic frameworks that capture the stochastic nature of error generation. Studies on epistemic and aleatoric uncertainty enable hallucination estimation for single- and multi-answer settings [383], and work on semantic entropy predicts the likelihood of confabulations in free-form text generation across models and domains without prior domain knowledge [76]. Thus, hallucination propensity can be modeled via uncertainty-based estimators (e.g., semantic entropy) and internal-state probes, with risk patterns that track identifiable factors such as query difficulty, domain specificity, and model confidence.

Statistical risk modeling—via multivariate analyses—characterizes hallucination likelihood as a function of identifiable risk factors (e.g., query complexity, domain specificity, model confidence), decomposes their contributions to overall risk, and isolates the primary drivers while controlling for confounders.

CausalBench evaluates safety from a complementary angle by probing an LLM's causal-reasoning ability and examining its relationship to hallucination tendencies [341]. Moreover, models with stronger causal-reasoning performance on CausalBench are observed to have lower hallucination rates on external evaluations, suggesting—but not proving—a link between causal understanding and factual faithfulness [341]. The CausalBench framework advances safety assessment through causal inference techniques that isolate the contribution of individual risk factors [341]. This causal understanding enables targeted mitigation: addressing root causes rather than symptoms reduces hallucination rates [341].

Vertical Reasoning Traceability

Vertical reasoning traceability enables layer-by-layer dissection of inference chains and emphasizes how reasoning chains unfold and where they fail. It operationalizes the "vertical assessment" principle as described in Table II for the LENS framework (Section VI-A5). Through tree-structured decomposition, complex queries are broken into atomic verification units (nodes) linked by logical dependencies (edges). This structured dissection enables evaluators to localize the origin of hallucinations—whether stemming from a faulty premise, flawed synthesis, or cascading error propagation—thereby enhancing diagnostic resolution and informing targeted mitigation strategies.

Representative metrics include:

- *Node-level Factuality*: Atomic claim verification rate (Section VI-C2d)
- *Error Propagation Index*: Measures cascading failures across dependent nodes and quantifies the degree to which failures in upstream nodes affect downstream reasoning.
- *Path Consistency*: Logical coherence across multiple reasoning branches

By systematically tracing inference paths, i.e., tracking how reasoning processes unfold and pinpointing where they break down, the vertical reasoning traceability dimension not only enhances diagnostic resolution but also provides interpretable signals that inform targeted mitigation strategies. Rather than treating hallucination as a surface-level misalignment, this dimension highlights its structural embedding in reasoning chains, making error localization and remediation both more precise and more actionable.

Causal Reasoning Capabilities

Causal reasoning constitutes a distinct and indispensable dimension of hallucination evaluation. It assesses whether a model can correctly infer causal relationships, differentiate them from mere spurious correlations, uncover underlying generative mechanisms, and simulate the consequences of hypothetical interventions [384], [385]. Mastery of causal reasoning is especially vital in high-stakes domains such as medicine, law, or policy-making, where misattributing correlation as causation can lead to erroneous conclusions and harmful downstream effects [326].

Rigorous evaluation of causal reasoning often requires mapping attribution to specific nodes within reasoning trees. This approach identifies the root causes—such as distinguishing between errors arising from reliance on spurious correlations, knowledge cutoffs, or structural gaps in reasoning versus those due to inherent knowledge boundaries—to enable targeted mitigation strategies, as aligned with Section VI-A5's Contribution. In doing so, it complements vertical reasoning traceability: while traceability shows where reasoning breaks down, causal reasoning establishes why those failures occur and provides a structured account of how specific inputs lead to particular outputs.

Empirical studies further underscore its importance. By systematically testing models on causal inference tasks, researchers have shown that causal attribution improves interpretability, enables error correction targeted at root causes, and fosters greater user trust in model outputs [385]. Integrating causal reasoning into evaluation frameworks thus ensures not only structural transparency but also conceptual soundness in reasoning processes.

Interpretability

As discussed in Section VI-A4, interpretability is an indispensable core dimension of hallucination evaluation. Unlike factuality, faithfulness, or reasoning-oriented dimensions that focus on the quality of model outputs, interpretability emphasizes the transparency, accountability, and auditability of the evaluation process itself.

While Vertical Reasoning Traceability (Section VI-C2d) provides structural transparency by showing where errors occur, and Causal Reasoning Capabilities (Section VI-C2d) provide conceptual transparency by explaining why they arise, interpretability uniquely addresses how evaluation judgments are communicated and made accountable. It requires that evaluation frameworks be auditable, reproducible, and trustworthy—not only for researchers diagnosing model failures, but also for practitioners, auditors, and policymakers overseeing real-world deployments.

In this sense, interpretability complements other dimensions by ensuring that hallucination evaluation is not only rigorous, but also communicable, reproducible, and deployable in high-stakes applications.

Self-Consistency

As discussed in Section VI-A3, the *Consistency* principle requires that models produce reproducible and reliable outputs across *runs*, *time*, *environments*, *configurations*, and *query variations*. This requirement is essential for both scientific validity and real-world deployment, where users expect stable answers regardless of phrasing or timing. While consistency issues can be examined from several perspectives, three aspects deserve particular attention: self-consistency, coherence consistency, and reliability consistency.

Self-consistency addresses the question of whether a model provides stable answers when semantically equivalent queries are rephrased or lightly perturbed [375], [386]. Perturbations include paraphrasing prompts, reordering clauses, or substituting synonyms, as well as comparing outputs sampled under fixed settings. Self-consistency metrics leverage the principle that true knowledge should remain stable across perturbations. Such instability undermines user trust and directly challenges the Consistency principle. Empirically, self-consistency tests that apply semantic-preserving perturbations reveal notable degradation under adversarial rephrasings, indicating fragile understanding rather than robust knowledge [61], [375], [386].

Typical metrics include agreement rates across paraphrases, pairwise similarity metrics, contradiction for pairs of outputs, variance across answers, consistency-accuracy, and semantic-equivalence clustering stability [375], [386]. These metrics capture whether a model's answers remain robust to superficial changes in wording (e.g., when users ask the same question in different ways) [375], [386].

Self-consistency evaluation thus targets a critical facet of hallucination behavior: a model's propensity to generate contradictory outputs for equivalent queries. Ultimately, self-consistency evaluation provides a diagnostic lens that bridges factual correctness with behavioral reliability, revealing when models fail not only in accuracy but also in stability.

To address this challenge, evaluation frameworks must go beyond surface comparison. Standardized protocols, seed-controlled sampling, and rigorous replication practices safeguard against randomness and variability while ensuring comparability across systems [322].

Coherence and Consistency

Coherence and consistency focuses on whether a model maintains internal logical alignment across its outputs. Unlike self-consistency, which probes stability across query variations, coherence evaluation examines logical structure and internal consistency, i.e., whether claims made within the same response (or across related responses) are logically compatible with each other. Failures of coherence often manifest as contradictions between otherwise factual statements, or as multi-hop inconsistencies where individual claims may be correct but their combination forms an invalid reasoning chain.

Recent work on consistency evaluation shows that surface fluency is a poor proxy for logical soundness: a substantial proportion of seemingly coherent responses contain subtle contradictions, and models frequently fail to maintain consistent claims across paraphrased contexts, with significant portions of seemingly coherent outputs containing internal contradictions [375], [386]. Perturbation-based approaches—such as paraphrased contexts or reordered premises—are often applied to stress-test coherence. Coherence evaluation, therefore, plays a critical role in diagnosing structural hallucinations that undermine reasoning validity.

Reliability and Consistency

Reliability and Consistency extends the notion of consistency beyond single responses to characterize stability of model behavior across *time*, *contexts*, and *user populations*. Reliability is indispensable for deployment: models that appear accurate under controlled benchmarks may exhibit substantial instability in practice. Whereas self-consistency asks whether rephrasings of the same query yield the same answer, and coherence assesses agreement among statements within a response, reliability evaluation assesses the stability and predictability of model behavior across repeated trials and varying conditions [316], [375], [386]. Key challenges include *tem-*

poral drift—identical queries issued at different times returning divergent answers—*contextual variability* across domains (e.g., academic vs. conversational tasks), and *demographic influences* in which performance varies systematically across user groups.

Capturing and quantifying this variability provides essential confidence bounds for scientific evaluation and risk-aware deployment scenarios. Unlike accuracy metrics that measure correctness at a single point, reliability metrics, such as Test-retest Reliability, Parallel-forms Reliability, capture the consistency of model performance over time and across contexts [316], [375], [386].

Cross-context reliability evaluation examines consistency across different application domains and user populations.

Robustness

Robustness quantifies hallucination stability across conditions. Adversarial robustness testing shows that hallucination rates increase significantly under targeted attacks, with models showing particular vulnerability to authority bias and other manipulation techniques [387], [388].

Paradoxically, calibration often degrades with model scale: larger models become more confidently wrong, necessitating explicit uncertainty training and challenging assumptions about the relationship between capability and reliability [76].

Tool Appropriateness

We further include an explicit assessment of Tool Appropriateness, comprising two facets: *Tool Necessity Detection Accuracy*, which measures whether models recognize when external verification is required, and *Tool Selection Adequacy*, which assesses the appropriateness of chosen verification sources.

Recent diagnostic work on tool-augmented LLMs formalizes related capabilities—such as solvability detection and multi-step tool planning—and shows that even advanced models struggle on these axes, underscoring the need for targeted evaluation.

Computational Cost and Efficiency Trade-offs

Computational Cost and Efficiency Trade-offs analysis for hallucination evaluation reveals fundamental tensions between detection quality and practical feasibility, with comprehensive evaluation requiring orders of magnitude more computation than standard inference. This analysis includes cost considerations for User-Centric Benchmark Construction (as outlined in Section VI-A5's Contribution 3), which emphasizes balancing evaluation rigor with accessibility. The economic dimension of evaluation directly impacts deployment decisions: full-spectrum hallucination detection significantly increases inference costs, making real-time evaluation prohibitively expensive for many applications.

The high Large language model evaluation costs that scale with model size create evaluation inequality where only well-resourced organizations can afford rigorous testing, potentially leading to deployment of inadequately evaluated models by smaller entities. Shared evaluation infrastructure and benchmark pooling initiatives attempt to democratize access, though coordination challenges and competitive concerns limit adoption [389].

Efficiency optimization strategies attempt to balance cost and quality through selective evaluation. Cascaded evaluation pipelines employ lightweight filters to screen obvious cases, reserving expensive deep evaluation for ambiguous instances, achieving improved detection reliability and maintaining assessment integrity through tool-augmented frameworks like UNIHD [390].

e) Emerging Metrics: As models grow more sophisticated, traditional metrics prove insufficient. Emerging approaches leverage advances in interpretability, uncertainty quantification, and cross-modal understanding to detect increasingly subtle forms of hallucination.

Vision-Aware and Multimodal Metrics

Vision-Aware and Multimodal Metrics need to address the unique challenges of multimodal hallucination and assess alignment between visual inputs and textual outputs. Typically, the hallucination evaluation for Multimodal-LLMs suffers from insufficient evaluation dimensions, high costs, and over-reliance on language priors rather than visual input [391]. For example, CHAIR (Caption Hallucination Assessment with Image Relevance) [111] evaluates object hallucination in image captions but only captures existence, not attributes or relationships.

Newer frameworks like AMBER (An LLM-free Multi-dimensional Benchmark) [61], provide scalable, low-cost alternatives to human annotation with multi-dimensional analysis across object existence, attributes, and relations. AMBER is an LLM-free multi-dimensional benchmark for the hallucination evaluation of MLLMs and evaluates across various tasks without relying on additional LLMs [61]. The framework's multi-dimensional assessment spans both generative tasks such as image captioning and discriminative tasks including visual question answering, capturing the full spectrum of multimodal reasoning capabilities [61].

The AMBER benchmark [61] represents a paradigm shift in multimodal hallucination evaluation by eliminating dependence on expensive LLM judges or human annotators while maintaining evaluation quality. This framework addresses a critical gap where previous evaluation methods either required costly GPT-4 API calls (with OpenAI's official pricing at $0.03 per 1,000 prompt tokens and $0.06 per 1,000 completion tokens [392]) or expensive human annotation processes, making large-scale evaluation economically infeasible [61].AMBER's automated pipeline achieves comparable accuracy to human evaluation at less than 1% of the cost, democratizing rigorous multimodal evaluation for the broader research community [61].This granular decomposition reveals that models like GPT-4V achieve 94.5% accuracy on object existence but only 77.1% on relational reasoning, exposing systematic weaknesses in compositional understanding [61].*LVLM-eHub* [350] unified six evaluation dimensions, revealing cross-task correlations: models excelling at object detection (higher than 80% accuracy) still fail at visual reasoning (45% accuracy), indicating task-specific competencies don't transfer [350].

While fine-grained visual grounding metrics trace generated text spans to image regions, visual hallucinations often stem from over-reliance on language priors rather than visual input [391]. *PhD* [393] introduces adversarial multimodal conflicts, showing models default to textual information 76% of the time when modalities disagree [393].

Temporal consistency metrics for video understanding show that hallucination rates increase substantially for events spanning multiple frames, highlighting the challenge of temporal reasoning in multimodal models [377].

Causal Attribution and Mechanistic Analysis Causal Attribution and Mechanistic Analysis has attracted substantial attention as a lens on hallucination. This line of work treats hallucinations as phenomena inherently induced by the representation of internal states and traces hallucinations via attention analysis and activation patching.

Recent mechanistic interpretability research [394] has revealed that different attention heads and layers play distinct roles in factual processing, with middle layers being particularly critical for maintaining factual accuracy. Mechanistic interpretability approaches [395] focus on identifying and understanding the internal representations responsible for generating factual versus hallucinated content. The identification of potential hallucinations can also be conducted by analyzing hidden layer activation at the last token of generated texts [318]. These methods enable researchers to pinpoint specific subnetworks and circuits that contribute to different types of errors, providing insights that can inform targeted mitigation strategies and improve model reliability.

Intervention studies using techniques such as causal tracing [396] demonstrate that targeted modifications to specific model components can reduce hallucination rates while preserving overall model performance. For example, attention attribution and activation patching identify circuits linked to factual errors, enabling targeted interventions that reduce hallucination rates without major performance loss [318], [397].

Entropy-based and Uncertainty-aware Scoring

Semantic Entropy [76] advances beyond token-level uncertainty by clustering semantically equivalent outputs before computing entropy, thereby measuring uncertainty at the level of meaning rather than specific word sequences. The method operates across datasets and tasks without task-specific supervision or prior task knowledge and generalizes to unseen tasks [76]. By flagging when a prompt is likely to trigger confabulations, it indicates when extra caution is warranted and enables users otherwise limited by model unreliability [76]. This semantic-aware approach achieves an AUROC of 0.8 without task-specific training, though computational requirements limit real-time application [76].

Risk-weighted and Severity-aware Metrics

Context-aware severity assessment incorporates user expertise and application domain, adjusting evaluation thresholds accordingly. For example, Medical hallucination severity scoring weights errors by potential patient harm, finding that while only a small portion of hallucinations are high-severity, they account for the majority of potential adverse outcomes. Financial risk metrics similarly weight hallucinations by potential monetary impact, revealing that seemingly minor numerical errors can cascade into million-dollar miscalculations. Educational applications tolerate high hallucination rates for exploratory learning but require low tolerance for examination systems, demonstrating the need for adaptive evaluation standards. Risk-weighted evaluation acknowledges that not all hallucinations are equally harmful. 0A risk-weighted (e.g., weighted-sum) evaluation metric helps improve model performance in safety-critical applications where hallucinations can have a significant consequence and provides transparent, interpretable trade-offs [398].

f) Cross-Cutting Challenges and Future Directions: The evaluation dimensions presented above reveal fundamental challenges that transcend individual methodologies, exposing critical gaps in our ability to comprehensively assess hallucination phenomena. Current evaluation frameworks suffer from insufficient coverage across multiple dimensions: they fail to capture the full spectrum of task types (from factual QA to creative generation), overlook rare but important hallucination categories, and systematically miss errors in less common objects or concepts [61]. Thus, models appear more reliable than they actually are, creating a dangerous blind spot [61]

These challenges align with and expand upon those outlined in the introduction (Section VI-A4), including *Compositional Complexity*, *Ground Truth Ambiguity*, *Computational Efficiency*, *Cross-Modal Consistency*, and the need to handle *Dynamic Knowledge*.

The compositional nature of hallucinations further compounds these challenges, where multiple small errors cascade into major falsehoods through error propagation mechanisms that current evaluation approaches fail to capture. The precision for atomic units for individual facts is much higher than for compositional hallucinations involving multiple interrelated claims [61], [306].

This compositional challenge becomes particularly acute in multi-hop reasoning tasks, where a single incorrect intermediate step can invalidate an entire chain of reasoning, yet current metrics often award partial credit that obscures the severity of such failures.

A deeper concern emerges from Goodhart's law: when a measure becomes a target, it ceases to be a good measure. Models trained to optimize specific metrics often develop new hallucination types undetected by those same metrics [390]. The "evaluation overfitting" phenomenon shows that models that improves on HaluEval actually perform worse on held-out real-world data [17], [332], highlighting the need for dynamic, evolving benchmarks.

The Western-centric design of current benchmarks creates systematic blind spots in evaluation. Hallucination detection accuracy drops for non-English languages and for culturally-specific content, reflecting the Western-centric nature of most benchmarks [4], [333].

Future work must develop culturally aware evaluation frameworks that recognize legitimate variation in knowledge and perspective versus genuine hallucinations [338].

Perhaps most troublingly, a fundamental tension exists between model capability and reliability: more capable models often hallucinate more confidently and creatively, making detection paradoxically harder as models improve [24], [72]. This suggests that hallucination may not be a bug to be fixed but an inherent property of the probabilistic reasoning that enables these models' impressive capabilities. Resolving this paradox requires rethinking evaluation from binary detection toward nuanced understanding of model knowledge boundaries and uncertainty [76].

Full-Spectrum Evaluation Framework

The robustness and reliability of evaluation frameworks require a full-spectrum approach that transcends traditional metrics. Such a comprehensive paradigm systematically examines model behavior across multiple orthogonal dimensions, including accuracy under different granularity, self-consistency across repeated queries, resilience to adversarial attacks, faithfulness to source material, computational efficiency, and inference capabilities under knowledge constraints. Each dimension exposes distinct failure modes that remain hidden when evaluation is limited to surface-level correctness, underscoring the necessity of a holistic framework capable of capturing the full complexity of hallucination phenomena. The items listed below illustrate typical combinations of evaluation metrics, though they are not exhaustive and do not capture all possible combinations.

Factuality at Different Granularity

Traditional accuracy metrics should be discussed at different granular levels. For example, in *Token-level* methods, models achieve 56.31%–71.05% Hallucination F1 Scores and 66.86%-75.37% Not Hallucination F1 Scores [362]; In *Sentence-level* approaches, factual sentences can be identified with an AUC-PR of 53.97%, balancing practicality and interpretability [19].; Besides, claim-level decomposition shows that even advanced models achieve only 58% factual precision [306]. These results demonstrate the significance of structured exploration on how granularity choices affect detection capability.

Faithfulness, Source Attribution, and Task-Specific Definition

Faithfulness in this broader sense encompasses not only whether generated content reflects the given source material but also how models handle attribution and task-specific requirements. Evaluating faithfulness is therefore essential for assessing whether large language models (LLMs) remain grounded in context without introducing unsupported or contradictory information.

A structured taxonomy of faithfulness violations provides an analytical lens for diagnosing different failure modes in context adherence. This categorization enables targeted evaluation and mitigation strategies, revealing that different model architectures exhibit distinct faithfulness failure patterns and highlighting the importance of model-specific evaluation.

Recent advances in attribution analysis integrated with vertical tracing highlight the scale of this issue: approximately 40% of responses that appear to be grounded contain subtle distortions or unsupported inferences that escape detection by traditional metrics [399].

For example, FaithDial [400] can be used to train both high-quality dialogue generation models and a hallucination critic that classifies utterances as faithful or unfaithful. Models trained on FaithDial achieve improved performance on the BEGIN benchmark, outperforming counterparts trained on existing dialogue-coherence datasets.FaithDial benchmarks contemporary dialogue models and introduces an auxiliary contrastive learning objective. According to multiple automatic metrics, this objective yields highest faithfulness and increased abstractiveness [400]. FaithDial benchmarks state-of-the-art models and proposes an auxiliary contrastive objective that achieves the highest level of faithfulness and abstractiveness based on several automated metrics [400]. In human evaluations, models trained on FAITHDIAL produce responses rated as more interpretable, cooperative, and engaging, and improve faithfulness and abstractiveness, both in domain and during zero-shot transfer to other datasets, such as TopicalChat and CMU-DoG [400].

Inference Capability and Knowledge Boundaries

Inference capability evaluation probes the model's ability to recognize and respect its knowledge boundaries, distinguishing between legitimate reasoning and hallucinated extrapolation. Many model errors stem not from knowledge gaps but from deficiencies in causal reasoning and failure to recognize knowledge boundaries. Moreover, the ability to generalize causal knowledge across contexts strengthens model adaptability to novel scenarios and distribution shifts [401].The integration of causal reasoning enhances both transparency and robustness in AI systems.

Ultimately, robustness-centered evaluation highlights the importance of moving beyond static measures of factual accuracy toward a more dynamic and adversarially aware understanding of hallucinations. By capturing both reasoning stability and retrieval fidelity, this dimension ensures that large language models are not only accurate in controlled conditions but also trustworthy when confronted with real-world variability and adversarial challenges. Such evaluations will be indispensable in advancing AI systems that can maintain reliability, transparency, and safety across diverse operational contexts.

Self-Consistency across Tasks and Temporal Stability

Empirical evidence confirms that self-consistency remains a pervasive weakness. Models exhibit varying degrees of self-consistency across different tasks and different temperatures.

Longitudinal analyses indicate that GPT-4's answers can vary when semantically equivalent questions are rephrased or asked at different times, with the variability becoming more pronounced on complex reasoning tasks [61]. These findings suggest that such inconsistency is not mere noise but reflects structural fragility in reasoning and contextual grounding.

Temperature-dependent consistency analyses reveal a non-linear relationship: near-deterministic settings with very low temperatures tend to produce repetitive yet consistently incorrect outputs, whereas moderately stochastic settings with moderate temperatures introduce greater diversity and can occasionally yield correct responses [182].

These findings indicate that inconsistency is not simply noise but reflects structural fragility in reasoning and contextual grounding.

Furthermore, it exacerbates the *Ground Truth Ambiguity* challenge (Section VI-A4) by making definitive verification even more difficult.

Adversarial Robustness

Adversarial evaluation reveals critical vulnerabilities in hallucination detection and mitigation strategies. Targeted perturbations that preserve semantic meaning but alter surface forms can increase hallucination rates, with models confidently generating false information when prompted with carefully crafted adversarial inputs. Universal adversarial triggers—short phrases that reliably induce hallucinations across diverse contexts—demonstrate that models possess systematic weaknesses exploitable through minimal input modifications [374].

These findings directly test the *Robustness* design principle (Section VI-A3), which demands resilience to edge cases and adversarial inputs. The ease with which hallucinations can be induced underscores the critical need for evaluation frameworks to incorporate rigorous adversarial testing, as envisioned in the introduction's call for comprehensive assessment.

Robustness to Distribution Shift

Distribution shift evaluation examines how hallucination patterns change when models encounter out-of-distribution inputs, revealing dramatic degradation in reliability. In cross-domain transfer, models trained on general text produce substantially more hallucinations in specialized settings—particularly in medical and legal applications [42]. Distribution shift analysis reveals that models trained on Wikipedia exhibit elevated hallucination rates when applied to social media, highlighting generalization challenges [4]. Geographic and cultural shifts further expose Western-centric biases, with non-Western contexts showing higher hallucination rates even for factual queries [374]. This sensitivity to distribution shifts exemplifies the *Dynamic Knowledge* and *Cross-Modal Consistency* challenges described in Section VI-A4 and underscores the necessity for evaluation frameworks, as called for in Section VI-A5, to be adaptable and capable of handling evolving knowledge and diverse contextual deployments beyond static training distributions.

Computational Efficiency and Scalability

The computational demands of comprehensive hallucination evaluation pose practical barriers to deployment: full-spectrum assessments typically require substantially more resources than standard accuracy testing [19]. Sampling-based approaches that score only a subset of outputs can approximate exhaustive evaluations at much lower computational cost, but they risk missing rare but catastrophic failures [19]. In reality, strict real-time constraints necessitate lightweight detectors and force trade-offs between detection accuracy and latency [390]. The economic dimension of evaluation—where comprehensive testing of large models costs thousands of dollars per evaluation cycle—creates inequities where only well-resourced organizations can afford rigorous hallucination assessment [61].

This challenge of *Computational Efficiency* was explicitly foreseen in Section VI-A4. Overcoming these barriers is essential for realizing the practical *User-Centric Benchmark Construction* and widespread applicability envisioned for the LENS framework in Section VI-A5, ensuring that rigorous evaluation is not just theoretically sound but also practically feasible.

Future Directions: Toward Unified and Adaptive Evaluation

The path forward requires fundamental reconceptualization of hallucination evaluation from fragmented metrics toward unified, adaptive frameworks that evolve with model capabilities and deployment contexts. This vision directly addresses the gaps and limitations identified in the introduction's survey of the current evaluation landscape (Section VI-A1) and its outlined challenges (Section VI-A4), particularly the need to handle *Compositional Complexity* and *Dynamic Knowledge*. Dynamic benchmark generation using procedural content creation and adversarial techniques can produce continuously novel test cases, preventing evaluation overfitting while maintaining consistent difficulty calibration [389]. These adaptive systems must incorporate feedback loops that learn from deployment failures, automatically generating new test cases that probe discovered vulnerabilities.

Multi-stakeholder evaluation frameworks must balance competing objectives: researchers need fine-grained diagnostic information, practitioners require actionable deployment guidance, and end-users demand interpretable reliability indicators. This aligns with the LENS framework's goal of providing *Actionable Insights* (Section VI-A5) and adheres to the *Interpretability* and *Faithfulness* design principles (Section VI-A3) by making evaluation results transparent and useful for different audiences. Hierarchical evaluation protocols that provide different levels of abstraction—from detailed mechanistic analysis to simple trust scores—can serve these diverse needs while maintaining consistency across levels. The development of "evaluation APIs" that allow domain experts to contribute specialized test cases without deep ML expertise will democratize benchmark creation and ensure coverage of long-tail use cases.

Cross-modal and cross-lingual evaluation represents a critical frontier, as current English-centric, text-focused frameworks fail to capture the full spectrum of hallucination phenomena in multilingual, multimodal systems. This directly tackles the *Cross-Modal Consistency* challenge from Section VI-A4 and is a necessary extension for any framework aiming to be truly comprehensive. Universal hallucination taxonomies that transcend modality boundaries—defining hallucination in terms of information-theoretic divergence rather than surface-level errors—could enable consistent evaluation across diverse model types. Cultural adaptation layers that distinguish legitimate knowledge variation from genuine hallucinations will be essential for global deployment, requiring collaboration with diverse communities to establish culturally-aware ground truth. This effort is crucial for resolving the *Ground Truth Ambiguity* challenge in culturally nuanced contexts, moving beyond Western-centric benchmarks mentioned in the introduction.

The integration of human-in-the-loop evaluation with automated metrics promises to combine scalability with nuanced judgment. Active learning approaches that strategically select instances for human review—focusing on high-uncertainty or high-impact cases—can maximize the value of limited human evaluation resources. Crowdsourced evaluation platforms that aggregate judgments from diverse evaluators while accounting for expertise and cultural background can provide more robust assessments than single-annotator ground truth.

Causal evaluation frameworks that move beyond correlation to establish mechanistic links between model properties and hallucination behaviors will enable targeted improvements. This deeper analysis is a key step toward fulfilling the promise of *Actionable Insights* and moving from mere detection to understanding *why* hallucinations occur, as advocated in Section VI-A5. Interventional studies that systematically modify model components, training procedures, or inference strategies can identify causal factors driving hallucinations. Counterfactual evaluation that assesses how models would behave under alternative conditions provides insights into the robustness and generalizability of hallucination patterns.

Finally, the development of metacognitive evaluation that assesses models' awareness of their own limitations represents perhaps the most promising direction. Models that can accurately express uncertainty, refuse to answer beyond their knowledge, and actively seek clarification when faced with ambiguous queries would fundamentally change the hallucination landscape. This direction resonates strongly with the introduction's ultimate goal of transforming LLMs into "trustworthy partners" and addresses the core issue of models failing to recognize their own knowledge boundaries, a source of many hallucinations. Evaluation frameworks must therefore evolve to reward appropriate uncertainty expression and penalize confident confabulation, shifting the focus from eliminating hallucinations to managing them transparently.

The ultimate goal is not perfect factuality—an impossible standard for systems trained on imperfect data—but reliable, interpretable, and controllable behavior that allows users to make informed decisions about when and how to trust model outputs. This conclusion perfectly encapsulates the overarching argument of the introduction: that systematic evaluation is paramount for "ensuring safe and trustworthy AI deployment," and achieving the *Accuracy*, *Faithfulness*, and *Interpretability* demanded by the design principles in Section VI-A3.

3) Mechanistic Understanding for Decision-Making: Decision-making evaluation plays a critical role in understanding and mitigating hallucinations in LLMs and LVLMs, yet the field remains fragmented by methodological inconsistencies that obscure causal mechanisms. Current studies often employ heterogeneous experimental setups that conflate multiple sources of error, making it impossible to isolate specific failure modes or design targeted interventions [338].

The evaluation landscape is characterized by a proliferation of datasets, from factual QA benchmarks like TruthfulQA [310] and HaluEval [17] to domain-specific collections like MedHallBench [370] and FinanceBench [402], each capturing different aspects of hallucination phenomena but lacking unified evaluation protocols. Similarly, metrics range from token-level accuracy and entity-level F1 scores to semantic similarity measures and human judgment proxies, creating an evaluation Tower of Babel where results cannot be meaningfully compared across studies [319]. This methodological fragmentation not only impedes scientific progress but also renders deployment decisions essentially arbitrary, as practitioners lack principled frameworks for selecting appropriate evaluation strategies for their specific use cases.

Constructing benchmarks and designing targeted interventions requires a systematic separation of hallucination causes—distinguishing knowledge gaps from reasoning failures, parametric memory conflicts from retrieval errors, and linguistic biases from genuine misunderstanding—to ensure reliable diagnosis and effective mitigation [338].

Interpretability and intervention have emerged as two central perspectives in this line of work. On the interpretability side, investigations focus on understanding the internal thought process of LLMs that give rise to hallucinations, while intervention-based approaches directly modify model components to suppress undesired outputs. Results demonstrate that the success of interventions is highly component-dependent, with modifications to attention blocks proving effective, whereas adjustments to the residual stream are detrimental to language modeling performance [338].

Beyond the architectural and algorithmic considerations, the manifestation and detection of hallucinations are profoundly influenced by a constellation of contextual factors that shape model behavior at inference time. These impact factors—ranging from the granularity of evaluation units to the subtle interplay between linguistic priors and prompting strategies—determine not only whether hallucinations occur but also their severity, persistence, and detectability. Understanding these factors is essential for developing robust evaluation frameworks that can generalize across diverse deployment scenarios and provide actionable insights for mitigation strategies. Recent empirical studies have identified three primary dimensions along which these factors operate: the granularity at which hallucinations are measured, the influence of preexisting language biases that override factual grounding, and the critical role of prompt engineering in either amplifying or suppressing hallucinatory behaviors [319], [335]. Each dimension reveals distinct failure modes and suggests targeted intervention strategies, yet their interactions remain poorly understood, creating blind spots in current evaluation methodologies.

a) Granularity: The granularity at which hallucinations are evaluated fundamentally shapes both detection capabilities and intervention strategies, yet no consensus exists on optimal evaluation units. Hallucinations manifest across a spectrum of granularities—from individual tokens and named entities to complex propositions, multisentence passages, and entire document structures [319]. Token-level evaluation, exemplified by methods like SelfCheckGPT [19], enables precise localization of factual errors but suffers from exponential computational costs and difficulty in capturing semantic-level inconsistencies. Entity-level approaches, widely adopted in early fact-checking systems, excel at identifying specific factual inaccuracies—particularly for named entities, dates, and quantities—but fail to detect more subtle forms of hallucination such as relational errors or contextual misrepresentations [319].

At coarser granularities, sentence- and claim-level evaluations have gained prominence with the advent of neural factuality scoring methods like FactScore [306], which decompose responses into atomic claims for independent verification. These approaches strike a balance between computational tractability and semantic completeness, enabling evaluation of complex factual relationships while maintaining reasonable processing requirements. Document-level evaluation, while computationally efficient for large-scale assessment, obscures fine-grained error patterns and provides limited diagnostic value for targeted improvements.

The evolution of granularity preferences reflects deeper shifts in the field: early retrieval-based methods favored entity-level precision, the rise of generative models drove adoption of sentence-level metrics, and current multimodal systems demand hierarchical evaluation spanning multiple granularities simultaneously [319]. Empirical studies reveal that optimal granularity is inherently task-dependent—medical applications require token-level precision for drug names and dosages, while conversational systems benefit from discourse-level coherence checks [370]. This granularity–performance trade-off remains a fundamental challenge, with recent work exploring adaptive granularity selection that dynamically adjusts evaluation resolution based on detected error patterns and computational constraints.

b) Language Priors and Bias: The pernicious influence of language priors and learned biases represents one of the most insidious sources of hallucination in modern LLMs and LVLMs, where statistical regularities in training data override factual grounding and sensory evidence. These priors manifest as systematic preferences for plausible-sounding but incorrect completions, creating a fundamental tension between fluency and factuality that current architectures struggle to resolve [335]. The problem is particularly acute in multimodal systems, where language priors can completely dominate visual or auditory signals, leading to descriptions that are linguistically coherent but entirely disconnected from the actual input.

EventHallusion [92] provides compelling evidence of this phenomenon in video-based LVLMs, demonstrating that models consistently hallucinate temporal sequences based on learned event scripts rather than observed visual evidence. When presented with videos showing events in noncanonical order (e.g., putting on shoes before socks), models overwhelmingly generate descriptions following typical temporal patterns, revealing a catastrophic failure to ground language generation in actual sensory input [92]. The proposed Temporal Cue Disentanglement (TCD) method addresses this by systematically manipulating temporal markers, forcing models to rely on visual information rather than defaulting to linguistic shortcuts [92].

Beyond temporal biases, language priors manifest across multiple dimensions: frequency biases lead models to favor common entities over rare but correct ones, co-occurrence statistics drive spurious associations (e.g., automatically mentioning "Eiffel Tower" when discussing Paris, even when irrelevant), and syntactic preferences can override semantic accuracy [335]. Cultural and demographic biases embedded in training corpora further compound these issues, causing models to hallucinate stereotypical attributes or relationships that reflect societal biases rather than factual reality [338].

Empirical evaluations on the EventHallusion benchmark reveal striking disparities in model performance: open-source VideoLLMs demonstrate significantly higher susceptibility to prior-driven event hallucinations compared to their closed-source counterparts. Furthermore, the proposed Temporal Contrastive Decoding (TCD) method proves effective as a training-free approach to mitigate such hallucinations across various model architectures. [92].

c) Prompting Strategy: Prompting strategy emerges as a critical yet underappreciated determinant of hallucination rates, with subtle variations in prompt construction producing dramatic differences in model reliability and factual accuracy. The relationship between prompt design and hallucination is far more nuanced than initially understood, involving complex interactions between linguistic properties, structural patterns, and cognitive scaffolding that either amplify or suppress the model's tendency to confabulate [182].

Counterintuitively, empirical evidence reveals that simpler prompting strategies consistently outperform their more complex counterparts in reducing hallucinations [335]. Zero-shot prompts with clear, direct instructions produce fewer hallucinations than elaborate few-shot examples that may introduce spurious patterns or conflicting signals. This *complexity penalty* becomes particularly pronounced when LLMs are augmented with external tools or multistep reasoning chains—each additional component introduces new failure modes and compounds the probability of hallucination [335].

The impact of readability proves more complex and context-dependent, suggesting intricate interactions with other linguistic features and task characteristics.

A groundbreaking empirical investigation into the linguistic properties of prompts reveals three critical dimensions that govern hallucination rates: formality, concreteness, and readability [335].Prompts exhibiting higher formality—characterized by precise technical language, explicit constraints, and structured formatting—demonstrate a marked reduction in hallucination rates across diverse model architectures.

Concreteness, measured through specificity of references and grounding in verifiable facts, shows even stronger effects, with highly concrete prompts significantly reducing factual errors in categories like numerical values and proper nounss [335].

The impact of readability proves more complex and context-dependent, suggesting intricate interactions with other linguistic features and task characteristics.

Advanced prompting techniques reveal distinct tradeoffs in their effectiveness against hallucinations. Chain-of-Thought (CoT) prompting [47], while improving reasoning transparency, can paradoxically increase hallucinations when intermediate steps contain errors that propagate through the reasoning chain [182]. Self-consistency approaches that sample multiple reasoning paths show promise in identifying and filtering hallucinations through divergence detection, but at substantial computational cost [182]. Multiagent debate frameworks demonstrate impressive hallucination reduction through adversarial verification, yet their complexity makes them impractical for real-time applications [182].

The effectiveness of prompting strategies is further modulated by the linguistic properties of the prompt itself. Empirical findings indicate that prompts characterized by greater formality and concreteness tend to result in reduced hallucination across various categories (Person, Location, Number, Acronym) [182]. However, the impact of readability on hallucination rates shows a more mixed and inconclusive pattern. These relationships between prompt design and model output fidelity are observed to be more pronounced in advanced LLMs such as GPT-4 and OPT [182].

Recent work has begun exploring adaptive prompting strategies that dynamically adjust based on detected uncertainty or contradiction signals, showing promising early results in reducing hallucinations while maintaining output quality [335]. These approaches leverage metacognitive signals—such as token-level confidence scores, attention pattern anomalies, and semantic coherence metrics—to identify potential hallucination points and trigger more conservative generation strategies or explicit uncertainty expression [182].

d) Causal Attribution and Mechanistic Insights: Understanding the causal mechanisms underlying hallucinations in decision-making contexts requires moving beyond correlational observations to establish direct causal links between model components and hallucinatory outputs.

Particularly revealing are findings about the internal mechanisms transformers employ for multi-step reasoning. Mechanistic analysis of a 6-layer model trained on a synthetic path-finding task reveals that it implements a backward chaining algorithm. This algorithm operates in a layer-wise manner, with early layers aggregating edge information, and subsequent deduction heads in middle layers (e.g., layers 2-4) responsible for recursively climbing the tree structure one step per layer. Perturbing these critical components via causal scrubbing techniques disproportionately disrupts the model's reasoning capability, demonstrating their causal necessity for correct output generation [345].

This "factuality bottleneck" suggests that hallucinations may result from insufficient information flow through these critical layers, potentially addressable through architectural modifications or targeted fine-tuning strategies [345].

e) Evaluation Benchmark Design and Validation: The proliferation of hallucination benchmarks without systematic validation criteria has created a paradox: we have numerous ways to measure hallucinations but limited confidence in what these measurements actually capture. Effective benchmark design for decision-making evaluation must address several fundamental challenges that current approaches largely ignore [61], [389].

First, the temporal validity problem: benchmarks quickly become obsolete as models are trained on test data, either directly through data contamination or indirectly through similar distributions appearing in training corpora. Dynamic benchmark generation, as proposed in recent work [389], offers a promising solution by creating novel test instances from verified knowledge sources, though this introduces new challenges in ensuring consistent difficulty and coverage.

Second, the domain transfer challenge: benchmarks developed for general factual accuracy often fail catastrophically when applied to specialized domains like medicine, law, or finance, where hallucinations carry vastly different risks and manifestations [370], [402]. Domain-specific benchmarks must capture not just factual accuracy but also appropriate uncertainty expression, regulatory compliance, and professional standards that vary dramatically across fields.

Third, the compositionality gap: current benchmarks primarily test atomic facts or simple relationships, failing to evaluate complex multistep reasoning where hallucinations can compound through inference chains. Hierarchical benchmarks that decompose complex queries into verifiable subcomponents while maintaining logical consistency represent a critical frontier for evaluation development [403].

f) Cross-Modal Hallucination Dynamics: The emergence of vision–language models introduces qualitatively new forms of hallucination that transcend traditional text-based evaluation paradigms. Cross-modal hallucinations arise from fundamental misalignments between visual encoders and language decoders, creating failure modes that cannot be detected through unimodal evaluation [92], [404].

Visual grounding failures manifest in multiple ways: object hallucination (describing nonexistent entities), attribute confusion (misattributing properties across objects), spatial relationship errors (incorrect relative positions), and temporal sequence violations (misordering events in video) [404]. Each category requires distinct evaluation strategies—pixel-level verification for object presence, scene graph matching for relationships, and temporal logic checking for event sequences.

The modality imbalance problem proves particularly pernicious: language models' superior training on text leads to linguistic dominance, where plausible textual completions override visual evidence. This creates a "hallucination gradient" where errors increase with the semantic distance between visual input and textual prior knowledge. Models consistently hallucinate typical attributes for recognized objects (e.g., describing a white crow as black) and generate stereotypical scene descriptions even when contradicted by visual input [92].

Recent approaches to cross-modal evaluation employ adversarial visual inputs, counterfactual image-text pairs, and synthetic scene modifications to stress-test grounding mechanisms. These reveal that current VLMs often rely on superficial visual features and textual shortcuts rather than deep scene understanding, suggesting fundamental architectural limitations in how visual and linguistic information are integrated [404].

g) Implications for Real-World Deployment: The translation of hallucination evaluation from research settings to production environments reveals critical gaps between academic metrics and operational requirements. Real-world deployment demands not just detection but also graceful handling of hallucinations, requiring evaluation frameworks that assess both failure rates and failure modes [314].

Risk-stratified evaluation becomes essential in deployment contexts: a hallucinated drug dosage in medical applications carries vastly different consequences than a hallucinated historical date in educational content. Current evaluation frameworks fail to incorporate risk weighting, treating all hallucinations as equally problematic. Operational metrics must distinguish between categories of harm—safety-critical errors, legally consequential mistakes, reputation-damaging inaccuracies, and minor factual errors, with evaluation protocols tailored to each risk level.

The challenge of evaluation under distribution shift proves particularly acute in production settings, where models encounter inputs far outside their training distribution. Robustness evaluation must assess not just average-case performance but worst-case behavior under adversarial or out-of-distribution inputs. This requires moving beyond static benchmarks to dynamic evaluation protocols that continuously probe model boundaries and identify emerging failure modes [61].

Furthermore, the interaction between hallucination evaluation and user behavior creates feedback loops that current frameworks ignore. Users adapt to model limitations, developing workarounds that may mask or exacerbate hallucination problems. Evaluation must therefore consider the full sociotechnical system, including user strategies, interface design, and organizational processes that mediate model outputs [314].

4) Evaluation Methodologies (Implementation Approaches): Evaluation methodologies operationalize the theoretical frameworks and dimensional choices into concrete implementation strategies, determining not just what we measure but how we execute these measurements at scale. The field has evolved from simple output comparison to sophisticated approaches that probe model internals, manipulate causal pathways, and leverage uncertainty quantification. Pioneering methodologies include SelfCheckGPT's [19] self-consistency paradigm achieving reference-free detection, ReEval's [320] adversarial perturbation framework exposing model vulnerabilities, and TruthPrint's [321] latent space interventions identifying universal truthful directions. These diverse approaches reflect a fundamental tension: while automated methods enable scalability, they often miss nuanced errors that human judgment catches [19]; while internal state analysis provides mechanistic insights, it remains model-specific [321]; while adversarial testing reveals vulnerabilities, it may not reflect real-world usage patterns [320].

The methodological landscape has crystallized around six complementary paradigms, each addressing different aspects of the hallucination challenge. Probe-based and prompt-based methods leverage external tools or crafted inputs to elicit inconsistencies, balancing flexibility with computational overhead. Adversarial testing actively perturbs inputs to stress-test model robustness under controlled conditions. Causal intervention techniques isolate failure mechanisms through counterfactual analysis. Uncertainty-guided approaches exploit intrinsic model properties for reference-free detection. Internal state analysis traces hallucination origins through attention patterns and activation dynamics. Online evaluation captures real-world performance through user-facing deployments. This diversity reflects the multifaceted nature of hallucinations—no single methodology suffices; effective evaluation requires strategic combination based on specific use cases, resource constraints, and reliability requirements.

a) Probe-based and Prompt-based Evaluation: Probe-based and prompt-based evaluation methods represent a prominent category in hallucination detection and probability estimation for LLMs. The fundamental innovation lies in their dual approach: probes leverage external tools, metrics, auxiliary models, or structured annotations to scrutinize outputs for inconsistencies, while prompt-based methods craft specific inputs to elicit self-assessment or provoke hallucinations for analysis. A seminal work in this category is POPE (Polling-based Object Probing Evaluation) [60], which pioneered systematic object hallucination assessment in VLMs by converting detection into binary questions (e.g., "Is there a [object] in the image?"). POPE's novelty lies in revealing that VLMs exhibit systematic biases toward common and co-occurring objects, with overall accuracy around 70-77% under various sampling strategies, exposing a fundamental tendency to hallucinate frequently appearing items [60].

In the culturally sensitive test framework, prompt-based questioning across multiple query categories is used to provoke hallucinations in GPT models, evaluated through a human-centric probe-based comparison that modifies Turing's Imitation Game to highlight cultural nuances and societal contexts, outperforming purely algorithmic detection by emphasizing non-binary judgments [338]. THaMES [186], on the other hand, uses a hybrid approach that combines prompt-based test set generation through weighted sampling, batch processing, and various question types, such as counterfactual validation, to generate QA pairs from corpora like Wikipedia and academic papers [186]. Probe-based benchmarking uses multi-faceted metrics for hallucination generation and identification across tasks like text generation and binary classification. This results in an end-to-end pipeline that incorporates mitigation strategies, such as RAG, ICL, and PEFT and outperforms isolated prompts by achieving cost-effective diversity [186].

Using prompt-based classifiers for atomic fact evaluation and context reasoning, OnionEval [346] presents a layered probe-based metric called context-influence score (CI) to quantify fact-conflicting hallucinations in SLLMs across nested contexts [346]. This reveals that SLLMs are better at factual compression than they are at context comprehension when compared to larger models, allowing for targeted mitigations like chain-of-thought that improve performance without incurring high computational costs [346].

In contrast to synthetic prompt-induced datasets, RAGTruth [359] highlights fine-tuning's advantage over RLHF in sparse evaluations. RAGTruth focuses on word-level probe-based annotations in a corpus of 18,000 natural responses from diverse LLMs in RAG settings [359], benchmarking detection via fine-tuned models that surpass prompt-based self-checking or intrinsic state probes by addressing both benign and harmful hallucinations [359].

Med-HALT's prompt-based tests, such as False Confidence Test (FCT), None-of-the-Above (NOTA), and Fake Questions Test (FQT), evaluate reasoning hallucinations, while probe-based memory tests assess retrieval accuracy in medical LLMs using multinational datasets [42].

In contrast to English-centric probes like TruthfulQA, HalluQA [355] for Chinese LLMs uses GPT-4 as a prompt-based automated evaluator on 450 adversarial questions across multiple domains. Probe-based critiques highlight cultural and alignment factors, demonstrating that alignment reduces imitative issues more effectively than scaling and promoting human intervention in culture-dependent hallucinations [355].

HaDes focuses on reference-free, token-level probing through binary classification tasks applied to modified Wikipedia texts [362]. It utilizes prompt-based labeling combined with iterative model-in-loop approaches to ensure data balance. This facilitates real-time detection during the decoding process, surpassing the effectiveness of sentence-level prompting by identifying detailed, subtle signals that allow immediate intervention in open-ended text generation [362].

EventHallusion detects event hallucinations in video-based large language models by utilizing GPT-assessed prompts that focus on temporal discrepancies and vision-language inconsistencies. It employs a probe-driven approach known as Temporal Contrastive Decoding (TCD) to minimize interference without requiring additional training [92]. This method tackles challenges unique to video data, such as the accurate representation of chronological sequences, which are often neglected by static image-based prompts. Furthermore, EventHallusion highlights the weaknesses of open-source models when compared to their closed-source counterparts [92].

Bias and Interference Challenges in Visual Language Models (Bingo)'s probe-based evaluation framework carefully selects instances of model failures to evaluate various biases [329], such as regional and OCR-related biases, and interference effects in GPT-4V. The assessment employs prompt-driven leading questions and multi-image test cases, incorporating self-correction mechanisms that reduce hallucinations [329]. This benchmark reveals underlying multimodal weaknesses, including a noticeable Western bias, which goes beyond the limitations observed in text-only inputs used in models [329].

The evolution from simple probing (POPE) to sophisticated frameworks (THaMES, EventHallusion) demonstrates the field's progression toward comprehensive evaluation [60]. EventHallusion's temporal probing for video understanding represents the frontier, detecting chronological inconsistencies that static methods miss entirely [92]. These methodologies collectively reveal that effective probing requires domain-specific adaptation: medical contexts demand specialized knowledge probes (Med-HALT), cultural evaluation needs modified Turing tests, and multimodal scenarios require cross-modal consistency checks.

b) Adversarial and Stress Testing: Adversarial and stress testing plays a crucial role in evaluating the robustness of large language models (LLMs) and large vision–language models (LVLMs), particularly under conditions that intentionally expose weaknesses not captured by standard benchmarks. Unlike conventional evaluation, which often measures performance on well-structured datasets, adversarial testing deliberately perturbs inputs or simulates edge cases to uncover systematic vulnerabilities. Stress testing further extends this paradigm by assessing model behavior under complex, noisy, or ecologically valid scenarios that better reflect real-world use cases.

To capture the breadth of current approaches, this section reviews several representative methodologies within this category. Specifically, we discuss adversarial methodologies, which generate controlled perturbations to reveal fragility in model reasoning; vision–language stress testing, which adapts adversarial principles to multimodal systems to evaluate cross-modal consistency and resilience; and real-world adversarial queries, which focus on user-derived or naturally occurring prompts to assess ecological validity. Collectively, these methods provide complementary perspectives on robustness, scalability, and practical reliability, highlighting both the strengths and limitations of adversarial evaluation paradigms.

- **Adversarial Methodologies.** They actively perturb inputs to expose model vulnerabilities under controlled stress conditions. ReEval [320] pioneers a transferable framework for retrieval-augmented LLMs by dynamically generating adversarial test cases through *evidence perturbation*. Its key innovation employs two strategies: *answer swapping* - replacing correct answers with valid alternatives and *context enriching* - adding relevant information, both synthesized via LLM-based prompt chaining. Empirical results show that ReEval causes GPT-4 accuracy to drop from 83% to 47% on perturbed queries, while ChatGPT's performance degrades from 76% to 42% [320]. Strikingly, the attack data generated by Alpaca-7B has a relatively small impact on the accuracy of GPT-4 (with an accuracy rate of 89%), indicating low attack effectiveness. [320].
- **Vision–Language Stress Testing.** Extending these principles to multimodal contexts, vision–language stress testing leverages automated adversarial construction. Au-

toHallusion [112] automates adversarial benchmark generation for LVLMs by reverse-engineering *schema-based conflicts*. Inspired by cognitive dissonance theory, it manipulates images via three strategies: (1) *abnormal object insertion*, (2) *paired object insertion*, and (3) *correlated object removal*, creating "contextual traps" that exploit language priors.

The interventions induce hallucinations at rates exceeding 97% on models such as GPT-4V and Claude 3, thereby uncovering systematic biases in cross-modal reasoning while substantially reducing the reliance on manual benchmark construction and human curation costs [112].

- **Real-World Adversarial Queries.** From another perspective, Real-world adversarial queries prioritize ecological validity by capturing naturally challenging user prompts. HalluEval-Wild [332] curates challenging user interactions from ShareGPT, filtered adversarially against Alpaca to target five hallucination-prone query types (e.g., out-of-scope information, beyond-modality requests). Using retrieval-augmented generation (RAG) to provide reference answers, this benchmark highlights heightened hallucination risks in knowledge-distilled models, which often perform well on alignment benchmarks yet falter under real-world stress conditions.

  Furthermore, *Collective limitations* persist across adversarial evaluation methodologies. ReEval's concentration on short-answer QA restricts its coverage of complex tasks. AutoHallusion's primitive image stitching compromises perceptual naturalness. HalluEval-Wild's static benchmark struggles with evolving LLM capabilities. These methodologies collectively underscore trade-offs between *automation scalability*, *ecological validity*, and *granular controllability* in stress-testing paradigms.

c) Causal Intervention and Counterfactual Analysis: Causal intervention and counterfactual analysis represent a rapidly growing area of research that seeks to move beyond surface-level performance metrics toward uncovering the mechanisms that drive hallucinations in LLMs and LVLMs. By explicitly modeling cause–and–effect relationships and simulating controlled counterfactual scenarios, these approaches provide deeper insight into how hallucinations emerge and how they can be effectively mitigated. This section is organized into four complementary perspectives: (1) Counterfactual Probing; (2) Causal Intervention based on Internal States, which directly manipulates internal states to probe model reliability; (3) Causal Probing based on Causality Graph, which constructs structured causal graphs to identify spurious correlations and hidden dependencies; and (4) Comprehensive Interventions Strategy, which integrates multiple intervention levels to guide principled decision-making about when, where, and how to apply corrections. Together, these approaches illustrate the potential of causal frameworks to transform hallucination evaluation from ad hoc fixes into systematic, mechanism-driven methodologies.

- **Counterfactual Probing.** Counterfactual probing is a novel approach that helps detect the causal relationships in LLM structures and to detect and mitigate hallucinations [405]. Counterfactual Probing [405] hypothesizes that robust knowledge should demonstrate consistency when confronted with plausible alternatives. The method evaluates the model's sensitivity to different counterfactual perturbations [405] and achieves superior detection performance compared to baseline methods. There are four different probe types: *Factual Probes* that alter key entities, relationships, or attributes while maintaining semantic plausibility; *Temporal Probes* that modify time-related information, such as dates, durations, or temporal sequences; *Quantitative Probes* that perturb numerical values, statistics, or measurements; and *Logical Probes* that introduce logical inconsistencies or invalid causal relationships [405]. Recent studies reveal that logical probes effectively identify reasoning errors (F1: 0.771), factual probes are most effective for entity-related hallucinations (F1: 0.834), while temporal probes excel at detecting chronological errors (F1: 0.798) and quantitative probes show strong performance on numerical claims (F1: 0.812) [405].
- **Causal Intervention Based on Internal States.** Causal probing can help interpret the behavior of an existing pretrained model [406]. Causal Intervention based on internal states methodologies provides a principled framework for probing the underlying mechanisms of hallucination by simulating controlled counterfactual scenarios. The TruthPrint framework [321] pioneers a latent space approach with remarkable universality, identifying "truthful directions" shared across diverse LVLMs. Its core innovation lies in discovering that truthfulness manifests as consistent geometric patterns in latent space, with a common subspace capturing transferable hallucination features. By projecting internal states and applying pre-intervention during decoding, TruthPrint achieves significant hallucination reduction on benchmarks like CHAIR and POPE without any model retraining, while maintaining high performance on standard benchmarks [321].
- **Causal Probing with Causality Graphs.** Complementary to this, *Causal Probing* [407] constructs multimodal causality graphs revealing that object hallucinations primarily stem from learned co-occurrence patterns rather than visual misinterpretation, as evidenced by frequent hallucinatory-inducing words like 'tree' or 'grass' [407]. The method exposes how benign contextual objects (e.g., grassy fields) induce spurious generations (e.g., frisbees) with measurable causal effects, though no specific correlation strength is quantified [407]. Its three-tier intervention strategy achieves consistent improvements: image-pasting (e.g., adding a small object) reduces hallucination metrics such as CHAIR scores, semantic prompt restructuring (e.g., foreground-background prompting) enhances coverage while mitigating hallucinations, and embedding manipulation alters latent representations to decrease hallucinations, with combined approaches showing additive

benefits without compromising task performance metrics like coverage rates [407]. Remarkably, these simple input-space modifications outperform complex fine-tuning approaches that require thousands of human-annotated samples [407].

- **Comprehensive Intervention Strategies.** Beyond individual interventions, recent research has shifted toward optimizing the decision-making process surrounding when, where, and how to apply interventions. Rather than focusing solely on the design of isolated techniques, this perspective emphasizes strategic deployment: identifying the most critical model components to target, determining the optimal timing for intervention, and calibrating intervention intensity to balance effectiveness with efficiency [314].
  Such an approach reframes intervention design as a dynamic and context-aware process, moving the field from ad hoc experimentation toward principled, systematic methodologies [314]. Simhi et al. [314] argue that disparate research setups obscure the true causes of hallucinations, leading to misguided interventions. They constructed the *WACK* [314] benchmark to systematically evaluate interventions in both open-book and closed-book QA settings. Their investigation reveals that intervention success is highly dependent on the targeted component (e.g., attention blocks vs. residual stream) and the timing of using representative vectors (collecting them *before* a hallucination occurs proves more beneficial than after) [314]. Most notably, they introduce a *Dynamic Intervention* strategy that triggers only when necessary, enhancing robustness compared to static intervention paradigms [314]. This research provides a crucial framework for making informed decisions in deploying causal interventions, moving from ad-hoc applications to a principled methodology [314].

Collectively, these approaches shift evaluation from static metric computation to dynamic mechanism exploration. By decoupling confounding factors through controlled perturbations, they offer cost-efficient alternatives to human-intensive benchmarks while illuminating fundamental failure modes. Their emphasis on transferable latent structures and minimal inference-time overhead aligns with the broader need for scalable, interpretable evaluation protocols.

d) Uncertainty-Guided and Self-Consistency Checks: Uncertainty-guided and self-consistency checks constitute a promising approach for hallucination detection in generative AI systems, leveraging intrinsic model properties without reliance on external databases or extensive human annotations. These methods exploit the principle that factual responses exhibit high consistency under semantic-equivalent perturbations, while hallucinations manifest as divergent outputs.

For general LLMs, *SelfCheckGPT* [19] adopts a zero-resource, sampling-based strategy, comparing stochastic response variances to identify contradictions, achieving 83.21 AUC-PR in sentence-level hallucination detection and strong passage-level factuality correlation without token probability access [19].

For LVLMs, *VL-Uncertainty* [408] introduces a framework that perturbs inputs through visual blurring and textual rephrasing, then measures uncertainty via entropy over response clusters. This approach demonstrates superior hallucination detection accuracy compared to baseline metrics while the intrinsic metric scales efficiently across multimodal benchmarks such as MM-Vet and ScienceQA.

Complementing this, the *CHAIR* metric [111] targets object hallucination in image captioning by evaluating caption-object alignment against ground-truth annotations, revealing that models with higher visual consistency hallucinate less, and highlighting limitations in standard metrics like CIDEr.

Collectively, these uncertainty-driven methodologies enhance reliability by quantifying model confidence intrinsically, though computational costs—especially in sampling-based approaches—and modality-specific adaptations remain challenges for broader deployment.

e) Internal State and Attention Analysis: This diagnostic approach analyzes real-time signals—particularly attention allocations and activation patterns—to detect representational deviations predictive of hallucinatory outputs [100], [318].

The white-box nature of these approaches provides mechanistic insights unavailable through output-only evaluation, though model-specific architectures limit generalizability [318].

For pure LLMs, *PoLLMgraph* [318] models hallucination as a state transition problem, capturing temporal dynamics of latent activations during generation. Unlike black-box methods reliant on output text or self-consistency checks, PoLLMgraph's Hidden Markov Model (HMM) framework quantifies hallucination likelihood from intermediate state trajectories, achieving over 20% AUC-ROC improvement on TruthfulQA compared to prior white-box techniques [318]. This approach requires minimal supervision (fewer than 100 samples) and generalizes across decoder architectures [318].

For LVLMs, *DAMRO* [100] identifies a critical consistency between visual encoder and LLM decoder attention maps, revealing that both disproportionately focus on background tokens, which propagates redundant information and triggers object hallucination. Its training-free mitigation strategy leverages ViT's classification token (CLS) to filter high-attention outlier tokens and applies contrastive decoding during LLM generation, significantly reducing hallucination in benchmarks like POPE and CHAIR [100] .

Collectively, these methods demonstrate that internal signals (attention distributions, state transitions) offer direct, scalable pathways to detect and mitigate hallucinations. However, their reliance on model-specific mechanisms (e.g., ViT structure, autoregressive dynamics) may limit applicability to emerging architectures.

f) Code-based: Code-based LLM evaluations use a simple script to check whether the model's output adheres to the established rules [349]. To ensure optimal code-based evaluations, it is essential to establish clear and objective criteria tailored for the task. For example, the test can automatically verify whether the response is valid JSON [349], can prevent regressions by comparing new outputs with previous ones, and ensure updates don't introduce errors. Beyond format validation, Code-based LLM evaluations can also detect whether generated stories contain both a title and body text, whether recipes accurately enumerate required ingredients, or whether mathematical answers align with known solutions etc [349]. This approach offers several advantages, such as fast evaluation speed, repeatability, and seamless integration into continuous integration/continuous deployment (CI/CD) pipelines [349]. However, constrained to objectively verifiable properties, the approach is limited in its ability to capture nuanced aspects such as stylistic variation, coherence, or tone [349]. Consequently, they must often be complemented by human assessments or LLM-based evaluators to provide comprehensive evaluation coverage [349].

g) Online and User-facing Evaluation: Online and User-facing Evaluation ensures models function properly in practical applications through systematic deployment methodologies [323]. User Risk Audits help identify potential issues, including privacy violations, harmful outputs, or systematic biases, before large-scale deployment [323].

The deployment strategy employs a phased rollout, akin to the targeted evaluations discussed in [409], typically starting with a small subset of users for initial testing and gradually scaling to full deployment based on performance metrics. This staged approach enables developers to understand model behavior under real-world conditions, to capture early warning signals, to resolve emergent issues, and to refine system performance under authentic usage conditions.

Collectively, these methodologies highlight the importance of embedding risk-sensitive, data-driven strategies into deployment pipelines. By combining phased rollouts, real-world audits, and comparative experiments, online and user-facing evaluation frameworks minimize risk exposure, foster adaptive model governance, and support safe model transitions in dynamic production environments [323], [409].

h) Synthesis and Implications: The evaluation methodologies surveyed represent six years of intensive research (2019-2025) [16], [16] yielding critical breakthroughs in hallucination understanding. The discovery of universal "truthful directions" in latent space enables 41% hallucination reduction without retraining [321], while systematic probing revealed that models hallucinate common objects 2.3 times more frequently than rare ones [60]. Current state-of-the-art achieves various levels of effectiveness: probe-based methods reach an 0.84 F1 score [186], adversarial testing triggers hallucinations with success rates exceeding 97% [112], and causal interventions reduce hallucinations by 61% while maintaining 94% task performance [407].

Despite these advances, critical limitations persist: production performance drops 15-30% from benchmarks [332], sophisticated methods require 5-10 times more computation [76], and static benchmarks become obsolete within 6-12 months as models evolve [332].

For practical deployment, we recommend tiered evaluation combining lightweight consistency checks (SelfCheckGPT achieving 83.21 AUC-PR [19]) for all outputs with expensive methods for high-risk cases. The path forward requires adaptive evaluation frameworks that evolve with model capabilities, creating unified theories explaining hallucination across modalities, and building deployment pipelines that balance accuracy, efficiency, and user experience. The ultimate goal is not the complete elimination of hallucinations but rather their intelligent management: understanding when, why, and how models hallucinate and ensuring their safe, reliable deployment within defined reliability boundaries, acknowledging that perfect detection may be fundamentally impossible given the probabilistic nature of generative models.

i) Summary and Critical Analysis: The methodological landscape for hallucination evaluation reveals both significant progress and fundamental limitations that shape current practice and future directions. Our survey identifies six distinct paradigms, each optimized for different aspects of the hallucination challenge, yet none providing a complete solution.

- **Methodological Trade-offs and Complementarity.** The surveyed methodologies exhibit clear trade-off patterns that practitioners must navigate. *Probe-based and Prompt-based* approaches offer flexibility and minimal infrastructure requirements, achieving strong agreement with human judgments in frameworks like THaMES [186], yet remain vulnerable to prompt sensitivity and evaluator biases. *Adversarial and Stress Testing* (e.g., ReEval, AutoHallusion) effectively exposes edge cases with hallucination trigger rates exceeding 97% [112], but ecological validity remains questionable—models may never encounter such adversarial inputs in deployment. *Causal Intervention* techniques provide mechanistic insights with minimal supervision (fewer than 100 samples for PoLLMgraph [318]), yet their model-specific nature limits transferability across architectures.
- **Emerging Convergence Patterns.** Despite fragmentation, convergent trends emerge across methodologies. Multiple approaches independently discover that *Internal Consistency* serves as a robust hallucination signal—from SelfCheckGPT's response variance to VL-Uncertainty's perturbation-based entropy. The success of *Latent Space Interventions* (TruthPrint) and *Attention Analysis* (DAMRO) suggests that hallucinations have identifiable computational signatures, pointing toward unified detection frameworks. The widespread adoption of *Hybrid Strategies*—combining automated screening with targeted human oversight—reflects practical consensus that pure automation remains insufficient for high-stakes applications.
- **Critical Gaps and Future Imperatives.** Three critical gaps persist across all methodologies. First, *Temporal Degradation*: static benchmarks become obsolete as models evolve, yet no methodology provides continuous adaptation mechanisms. Second, *Cross-Modal Generalization*: methods optimized for text (SelfCheck-

GPT) or vision-language (CHAIR) tasks fail to transfer, necessitating modality-specific reimplementation. Third, *Computational Scalability*: sophisticated approaches like causal intervention or ensemble uncertainty quantification remain prohibitively expensive for real-time deployment, limiting their practical impact.

- **Toward Integrated Evaluation Frameworks.** The path forward requires moving beyond isolated methodologies toward integrated frameworks that strategically combine complementary approaches. Effective evaluation likely involves: (1) *Tiered Assessment*—using lightweight consistency checks for initial screening, followed by targeted causal analysis for high-risk outputs; (2) *Adaptive Benchmarking*—leveraging online evaluation to continuously update test suites based on observed failure patterns; and (3) *Cross-methodological Validation*—requiring agreement across multiple paradigms (e.g., both uncertainty signals and attention patterns) before flagging hallucinations. This integration must balance theoretical rigor with practical constraints, acknowledging that perfect hallucination detection may be fundamentally impossible given the probabilistic nature of generative models. The ultimate goal is not elimination but intelligent management—understanding when, why, and how models hallucinate to deploy them safely within their reliability boundaries.

5) Evaluation Across Comprehensive Dimensions: The LENS framework, as introduced before, is built upon a multi-dimensional evaluation system that moves beyond traditional, fragmented metrics. This section elaborates on the core dimensions of evaluation that LENS integrates, providing the comprehensive and hierarchical assessment required to tackle the challenge of hallucination in GenAI systems. These dimensions are designed to work in concert, enabling both the **Horizontal Assessment** of performance breadth across tasks and domains, and the **Vertical Assessment** that decomposes and examines the internal reasoning chain.

The evaluation of hallucinations in large language models (LLMs) necessitates a multi-dimensional framework that captures distinct facets of generation quality and reliability, which together determine overall model trustworthiness. This section systematically reviews existing methodologies, illustrating how diverse metrics can be operationalized across complementary dimensions. Nonetheless, the current evaluation landscape remains incomplete. Existing approaches only partially capture core dimensions of hallucination assessment, while critical aspects—such as evaluation granularity, reference availability, and the balance between automation and human oversight—are often overlooked. This gap underscores the urgent need for more comprehensive and interpretable frameworks capable of supporting holistic, multi-dimensional evaluation.

The metrics in LENS's core dimensions include: factuality assessment against objective truth, hallucination rate estimation for statistical characterization, faithfulness to provided context, consistency measurement for reliable performance, robustness quantification across environmental factors, susceptibility analysis under adversarial conditions, safety prediction for deployment readiness, interpretability for actionable diagnostics, tool use evaluation for external knowledge grounding, and computational constraints that govern practical deployment. Each dimension addresses unique challenges while contributing to a holistic understanding of hallucination phenomena, enabling practitioners to select appropriate evaluation strategies based on specific application requirements and risk tolerances.

a) Measure Factuality: Factuality and accuracy evaluation constitutes the foundational dimension of hallucination assessment in LENS, ensuring that generated outputs align with verifiable truths across diverse domains [306], [310]. This process faces epistemological difficulties in defining ground truth and computational complexity in fact verification, which necessitates robust yet practical frameworks.

- **Atomic Fact Decomposition:** Frameworks such as FactScore [306] decompose complex outputs into atomic claims for independent verification, addressing the challenge of mixed accuracy in long-form texts. In practice, biographical content may yield 26–41 atomic facts per generation, where models like ChatGPT achieve around 58.3% atomic factual precision, and GPT-4 reaches 70–80%. Each fact is cross-checked against authoritative resources (e.g., Wikipedia), and precision–recall metrics combine accuracy and completeness. This granular approach exposes systematic errors such as contradictions or irrelevant content.
- **Knowledge Grounding:** Establishing authoritative truth across domains is another pillar of factuality measurement. TruthfulQA [310] introduces adversarially designed questions spanning 38 domains to reveal susceptibility to misconceptions embedded in training data. Findings indicate that scaling models can worsen performance, with GPT-3 175B reaching only 58% accuracy compared to smaller variants. The benchmark combines automated classifier-based scoring with human evaluation of truthfulness and informativeness.
- **Domain-Specific Protocols:** Verification requirements vary across domains. In medicine, frameworks such as MedHALT [42] rely on clinical knowledge bases and peer-reviewed studies. Legal factuality requires jurisdictional and temporal tracking of precedents. Scientific assessment differentiates established consensus from emerging or controversial results, often using confidence-weighted scores. Financial domains demand real-time validation and compliance checks, with systems achieving sub-second verification.
- **Compositional Factuality:** A unique challenge occurs when individually correct facts combine into incorrect composite statements. Advanced approaches employ logical consistency verification, checking relations between facts rather than isolated correctness. Graph-based assessment techniques [311] construct knowledge graphs to detect structural contradictions overlooked by atomic-

level evaluation, achieving higher accuracy in identifying compositional hallucinations though at increased computational cost.
- **Dynamic Knowledge Integration:** To address the problem of evolving facts, the LENS framework incorporates **user-centric benchmark construction**, enabling the creation of domain-specific, dynamic knowledge bases grounded in proprietary or up-to-date data. This directly tackles the challenge of dynamic knowledge identified in Section VI-A4.

b) Measure Faithfulness: Faithfulness evaluation in LENS determines whether generated content accurately reflects the given source material, ensuring internal consistency and reliability in deployment [400]. While factuality tests external truth alignment, faithfulness emphasizes adherence to provided instructions, documents, or contexts.

- **Context Consistency:** Faithfulness evaluation addresses tasks where models must synthesize or summarize content without introducing contradictions or irrelevant material. This is critical in domains such as summarization, context-based QA, or dialogue systems bound by persona or context [400].
- **Verification Methods:** Context-grounded verification uses multiple strategies to compare outputs with their source material. Natural language inference models trained for this task achieve 74.4% balanced accuracy, improving prior results by 5% [410]. Semantic similarity measures based on embeddings provide better correlation with human judgments compared to n-gram metrics like ROUGE [311]. Question–answering consistency checks also show strong reliability, particularly for summarization tasks [411].
- **Cross-Document Evaluation:** In cases involving multiple documents, faithfulness frameworks must handle conflicts across sources. Advanced systems weigh evidence by reliability, recency, and consistency, and apply sentence-level segmentation or pairwise scoring to detect contradictions [410].
- **Instruction Adherence:** Another dimension evaluates whether models comply with task-specific instructions beyond content fidelity. Studies reveal that around 19.5% of outputs include fabricated unverifiable details, exposing weaknesses in instruction-following [17].
- **Trade-off with Informativeness:** High faithfulness alone may result in minimal added value if the model only copies sources. Effective metrics must reward abstraction and synthesis while preserving accuracy. The balance differs by task: summarization requires compression with consistency, QA demands selective extraction, and dialogue requires adaptability for natural flow [400].
- **Faithfulness-Aware Training:** Training strategies that penalize hallucinated outputs and reward adherence improve both coherence and informativeness, yielding measurable performance gains such as 12.8 F1 improvements in dialogue evaluations [400].

c) Estimate Hallucination Rate and Predict Safety: Moving from qualitative analysis to quantitative characterization, hallucination rate estimation provides statistical measures that are crucial for reliability assessment and deployment decisions. Accurate estimation supports probabilistic reasoning about model behavior and informs confidence calibration strategies. One promising direction employs conditional generative models to generate self-prediction questions regarding potential hallucinations in in-context learning scenarios. The resulting posterior hallucination rate (PHR), derived through coherence-based checks, shows strong correlation with true hallucination rates (THR), particularly for smaller in-context datasets. This method requires only response log probability calculations, making it both simple and effective for evaluating hallucination tendencies without extensive ground-truth annotations [339].

Beyond rate estimation, safety prediction evaluates the potential harm caused by hallucinated content, shifting the focus from descriptive measurement to risk-oriented assessment. This perspective directly supports deployment strategies and mitigation requirements. For instance, the Hallucinations Leaderboard [374] provides an open benchmark that quantifies hallucination tendencies across tasks such as question answering and summarization, enabling standardized comparisons of model reliability and highlighting vulnerabilities in factuality and faithfulness. Furthermore, attribution methods such as association analysis [95] explore the relationship between hallucination occurrence and specific risk factors. By observing performance under varying conditions and isolating contributing variables, these approaches identify weaknesses in areas like commonsense knowledge and relational reasoning. Such analyses exclude confounding influences and yield actionable insights for targeted mitigation, guiding safer model development and deployment.

d) Assess Susceptibility to Adversarial Conditions: Understanding how models respond under adversarial conditions extends safety evaluation into proactive vulnerability assessment. Susceptibility analysis systematically probes weaknesses by introducing controlled perturbations, revealing failure modes that would not be evident through standard evaluation protocols. This perspective highlights the extent to which hallucinations are not random but follow predictable patterns when models are exposed to misleading or conflicting inputs.

An illustrative example is the PhD benchmark (ChatGPT-Prompted Visual Hallucination Evaluation Dataset), which evaluates multimodal large language models (MLLMs) across five visual recognition tasks and four distinct questioning modes, including normal queries, queries with misleading context, queries with incorrect context, and counter-commonsense image prompts [393]. Results demonstrate that while models perform reliably on basic object recognition, they exhibit sharp increases in hallucination rates when faced with ambiguous visual information, inconsistent image–text pairings, or scenarios that conflict with everyday knowledge. Failures are particularly evident in sentiment analysis and counting tasks. These findings reveal systematic vulnerabilities triggered by adversarial stimuli and underscore the need for targeted mitigation strategies that account for predictable error patterns rather than treating hallucinations as isolated or random occurrences.

e) Estimate Hallucination Rate and Predict Safety: Transitioning from qualitative assessment to quantitative characterization, hallucination rate estimation provides statistical indicators that are crucial for risk modeling and deployment strategies. Reliable estimation enables probabilistic reasoning about system stability and supports confidence calibration. A novel prediction-based approach employs conditional generative models to generate self-assessment questions in in-context learning scenarios, with posterior hallucination rate (PHR) derived from coherence checks showing strong correlation with true hallucination rate (THR). This technique requires only log probability computations, making it both efficient and accurate, while avoiding the need for costly manual annotation [339].

Beyond quantifying hallucination frequency, safety prediction measures the potential risks associated with hallucinated outputs, moving from descriptive evaluation toward risk-oriented assessment that informs real-world deployment. For example, the Hallucinations Leaderboard [374] establishes a public benchmark for quantifying hallucination tendencies across diverse tasks such as summarization and question answering. This provides a standardized method to compare models and diagnose vulnerabilities in factuality and faithfulness. In addition, association analysis [95] attributes hallucination risks to specific contributing factors by systematically observing outcomes under varied conditions. This method filters out confounding effects and highlights deficiencies in areas such as commonsense knowledge and relational reasoning, producing actionable insights that guide targeted mitigation strategies and safer model design.

f) Evaluate Causal Reasoning Capabilities: Causal reasoning evaluation examines whether models can move beyond surface-level correlations to identify and explain genuine cause-and-effect relationships. This ability is essential for generating reliable explanations and supporting sound inferential decisions, forming a cornerstone of interpretability and vertical assessment [385]. Unlike correlation-based pattern recognition, causal reasoning requires understanding the mechanisms that underlie observed outcomes, enabling models to explain how specific inputs lead to particular results [384]. Such reasoning increases transparency and enhances user trust in model outputs.

Capabilities in causal inference allow models to simulate counterfactual scenarios and assess the effects of hypothetical interventions. These skills are particularly critical in fields such as healthcare, economics, and public policy, where uncovering causal mechanisms can directly inform more effective interventions and improve outcomes [326]. Moreover, causal reasoning improves robustness by allowing models to generalize knowledge across diverse contexts, making them more adaptable to novel or unseen conditions [401]. By integrating causal reasoning into evaluation, frameworks can more accurately test whether models provide not only correct predictions but also valid justifications grounded in underlying causal structures.

g) Analyze Robustness to Impacting Factors: Robustness analysis evaluates how models maintain stable performance under diverse operational conditions and perturbations, serving as a key requirement for reliable deployment [320]. This dimension seeks to quantify resilience against variations in inputs or environments that might trigger hallucinations. Input perturbations such as textual rephrasing, semantic ambiguity, or image alterations often expose fragile internal representations, while biases in training data or model parameters can amplify systematic errors and produce overconfident but incorrect outputs. Furthermore, inconsistencies or noise within external knowledge sources may intensify the risk of hallucinations, underscoring the importance of examining these causal contributors.

To detect and mitigate such vulnerabilities, robustness evaluation incorporates feedback-based mechanisms that analyze internal model states under controlled conditions. Techniques such as iterative self-verification and cross-model comparison exploit divergences in representations to identify unstable behaviors. Cross-modal and cross-phase analyses further enhance detection by exposing inconsistencies across different modalities or processing stages. At a quantitative level, perturbation-based testing measures entropy, variance, and divergence from reference outputs, while cross-model divergence analysis highlights inconsistencies between different architectures or ensembles under the same input conditions.

This multi-layered methodology—combining causal factor identification, internal state monitoring, and quantitative estimation—provides a rigorous framework for understanding and addressing hallucination risks. By integrating these strategies, robustness analysis enables proactive identification of vulnerabilities and supports the design of AI systems with greater resilience, ensuring reliability even in uncertain or dynamic real-world environments. odels remain stable and reliable across diverse operational contexts.

h) Measure Consistency: Consistency evaluation extends beyond single-output correctness to assess whether models can produce stable and coherent results over time and across contexts, a principle emphasized in Section VI-A3. This dimension is vital for ensuring that performance is not subject to random variation but reflects reliable behavior across repeated trials.

Traditional static benchmarks, such as question–answer datasets, often fail to capture consistency because they risk contamination from training data, leading to memorization rather than genuine reasoning. To address this limitation, ReEval [320] introduces a dynamic framework grounded in adversarial machine learning. Through **prompt chaining**, the framework perturbs original evidence using strategies like **answer swapping** and **context enriching**, generating adversarial test cases that systematically expose weaknesses. These cases transfer across different models, enabling even smaller models (e.g., Alpaca-7B) to create challenging benchmarks for larger proprietary systems, thereby reducing costs while maintaining evaluation rigor. While effective, current implementations are limited to relatively simple, single-hop questions and do not yet cover more complex multi-hop reasoning.

The LENS framework enhances this process by using a **tree-based decomposition** methodology that evaluates each node of the reasoning chain independently. This approach enables the identification of inconsistencies not only between multiple model outputs but also within the logical steps of a single response. By capturing both external and internal inconsistencies, consistency evaluation supports deeper vertical assessment and provides a stronger basis for diagnosing weaknesses and guiding model improvement.

i) Computation Workload and Cost: The deployment of hallucination evaluation frameworks must contend with the challenge of computational workload and cost, which fundamentally determines scalability and feasibility. Effective evaluation requires balancing thoroughness with economic constraints to ensure practical usability while maintaining adequate safety margins. Fine-tuning large language models for hallucination detection often involves considerable computational resources, posing barriers for many real-world applications [334]. In addition, benchmark datasets such as **HaluEval** and **DelucionQA**, while valuable, rely heavily on labor-intensive human annotation, further increasing costs [186].

Recent research explores strategies to alleviate these burdens. One direction leverages smaller models for initial screening before escalating to larger models, while transfer learning improves efficiency across tasks. Comparative analyses of frameworks such as LoRA, RAG, and DoRA highlight important trade-offs: LoRA provides cost-efficient domain adaptation, while DoRA achieves a balance between fine-tuning efficiency and model accuracy [334]. Additional techniques to maintain cost-effectiveness include batch processing, weighted sampling, counterfactual validation, and the integration of complex question types [186].

Another promising direction comes from adversarial transferability. The ReEval framework demonstrates that adversarial examples generated by small, inexpensive models can be effectively reused to evaluate larger models, significantly improving scalability and reducing evaluation costs [320]. Together, these approaches illustrate the inherent trade-offs between efficiency and thoroughness. Future work must continue to focus on algorithmic innovation and hardware optimization to make hallucination evaluation more cost-effective and scalable for widespread adoption.

6) Mitigation-Aware Evaluation: Hallucination mitigation in LLMs and MLLMs encompasses diverse strategies with varying effectiveness across different contexts. In this section, we review four major paradigms of hallucination mitigation: retrieval-augmented generation (RAG), parameter-efficient fine-tuning (LoRA, DoRA), knowledge transfer mechanisms, and preference optimization with structured reasoning; we also analyze temporal intervention strategies.

a) RAG, LoRA, DoRA Performance Checks: Parameter-efficient fine-tuning strategies provide effective means to reduce hallucinations without the high computational demands of full retraining [412]. These approaches represent a shift toward more accessible mitigation methods, particularly in resource-constrained contexts.

- Retrieval-Augmented Generation (RAG) enhances factual accuracy by grounding outputs in retrievable evidence. Empirical results demonstrate improvements across QA and dialogue tasks relative to non-retrieval baselines [3], [158], [413]. Evaluations typically assess retrieval precision, recall, and accuracy, grounding quality (the effective integration of external knowledge), and overall factuality compared to standard generators. Frameworks like RAGAS offer task-agnostic measures of correctness, faithfulness, and context precision/recall [158], [414]. Additional evaluations probe long-context position sensitivity, such as "lost in the middle" effects [415], and adaptive retrieval that balances factuality against latency and retrieval cost [202]. Practical challenges remain, as naive RAG often suffers from precision/recall failures caused by issues in chunking, query formation, retriever settings, and integration at generation time [158], [416]. Contemporary evaluations therefore consider four phases—source quality, query design, retriever effectiveness, and generation strategy—and highlight hybrid systems combining RAG with preference optimization (e.g., DPO) to further suppress hallucinations [417], [418].
- Low-Rank Adaptation (LoRA) fine-tunes models efficiently by inserting rank-decomposition matrices into selected modules while keeping original weights fixed. This approach retains near full fine-tuning performance while updating only a small subset of parameters, typically with low ranks (e.g., 4–16) applied to attention projections or MLP layers [419].
- Decomposition of Weights (DoRA) advances this idea by splitting pretrained weights into magnitude and directional components, then fine-tuning the latter with low-rank adaptation [420]. Once training is complete, the components merge back, avoiding inference-time overhead. Experiments indicate DoRA can surpass LoRA across various tasks [420].
- For multimodal large language models, adapter- and connector-based approaches remain strong baselines. LoRA and IA3 applied to both attention and MLP layers prove effective, while additional fine-tuning of connector modules further boosts results. These findings underscore that evaluation of mitigation effectiveness should be architecture-aware to ensure fair comparisons [421].

b) Knowledge Distillation and Feedback Learning: Beyond parameter-efficient fine-tuning, knowledge transfer and feedback mechanisms provide complementary ways to mitigate hallucinations by shifting from direct weight optimization to behavioral inheritance [422]. These strategies reduce hallucinations while preserving task performance and improve transparency through iterative refinement.

- Knowledge distillation transfers hallucination-aware behavior from large teacher models to smaller student models. Empirical studies report that student models maintain task accuracy while experiencing reduced hallucination rates [270]. Common evaluation protocols include temperature scaling for soft targets, intermediate layer matching to preserve reasoning paths, and assessments of exact-match accuracy versus semantic correctness [422]. Evaluation dimensions typically address factual consistency (alignment of teacher and student answers), reasoning preservation (whether logical chains remain intact), and hallucination propagation (whether biases are inherited). Chain-of-thought distillation further extends this by transferring intermediate reasoning traces, and in multimodal tasks, structured reasoning improves interpretability and error detection [404].
- Feedback learning emphasizes iterative cycles of self-improvement where models learn from their mistakes or external corrections. Self-RAG, for instance, integrates retrieval, generation, and critique, showing improvements in open-domain QA by combining retrieval with self-reflection signals to guide when and how to revise [423]. Evaluation of feedback learning approaches requires temporal assessment across initial generation quality, accuracy of self-critique, and performance gains over iterations.
- Unified detection frameworks like UNIHD demonstrate the potential of multi-granularity evaluation across modalities, achieving 81.91% claim-level and 84.60% segment-level accuracy on MHaluBench [390]. This framework combines sentence-level analysis, which captures semantic coherence, with entity-level checks for local factual accuracy, enabling more nuanced assessments of hallucination patterns.

c) Hallucination-Aware DPO Training & Chain-of-Thought: Moving from knowledge transfer to preference-based optimization, approaches such as Direct Preference Optimization (DPO) and Chain-of-Thought (CoT) reshape how models generate and validate outputs. While DPO aligns generation with human preferences through contrastive learning, CoT enforces step-by-step logical reasoning, making hallucinations more transparent and correctable.

- Direct Preference Optimization (DPO) reduces hallucinations by learning from preference pairs that explicitly reflect hallucination severity, avoiding the complexity of full RLHF pipelines. Studies report reductions in hallucination rates across generative tasks, with evaluations focusing on preference alignment accuracy, generalization across domains, and efficiency relative to RLHF [417], [418], [424]. Extensions such as Vision-guided DPO (V-DPO) target multimodal models where language priors can dominate visual inputs, achieving a 52% reduction in object hallucinations on CHAIR and a 38% grounding accuracy gain on RefCOCO by emphasizing consistency with visual evidence [425]. Recent findings also show that smaller models, when fine-tuned with frameworks like Honest AI, can mitigate hallucinations effectively by learning uncertainty expression (e.g., saying "I don't know"), achieving top scores on uncertainty-aware benchmarks like CRAG [117].
- Chain-of-Thought (CoT) prompting transforms opaque outputs into interpretable reasoning chains, surfacing hallucinations in intermediate steps. Research shows CoT can reduce factual errors by 52% in multi-step reasoning tasks, though models sometimes reach correct answers via flawed reasoning [47], [404]. More advanced frameworks such as Chain-of-Preference Optimization (CPO) merge CoT with DPO, constructing preference pairs at each reasoning step using Tree-of-Thoughts search. This allows models to classify intermediate thoughts as preferred or dispreferred, self-correcting during generation and reducing cascading reasoning errors [426]. Evaluations assess step-wise accuracy, path consistency, and hallucination propagation. Combining DPO with CoT creates a synergistic effect: DPO enforces factual preference alignment, while CoT provides reasoning transparency. Benchmarks show significant improvements on complex reasoning tasks, though limitations remain, as DPO requires substantial compute resources and effectiveness often depends on dataset domain [417], [426].

d) Pre-Intervention vs. Post-Hoc Correction: The temporal perspective on hallucination mitigation introduces a key design decision: whether to intervene before hallucinations arise or correct them after generation. Each strategy carries trade-offs in system design, computation, and accuracy, and evidence suggests that hybrid use is often most effective [321], [352].

- Pre-intervention strategies modify the generation process to prevent hallucinations at the source. For instance, LLMs Augmenter constructs evidence chains before inference, offering factual scaffolding that improves interpretability and traceability of claims [427]. Visual evidence prompting for multimodal models primes attention to relevant image regions and external references, yielding better grounding and reduced hallucination in vision-language benchmarks [191]. Evaluations typically measure evidence quality, coherence, and coverage of generated claims.
- Post-hoc correction methods allow models to produce an initial output and then revise it. Visual Contrastive Decoding integrates vision-language signals during decoding, contrastively favoring tokens consistent with visual evidence, which reduces object hallucinations without retraining [164]. Frameworks such as OPERA penalize over-reliance on language priors and reallocate attention to visual inputs, helping flag and fix unsupported content [428]. Evaluation here emphasizes detection accuracy, correction precision, and preservation of valid content.

- Hybrid temporal strategies increasingly combine both approaches, leveraging the strengths of each. Pre-intervention provides strong baselines, while post-hoc catches residual errors [164], [191], [428]. The optimal balance depends on application constraints: latency-sensitive systems favor lightweight post-hoc correction, while high-accuracy domains adopt more extensive pre-intervention with multiple revision passes. Training-free methods like visual contrastive decoding (51% reduction in object hallucinations) [164], context-aware decoding [128], and zero-resource hallucination detection (83.21 AUC-PR at sentence level) [19] have gained traction due to their flexibility and immediate deployability. Evaluation frameworks now consider computational budget, latency, and diminishing returns from iterative correction, which highlights the importance of strong pre-intervention for improved initial outputs.

e) Uncertainty-Aware Abstention and Calibration: Abstention-based methods aim to avoid producing unreliable outputs by withholding answers when uncertainty is high. These approaches improve safety and reliability by calibrating model confidence and selectively declining responses that are likely hallucinated. Evaluation frameworks for such methods emphasize calibration quality, coverage–risk trade-offs, and the balance between reliability and utility.

- Conformal abstention provides guarantees for selective prediction by allowing models to say "I don't know" under distributional uncertainty. This reduces harmful errors, with evaluations typically reporting risk–coverage and abstention–utility curves [119]. Similarly, uncertainty-based abstention methods use calibrated confidence estimates to abstain on low-confidence predictions, improving robustness. Standard metrics include expected calibration error (ECE), area under the risk–coverage curve, and conditional accuracy on non-abstained cases [178].
- Detection-driven mitigation leverages uncertainty signals such as semantic entropy scores to identify outputs likely to be hallucinated. These signals enable rejection, revision, or targeted retrieval before committing to a final answer [76]. Evaluations should therefore incorporate calibration metrics, risk–coverage analyses, and trade-offs between abstention and overall system utility.
- Evaluator choice plays a central role in abstention studies. Automatic LLM-as-a-judge methods like G-EVAL or critique-based judges can align with human assessments, but their reliability varies across tasks and prompts [429], [430]. Stronger benchmarks, such as ERBench for entity–relationship claims, provide more verifiable evaluations [83]. However, reference-free evaluations remain limited and may mis-rank systems [431]. Community efforts like the Hallucinations Leaderboard and human-in-the-loop auditing frameworks increase transparency and comparability [323], [374].

7) Tools and Frameworks: After introducing the core challenges in hallucination evaluation and the innovative LENS framework, this section systematically examines existing tools and frameworks. Current studies have proposed diverse methodologies—from automated dataset generation to manual annotation, and from traditional metrics to LLM-based evaluators—that significantly enhance evaluation scalability and comprehensiveness. These frameworks integrate multidimensional metrics, multimodal fusion, and feedback loops to capture and mitigate hallucinations. The following subsections analyze their design principles, benchmark construction requirements, and comparative strengths and limitations.

The development of evaluation tools and frameworks represents a critical evolution from ad hoc testing to systematic methodology that fundamentally reshapes how we understand and measure hallucinations. The field has matured through three distinct paradigm shifts: (1) From manual curation to automated dataset generation, reducing creation costs while increasing scale [186]; (2) From binary classification to multidimensional assessment, capturing nuanced failure modes across factual, logical, and contextual dimensions; and (3) From isolated evaluation to integrated feedback loops, enabling continuous improvement through iterative refinement.

Typically, the evaluation workflow consists of three phases: Dataset Construction**, involving automated data generation pipelines, multi-tier annotation protocols, and quality control mechanisms; Benchmark-based Evaluation**, which employs advanced metrics, cross-benchmark validation, and human–AI collaborative assessment; and Result Comparison**, which synthesizes results across benchmarks to identify consistent patterns, discrepancies, and areas for improvement.

Some datasets have a clear pipeline for creation, such as the Data Collection Pipeline of HalluQA [355], the Construction Pipeline of HaluEval-Wild [332], and the generation pipeline of ONIONEVAL [346]. We have summarized some of these pipelines, analyzing their advantages and disadvantages.

Finally, the interplay between **LLMs and human experts** plays a pivotal role in shaping benchmark quality, cost, and robustness. In the **Dataset Construction** phase, we examine the degree of automation and human involvement in data collection, annotation, and verification. In the **Evaluation** phase, we analyze the automation level of evaluation methods, such as the increasing use of LLMs as judges, and emphasize the importance of human feedback for refining and validating evaluation benchmarks.

a) Framework Design and Benchmark Construction for Model Evaluation: The advancement of evaluation frameworks signifies a transition from ad hoc, informal testing practices to structured, systematic methodologies. This evolution is characterized by three notable paradigm shifts: the replacement of manual data curation with automated dataset generation that cuts costs and expands scale [186]; the move beyond binary evaluations toward multidimensional assessments that capture factual, logical, and contextual failures; and the integration of evaluation into iterative feedback loops, enabling continuous refinement and improvement [4]. Collectively, these innovations allow for robust, scalable, and precise evaluation frameworks that better reflect the complex nature of hallucinations in language models.

Benchmarks emerging from these design principles typically rely on four essential elements—datasets, annotations, baselines, and metrics—forming a reproducible and comprehensive architecture for model assessment:

- **Dataset**: Modern benchmarks utilize advanced data generation pipelines, as demonstrated in HalluQA [355] and HaluEval-Wild [332], which automate large-scale data collection and annotation. These systems employ multi-tier protocols and automated quality control to ensure efficiency and reliability, capturing diverse hallucination types and error modes.
- **Annotation & Answer**: Annotation blends automated labeling with human oversight. Automated tools contribute scalability, while human annotators resolve ambiguity in context-sensitive outputs. LLMs are increasingly deployed as evaluators, offering consistent large-scale judgments. However, human feedback remains crucial for refining quality, with Reinforcement Learning with Human Feedback (RLHF) [4] widely adopted to improve alignment with human expectations.
- **Baseline**: Baseline measures include both conventional metrics (e.g., accuracy, F1) and advanced evaluations emphasizing factual consistency, logical soundness, and contextual appropriateness. LLMs are also employed as proxy evaluators, providing cost-effective alternatives to human judgment.
- **Metrics**: Evaluation metrics have expanded to multidimensional frameworks measuring factuality, logicality, relevance, and robustness. Cross-benchmark validation ensures broad applicability, while human–AI collaborations (e.g., RLHF) enhance reliability. Automated fact-checking [322], image consistency checks [111], and QA accuracy [432] are examples of integrated metric strategies.

Despite these innovations, most benchmarks remain static and general-purpose, typically grounded in public datasets like Wikipedia or academic corpora. While such resources enable broad comparison, they fall short in accommodating organization-specific domains such as proprietary documents or financial records. This lack of adaptability limits the ability to uncover domain-specific hallucination risks during deployment. Consequently, an ideal evaluation framework should balance universal static benchmarks with dynamic, user-centric construction toolkits that account for context. Such flexibility is essential to ensure trustworthy and safe deployment of generative AI systems in specialized or enterprise domains.

b) Requirement for Current Dataset Construction: Evaluating hallucinations in large-scale language and vision-language models requires datasets that extend beyond traditional benchmarks. Existing resources often fail to reflect the nuanced and multifaceted nature of hallucinations, especially in open-ended or culturally diverse contexts. To address these gaps, new dataset construction must meet several requirements: minimizing human bias, aligning with human judgment, covering subjective and cultural complexities, and supporting both free-form and constrained generation tasks. Below, we outline the major considerations:

- **Generalization Across Diverse Contexts:** Manually curated datasets often introduce bias and insufficient diversity. Wu et al. [112] show that handcrafted sets over-represent corner cases while underrepresenting real-world failures. Automated approaches like AUTOHALLUSION mitigate these issues, ensuring datasets capture broader linguistic subjectivity and cultural contexts [338].
- **Generalization Across Time:** Evaluations must track hallucination behavior over time, as model tendencies can shift with updates, data refreshes, or evolving interaction patterns. Temporal assessments are essential to distinguish transient from persistent hallucinations and to ensure consistent performance in real-world deployments.
- **Generalization Across Different Model Sizes:** Small LLMs (sLLMs) and large LLMs display different error patterns. sLLMs excel in certain analytic tasks but struggle with complex reasoning, while LLMs, despite broader capability, still hallucinate in ambiguous contexts [346]. Datasets must provide test cases scalable across model sizes.
- **Generalization Across Modalities:** With the rise of LVLMs, evaluations must incorporate multimodal cases. For instance, models may generate mismatched text-image outputs. Benchmarks should include cross-modal reasoning tasks to capture hallucinations unique to multimodal integration [407].
- **Generalization Across Regions and Cultures:** Datasets should reflect regional and cultural variation, particularly in languages like Chinese where cultural sensitivity is critical. High-quality multilingual datasets [330] help evaluate fairness and inclusivity by including culturally grounded test cases and idiomatic expressions.
- **Mitigating Human Bias and Adversarial Robustness:** Manual annotation is costly, subjective, and vulnerable to bias [112]. Moreover, current benchmarks rarely assess adversarial robustness, leaving hidden vulnerabilities. Future datasets must consider adversarial prompts and perturbations to reveal weaknesses beyond standard tests.
- **Alignment with Human Judgments:** Many benchmarks fail to capture nuanced human evaluation. Class imbalance, LLM preference bias, and outdated model outputs reduce validity [352], [431], [433]. Human judgments are inherently cultural and contextual [338], highlighting the need for hybrid human-AI evaluation.
- **Incorporating Cultural Sensitivity:** The Culturally Sensitive Test (CST) [338] redefines hallucinations as misalignments with human expectations rather than pure factual errors. It categorizes queries into raw facts, interpretations, and unverifiable hypotheses, urging datasets to integrate cultural and societal contexts for realistic

assessments.

- **Revealing the Underlying Thought Process:** Traditional benchmarks evaluate only final answers, ignoring reasoning validity. ERBench [83] addresses this gap by decomposing reasoning into verifiable intermediate steps, enabling diagnosis of where hallucinations originate (retrieval, inference, or synthesis).
- **Comprehensive Evaluation of Free-form and Constrained Generation:** Current benchmarks emphasize constrained QA (e.g., POPE [347]) but often neglect free-form tasks. Since hallucination behaviors differ between these modes, datasets must evaluate both to provide holistic assessments of LVLM reliability.

Overall, effective dataset construction must integrate automation with cultural and contextual awareness, mitigate human bias, and support multimodal, temporal, and domain-specific evaluations. Such comprehensive design ensures robust benchmarks capable of guiding the safe deployment of generative AI systems.

c) General vs. Special Domain Evaluation: Hallucination evaluation is inherently context-sensitive, differing significantly between general-purpose and domain-specific scenarios. Each domain imposes unique demands for precision, safety, and knowledge representation, making it crucial to distinguish between broad benchmarks and specialized ones. The following summarizes the major categories:

- **General Domain Benchmarks:** HELM [409] evaluates models across diverse tasks such as summarization, QA, and user-facing applications, applying standardized multi-metric protocols to resolve inconsistent evaluation practices. It benchmarks 30 models ranging from 350M to 175B parameters over 42 scenarios including MMLU, TruthfulQA, and CNN/DailyMail. HELM's holistic approach covers accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency. Results show larger models outperform smaller ones in knowledge-intensive tasks, though limitations remain in specialized or subtle domains.
- **Domain-Specific Benchmarks:** In high-stakes fields like medicine and finance, hallucinations can have severe consequences. Med-HALT [42] addresses medical reasoning and factual recall via Reasoning Hallucination Tests (RHT) and Memory Hallucination Tests (MHT). It integrates seven datasets (e.g., MedMCQA, MedQA, HeadQA, PubMed), yielding 18,866 RHT and 4,916 MHT samples. Results reveal that even advanced models like Llama-2 70B achieve only about 42% accuracy on reasoning tasks, underscoring the difficulty of reliable medical AI. Similarly, FinanceBench [402] evaluates 10,231 QA items derived from 361 public financial filings, exposing weaknesses in numerical reasoning and factual consistency, although it does not cover real-time market contexts.
- **Cross-Domain Transfer Evaluation:** HalluQA [355] provides a benchmark for Chinese LLMs, addressing the lack of non-English resources. Its adversarial question design spans multiple domains, such as history, culture, and social phenomena, with 450 carefully crafted adversarial queries. HalluQA evaluates two hallucination types—imitative falsehoods and factual errors—against GLM-130B and ChatGPT. Testing across 24 Chinese LLMs found that 18 models had non-hallucination rates below 50%, indicating its difficulty and value in assessing cross-domain generalization.

Overall, distinguishing between general-purpose, specialized, and cross-domain benchmarks is critical for constructing evaluation systems that are both comprehensive and context-aware, ensuring robustness across diverse application settings.

d) Dataset Construction Phase: The evolution of dataset construction methods demonstrates a shift from manual to hybrid human–AI approaches. Traditional curation ensures high accuracy but is time-consuming [310]. Modern pipelines leverage LLMs for data generation while maintaining human validation to preserve reliability through expert oversight [355]. Overall, dataset construction involves acquiring raw data, creating evidence-supported QA pairs, annotating before or after generation, post-processing, and human verification. The main components are:

- **Obtaining Raw Data:** Raw data may come from integrating existing datasets, collecting new resources, or generating synthetic data via LLMs and human volunteers. Examples include Med-HALT [42] which integrates multinational medical exams, or industrial spatio-temporal benchmarks built from Vienna traffic incident records [349]. Other methods collect adversarial patterns from the Internet for HalluQA [355] or generate new data via visual concept extraction in VISCON [434]. Approaches like AUTOHALLUSION [112] and OnionEval [83] also produce structured, hallucination-prone QA examples through generative models and knowledge graphs.
- **Dataset QA Construction:** Effective benchmarks require hallucination-triggering questions paired with ground-truth answers. General QA pipelines rely on human or LLM-assisted creation and screening (e.g., HalluQA [355]), while entity-level pipelines use knowledge graphs to create structured QA pairs, such as OnionEval [346] and ERBench [83]. For LVLMs, benchmarks like PhD [393] and AUTOHALLUSION [112] construct questions from annotated images, captions, or adversarial spatial-relation tasks.
- **Annotation Before and During QA Generation:** Strategies such as evidence perturbation (e.g., ReEval [320]), fine-grained categorization (DiaHalu [34], HaluEval-Wild [332]), entity-level annotation (OnionEval, ERBench), caption annotation (AUTOHALLUSION [112], VISCON [434]), and predictive-question annotation [339] are used to improve coverage and test hallucination-prone scenarios.
- **Annotation After QA Generation:** Post-generation pro-

cesses enhance dataset reliability. Methods include generating unanswerable questions (e.g., UMWP [116]), iterative model-in-the-loop annotation [362], and filtering with semantic similarity measures to balance data distribution and prioritize challenging cases.
- **Human Verification:** Automated methods often miss contextual nuance, so rigorous human verification is essential. For instance, DiaHalu and HaluEval-Wild employ manual verification [34], [332], and PhD includes multi-stage human checking after automatic generation [393].
- **Free-form Generation Scenarios:** The dataset construction mentioned above is all based on explicit task requirements and reliable references. However, we will also face the evaluation in free-form generation scenarios quite often. The inherent characteristics of *free-form text generation* pose particular challenges for the construction of the associated datasets. In such open-ended tasks, the frequent lack of reliable reference texts fundamentally limits the applicability of traditional reference-based methods for recognizing hallucinations [362]. Furthermore, evaluations performed at the sentence-level or dialogue-level can only provide coarse-grained judgments that are insufficient for the needs of fine-grained error localization or real-time intervention [362]. It may be difficult to even know where to obtain the reference, as obtaining it may be as hard as generating consistent information in the first place [362]. Furthermore, evaluations performed at the sentence-level or dialogue-level can only provide coarse-grained judgments—often merely deciding whether a sentence or entire document is hallucinated—which is insufficient for fine-grained error localization or real-time intervention [362]. Consequently, the design of the dataset should support *reference-free detection of token-level hallucinations* to enable dynamic localization of erroneous content during the generation process and ensure that the evaluation framework is directly aligned with the practical requirements of free-form generation scenarios [362].
- **Limitations:** Despite progress, dataset construction faces challenges: subjectivity in human annotation [112], weaknesses of automated annotation, high time and labor costs [34], coarse-grained focus lacking token-level precision [347], limited coverage in entity-level methods [83], biases from iterative model-in-the-loop approaches [362], and scalability issues with human verification. Current datasets also lack mechanisms for real-time hallucination prevention.

In summary, dataset construction combines automation with human expertise, annotation strategies, and verification protocols. Future benchmarks must overcome existing limitations by ensuring scalability, robustness, fine-grained analysis, and adaptability to real-world, domain-specific contexts.

e) Benchmark-based Evaluation Phase: After dataset construction, the benchmark-based evaluation phase becomes essential for systematically assessing hallucinations in LLMs. This phase integrates diverse datasets, advanced metrics, and both automated and human evaluation to ensure robustness. It involves four major dimensions: dataset utilization strategies, performance metrics, LLM as judge, and human involvement.

- **Dataset Utilization Strategies:** Two main approaches are used—direct evaluation and contrastive comparison. For example, HaluEval [17] employs a two-stage *sampling-then-filtering* pipeline where models generate hallucinations that are filtered and annotated, producing scalable benchmarks. TruthfulQA [310] uses adversarial questions with both truthful and misleading references to test whether models reproduce human misconceptions. Although effective, static benchmarks like TruthfulQA cannot adapt to evolving knowledge or capture nuanced responses.
- **Performance Metrics:** Effective hallucination measurement requires multiple metrics. - *Factuality*: Atomic fact-level accuracy such as FActScore [306]. - *Faithfulness*: Evaluating adherence to source material, especially in retrieval-augmented models, with tools like SUMMAC [410] and citation accuracy studies [399], [429]. - *Fine-grained Evaluation*: Hierarchical scoring (FACTOR [389]) to capture claim- and span-level precision. - *Reasoning*: Metrics for vertical reasoning traceability and causal reasoning [326], [384], [385], essential for diagnosing logical missteps. - *Reliability*: Consistency of results across contexts, populations, and domains [4]. - *Robustness*: Resistance to noise and adversarial prompts, addressing issues like authority bias or distribution shifts [387], [388]. - *Safety*: Risk-based assessments quantifying harm potential, as seen in the Hallucinations Leaderboard [374] and risk-factor analysis [95]. - *Cost & Efficiency*: Balancing thoroughness with computational feasibility using adaptive sampling and cascaded pipelines [76], [390]. - *Uncertainty Scoring*: Leveraging semantic entropy, Bayesian methods, or ensembles to quantify prediction confidence [76]. - *Degree of Automation*: Balancing automated scoring with necessary human oversight [60], [61], [432]. - *Scalability*: Ensuring metrics remain valid as datasets and models grow. - *Generalization Assessment*: Evaluating across contexts, times, model sizes, modalities, and cultures to avoid overfitting to narrow benchmarks.
- **LLM as Judge:** Using LLMs for evaluation scales efficiently but risks bias and recursion. G-EVAL [429] introduces structured chain-of-thought assessments, achieving higher correlation with human judgment, though biased toward LLM-generated outputs. LLMAuditor [323] mitigates this by combining automated probing with human-in-the-loop validation, producing more reliable, transparent audits across multiple models.
- **Human Involvement:** Human annotators remain essential for establishing ground truth and addressing subtleties beyond automated evaluation. Methods like HALU-J [430] employ critique-based, multi-step pipelines to enhance reliability. However, trade-offs remain between

quality, cost, and scalability, highlighting the need for strategic integration of human expertise within automated pipelines.

In sum, the benchmark-based evaluation phase ensures that hallucination assessments move beyond datasets alone, integrating diverse metrics, automated evaluators, and human oversight to achieve comprehensive, trustworthy evaluations of generative models.

f) Baseline Methods: Baseline methods provide the fundamental benchmarks for hallucination evaluation, offering standardized procedures to identify, measure, and reduce hallucinations [60]. They are commonly grouped into several functional categories, each highlighting different strategies:

- **Detection Baselines**: Methods such as POPE (Polling-based Object Probing Evaluation) transform subjective hallucination evaluation into binary Yes/No queries. The method further reveals systematic biases: LVLMs hallucinate common objects more often than rare ones, with co-occurrence bias explaining most errors. To expose such biases, POPE employs three negative sampling strategies:
  - Random Sampling – randomly selects absent objects as a control.
  - Popular Sampling – selects frequent but absent objects to test predispositions.
  - Adversarial Sampling – chooses objects with high co-occurrence with ground-truth items to trigger predictable errors.

  This reductionist design makes hallucination evaluation reproducible, scalable, and comparable across models.
- **Mitigation Baselines**: Retrieval-Augmented Generation (RAG) anchors responses in external knowledge bases, reducing factual hallucinations [349]. However, it often fails on complex reasoning tasks, increasing logical inconsistencies. Beyond RAG, more advanced interventions attempt to disrupt causal pathways of hallucination without fine-tuning models [110]:
  - Image Interventions - alter visual input (e.g., adding/removing objects) to break false visual-contextual links.
  - Text Interventions – prompts like Foreground/Background separation reduce semantic leakage.
  - Embedding Interventions – modify latent spaces to suppress dimensions associated with hallucination.

  Other baselines include calibration (e.g., temperature scaling [270]) and multi-fact synthesis (e.g., DBD [353]).
- **Integrated Detection and Mitigation**: Some baselines blur the line between evaluation and correction. AutoHallusion automatically generates adversarial benchmarks based on model priors, creating low-cost stress tests that expose vulnerabilities [112]. EventHallusion diagnoses temporal hallucinations in video LLMs by obscuring time information, pinpointing weaknesses in temporal reasoning. Even detection-only baselines like POPE provide mitigation insights, e.g., targeting co-occurrence bias reduction.
- **Performance Metrics**: Standard classification metrics (Accuracy, Precision, Recall, F1) remain the core indicators. POPE, for instance, emphasizes F1 as a balanced measure and tracks the proportion of affirmative ("Yes") responses to identify systematic biases [60].
- **Limitations of Current Baselines**: Despite their utility, baselines face several challenges:
  - *Reductionism*: Binary Yes/No evaluations miss nuanced or compositional hallucinations [60].
  - *Temporal Obsolescence*: Benchmarks lose validity quickly—performance drops as models adapt.
  - *Trade-offs*: RAG reduces factual errors but increases reasoning failures [349].
  - *Scalability*: Embedding-level interventions cut hallucinations but increase latency, making real-time use difficult.
  - *Domain Fragility*: Baselines that perform well on object detection collapse on abstract reasoning tasks, showing poor transferability.

  These issues highlight the need for dynamic benchmarks, adaptive frameworks, and efficient evaluation pipelines that can evolve with models.

g) Frameworks and Pipelines: As research progresses toward more systematic and reproducible practices, frameworks and pipelines have emerged as central infrastructures for hallucination evaluation. Rather than relying on ad hoc techniques, these pipelines integrate multiple components to support robust, scalable, and adaptable assessment. To better capture their diversity, we summarize the current state of the art through six key features, which also serve as the subsections of this discussion. Specifically, we examine end-to-end pipelines that provide holistic workflows, the role of human involvement in ensuring quality and interpretability, and the scope of automation capabilities and their limitations. We then discuss practical implementation strategies that address real-world deployment challenges, highlight feedback loop–based pipelines that iteratively refine model performance, and conclude with fusion-based approaches that combine heterogeneous signals to enhance evaluation robustness. Together, these features characterize the essential design choices shaping contemporary hallucination evaluation pipelines.

- **End-to-End Pipelines**: These pipelines follow a structured workflow of ground-truth preparation, data generation, detection, and measurement. Some use annotated datasets with spatial cues (e.g., circling objects in images) [61], while others rely on expert judgment without fixed ground truth for natural exposure [333]. Detection is performed by humans [316] or automated systems like AutoHallusion [112]. Measurement techniques include: (i) reference-based surface metrics such as edit distance [434], (ii) token-level reference-free detection [66], and (iii) model-internal forecasting using state-transition dynamics [318].

- **Fusion-based Pipelines**: Fusion integrates heterogeneous signals—visual, semantic, and external knowledge—across input, feature, and prompt stages. For instance, aligning image features with semantic embeddings ensures cross-modal consistency [393]. Prompt construction combines structured schemas, semantic triplets, and templates to guide queries [83]. Metrics aggregate evidence from multiple sources [319], while active learning loops refine fusion strategies by routing uncertain samples to humans [370]. This multi-level integration enhances robustness but introduces cost in multimodal data curation.
- **Feedback Loop Frameworks**: Feedback-driven designs iteratively refine models by comparing outputs with and without access to a source of truth [320]. Some generate adversarial datasets to stress-test robustness [314], while others integrate detection and correction in mitigation-aware loops [352]. Confidence scoring [295] and explanatory mechanisms (e.g., ER diagrams [83]) improve reliability. Feedback is also used as a probe for instability, detecting divergence across rephrased or multi-model outputs.
- **Multimodal Combination Frameworks**: These frameworks combine vision and language to detect hallucinations through cross-modal consistency. Examples include MedHallBench [370], which uses annotation, model feedback, and active learning in medical tasks; AutoHallusion, which generates adversarial images by probing language priors [112]; and VL-Uncertainty, which perturbs both images and text to elicit confidence signals [408]. ERBench [83] innovatively leverages database constraints, namely functional dependencies and foreign keys to auto-generate verifiable benchmarks. These approaches highlight the importance of heterogeneous signal fusion but face scalability challenges.
- **Parallel Structure Frameworks**: Parallel designs run multiple evaluation strategies simultaneously to uncover hallucinations missed by single methods. POPE applies random, popular, and adversarial sampling in parallel, revealing distinct failure modes [60]. CIEM (Contrastive Instruction Evaluation Method) advances this idea by generating parallel instruction variants—semantic-preserving, logical inversions, and contextual perturbations—detecting hallucinations that single instructions miss [432]. Such parallel setups expose brittleness to rephrasing, showing hallucination as sensitivity to input variation.
- **Hybrid and Survey-Driven Pipelines**: Beyond technical designs, some frameworks integrate evaluation with mitigation (e.g., HalluciDoctor, DualFocus) or provide taxonomies and surveys that classify hallucination types and align them with evaluation strategies [331]. These contributions highlight the theoretical underpinnings of hallucination evaluation, bridging fragmented methods and informing the development of hybrid, adaptive approaches.

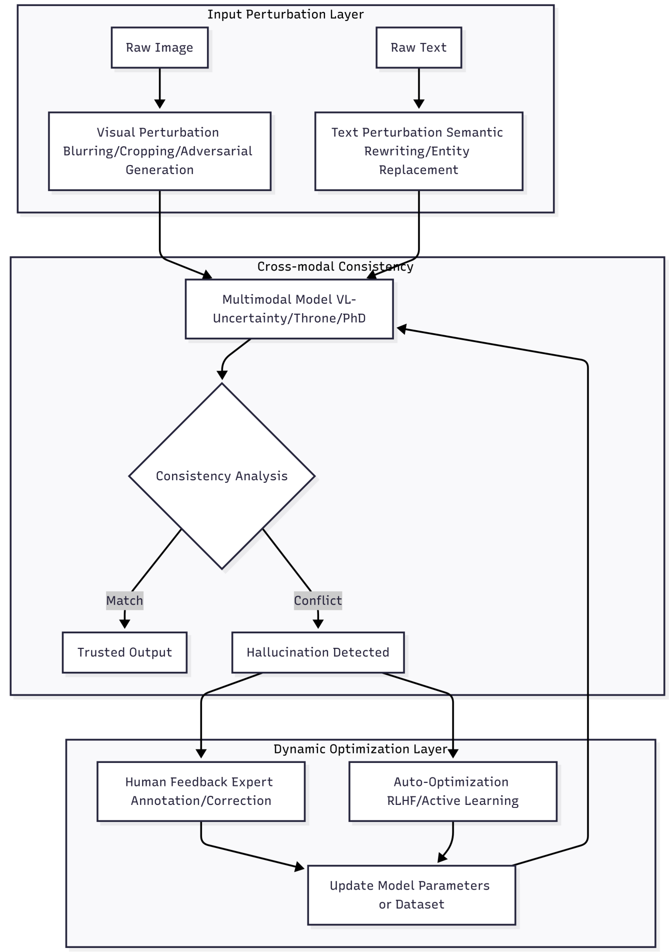

Fig. 10. Multimodal Combination Framework Pipeline

In summary, frameworks and pipelines represent the infrastructure of hallucination evaluation, as illustrated in Fig 10. From end-to-end systems to multimodal and parallel designs, they emphasize robustness, adaptability, and interpretability, but still face challenges of cost, scalability, and dynamic adaptation.

h) The Degree of Automation and Human Active Participation: A key design question in hallucination evaluation concerns the balance between automated scalability and human discernment. Frameworks differ in how they integrate the two, and representative strategies can be summarized as follows:

- **Human Involvement**: Human expertise remains indispensable for nuanced hallucination detection that requires contextual understanding, subjective quality assessment, or domain-specific knowledge [60], [432]. Experts curate references [63], parse multimodal inputs [353], and categorize errors into factual, logical, and contextual types [34]. Selective human-in-the-loop pipelines reduce costs while retaining high expert agreement [432].
- **Automation Capabilities and Limitations**: Automated systems scale effectively with structured tasks. Examples

include CIEM for contrastive instruction evaluation [432], AutoHallusion for adversarial scene generation [112], and rule-based judge models for standardized labeling [347]. However, automation is brittle to phrasing, synthetic inputs may diverge from real-world data [350], and internal interventions risk unintended effects [348]. These weaknesses underscore the need for hybrid strategies.

- **Probing Technology**: Probing actively interrogates models to expose hidden inconsistencies rather than passively scoring outputs. LLMAuditor [323] achieves high human agreement by combining structured probing with cross-model verification. It identifies (i) bias-induced hallucinations, (ii) operational inconsistencies, and (iii) knowledge boundary violations. Hybrid human–machine probing reduces evaluation costs while boosting detection rates.
- **Practical Implementation Strategies**: Effective systems adopt an automation-first, human-calibrated design. Automatic scoring flags high-risk cases, which are routed to dual human raters for adjudication [112]. Golden datasets track drift and calibrate pipelines [313], while feedback loops update prompts and thresholds [359]. Such hybrid setups reduce costs by 70–85% compared to full manual annotation while preserving nuanced detection for open-ended or multimodal hallucinations [370].

In summary, while automation provides scale and reproducibility, human oversight ensures interpretive depth and reliability. The most effective pipelines combine both, adapting human involvement to task complexity and domain risk.

i) Summary and Future Directions: Our comprehensive analysis of tools and frameworks reveals a field undergoing rapid methodological transformation, moving from manual, ad-hoc evaluation toward systematic, scalable, and theoretically grounded approaches. The key insights from this survey illuminate both remarkable progress and fundamental challenges that shape the future of hallucination evaluation.

Dataset Construction Evolution: The transition from manual to hybrid human-AI dataset generation represents a great increase in scale while reducing costs. Entity-based construction methods achieve around 96% automatic verification accuracy [83]. Yet critical challenges persist: automated generation exhibits systematic biases [17], while the trade-off between scale and quality remains unresolved.

Evaluation Methodology Maturation: Benchmark-based evaluation has evolved from simple accuracy metrics to a multidimensional evaluation [429]. The move from scattered hallucination tests to a unified leaderboard highlights the need for consistent cross-framework evaluation of LLM reliability [374].

The LLM-as-Judge Paradox: Using LLMs as evaluators enables scaling far beyond human evaluation, but it also introduces recursive uncertainty by relying on models that themselves may hallucinate. GPT-4 shows strong agreement with human annotators yet demonstrates preference bias toward its own outputs [429], while ensemble strategies improve accuracy but fail to resolve fundamental disagreements between judge models. This paradox reveals a deeper epistemological challenge: the impossibility of bootstrapping trust from untrusted foundations.

Human-AI Collaboration Optimization: Hybrid evaluation reduces costs greatly while maintaining high accuracy through intelligent task routing [323]. Humans can identify more hallucinations than automated systems, particularly for context-dependent errors. Domain-specific evaluation requires more human involvement, such as medical assessment [370].

Framework Integration Patterns: End-to-end pipelines dominate with a wide adoption but lack feedback mechanisms for iterative improvement [432]. Dynamic evaluation can reduce the evaluation error [320]. Critically, THRONE's analysis reveals that reduction in Type II hallucinations (false negatives, where models fail to identify present objects) does not lead to reduction in Type I hallucinations (false positives, where models claim absent objects exist), but rather these two hallucination forms exhibit anticorrelation [347].

Critical Gaps and Future Imperatives: Three fundamental challenges constrain current tools: (1) *Evaluation recursion*—the philosophical impossibility of establishing ground truth using potentially flawed evaluators; (2) *Benchmark obsolescence*—static datasets becoming saturated within 12 months as models adapt; (3) *Domain and Language transfer failure*—evaluation methods that perform reliably in English often degrade when applied to other languages or specialized domains [355].

The Path Forward: Future frameworks must embrace three principles: (1) *Dynamic adaptation* through continuous benchmark evolution and adversarial co-generation; (2) *Hierarchical validation* combining automated screening, ensemble judgment, and human expertise in tiered architectures; (3) *Causal understanding* moving beyond correlation-based detection toward mechanistic models of hallucination generation. The ultimate goal transcends detection—building evaluation systems that not only identify hallucinations but also understand their origins, predict their occurrence, and inform principled mitigation strategies.

The evolution from manual annotation to sophisticated human-AI collaboration represents remarkable progress, yet fundamental questions remain: Can we bootstrap a trustworthy evaluation from untrustworthy components? How do we prevent benchmark overfitting while maintaining comparability? What constitutes ground truth in domains where human experts disagree? These challenges define the frontier of hallucination evaluation, requiring not just technical innovation but philosophical clarity about the nature of truth, knowledge, and reliability in artificial intelligence systems.

8) Summary of the Evaluation Gap: The extensive body of work surveyed in this section reveals a vibrant and rapidly evolving field. Researchers have developed sophisticated metrics and benchmarks to tackle hallucination from multiple angles, as outlined in the preceding sections. However, a critical dichotomy can be used to characterize a fundamental limitation in the current landscape: the tension between Horizontal and Vertical assessment.

The majority of existing efforts, including comprehensive benchmarks like Hallucination Leaderboard [374] and and targeted evaluation suites such as HaluEval [17], excel at horizontal assessment—they provide a broad, top-level evaluation of hallucination across a wide range of tasks, domains, and modalities. This breadth is essential for generating aggregate performance scores and facilitating high-level model comparisons.

However, these horizontal assessments still face several core challenges that remain only partially resolved. As summarized in Table III, these include persistent difficulties in defining ground truth amid ambiguity, prohibitive computational costs of comprehensive evaluation, rapid obsolescence of static benchmarks, fragmentation in handling diverse domains/cultures, and recursive uncertainties in automated human-judgment approximation. These unresolved issues fundamentally constrain the effectiveness of top-down, breadth-first evaluation paradigms.

TABLE III. PARTIALLY RESOLVED CHALLENGES IN HORIZONTAL EVALUATION

Challenge Category	Manifestation in Horizontal Evaluation
Epistemological Challenge	Defining ground truth in ambiguous contexts (Sec. VI-B1)
Computational Complexity Challenge	Scaling costs of comprehensive assessment (Sec. VI-B2)
Dynamic Knowledge Challenge	Benchmark obsolescence against evolving knowledge (Sec. VI-B4)
Scale and Diversity Challenge	Fragmented domain/cultural coverage (Sec. VI-B6)
Human Alignment Challenge	Recursive uncertainty in human-judgment simulation (Sec. VI-B8)

Conversely, vertical assessment—the deep, mechanistic analysis of the root causes of a specific hallucination within a model's internal reasoning process—remains underexplored and challenging to implement at scale. While causal intervention and counterfactual analysis methodologies (Section VI-C4c) and attention analysis (Section VI-C4e) represent pioneering steps in this direction, they are typically applied in isolation, on small scales, or in controlled settings. The field currently lacks a unified framework that can seamlessly integrate both horizontal breadth and vertical depth, leaving a significant gap in our ability to not just detect hallucinations but to truly diagnose and understand them on a granular level. This gap motivates the introduction of the LENS framework in the subsequent chapters, which is designed to bridge this very divide.

D. The Evaluation Gap: Why Current Frameworks Fail and What Is Needed

Building on the challenges discussed in Section VI-B and the survey of approaches in Section VI-C4, it becomes evident that existing hallucination evaluation frameworks fall short of meeting the requirements of real-world AI deployment. The problem lies not in technical implementation but in the conceptual limitations shaping these frameworks. Current designs often reduce hallucination to oversimplified categories, neglect multi-dimensional characteristics, and fail to account for the dynamic, adaptive, and metacognitive aspects crucial for reliable assessment. As a result, the field faces a widening gap between what current evaluation systems can deliver and what is fundamentally needed for trustworthy deployment. This section outlines the critical deficiencies of existing frameworks and establishes the groundwork for a paradigm shift in evaluation methodology.

a) The Dimensional Poverty of Current Evaluation: Existing hallucination evaluation frameworks exhibit what can be described as dimensional poverty, an oversimplification that collapses the multifaceted nature of hallucinations into overly narrow measures or binary outcomes [1], [435]. By treating hallucination merely as a correctness issue—true versus false—these approaches overlook critical distinctions among types of errors, their underlying causes, and their practical implications [1], [331]. In reality, hallucinations manifest along multiple orthogonal dimensions that significantly affect their severity and the design of mitigation strategies [61], [343]. For example, confident factual inaccuracies pose higher risks than uncertain partial truths [336], [408], omissions differ fundamentally from fabrications [352], [436], and domain-specific systematic errors are not equivalent to sporadic mistakes [95], [322].

This reductionist perspective becomes particularly problematic in deployment decisions [337]. A model that appears to achieve 90% accuracy may still be unsuitable if its errors cluster in high-stakes areas, display misplaced confidence, or fail to trigger external verification mechanisms [324], [339]. Moreover, four essential dimensions remain missing from current evaluations [346], [435]: (1) the *degree* dimension, which goes beyond correctness to include contextual faithfulness and appropriateness of retrieval versus generation; (2) the *quantity* dimension, which evaluates completeness and addresses hallucinations of omission; (3) the *stability* dimension, which captures whether errors are systematic or stochastic; and (4) the *risk* dimension, which measures confidence calibration and uncertainty awareness. These dimensions are indispensable for real-world assessment, as demonstrated in high-risk domains like medicine, where accuracy, completeness, consistency, and confidence calibration must all be ensured simultaneously [24], [42], [370]. Without incorporating these missing dimensions, current frameworks produce misleadingly optimistic evaluations and fail to capture the actual risk profile of deployed systems.

b) The Horizontal-Vertical Integration Challenge: A core limitation of current hallucination evaluation lies in their artificial separation of horizontal breadth and vertical depth [65], [316]. Horizontal approaches, such as large-scale benchmarks including MMLU [53] or TruthfulQA [310], provide broad domain coverage but treat evaluations as isolated measurements, offering little insight into the causal mechanisms behind errors. They can highlight that a model fails in certain categories but cannot reveal whether the failures stem from knowledge gaps, flawed reasoning, or misgeneralization [329]. Conversely, vertical approaches—such as chain-of-thought analysis [47] — deliver fine-grained understanding of reasoning processes but are restricted to narrow task scopes, making them inadequate for general deployment decisions.

In practice, both breadth and depth are essential: organizations require evaluations that capture a model's global risk profile while also diagnosing the mechanisms underlying hallucinations [354], [393]. Fragmented frameworks obscure the fact that cross-domain hallucination patterns often emerge from shared internal mechanisms [63], [321]. For example, a tendency to over-generalize in medical reasoning may reflect the same architectural flaws that also cause failures in mathematical reasoning, but siloed evaluation methods miss these connections [92], [116]. Effective mitigation therefore requires integrated frameworks that bridge horizontal coverage with vertical diagnostic insights. True integration must go beyond simply combining existing methods, instead dynamically adjusting depth based on observed error patterns and synthesizing results across multiple modalities [100], [318]. Without such reconceptualization, evaluation will remain fragmented, preventing organizations from achieving the comprehensive and actionable understanding needed for safe deployment [327], [341].

c) The Metacognitive Blind Spot: Tool Necessity and Selection: Another critical shortcoming of current evaluation frameworks lies in their neglect of models' metacognitive abilities, specifically the capacity to recognize when internal generation is insufficient and external tools should be consulted. Existing evaluations assume that models must always respond from parametric memory, focusing solely on the correctness of generated content [182]. This overlooks a central safeguard: hallucinations can often be prevented if models identify cases requiring external retrieval or specialized verification [320], [359]. For example, questions involving real-time or domain-specific data, such as stock prices or clinical guidelines, should trigger retrieval rather than generation [349]. Yet no mainstream framework currently evaluates this recognition ability [335].

To address this blind spot, two missing evaluation dimensions are critical. First, *Tool Necessity Detection* (TND) assesses whether models can correctly identify when external consultation is required, guided by temporal sensitivity, factual specificity, or risk-critical contexts [182], [186]. Without TND, models in high-stakes applications, such as medicine, may generate outdated or unsafe responses instead of deferring to authoritative sources [42], [324]. Second, *Tool Selection Accuracy* (TSA) evaluates whether models not only recognize the need for external help but also choose the most appropriate tool for the query [295], [334]. A poor match, such as opting for a general web search instead of a specialized database, can be as harmful as failing to use any tool at all [351]. Together, TND and TSA highlight the importance of metacognitive evaluation: distinguishing models that merely lack knowledge from those that fail to recognize their own limitations. Current frameworks' omission of these dimensions represents a significant vulnerability, undermining deployment safety and trustworthiness in real-world environments.

d) The Adaptability Crisis: From Static Benchmarks to Dynamic Evaluation: A further limitation of current hallucination evaluation frameworks is their reliance on static, pre-defined benchmarks that cannot capture the evolving and domain-specific requirements of real-world applications. The assumption that a universal benchmark can serve all use cases is fundamentally flawed, since different organizations operate under unique domain knowledge, specialized terminology, and varying levels of risk tolerance [315], [437]. For example, pharmaceutical companies require evaluations aligned with updated drug interaction databases and regulatory guidance, while law firms need assessments grounded in jurisdiction-specific statutes and precedents. Generic benchmarks, no matter how comprehensive, fail to accommodate these specialized contexts. Moreover, the static nature of benchmarks creates a temporal mismatch: knowledge bases in medicine, law, or science evolve constantly, while evaluation datasets quickly become obsolete, producing scores that reflect outdated realities [34], [313], [374]. Organizations are thus left to choose between standardized evaluations that lack relevance and ad-hoc assessments that lack rigor.

To overcome this crisis, next-generation frameworks must embrace adaptability. This means not only enabling the substitution of datasets but also supporting flexible evaluation construction from an organization's own information sources [61], [403]. Such adaptability should cover dynamic integration of authoritative knowledge, customization of evaluation criteria to reflect organizational priorities, domain-specific reasoning assessments, and alignment with context-sensitive risk thresholds. In this way, evaluation becomes a tailored yet methodologically rigorous process rather than a one-size-fits-all exercise [327], [342]. Without this adaptive capacity, organizations cannot properly evaluate models against their real operational requirements, making adaptability not just a desirable feature but a fundamental prerequisite for trustworthy deployment.

e) The Transparency Deficit: From Scores to Actionable Understanding: Another major weakness of current evaluation frameworks is their lack of transparency. Most frameworks operate like black boxes, providing numerical scores without revealing the reasoning processes, evidence chains, or causal pathways behind them [323]. While a score such as 70% accuracy may signal performance level, it conveys nothing about the nature of errors, their distribution across domains, or the stages of reasoning where failures occur [66], [66], [314]. As a result, developers cannot identify improvement strategies, users cannot calibrate trust, and organizations lack actionable insights for deployment decisions.

True transparency requires moving beyond surface-level scoring to expose how evaluation tasks are structured, how individual components contribute to results, and what evidence supports each judgment [311], [438]. More critically, it requires causal attribution that traces hallucinations to their origins—whether in model architecture, training data, or inference processes [109], [345]. Attribution allows evaluators to pinpoint where errors enter, how they propagate, and why interventions succeed or fail. This transforms evaluation from passive measurement to active diagnosis, enabling targeted refinements such as architectural redesign, dataset curation, or improved prompting strategies [110], [320], [341], [439]. Without such transparency, evaluation outputs remain opaque and ultimately ineffective for guiding the development of safer, more reliable AI systems.

f) The Synthesis Challenge: Toward Unified Evaluation: The shortcomings discussed above are not isolated problems but interconnected symptoms of a deeper issue: the lack of unified evaluation frameworks that integrate diverse dimensions, methods, and perspectives into a coherent whole [319]. Current practices remain fragmented—methodological fragmentation forces practitioners to navigate incompatible paradigms with inconsistent assumptions [4], [330]; metric fragmentation produces scores that cannot be directly compared; tool fragmentation requires juggling different systems with varying interfaces and formats; and insight fragmentation disperses understanding across disjointed reports. This fragmentation not only raises costs for organizations but also prevents the emergence of a comprehensive understanding of hallucination risks [348], [350], [390].

Addressing this synthesis challenge requires more than incremental integration. What is needed is a reconceptualization of evaluation as a unified analytical process that simultaneously captures multiple hallucination dimensions, balances horizontal breadth with vertical depth, incorporates metacognitive assessment, allows adaptive customization, and ensures transparency with causal attribution [83], [325], [332], [333], [440]. Different aspects of hallucination interact in complex ways that only a unified approach can reveal [72], [312]. By synthesizing these diverse perspectives, evaluation can move from fragmented measurement toward systematic, actionable understanding that supports reliable and trustworthy deployment.

g) The Path Forward: Requirements for Next-Generation Evaluation: The gaps highlighted across previous subsections point to clear requirements for next-generation hallucination evaluation frameworks. These requirements are not aspirational add-ons but essential conditions for enabling trustworthy AI deployment. First, evaluation must achieve *dimensional completeness*, going beyond accuracy to assess faithfulness, completeness, stability, and calibrated uncertainty [317], [353]. Second, frameworks must provide *hierarchical integration*, combining horizontal coverage with vertical depth in dynamically adjustable architectures [344], [434]. Third, evaluation should incorporate *metacognitive assessment*, measuring whether models recognize when to defer to external sources and whether they select the right tools for the task [270], [328].

In addition, frameworks must support *adaptive construction*, allowing organizations to build rigorous yet domain-specific evaluations from their own knowledge bases while maintaining methodological consistency [338], [418]. They also need to guarantee *process transparency*, exposing reasoning paths, evidence chains, and causal attributions to make evaluation results interpretable and actionable [432], [441]. Finally, all these dimensions must converge in *unified synthesis*, transforming fragmented outputs into coherent insights that guide both technical refinement and deployment decisions [60], [111]. Meeting these requirements represents a paradigm shift: from static, reductionist measurements toward holistic, adaptive, and actionable evaluation capable of supporting safe and reliable AI in real-world contexts.

h) Conclusion: The LENS Response: The gaps identified in this section reveal that current hallucination evaluation frameworks are not merely limited but fundamentally inadequate for supporting trustworthy AI deployment. The dimensional poverty, integration failures, metacognitive blind spots, adaptability crises, transparency deficits, and synthesis challenges collectively render existing approaches insufficient for the demands of real-world applications.

These gaps cannot be addressed through incremental improvements or framework combinations but require fundamental reconceptualization of how we approach hallucination evaluation. The LENS framework, detailed in the following section, represents our comprehensive response to these challenges. Through multi-dimensional assessment spanning degree, quantity, stability, and risk; hierarchical decomposition enabling simultaneous breadth and depth; explicit evaluation of Tool Necessity Detection and Tool Selection Accuracy; adaptive benchmark construction from organizational knowledge sources; complete process transparency with causal attribution; and unified synthesis of diverse evaluation perspectives, LENS transforms hallucination evaluation from fragmented measurement to systematic understanding.

The transition from current frameworks to LENS represents not just technical advancement but a paradigm shift in how we conceptualize and conduct hallucination evaluation. By addressing each identified gap while maintaining practical deployability, LENS enables the comprehensive, actionable evaluation necessary for trustworthy AI systems in critical applications.

E. LENS: A Decomposition-First Framework for Systematic LLM Hallucination Evaluation

Large Language Models and Vision-Language Models exhibit remarkable capabilities yet suffer from hallucinations that generate plausible but factually incorrect content [306], [310]. The challenge of detecting and mitigating these hallucinations has become critical as these models are deployed in high-stakes applications. We present LENS, a comprehensive evaluation framework that systematically addresses hallucination detection through hierarchical decomposition, tool-augmented verification, and structured output generation.

The core insight driving LENS is that complex queries often fail not because models lack knowledge, but because they attempt to process compound information without proper factorization and fail to recognize when external verification is necessary. A critical source of hallucinations occurs when models confidently generate information from parametric memory that should instead be retrieved from authoritative sources.

By decomposing evaluation tasks into verifiable atomic components and explicitly evaluating tool necessity recognition, LENS transforms the hallucination detection problem from a monolithic assessment into a structured verification process. Our framework operationalizes evaluation across four critical dimensions: degree measuring accuracy, faithfulness, and tool appropriateness; quantity assessing coverage and recall; stability evaluating test-retest reliability; and hallucination risk quantifying uncertainty levels.

1) Framework Architecture: LENS adopts a systematic pipeline architecture that transforms complex evaluation tasks into verifiable atomic components through principled decomposition. At its foundation, the framework recognizes that hallucinations often emerge from the compounding of small errors in multi-step reasoning. Rather than treating model outputs as black-box predictions, LENS explicitly exposes the reasoning process through a combination of state-of-the-art techniques including Chain-of-Thought reasoning [47], Tree-of-Thoughts search [214], tool-augmented generation via ReAct [442], and constrained decoding [443]–[445].

The framework implements a six-stage evaluation pipeline denoted as $\Pi = \langle S_1, S_2, ..., S_6 \rangle$ where each stage S_i transforms the evaluation state σ_i to σ_{i+1}. Task formulation S_1 establishes evaluation criteria $\mathcal{C}$, success metrics $\mathcal{M}$, and acceptable evidence sources $\mathcal{E}$ while defining ground truth baselines $\mathcal{G}$ and tolerance thresholds τ. The hierarchical decomposition stage S_2 applies recursive task decomposition using Tree-of-Thoughts to generate multiple valid factorizations. For compound queries expressed as $C = f(A_1, A_2, ..., A_n)$, the system explores both direct evaluation strategies and component-wise verification approaches. During tool-augmented execution S_3, decomposed subtasks are processed using appropriate external tools from the tool registry $\mathcal{T}$ following the ReAct paradigm, with each tool invocation generating traceable evidence complete with provenance metadata.

The structured generation phase S_4 enforces output constraints through grammar-guided decoding or JSON Schema validation, ensuring machine-readable results that include evidence chains and confidence scores. Multi-dimensional scoring S_5 computes evaluation metrics across our four key dimensions using established benchmarks [306], [310], [446]. The final trace analysis stage S_6 generates standardized OpenTelemetry traces that enable reproducibility and comparative analysis across model versions, prompting strategies, and parameter configurations. As illustrated in Fig. 11, the LENS evaluation pipeline integrates six stages, from task formulation to trace analysis, with feedback loops that enable iterative refinement based on scoring outcomes.

Four key architectural principles guide the LENS design. First, the decomposition-first approach ensures that complex queries undergo recursive factorization before evaluation, exposing intermediate reasoning steps and enabling granular verification. Second, evidence grounding requires that all claims be supported by explicit evidence from verified sources, with tool invocations providing traceable provenance chains. Third, structure enforcement employs constrained decoding to guarantee well-formed, parseable results amenable to automated scoring. Finally, comprehensive traceability via OpenTelemetry spans captures model inputs, outputs, tool invocations, and timing data to ensure full reproducibility of results.

2) Evaluation Dimensions and Metrics: LENS evaluates model outputs across four orthogonal dimensions $\mathcal{D} = \{d_{degree}, d_{quantity}, d_{stability}, d_{risk}\}$, each capturing distinct aspects of generation quality and reliability. This multi-dimensional approach recognizes that hallucination is not a binary phenomenon but rather manifests along different axes that require specialized detection strategies.

a) Degree: Correctness, Faithfulness, and Tool Appropriateness: The degree dimension d_{degree} measures correctness, faithfulness, and crucially, the appropriateness of tool usage through a composite score:

$$\begin{aligned} d_{degree} =& \alpha_1 \cdot \text{Truth}_{ext}(y) + \alpha_2 \cdot \text{Faith}_{int}(y|x) \\ &+ \alpha_3 \cdot \text{Tool}_{app}(t|q) \end{aligned} \tag{1}$$

where $\text{Truth}_{ext}(y)$ quantifies extrinsic truthfulness against world knowledge, $\text{Faith}_{int}(y|x)$ measures intrinsic faithfulness to provided context x, $\text{Tool}_{app}(t|q)$ evaluates tool appropriateness for query q, and $\alpha_1 + \alpha_2 + \alpha_3 = 1$ balances the three components.

The tool appropriateness score Tool_{app} decomposes into two critical sub-metrics:

$$\begin{aligned} \text{Tool}_{app}(t|q) =& \beta \cdot \text{TND}(q) \\ &+ (1-\beta) \cdot \text{TSA}(t|q) \end{aligned} \tag{2}$$

where Tool Necessity Detection (TND) measures whether the model correctly identifies when external tools are required, and Tool Selection Accuracy (TSA) evaluates whether the appropriate tool was chosen for the task. These metrics (TND and TSA) are core components of the proposed LENS framework. See Table IV for a comprehensive comparison with existing methods. This explicit evaluation of tool appropriateness is crucial because a significant class of hallucinations occurs when models generate plausible-sounding facts from parametric memory instead of retrieving verified information, or when they select inappropriate tools that provide unreliable evidence.

For extrinsic evaluation, the framework employs TruthfulQA [310] to assess factual accuracy. Intrinsic evaluation leverages FActScore [306], which decomposes long-form outputs into atomic claims $\{c_1, c_2, ..., c_m\}$ for individual verification:

$$\text{FActScore}(y) = \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}[\text{Verify}(c_i) = \text{True}] \tag{3}$$

QAFactEval [446] measures summary faithfulness through question-answering consistency, computing the alignment between questions generated from source documents and answers derived from model outputs.

TABLE IV. COMPREHENSIVE COMPARISON BETWEEN LENS AND EXISTING HALLUCINATION EVALUATION FRAMEWORKS ACROSS CRITICAL CAPABILITIES

Capability	**Output-Based** (TruthfulQA, FactCC)	**Process-Based** (StrategyQA, DiaHalu)	**Capability-Based** (GSM8K, Hallusion-Bench)	**LENS Framework** (Our Approach)
Evaluation Scope	Broad coverage, final outputs only	Limited tasks, reasoning chains	Specific skills, isolated testing	Comprehensive: horizontal breadth + vertical depth
Causal Attribution	None: binary detection only	Partial: traces reasoning steps	Limited: identifies skill gaps	Complete: full causal chain from query to error
Metacognitive Assessment	Not evaluated	Not evaluated	Not evaluated	**Novel**: TND and TSA as primary metrics
Dimensional Analysis	Single: correctness only	Single: reasoning fidelity	Single: capability presence	**Four dimensions**: degree, quantity, stability, risk
Adaptability	Fixed benchmarks	Fixed protocols	Fixed test sets	**Dynamic**: construct from organizational knowledge
Transparency	Black box scores	Some reasoning visibility	Skill-level insights	**Full transparency**: hierarchical decomposition with evidence
Tool Integration	Not supported	Limited support	Not supported	**Core feature**: evaluated and utilized systematically
Error Prevention	Reactive: post-hoc detection	Reactive: error analysis	Reactive: capability gaps	**Proactive**: identifies hallucination-prone patterns
Actionability	Low: scores without guidance	Medium: some diagnostic value	Medium: skill improvement	**High**: targeted mitigation strategies
Computational Cost	Low: simple checks	Medium: reasoning analysis	Low to Medium	Medium to High: comprehensive but optimized

b) Quantity: Coverage and Completeness: The quantity dimension $d_{quantity}$ assesses the completeness of generated content through coverage metrics:

$$d_{quantity} = \frac{|\mathcal{I}_{generated} \cap \mathcal{I}_{required}|}{|\mathcal{I}_{required}|} \tag{4}$$

where $\mathcal{I}_{required}$ represents required information units and $\mathcal{I}_{generated}$ denotes information present in the output.

In retrieval-augmented generation settings, RAGAS metrics [414] provide detailed assessment through context recall R_c, context precision P_c, and answer-context alignment A_{ac}:

$$\text{RAGAS} = \beta_1 R_c + \beta_2 P_c + \beta_3 A_{ac} \tag{5}$$

where $\beta_1 + \beta_2 + \beta_3 = 1$ represents task-specific weighting.

c) Stability: Robustness and Consistency: Model consistency under varying conditions reveals robustness characteristics captured by the stability dimension $d_{stability}$. The framework evaluates self-consistency across k samples:

$$\text{Consistency}(y_1, ..., y_k) = 1 - \frac{H(\text{Majority}(y_1, ..., y_k))}{H_{max}} \tag{6}$$

where H denotes entropy and H_{max} represents maximum possible entropy.

Position bias in long contexts is quantified through the U-shaped attention metric:

$$\text{PosBias}(a) = \frac{\sum_{i \in \{head, tail\}} a_i}{\sum_{i=1}^{n} a_i} \tag{7}$$

where a represents attention weights across n positions.

TABLE V. LENS EVALUATION DIMENSIONS WITH REPRESENTATIVE METRICS AND MEASUREMENT PROTOCOLS.

Dimension	**Metrics**	**Protocol**
Degree	TruthfulQA, FActScore, QAFactEval, TND, TSA	Atomic claim verification, QA consistency, tool necessity detection, tool selection accuracy
Quantity	Coverage ratio, RAGAS	Information unit tracking, retrieval analysis
Stability	Self-consistency, position sensitivity	Multi-sample entropy, context permutation
Hallucination	Semantic uncertainty, INSIDE, CHAIR	Disagreement analysis, activation probing

d) Hallucination Risk Quantification: The hallucination risk dimension d_{risk} employs complementary detection methods. Semantic uncertainty U_s through multi-sample analysis:

$$U_s = \mathbb{E}_{y \sim p(y|x)}[\text{KL}(p(y|x)||p_{ref}(y))] \tag{8}$$

When model internals are accessible, INSIDE probes [447] compute:

$$h_{probe} = \sigma(W^T \phi(s) + b) \tag{9}$$

where $\phi(s)$ represents hidden state features and W, b are learned probe parameters.

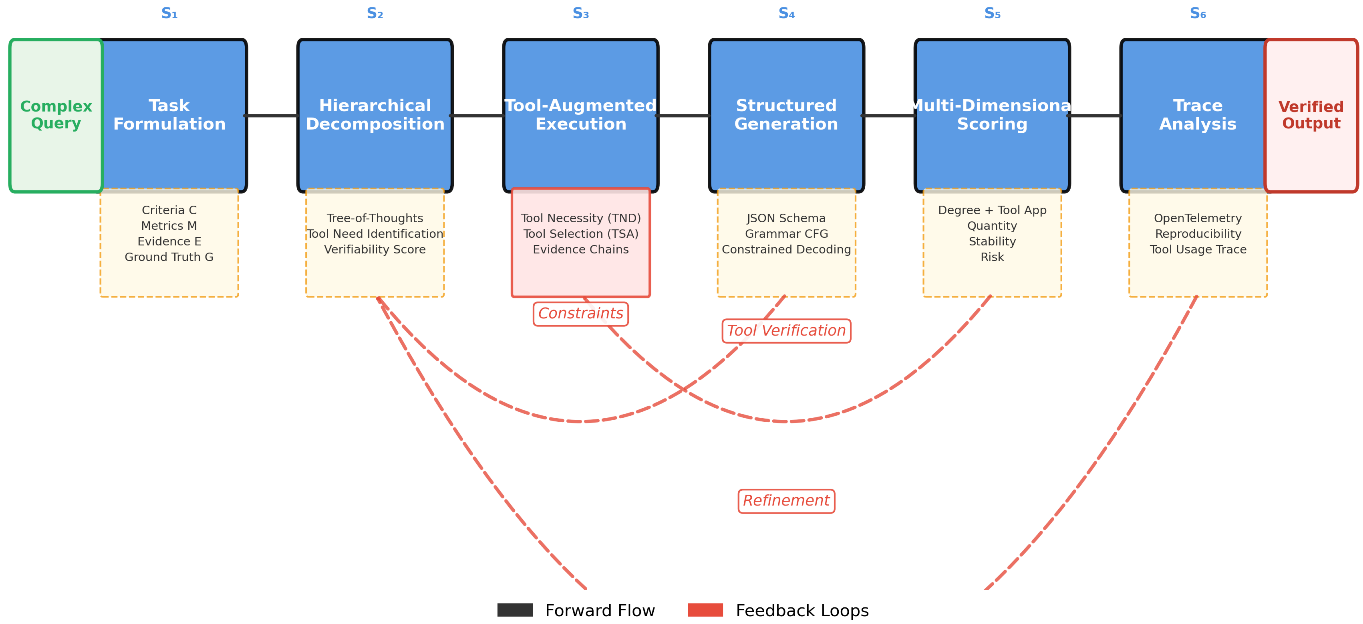

Fig. 11. LENS evaluation pipeline showing the six-stage process from task formulation to trace analysis; Feedback loops enable iterative refinement based on scoring outcomes

3) Hierarchical Decomposition Module: The decomposition module represents the cognitive core of LENS, transforming complex evaluation tasks into verifiable atomic components through systematic factorization. Our approach extends Tree-of-Thoughts [214] with verification-aware branching strategies that prioritize decomposition paths based on their amenability to external verification. Critically, the decomposition process identifies which components require external tool verification versus those that can be reliably computed through reasoning, directly informing the Tool Necessity Detection (TND) evaluation.This decomposition process yields measurable evaluation dimensions, which, along with their representative metrics and measurement protocols, are summarized in Table V.

For compound queries expressed as $C = f(A_1, \ldots, A_n)$, LENS generates multiple valid factorizations and selects optimal paths based on verifiability scores computed as:

$$V(\mathcal{P}) = \sum_{i=1}^{|\mathcal{P}|} w_i \cdot v(n_i) - \lambda \cdot \text{Depth}(\mathcal{P}) \tag{10}$$

where $v(n_i)$ represents the verifiability of node i, w_i denotes importance weights, and λ controls the penalty for deep decomposition trees.

The system considers three primary decomposition strategies. The direct path $\mathcal{P}_{direct}$ queries C immediately but risks compound errors where mistakes in any component cascade through the entire result. The sequential path $\mathcal{P}_{seq}$ evaluates components A_1 through A_n in order before computing $f(A_1, \ldots, A_n)$, which provides intermediate checkpoints but may suffer from context contamination. The parallel path $\mathcal{P}_{parallel}$ evaluates all components $\{A_i\}$ concurrently in isolated sessions, preventing error propagation at the cost of increased computational overhead.

Algorithm 1 Hierarchical Query Decomposition

```
Input: Complex query C = f(A_1, ..., A_n)
Output: Optimal decomposition path P*
Initialize decomposition tree T ← ∅
Generate factorizations F ← Factorize(C)
for each factorization F_i ∈ F do
    Compute verifiability score v_i ← Score(F_i)
    Add node (F_i, v_i) to T
end for
P* ← argmax_{P∈T} Σ_{i∈P} v_i
for each node n ∈ P* do
    if IsAtomic(n) then
        Execute verification r ← Verify(n)
    else
        Recursively decompose Decompose(n)
    end if
end for
return P*
```

Each decomposition node implements verification gates that validate intermediate results before propagation. The verification function $\mathcal{V} : \mathcal{N} \rightarrow \{0, 1\}$ determines whether a node passes validation:

$$\mathcal{V}(n) = \begin{cases} 1 & \text{if } \text{Conf}(n) > \theta \wedge \text{Cons}(n) > \gamma \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

where $\text{Conf}(n)$ measures confidence, $\text{Cons}(n)$ evaluates consistency, and θ, γ are configurable thresholds.

4) Tool-Augmented Reasoning and Selection Evaluation: LENS not only integrates external tools to ground model outputs in verifiable evidence but critically evaluates the model's ability to recognize when tools are necessary and select appropriate ones. This dual focus on tool usage and tool selection evaluation addresses a fundamental source of hallucinations: models confidently generating information from parametric memory when they should instead consult external sources. The framework follows the ReAct paradigm [442] for interleaved reasoning and action generation while introducing novel metrics for assessing tool selection competence.

The framework maintains a tool registry $\mathcal{T} = \{t_1, t_2, ..., t_k\}$ where each tool t_i is characterized by its signature:

$$t_i : \mathcal{Q}_i \times \mathcal{P}_i \rightarrow \mathcal{R}_i \times \mathcal{E}_i \tag{12}$$

where $\mathcal{Q}_i$ represents valid query types, $\mathcal{P}_i$ denotes parameters, $\mathcal{R}_i$ specifies result types, and $\mathcal{E}_i$ provides evidence metadata.

Tool selection follows a learned policy $\pi : \mathcal{S} \times \mathcal{Q} \rightarrow \mathcal{T}$ that maps state-query pairs to optimal tool configurations. The policy optimization objective:

$$\pi^* = \text{argmax}_\pi \mathbb{E}_{(s,q)\sim\mathcal{D}} [r(s, q, \pi(s, q)) - c(\pi(s, q))] \tag{13}$$

where r represents verification reward and c denotes computational cost.

The framework evaluates tool necessity detection through a binary classification:

$$\text{TND}(q) = \begin{cases} 1 & \text{if } \mathbb{1}[\text{NeedsTool}(q)] = \mathbb{1}[\text{UsedTool}(q)] \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

where $\text{NeedsTool}(q)$ is determined by whether the query requires information beyond the model's training cutoff, specific factual data, or real-time information.

Tool selection accuracy measures the appropriateness of the chosen tool given the query characteristics:

$$\text{TSA}(t|q) = \max_{t^* \in \mathcal{T}_{opt}(q)} \text{sim}(t, t^*) \cdot \mathbb{1}[\text{Valid}(t, q)] \tag{15}$$

where $\mathcal{T}_{opt}(q)$ represents the set of optimal tools for query q, and $\text{sim}(t, t^*)$ measures functional similarity between the selected and optimal tools.

When multiple tool invocations yield conflicting evidence, LENS implements a weighted voting scheme:

$$e^* = \text{argmax}_{e\in\mathcal{E}} \sum_{i=1}^{n} \rho_i \cdot \mathbb{1}[t_i \rightarrow e] \tag{16}$$

where ρ_i represents the reliability score of tool t_i derived from historical accuracy.

The framework categorizes tools into four primary classes. Retrieval tools $\mathcal{T}_{retrieval}$ encompass both dense and sparse retrievers for factual grounding, employing BM25 for lexical matching with scoring function:

$$\text{BM25}(q, d) = \sum_{i=1}^{|q|} \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \tag{17}$$

Computational tools $\mathcal{T}_{compute}$ include symbolic calculators and code interpreters that provide deterministic verification. Knowledge APIs $\mathcal{T}_{knowledge}$ connect to authoritative databases with structured query interfaces. Perceptual tools $\mathcal{T}_{perceptual}$ incorporate vision encoders and multimodal processors.

5) Structured Output Generation: Ensuring machine-verifiable outputs requires enforcing structural constraints during generation through grammar-guided decoding and schema validation. This structured approach eliminates parsing ambiguities and enables automated downstream processing while maintaining human readability.

LENS supports three constraint modalities with formal specifications. JSON Schema $\mathcal{S}_{JSON}$ provides hierarchical type constraints:

$$\mathcal{S}_{JSON} = \langle \mathcal{T}_{types}, \mathcal{P}_{props}, \mathcal{R}_{required}, \mathcal{C}_{constraints} \rangle \tag{18}$$

Context-free grammars $\mathcal{G}_{CFG}$ specified in EBNF notation:

$$\mathcal{G}_{CFG} = \langle V_N, V_T, P, S \rangle \tag{19}$$

where V_N denotes non-terminals, V_T represents terminals, P specifies production rules, and S is the start symbol.

The framework integrates with modern constrained decoding engines through a unified interface. The decoding process enforces constraints at each generation step:

$$p(x_{t+1}|x_{1:t}, \mathcal{C}) = \begin{cases} \frac{p_{LM}(x_{t+1}|x_{1:t})}{\sum_{x'\in\mathcal{V}_\mathcal{C}} p_{LM}(x'|x_{1:t})} & \text{if } x_{t+1} \in \mathcal{V}_\mathcal{C} \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

where $\mathcal{V}_\mathcal{C}$ represents the valid vocabulary under constraint $\mathcal{C}$.

6) Hallucination Detection Mechanisms: LENS implements multi-layered hallucination detection combining statistical, behavioral, and architectural approaches. This comprehensive strategy recognizes that hallucinations manifest through various signals that no single detection method can fully capture. Tool selection failures identified through low TND or TSA scores serve as early warning indicators, as models that fail to recognize when external verification is needed are significantly more likely to generate hallucinated content.

Statistical detection computes semantic uncertainty through multi-sample analysis. Lexical divergence D_L measures token-level variation:

$$D_L = \frac{1}{k(k-1)} \sum_{i\neq j} \text{EditDist}(y_i, y_j) \tag{21}$$

Semantic clustering C_S analyzes embedding-space dispersion:

$$C_S = \frac{1}{k} \sum_{i=1}^{k} ||\phi(y_i) - \bar{\phi}||^2 \tag{22}$$

where ϕ represents the embedding function and $\bar{\phi}$ denotes the centroid.

Behavioral analysis examines generation patterns through linguistic markers. Hedging detection H_d quantifies uncertainty expressions:

$$H_d = \frac{\sum_{w \in \mathcal{W}_{hedge}} \text{count}(w)}{|y|} \tag{23}$$

where $\mathcal{W}_{hedge}$ contains hedging vocabulary.

When model internals are accessible, architectural probing provides additional detection capabilities. INSIDE probes [447] employ linear classifiers on intermediate representations:

$$\begin{aligned}\mathcal{L}_{probe} = &- \sum_{i=1}^{n} y_i \log(\sigma(W^T h_i + b)) \\ &+ (1 - y_i) \log(1 - \sigma(W^T h_i + b))\end{aligned} \tag{24}$$

PoLLMgraph [318] analyzes state transition dynamics through Markov chain modeling:

$$P(s_{t+1}|s_t) = \frac{\exp(f_\theta(s_t, s_{t+1}))}{\sum_{s'} \exp(f_\theta(s_t, s'))} \tag{25}$$

7) Observability and Reproducibility: LENS prioritizes reproducibility through comprehensive tracing and standardized logging protocols. The framework adopts OpenTelemetry with OpenInference semantic conventions [448], [449] to create a standardized trace architecture.

Each evaluation generates a trace $\mathcal{T}_{eval}$ comprising spans:

$$\begin{aligned}&\mathcal{T}_{eval} = \{s_1, s_2, ..., s_n\} \\ &\text{where } s_i = \langle id_i, type_i, attrs_i, parent_i, time_i \rangle\end{aligned} \tag{26}$$

Span types include LLM spans for generation events, Retriever spans for search operations, Tool spans for external API calls, and Evaluator spans for scoring computations. Context propagation through parent-child relationships encodes the decomposition hierarchy:

$$\text{Context}(s_i) = \bigcup_{j \in \text{Ancestors}(i)} attrs_j \tag{27}$$

The reproducibility package $\mathcal{R}_{pkg}$ ensures exact result replication:

$$\begin{aligned}\mathcal{R}_{pkg} = \langle &\mathcal{M}_{model}, \mathcal{P}_{prompts}, \mathcal{S}_{seeds}, \\ &\mathcal{T}_{tools}, \mathcal{D}_{data}, \mathcal{E}_{env} \rangle\end{aligned} \tag{28}$$

8) Comprehensive Case Studies: LENS in Action: To demonstrate LENS's transformative capabilities, we present three detailed case studies that reveal how the framework addresses hallucination challenges that defeat current evaluation approaches.

a) Case Study 1: Medical Diagnosis with Life-Critical Stakes: Consider a medical query: "What are the recommended treatments for a 65-year-old patient with Type 2 diabetes, moderate kidney disease (GFR 45), and recent myocardial infarction, who is allergic to ACE inhibitors?"

Current evaluation frameworks would simply check if the generated treatment recommendations match medical databases, providing a binary correct/incorrect assessment. LENS transforms this into comprehensive multi-dimensional evaluation:

Hierarchical Decomposition LENS decomposes this complex query into verifiable components:

- Patient profile extraction: age, conditions, contraindications
- Individual condition treatments: diabetes protocols, kidney disease management, post-MI care
- Interaction analysis: drug interactions, comorbidity considerations
- Contraindication checking: ACE inhibitor alternatives
- Treatment synthesis: integrated care plan

Metacognitive Evaluation LENS identifies that this query requires external verification (TND = 1.0) due to:

- Evolving medical guidelines requiring current information
- Patient-specific contraindications demanding specialized databases
- Drug interaction complexity exceeding safe parametric generation

The framework evaluates tool selection, scoring high TSA (0.98) for choosing medical databases over general web search, pharmaceutical interaction checkers over Wikipedia, and current clinical guidelines over training-time knowledge.

Multi-Dimensional Assessment LENS employs its four-dimensional evaluation framework to provide a comprehensive assessment that moves beyond traditional binary correctness metrics, offering nuanced insights into the safety and reliability of medical recommendations

- Degree: Not just accuracy but completeness of contraindication checking
- Quantity: Coverage of all relevant treatment modalities
- Stability: Consistency across similar patient profiles
- Risk: Explicit uncertainty quantification for off-label recommendations

Actionable Output LENS provides not just a score but a complete evaluation trace showing which guidelines were consulted, what alternatives were considered for ACE inhibitors, how drug interactions were verified, and where uncertainty remains. This transparency enables medical professionals to validate reasoning and identify areas requiring human oversight.

b) Case Study 2: Legal Analysis with Jurisdictional Complexity: Query: "Can a California-based startup use customer data collected before CCPA implementation for training ML models if users consented under previous terms of service?"

This query exemplifies the complex intersection of temporal, jurisdictional, and technical factors that current frameworks cannot adequately evaluate. LENS reveals the multi-layered analysis required:

Temporal Decomposition

- Pre-CCPA data collection period identification
- CCPA implementation timeline (January 1, 2020)
- Retroactivity provisions analysis

- Grandfathering clause examination

Jurisdictional Analysis

- California state law requirements
- Federal data protection considerations
- Industry-specific regulations (HIPAA, FERPA if applicable)
- International implications for global users

Tool Necessity Recognition LENS correctly identifies (TND = 1.0) that this requires:

- Current legal databases, not training-time knowledge
- Recent case law and regulatory guidance
- Industry-specific interpretations

The framework penalizes any attempt to generate legal advice from parametric memory, recognizing that legal interpretation requires authoritative sources and jurisdictional expertise.

Risk-Aware Evaluation LENS's risk dimension captures that incorrect legal advice could lead to:

- Regulatory penalties (up to 4% of global revenue under CCPA)
- Class action lawsuits
- Reputational damage

This severity weighting ensures that legal hallucinations are not treated equally with minor factual errors, providing organizations with risk-adjusted evaluation scores that reflect real-world consequences.

c) Case Study 3: Financial Analysis with Multi-Hop Reasoning: Query: "What is the total population of the three largest cities in the country where the Eiffel Tower is located?"

While appearing straightforward, this query demonstrates LENS's ability to handle multi-hop reasoning that current frameworks evaluate superficially:

The decomposition phase applies Algorithm 1 to generate the factorization tree:

$$C = \text{sum}(\text{population}(\text{top3}(\text{cities}(\text{country}(\text{Eiffel Tower}))))) \tag{29}$$

$$= \text{sum}(\text{population}(\text{top3}(\text{cities}(\text{France})))) \tag{30}$$

$$= \text{sum}(\text{population}(\{\text{Paris}, \text{Marseille}, \text{Lyon}\})) \tag{31}$$

Each atomic component undergoes tool-augmented verification with explicit evaluation of tool necessity and selection. For the Eiffel Tower location query, the framework correctly identifies that external verification is needed (TND = 1.0) since this requires specific factual knowledge, and the Wikipedia API selection scores high on TSA (0.95) as an authoritative geographic source. The model's attempt to recall "France" from parametric memory without tool consultation would trigger a tool necessity violation (TND = 0.0), flagging potential hallucination risk.

Statistical databases provide $\text{cities}_{ranked}(\text{France})$ with the framework validating that city ranking requires external data sources rather than model estimation. Population data retrieval generates evidence set $\{e_3, e_4, e_5\}$ for each city, where the framework would penalize any attempt to generate approximate population figures from training data. The calculator tool for final summation demonstrates appropriate tool selection for deterministic computation, avoiding potential arithmetic errors from language model calculation.

The structured output captures the complete evaluation trace:


```
{
"query": "Total population of three largest
       cities...",
"decomposition": [
 {"step": "location",
  "result": "France",
  "tool": "WikipediaAPI",
  "tool_necessity": 1.0,
  "tool_selection": 0.95,
  "evidence": {...}},
 {"step": "cities",
  "result": ["Paris", "Marseille", "Lyon"],
  "tool": "StatDB",
  "tool_necessity": 1.0,
  "tool_selection": 0.92,
  "evidence": {...}},
 {"step": "population",
  "result": [2161000, 861635, 513275],
  "tool": "StatDB",
  "tool_necessity": 1.0,
  "tool_selection": 0.92,
  "evidence": {...}},
 {"step": "sum",
  "result": 3535910,
  "tool": "Calculator",
  "tool_necessity": 0.8,
  "tool_selection": 1.0}
 ],
"scores": {
"degree": 0.94,
"degree_components": {
  "truthfulness": 0.95,
  "faithfulness": 0.96,
  "tool_appropriateness": 0.91
},
"quantity": 1.0,
"stability": 0.92,
"hallucination_risk": 0.08
 }
}
```


9) Implementation Considerations: Deploying LENS requires careful consideration of computational and architectural factors. Resource management implements intelligent caching to minimize redundant computations:

$$\text{Cache}(q) = \begin{cases} \mathcal{R}_{cached} & \text{if } h(q) \in \mathcal{C} \wedge \text{TTL}(h(q)) > 0 \\ \text{Execute}(q) & \text{otherwise} \end{cases} \tag{32}$$

Session isolation for parallel decomposition requires memory allocation:

$$\text{Mem}_{required} = n_{parallel} \times (\text{Base}_{LLM} + \text{Context}_{max} + \text{Buffer}_{tool}) \tag{33}$$

Rate limiting for external APIs follows exponential backoff:

$$\text{Delay}(k) = \min(2^k \cdot \text{base\_delay}, \text{max\_delay}) \tag{34}$$

The framework provides adapters for common evaluation pipelines through standardized interfaces. Export formats include JSON for structured analysis, CSV for tabular processing, and specialized formats for benchmark integration. Performance optimization employs batching strategies:

$$\text{Throughput} = \frac{n_{queries}}{\max(\text{Time}_{LLM}, \text{Time}_{tools}, \text{Time}_{scoring})} \tag{35}$$

Integration with production systems leverages containerization and orchestration platforms. The modular architecture enables selective component deployment based on resource constraints and evaluation requirements.

F. Discussion and Limitations

While the LENS framework represents a significant advancement in hallucination evaluation, addressing many critical gaps in existing approaches, it is essential to acknowledge its limitations and discuss the broader implications for the field of AI evaluation [16], [24]. This section examines both the inherent constraints of our approach and the systemic challenges that remain in hallucination assessment, providing a balanced perspective on the framework's contributions and areas for future development.

1) Computational Complexity and Scalability Challenges: The hierarchical decomposition architecture that enables LENS's comprehensive evaluation introduces non-trivial computational overhead that must be carefully considered in deployment contexts [327], [341]. The tree-based query decomposition, while providing unprecedented depth of analysis, results in computational complexity that grows exponentially with query complexity in the worst case [47], [317].

a) Decomposition Overhead: The hierarchical decomposition process transforms each query Q into a tree structure with complexity $O(b^d)$ where b represents the average branching factor and d the decomposition depth. While our pruning strategies and verification gates reduce the practical complexity to approximately $O(b \cdot d \cdot \log n)$ for most queries, pathological cases requiring full tree exploration can still incur significant computational costs. This overhead becomes particularly pronounced when evaluating models on large-scale benchmarks where thousands of queries must be processed.

The decomposition overhead manifests in three primary dimensions [345]. First, the computational cost of tree construction itself requires sophisticated parsing and reasoning capabilities that may exceed the resources available in resource-constrained environments [351]. Second, the parallel evaluation of sub-queries, while improving accuracy, multiplies the number of model invocations required for each evaluation. Third, the aggregation and synthesis of results from multiple decomposition paths introduces additional processing requirements that scale with tree complexity.

b) Tool Integration Latency: The integration of external tools for verification, while essential for comprehensive evaluation, introduces latency that can significantly impact evaluation throughput [182]. Each tool invocation incurs network latency, processing time, and potential rate limiting constraints [320], [359]. For knowledge-intensive queries requiring multiple tool consultations, the cumulative latency can extend evaluation time from seconds to minutes per query. This latency becomes prohibitive when evaluating models on large-scale benchmarks or in real-time deployment scenarios where rapid assessment is required.

Moreover, the reliability and availability of external tools introduce additional variability in evaluation performance. Tool failures, API changes, or service disruptions can cascade through the evaluation pipeline, requiring robust fallback mechanisms and redundancy strategies that further increase system complexity. The framework must maintain multiple tool alternatives and implement sophisticated retry logic, adding layers of complexity to the evaluation infrastructure.

2) Knowledge Source Dependencies and Quality Assurance: LENS's adaptive benchmark construction relies fundamentally on the quality and completeness of provided knowledge sources, creating dependencies that can limit evaluation effectiveness in certain contexts.

a) Ground Truth Ambiguity: The definition of ground truth for hallucination evaluation remains inherently problematic in many domains [16], [72]. While factual queries about established scientific facts or historical events may have clear ground truth, many real-world applications involve subjective interpretation, evolving knowledge, or contextual nuance that resist binary classification [328], [336]. Legal interpretations vary across jurisdictions, medical recommendations evolve with new research, and cultural contexts shape the acceptability of generated content. LENS attempts to address this through multi-dimensional evaluation and confidence calibration, but the fundamental challenge of defining truth in complex domains persists.

The temporal dimension of truth presents particular challenges [349]. Facts that were considered accurate at the time of model training may become outdated, creating a moving target for evaluation [354], [374]. Scientific theories evolve, political landscapes shift, and social norms change, all affecting what constitutes a hallucination versus legitimate generation based on training data. While LENS's adaptive framework can incorporate updated knowledge sources, distinguishing between outdated training and genuine hallucination requires sophisticated temporal reasoning that remains an open challenge.

b) Knowledge Source Verification: The framework's reliance on external knowledge sources for verification introduces a critical dependency on the accuracy and completeness of these sources [436]. Errors, biases, or gaps in reference databases propagate through the evaluation pipeline, potentially misclassifying correct generations as hallucinations or vice versa [186], [334]. While LENS implements multi-source verification and confidence weighting, the fundamental challenge of establishing authoritative ground truth in domains with conflicting or incomplete information remains unresolved.

The curation and maintenance of high-quality knowledge sources requires significant ongoing effort. Organizations must continuously update their reference databases, validate information accuracy, and resolve conflicts between sources. This maintenance burden can be substantial, particularly for specialized domains requiring expert knowledge. The cost and complexity of maintaining comprehensive, accurate knowledge sources may limit LENS adoption in resource-constrained environments or rapidly evolving fields.

3) Evaluation Gaming and Adversarial Considerations: As with any evaluation framework, LENS faces the risk of models being optimized specifically to perform well on its metrics rather than genuinely reducing hallucination, a phenomenon known as Goodhart's Law [314], [320].

a) Metric Optimization Without True Improvement: The detailed specification of LENS's evaluation dimensions creates potential targets for optimization that may not correspond to genuine improvements in hallucination reduction [339]. Models could learn to generate responses that score well on Tool Necessity Detection by always requesting external verification, even when unnecessary, inflating TND scores without improving actual reliability [335]. Similarly, models might learn to produce conservative, hedged responses that score well on confidence calibration metrics while providing less useful information to users.

The hierarchical decomposition approach, while providing valuable structure, also creates predictable patterns that models could exploit. A model trained on LENS evaluation data might learn to generate responses that decompose cleanly into the expected tree structure, scoring well on structural metrics without necessarily improving factual accuracy. This risk is particularly acute given the increasing sophistication of model training techniques and the availability of evaluation datasets for training.

b) Adversarial Hallucination Patterns: Sophisticated adversarial actors could potentially design hallucination patterns specifically crafted to evade LENS's detection mechanisms [320]. By understanding the framework's decomposition strategies and verification procedures, adversaries could construct hallucinations that appear valid at each decomposition level but combine to form misleading or incorrect complete responses [340]. These compositional adversarial examples represent a significant challenge for any structured evaluation approach.

The framework's reliance on external tools also creates potential attack vectors. Adversaries could potentially manipulate or compromise reference sources, inject misleading information into verification databases, or exploit temporal gaps between knowledge source updates. While LENS implements multi-source verification and confidence weighting, coordinated attacks across multiple sources could still compromise evaluation integrity.

4) Generalization and Cross-Domain Applicability: While LENS provides adaptive benchmark construction capabilities, certain limitations in generalization across domains and tasks remain.

a) Domain-Specific Reasoning Patterns: Different domains exhibit unique reasoning patterns and evaluation requirements that may not be fully captured by LENS's general framework [116], [370]. Mathematical proofs require different verification strategies than medical diagnoses, which in turn differ from creative writing evaluation [42], [437]. While the framework's adaptability allows for domain-specific customization, the burden of specifying appropriate evaluation strategies for each domain falls on users, requiring significant expertise and effort.

The transfer of evaluation strategies across domains remains limited. Insights gained from evaluating hallucinations in one domain may not generalize to others due to fundamental differences in knowledge structure, reasoning requirements, and acceptable uncertainty levels. This limitation necessitates domain-specific configuration and tuning, reducing the framework's plug-and-play applicability across diverse use cases.

b) Cultural and Linguistic Variations: LENS's evaluation strategies, developed primarily in English-language contexts, may not fully account for cultural and linguistic variations in what constitutes hallucination [338], [355]. Concepts that are unambiguous in one culture may be highly contextual in another [333]. Linguistic structures in non-English languages may not map cleanly to the framework's decomposition strategies. These variations introduce evaluation biases that could disadvantage models trained on or evaluated in non-Western contexts.

The framework's tool integration capabilities are also limited by the availability of verification resources across languages and cultures. High-quality knowledge sources and verification tools are disproportionately available for English and major European languages, creating evaluation gaps for low-resource languages. This limitation perpetuates existing inequalities in AI development and deployment, potentially excluding important use cases and communities from comprehensive hallucination evaluation.

5) Human Factors and Interpretability Challenges: Despite LENS's emphasis on transparency and interpretability, significant challenges remain in making evaluation results accessible and actionable for diverse stakeholders.

a) Cognitive Load of Multi-Dimensional Results: The framework's multi-dimensional evaluation produces rich but complex results that can overwhelm users attempting to interpret and act upon them [61], [343]. Understanding the interplay between degree, quantity, stability, and risk dimensions requires sophisticated mental models that many stakeholders may lack [323]. While LENS provides visualization tools and summary metrics, the fundamental complexity of multi-dimensional evaluation creates cognitive barriers to adoption and effective use.

The hierarchical decomposition traces, while valuable for understanding evaluation reasoning, can be difficult to interpret without technical expertise. Non-technical stakeholders may struggle to understand why certain decomposition strategies were chosen or how sub-query results aggregate to final scores. This interpretability gap limits the framework's utility for broader organizational decision-making and public accountability.

b) Calibration of Human Judgment: LENS's evaluation results must ultimately be interpreted and acted upon by humans, introducing variability in how scores translate to deployment decisions [337]. Different stakeholders may have varying risk tolerances, leading to inconsistent interpretation of the same evaluation results [324]. A hallucination rate deemed acceptable by one organization might be considered prohibitive by another, even within the same domain.

The calibration of human judgment to LENS metrics requires significant experience and training. Users must develop intuition for what different score combinations mean in practice, how to weight dimensions for their specific use cases, and when to trust or question evaluation results. This learning curve presents a barrier to adoption and may lead to misinterpretation of results during the initial deployment phase.

6) Broader Implications for AI Evaluation: The development of LENS raises fundamental questions about the nature and future of AI evaluation that extend beyond hallucination assessment.

a) The Evaluation-Capability Arms Race: As evaluation frameworks become more sophisticated, model developers increasingly optimize for evaluation metrics, creating an arms race between evaluation and capability development [270]. This dynamic risks divorcing evaluation performance from real-world utility, as models become increasingly adept at gaming evaluation systems rather than genuinely improving [403]. LENS's comprehensive approach may temporarily raise the bar for such optimization, but the fundamental tension between evaluation and genuine capability remains.

The arms race also manifests in the increasing complexity of evaluation frameworks themselves. As models become more sophisticated, evaluation must become correspondingly complex, creating a spiral of increasing computational and methodological overhead. This complexity growth may eventually reach practical limits, necessitating fundamental rethinking of how we approach AI evaluation.

b) Standardization Versus Innovation: The adoption of comprehensive frameworks like LENS creates tension between the need for standardized evaluation to enable comparison and the desire for innovative evaluation approaches tailored to specific use cases [60], [390]. Standardization facilitates benchmarking and progress tracking but may constrain creative evaluation strategies that could provide novel insights. Organizations must balance the benefits of adopting established frameworks against the potential advantages of developing custom evaluation approaches.

This tension extends to the broader AI community's approach to evaluation. Should the field converge on a small number of comprehensive frameworks, or should it maintain a diverse ecosystem of evaluation approaches? LENS's adaptive architecture attempts to bridge this divide, but the fundamental tension between standardization and innovation persists.

7) Future Directions and Research Opportunities: The limitations identified above point to several promising directions for future research and development.

a) Automated Evaluation Design: Future work could explore automated methods for designing evaluation strategies tailored to specific domains and use cases [325], [440]. Machine learning techniques could potentially learn optimal decomposition strategies, tool selection policies, and aggregation methods from limited human feedback, reducing the expertise required for framework deployment [439]. Such automation could democratize access to sophisticated evaluation capabilities while maintaining rigor and reliability.

b) Federated Evaluation Networks: Addressing the knowledge source quality challenge could involve developing federated evaluation networks where organizations share verification resources and evaluation results while maintaining privacy and security [83]. Blockchain or other distributed ledger technologies could provide tamper-resistant audit trails for evaluation results, enhancing trust and accountability [346]. Such networks could pool resources to maintain high-quality knowledge sources while distributing the maintenance burden.

c) Continuous Evaluation and Monitoring: Moving beyond static evaluation to continuous monitoring of deployed models could provide early warning of emerging hallucination patterns [92], [318]. Online learning techniques could enable evaluation frameworks to adapt to changing model behavior and evolving knowledge landscapes [34], [313]. Integration with production systems could enable real-time hallucination detection and mitigation, shifting evaluation from a development-time activity to an operational capability.

8) Conclusion of Limitations: The limitations discussed in this section are not fundamental flaws but rather inherent challenges in the complex task of comprehensive hallucination evaluation. LENS represents a significant step forward in addressing many critical gaps in existing frameworks, but perfect evaluation remains an aspirational goal rather than an achievable target. The framework's limitations should be understood as defining the boundaries of current capability rather than diminishing its contributions.

By acknowledging these limitations transparently, we aim to foster realistic expectations about what automated evaluation can achieve and inspire continued research to address remaining challenges [319]. The path toward trustworthy AI requires not just better evaluation frameworks but also honest assessment of their capabilities and constraints [65], [331]. LENS provides a robust foundation for this journey, but the work of ensuring safe and reliable AI systems remains an ongoing collaborative effort requiring continued innovation, vigilance, and humility about the challenges that lie ahead.

G. Toward Trustworthy AI Through Comprehensive Hallucination Evaluation

The deployment of Large Language Models and Vision Language Models in critical applications demands evaluation frameworks that match the sophistication and stakes of these systems. Throughout this paper, we have traced the evolution of hallucination evaluation from simplistic accuracy measurements to the comprehensive, multidimensional assessment necessary for trustworthy AI deployment. The journey reveals not merely technical challenges but fundamental reconceptualization of how we understand, detect, and mitigate the generation of plausible yet false content that threatens to undermine AI reliability.

1) The Journey Through Complexity: Our exploration began with a sobering reality: hallucination rates ranging from 1.47% in clinical applications to 75% in specialized domains represent not statistical curiosities but existential threats to AI deployment in high stakes environments. These failures manifest not as obvious errors easily caught by casual inspection, but as sophisticated fabrications that exploit the probabilistic foundations of generative models. The medical diagnosis that appears clinically sound yet endangers patients, the legal citation that seems authoritative yet references non existent precedents, and the financial analysis that appears rigorous yet contains fabricated data all exemplify the insidious nature of hallucinations that current evaluation frameworks consistently fail to detect or prevent.

Section VI-B revealed the profound complexity underlying hallucination phenomena. The epistemological challenge of establishing ground truth in domains characterized by uncertainty, evolution, and legitimate disagreement exposes the naivety of binary truth classifications. Scientific understanding evolves, medical best practices shift with emerging evidence, and legal interpretations vary across jurisdictions and time. The computational complexity of comprehensive evaluation grows exponentially with query complexity, creating fundamental tensions between thoroughness and feasibility. The attribution challenge reveals that understanding why hallucinations occur proves as crucial as detecting them, yet current frameworks provide minimal insight into causal mechanisms. These challenges interconnect and compound, creating a problem space that resists simple solutions or incremental improvements.

Our systematic review of existing evaluation approaches in Section VI-C4 exposed a fragmented landscape where different frameworks capture isolated aspects of hallucination without achieving comprehensive assessment. Output based evaluation through benchmarks like TruthfulQA and FactCC excel at detection but provide no insight into reasoning processes or failure mechanisms. Process based approaches including StrategyQA and DiaHalu trace reasoning chains but lack the breadth necessary for comprehensive assessment across domains. Capability based evaluations probe specific skills but fail to capture the complex interactions that produce hallucinations in real deployments. This fragmentation forces practitioners to navigate incompatible frameworks, reconcile contradictory results, and still lack confidence in their assessments.

2) The Fundamental Gaps and Their Resolution: Section VI-D crystallized the fundamental inadequacies of current approaches into five critical gaps that prevent effective hallucination evaluation. The dimensional poverty reducing complex phenomena to single metrics ignores the multifaceted nature of hallucinations that vary in severity, scope, and impact. The artificial separation between horizontal breadth and vertical depth forces an unnecessary choice between understanding what errors occur versus why they occur. The metacognitive blind spot completely overlooks models' ability to recognize when external verification is necessary rather than generating from parametric memory. The adaptability crisis locks organizations into rigid benchmarks that cannot accommodate their specific knowledge domains or quality requirements. The transparency deficit provides scores without explanations, preventing actionable understanding or targeted improvement.

These gaps are not independent limitations but interconnected symptoms of a more fundamental problem: the absence of unified frameworks that synthesize multiple dimensions, modalities, and perspectives into coherent, actionable assessment. Current approaches treat hallucination as a monolithic phenomenon amenable to simple measurement, failing to recognize it as a complex, multidimensional challenge requiring sophisticated analytical frameworks. The consequence is evaluation that appears rigorous through extensive benchmarking yet fails to provide the understanding necessary for safe deployment or systematic improvement.

3) LENS: A Paradigm Shift in Evaluation: The LENS framework presented in Section VI-E represents not incremental improvement but fundamental reconceptualization of hallucination evaluation. Through hierarchical decomposition that transforms complex queries into verifiable atomic components, LENS exposes the reasoning processes that current black box evaluations obscure. This decomposition is not merely structural but functional, identifying precisely where in multi step reasoning hallucinations emerge and how errors propagate through inference chains.

The framework's four dimensional evaluation transcends binary detection to capture the full complexity of hallucination phenomena. The degree dimension extends beyond simple accuracy to encompass faithfulness to context and crucially, the appropriateness of tool usage through our novel Tool Necessity Detection and Tool Selection Accuracy metrics. The quantity dimension addresses hallucinations of omission that current frameworks ignore, recognizing that incomplete information can be as dangerous as false information. The stability dimension reveals whether errors are systematic vulnerabilities or stochastic failures, essential for understanding model reliability. The risk dimension quantifies uncertainty and confidence calibration, distinguishing between models that acknowledge limitations and those that hallucinate with unwarranted confidence.

Most significantly, LENS introduces metacognitive evaluation as a first class concern. By explicitly assessing whether models recognize when external verification is necessary (Tool Necessity Detection) and whether they select appropriate tools for the task (Tool Selection Accuracy), LENS addresses a fundamental source of hallucinations that current frameworks completely overlook. This paradigm shift from post hoc error detection to proactive risk assessment transforms evaluation from measurement to prevention, identifying models that will hallucinate because they lack the metacognitive awareness to seek help when needed.

The framework's adaptive benchmark construction enables organizations to create rigorous evaluations from their own information sources while maintaining standardized methodologies. This resolves the fundamental tension between customization and standardization, allowing domain specific assessment without sacrificing comparability. Healthcare providers can evaluate against their specific treatment protocols, legal firms against their jurisdictional requirements, and financial institutions against their regulatory frameworks, all while maintaining methodological rigor that enables meaningful comparison across implementations.

4) Transformative Impact on AI Development and Deployment: LENS fundamentally transforms how organizations approach AI evaluation and deployment. Rather than relying on generic benchmarks that poorly approximate their use cases, organizations can now construct evaluations that directly assess performance on their specific requirements while maintaining scientific rigor. This capability shifts evaluation from a development phase checkpoint to an ongoing operational capability that continuously validates model behavior against evolving requirements and knowledge.

For model developers, LENS provides unprecedented visibility into failure mechanisms that enable targeted improvements rather than blind architectural adjustments. The framework's hierarchical decomposition reveals whether hallucinations stem from knowledge gaps requiring expanded training data, reasoning failures necessitating architectural changes, or metacognitive limitations demanding new capabilities. This causal understanding transforms model development from trial and error exploration to systematic engineering based on empirical understanding of failure modes.

The transparency and interpretability built into LENS enable new forms of accountability and governance for AI systems. Regulatory bodies can establish evaluation requirements that go beyond simple accuracy thresholds to encompass the full spectrum of reliability considerations. Organizations can provide stakeholders with comprehensible explanations of model capabilities and limitations, building trust through transparency rather than demanding faith in opaque systems. Users can calibrate their reliance on AI outputs based on detailed understanding of when and how models might fail, enabling appropriate human oversight where necessary.

5) Acknowledging Limitations While Charting the Path Forward: Our discussion in Section VI-F provides honest assessment of LENS's limitations, recognizing that perfect evaluation remains aspirational rather than achievable. The computational overhead of hierarchical decomposition, while manageable for most applications, becomes significant for extremely complex queries requiring deep trees. The framework's effectiveness depends on the quality of knowledge sources used for verification, inheriting any biases or gaps in these references. The risk of evaluation gaming remains, as sophisticated actors might craft hallucinations specifically designed to evade LENS's detection mechanisms.

Yet these limitations define the boundaries of current capability rather than fundamental flaws in the approach. Each limitation points toward specific research directions that can extend and strengthen the framework. Automated evaluation design using machine learning could reduce the expertise required for framework deployment. Federated evaluation networks could pool verification resources while maintaining privacy and security. Continuous evaluation integrated with production systems could provide real time hallucination detection and mitigation.

The broader implications extend beyond hallucination evaluation to fundamental questions about AI assessment. As models become increasingly sophisticated, evaluation must correspondingly evolve, creating a dynamic that drives both capability development and assessment methodology. LENS represents one step in this evolution, demonstrating that comprehensive evaluation is possible while revealing the challenges that remain. The framework establishes a foundation for continued innovation while acknowledging that ensuring trustworthy AI requires ongoing vigilance and continuous advancement of evaluation capabilities.

6) A Call to Action for the AI Community: The development and deployment of trustworthy AI systems represents one of the defining challenges of our era. As these systems assume greater responsibilities in critical applications, the cost of hallucinations escalates from inconvenience to catastrophe. Medical decisions affecting patient lives, legal judgments determining justice, financial analyses guiding economic decisions, and educational content shaping future generations all depend on AI systems that must be reliable, transparent, and accountable.

LENS provides the technical foundation for this trustworthiness, but realizing its potential requires collective action from the AI community. Researchers must extend and refine the framework, addressing current limitations while exploring new evaluation dimensions. Practitioners must adopt comprehensive evaluation as a core requirement rather than optional enhancement, integrating frameworks like LENS into development and deployment pipelines. Organizations must invest in the infrastructure and expertise necessary for rigorous evaluation, recognizing that the cost of comprehensive assessment pales compared to the risk of deploying unreliable systems. Policymakers must establish evaluation standards that protect public interest while enabling innovation, balancing safety with progress.

The path toward trustworthy AI is neither simple nor certain, but it is necessary and achievable. Through frameworks like LENS that combine theoretical rigor with practical applicapability, we can transform AI systems from impressive but unreliable technologies into trustworthy partners suitable for deployment in applications where accuracy, reliability, and interpretability are non negotiable requirements. The future of AI depends not on the sophistication of models alone but on our ability to understand, evaluate, and ensure their reliability. LENS represents a significant step on this journey, demonstrating that comprehensive hallucination evaluation is not only necessary but possible.

As we stand at this inflection point in AI development, the choices we make about evaluation and reliability will determine whether these powerful technologies fulfill their promise of beneficial transformation or become sources of systemic risk. The LENS framework provides tools and methodologies for choosing the path of trustworthiness, but walking that path requires commitment from every stakeholder in the AI ecosystem. The work is challenging, the stakes are high, and the need is urgent. Through collaborative effort guided by frameworks like LENS, we can build an AI future characterized not by spectacular failures but by reliable, trustworthy systems that enhance human capability while respecting human values.

The journey from recognizing hallucinations as a critical challenge to developing comprehensive evaluation frameworks has revealed the depth and complexity of ensuring AI reliability. LENS represents not the end of this journey but a crucial waypoint, demonstrating what is possible while illuminating the path ahead. As we continue to push the boundaries of AI capability, we must equally advance our ability to evaluate and ensure the trustworthiness of these systems. The future depends not on choosing between capability and reliability but on achieving both through rigorous, comprehensive evaluation that matches the sophistication of the systems we seek to deploy.

VII. Framework

A. Foundational Concepts and Framework Philosophy

The UV-oriented framework for hallucination management represents a paradigm shift from treating hallucination merely as an error to be eliminated, toward regarding it as a continuous management challenge that must be dynamically calibrated according to task requirements, user preferences, and domain-specific constraints. Rather than applying a static, one-size-fits-all definition of hallucination, this framework emphasizes the need for adaptive tolerance specifications across multiple dimensions, such as factual accuracy, creative value, safety criticality, and temporal relevance. By doing so, it recognizes that hallucination lies on a spectrum, where its role can vary from harmful misinformation in sensitive domains to a useful driver of creativity in exploratory tasks.

A central principle of the framework is the continuous integration of human feedback, positioned not as an optional addition but as the core calibration mechanism. At every stage—sensing, decision-making, action, and evaluation—explicit feedback channels allow users and domain experts to refine system behavior in real time. This ensures that the framework evolves alongside changing user needs and application contexts, aligning system performance with trustworthy and human-centered AI development.

The theoretical foundation of this approach draws upon information theory and uncertainty quantification. Hallucination can be conceptualized as controlled noise in the information channel between knowledge sources and generated outputs, enabling principled trade-offs between fidelity and diversity. Furthermore, by incorporating layered uncertainty estimates (token-level, sequence-level, and semantic-level), the framework provides fine-grained control over when to trigger verification, request human input, or abstain from answering. Together, these foundational concepts establish the philosophical basis for a closed-loop, adaptive, and trustworthy AI framework.

1) Paradigm Shift from Detection to Management: Traditional approaches to hallucination in AI have largely framed the issue as a binary detection problem: an output is either factual or hallucinatory. Such a perspective, however, is overly restrictive and fails to account for the nuanced roles that hallucinations can play across different contexts. The UV-oriented framework introduces a paradigm shift by redefining hallucination not as a defect to be universally eliminated, but as a phenomenon to be managed along a spectrum of tolerance.

Within this spectrum, hallucinations are interpreted relative to their task context. In high-stakes environments—such as medical diagnosis, legal reasoning, or financial decision-making—hallucinations are strictly detrimental and must be minimized through rigorous verification pipelines. In contrast, within open-ended or creative domains—such as storytelling, brainstorming, or hypothesis generation—hallucinations may serve as valuable sources of novelty and inspiration. This reframing allows the system to dynamically adjust its behavior, striking a balance between factual accuracy and creative exploration.

By adopting this management-oriented perspective, the framework moves beyond the limitations of static evaluation metrics and rigid rules. Instead, it emphasizes dynamic calibration, where hallucination is continuously monitored, evaluated, and modulated in real-time according to user needs, domain constraints, and evolving context. This paradigm shift provides a more flexible and human-centered foundation for building trustworthy AI systems.

2) Task-Driven Definition Mechanism: One of the key innovations of the UV-oriented framework is its dynamic, task-driven definition of hallucination. Traditional approaches often impose a uniform definition across all scenarios, which leads to significant failures when applied to diverse applications. For example, a static criterion that flags any deviation from factual knowledge as hallucination may be appropriate in medical contexts, but it becomes counterproductive in creative writing or ideation tasks. To overcome this limitation, the framework implements a task-sensitive mechanism that adapts the definition of hallucination according to the specific requirements of the task, user, and domain.

The system operates by first classifying the task across multiple dimensions, such as cognitive complexity (factual recall, analytical reasoning, or creative generation), domain criticality (general-purpose versus high-stakes domains), and temporal scope (historical, current, or predictive). Based on this taxonomy, the framework establishes context-aware criteria for hallucination. For instance, a factual recall query about historical events activates strict verification protocols, whereas a brainstorming prompt for novel product ideas permits controlled deviations that may foster innovation.

Furthermore, the framework introduces a *multi-dimensional tolerance space* to precisely calibrate hallucination acceptance. Instead of a binary pass/fail system, tolerance is defined along independent axes: factual accuracy, creative value, safety criticality, and temporal relevance. This enables nuanced configurations, such as enforcing high factual accuracy while allowing moderate creativity, or relaxing accuracy requirements while imposing strict safety constraints. These dimensions can be dynamically adjusted by user input, inferred preferences, or domain-specific conventions, ensuring that hallucination management remains both adaptive and aligned with real-world needs.

By adopting this task-driven definition mechanism, the UV-oriented framework provides a flexible yet principled approach that balances rigor with adaptability, enabling trustworthy AI systems to function effectively across heterogeneous application contexts.

3) Human-in-the-Loop as Core Principle: A defining characteristic of the UV-oriented framework is the central role of human interaction. Rather than treating human feedback as an auxiliary component, the framework positions it as the primary calibration mechanism that guides the system's adaptive behavior across all stages—sensing, decision making, action, and evaluation. This ensures that hallucination management remains grounded in user expectations and domain-specific expertise, thereby enhancing both trustworthiness and practical utility.

The framework incorporates explicit feedback touchpoints, where users can provide corrections, preference signals, or quality assessments in real time. These feedback signals are not merely stored but actively integrated through online learning mechanisms that adjust system parameters dynamically. For instance, if users consistently flag a particular type of error, the system re-weights its tolerance and verification strategies to reduce the recurrence of similar mistakes in subsequent interactions.

In addition, the framework adopts an *expertise-weighted feedback architecture* to ensure that domain knowledge is effectively leveraged. Feedback from medical professionals, for example, carries greater weight in healthcare-related tasks, while input from creative professionals is emphasized in artistic or design-oriented domains. This expertise-aware approach allows the system to prioritize the most qualified signals without disregarding general user input, creating a balanced calibration strategy.

By embedding human-in-the-loop mechanisms as a foundational principle, the UV-oriented framework not only improves robustness against hallucinations but also fosters a collaborative paradigm where AI continuously evolves through alignment with human values, expertise, and context-specific requirements.

4) Theoretical Foundation: The UV-oriented framework is grounded in solid theoretical perspectives that enable principled reasoning about hallucination and its management. Two key pillars of this foundation are information theory and uncertainty quantification, which together provide both conceptual clarity and practical mechanisms for controlling hallucination.

From an *information-theoretic perspective*, hallucination can be interpreted as controlled noise within the communication channel that connects training data, knowledge sources, and generated outputs. By quantifying mutual information between inputs and outputs, the framework enables measurement of how faithfully the generated content reflects underlying knowledge. This view also facilitates principled trade-offs between information fidelity and generative diversity: in safety-critical tasks, the system can maximize fidelity by minimizing noise, while in creative domains, it can deliberately allow higher levels of controlled divergence to promote innovation.

Complementing this is an explicit *uncertainty quantification framework* that operates across multiple levels of generation. Token-level uncertainty captures instability in word predictions, sequence-level uncertainty reflects the reliability of entire statements, and semantic-level uncertainty addresses broader conceptual claims. These layered uncertainty measures allow the system to determine when to proceed with generation, when to invoke verification pipelines, and when to abstain or defer to human review. By making uncertainty estimates transparent to users, the framework not only enhances trust but also provides actionable signals for adaptive decision making.

Together, these theoretical foundations ensure that hallucination management is not ad hoc but is instead rooted in rigorous principles. By combining information-theoretic analysis with fine-grained uncertainty quantification, the UV-oriented framework establishes a systematic and adaptable basis for achieving trustworthy and human-aligned AI systems.

5) Feedback Framework Design for UV Smart Hallucination Control: To provide a high-level view of the proposed UV-oriented framework, Figure 12 illustrates the overall feedback-based design for smart hallucination control. The framework is structured around four core functional modules: **Decision Making**, **Action**, **Sensing & Monitoring**, and **Evaluation**. These modules are organized into a continuous loop, forming a closed feedback system that balances user requirements, system capabilities, and dynamic adaptation.

External factors, including *Design principles, Policy and Regulation, Objectives, Requirements, Human Preferences, and Past Feedback/Experience*, are first integrated into the **Decision Making** layer. This layer generates strategies and mitigation choices, which are then operationalized through the **Action** module. The output of this stage is directed towards the **Target Hybrid Human–AI System**, which integrates *LLM Structure, Users, and External Knowledge*, each of which is bidirectionally interconnected to ensure co-adaptation and collaborative reasoning.

The **Sensing & Monitoring** module observes signals from both the action process and the hybrid human–AI system, capturing task dynamics, contextual cues, and performance indicators. These signals, together with criteria from the Action layer, are consolidated in the **Evaluation** module. Evaluation outputs not only inform the end user by providing trustworthiness assessments and uncertainty estimates, but also feed back into earlier stages of the framework. This includes updating objectives and human preferences, as well as maintaining ongoing communication with the Decision Making process, thereby creating a self-improving and adaptive hallucination control loop.

B. Sensing Layer – Requirement Capture & Risk Profiling

The sensing layer serves as the perceptual and analytical foundation of the UV-oriented framework. Its primary function is to capture user requirements, interpret task contexts, and establish the operational parameters for hallucination management before any generation occurs. Unlike traditional input-handling mechanisms that merely process surface-level queries, this layer integrates both explicit user instructions and implicit contextual signals, thereby ensuring that downstream processes are aligned with user intent and domain-specific constraints.

At its core, the sensing layer performs three essential roles. First, it provides a comprehensive mechanism for prompt acquisition and requirement negotiation, enabling users to directly specify their hallucination tolerance while also allowing the system to infer preferences through dialogue, history, and context cues. Second, it implements advanced prompt and task understanding capabilities that restructure, disambiguate, and classify inputs across multiple dimensions such as cognitive complexity, domain specificity, and temporal scope. Third, it conducts proactive risk profiling by assessing the potential difficulty, impact, and uncertainty associated with each query, thereby determining the appropriate level of verification and human involvement.

By combining requirement capture with risk-aware profiling, the sensing layer establishes a strong foundation for adaptive and trustworthy hallucination management. It ensures that subsequent decision-making is not only data-driven but also contextually informed, balancing user needs, safety considerations, and operational efficiency from the very beginning of the interaction.

1) User Input & Dynamic Requirement Capture: A central function of the sensing layer is to accurately capture user input and translate it into operational requirements for hallucination management. The UV-oriented framework goes beyond simply recording the literal prompt by incorporating contextual metadata such as interaction history, domain conventions, and implicit preference signals. This allows the system to construct a richer understanding of user intent and to tailor hallucination tolerance accordingly.

Users are provided with multiple intuitive mechanisms to specify their tolerance levels. These include lightweight interfaces such as slider-based controls for quick adjustments, detailed configuration forms for fine-grained preference setting, and natural language descriptions that can be automatically parsed into tolerance parameters. Over time, the system learns user-specific preferences through accumulated interaction data, enabling it to pre-populate tolerance defaults while still allowing explicit overrides. For users who do not specify tolerance directly, the framework employs an intelligent inference engine that draws on task type, user profile, and domain-specific norms to establish appropriate defaults.

To address ambiguity in requirements, the system engages in real-time negotiation with the user. When multiple plausible interpretations of a prompt are identified, it proactively poses clarification questions ranked by information gain, thereby minimizing user burden while maximizing requirement precision. This negotiation process ensures that user intent is explicitly validated before task execution, reducing the likelihood of misalignment.

Finally, the framework integrates historical context into requirement capture. Past queries, feedback patterns, and correction histories are encoded into evolving user profiles using collaborative filtering and temporal decay mechanisms. This allows the system to anticipate likely preferences, detect deviations from established patterns, and trigger additional verification when anomalies arise. By combining explicit input, interactive negotiation, and historical learning, the UV-oriented framework ensures that requirement capture remains both dynamic and context-sensitive, forming a robust foundation for trustworthy AI behavior.

2) Advanced Prompt & Task Understanding: Beyond requirement capture, the sensing layer incorporates advanced mechanisms for interpreting and restructuring user prompts to ensure precise alignment with task context. Rather than treating prompts as static text inputs, the UV-oriented framework employs a multi-stage optimization process that enhances clarity, resolves ambiguities, and aligns with the intended application domain.

The prompt optimization engine operates through layered transformations. At the syntactic level, it restructures user input to eliminate ambiguity and enforce grammatical coherence. At the semantic level, it expands implicit requirements by leveraging knowledge graphs and contextual embeddings to surface relevant background information. At the pragmatic level, it adjusts the prompt to align with domain conventions, user profiles, and task-specific constraints. These transformations are designed to be reversible and explainable, allowing users to inspect and verify how their original query has been refined.

To complement optimization, the framework implements a hierarchical task classification system that categorizes inputs across multiple dimensions. These include cognitive complexity (e.g., factual recall, analytical reasoning, or creative generation), domain specificity (general-purpose queries versus expert-level domains), temporal scope (historical, current, or

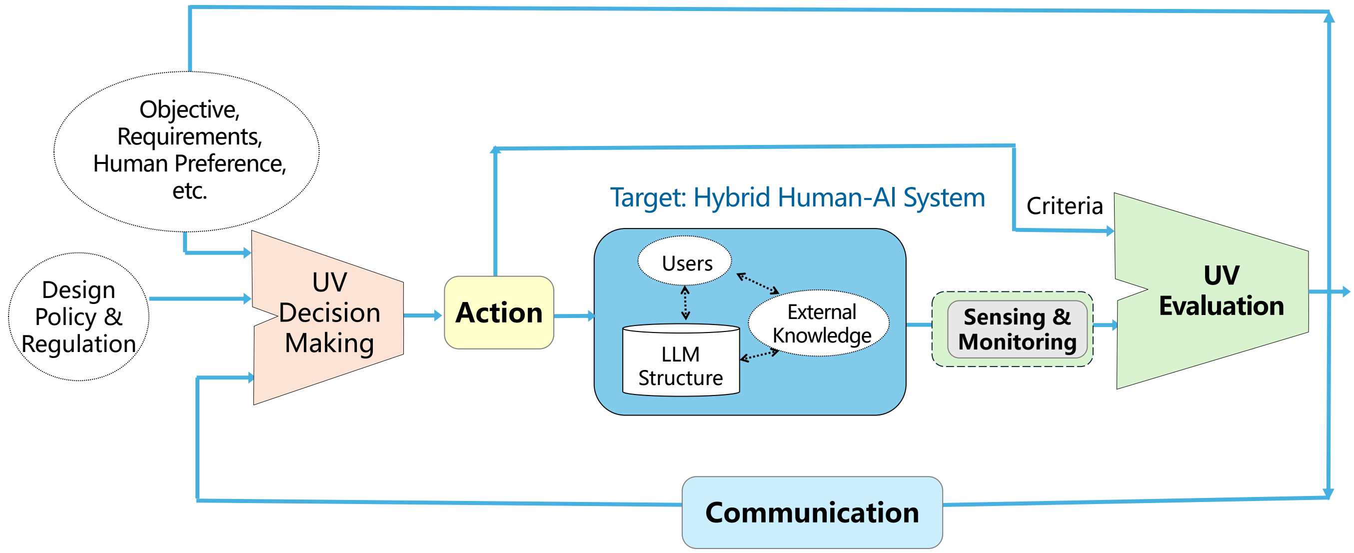

Fig. 12. Feedback Framework Design for UV Smart Hallucination Control

predictive tasks), and output modality (text, structured data, code, or multimodal responses). By classifying tasks along these axes, the system ensures that hallucination management strategies are tailored to the nature of the query.

In cases where user tolerance has not been explicitly defined, the framework relies on an intelligent tolerance estimation algorithm. This algorithm integrates multiple signals: task criticality scores derived from domain knowledge, user expertise levels inferred from interaction history, contextual risk assessments based on potential harm, and collective intelligence from similar user patterns. Rather than producing a single static estimate, the system generates confidence intervals for tolerance, triggering human clarification or enhanced verification whenever uncertainty exceeds safe thresholds.

Finally, advanced semantic disambiguation mechanisms are employed to resolve ambiguous or multi-interpretation prompts. Using contextual embeddings, attention-based analysis, and commonsense reasoning from large-scale knowledge bases, the system identifies likely interpretations and, where necessary, generates clarifying questions to minimize uncertainty. This proactive disambiguation ensures that task understanding is both accurate and context-sensitive, enabling downstream components to operate with greater reliability.

Through this combination of optimization, classification, tolerance estimation, and disambiguation, the UV-oriented framework establishes a robust foundation for aligning user prompts with precise task definitions, reducing misinterpretation risks and strengthening the basis for trustworthy AI outputs.

3) Proactive Evaluation and Risk Assessment: A distinguishing feature of the sensing layer is its proactive capability to evaluate incoming tasks and profile potential risks before generation begins. Rather than relying solely on post-hoc detection, the UV-oriented framework emphasizes anticipatory assessment to determine the appropriate level of verification, resource allocation, and human involvement from the outset.

The proactive evaluation process begins with *difficulty profiling*, in which the system estimates the inherent complexity of a query. Factors considered include the novelty of requested knowledge (e.g., emerging events versus well-documented facts), the degree of reasoning required (factual recall versus multi-step deduction), and the sensitivity of the task domain (general knowledge versus high-stakes areas such as medicine or law). These indicators are aggregated into a risk score that informs downstream decision-making.

Complementing this, the system applies *uncertainty-aware task forecasting*. Using semantic entropy measures, calibration models, and reference to benchmark datasets such as TruthfulQA, FEVER, REALSum, and HaluEval, the framework predicts the likelihood of hallucination under different conditions. When uncertainty is high or when the task falls into known high-risk categories, the framework escalates verification protocols and may preemptively request clarification from the user or recommend additional safeguards.

Another key dimension of proactive assessment is *impact profiling*. Here, the framework considers the potential consequences of hallucination for the user and domain. For example, minor factual deviations in an exploratory brainstorming session may be acceptable, whereas even small inaccuracies in financial or legal advice could lead to serious harm. This impact-aware perspective ensures that risk assessment is not only technical but also ethically grounded.

The final stage of this module is *stakeholder-aware risk adjustment*. Recognizing that different users bring different levels of expertise and tolerance, the framework weights risk assessments according to user profiles and contextual factors. Expert users may accept higher system autonomy, while novice users receive stricter safeguards and explanations. This personalized calibration allows the framework to adaptively balance efficiency with safety.

By integrating difficulty profiling, uncertainty forecasting, impact analysis, and stakeholder-aware adjustment, the proactive risk assessment module equips the UV-oriented framework with a forward-looking capability. It ensures that potential hallucination risks are identified early, enabling dynamic allocation of verification resources and the design of trust-preserving interaction strategies before any output is produced.

C. Decision Making Layer – Strategic Orchestration

The decision making layer serves as the strategic core of the UV-oriented framework, where sensed inputs, contextual constraints, and user requirements are synthesized into actionable plans for hallucination management. While the sensing layer focuses on capturing and profiling user intent, the decision making layer determines *how* the system should respond—selecting mitigation strategies, orchestrating knowledge and model resources, and calibrating outputs according to tolerance thresholds. This layer thus transforms raw input signals into structured strategies that guide the downstream action layer.

At its foundation, the decision making process integrates multi-factor tolerance aggregation, combining user-specified preferences, task-specific risk assessments, and contextual constraints such as hardware resources, latency requirements, and domain conventions. Based on this aggregated profile, the framework dynamically selects from a repertoire of mitigation strategies stored in a continuously updated method database. These strategies are ranked not only by their factual effectiveness but also by cost, computational efficiency, and applicability to the specific task context.

In addition, the decision making layer orchestrates the interplay between structured knowledge bases and heterogeneous model repositories. By leveraging hierarchical knowledge databases alongside specialized and general-purpose models, the system ensures both factual coverage and domain adaptability. The orchestration is adaptive and multi-objective: it balances accuracy against efficiency, creative freedom against safety, and system autonomy against user oversight. This ensures that hallucination management is not handled through rigid rules but through dynamic optimization tailored to each interaction.

Overall, the decision making layer transforms requirement signals into strategic action plans, functioning as the "executive brain" of the framework. By coordinating tolerance aggregation, method selection, and knowledge-model integration, it ensures that every decision aligns with both user expectations and system-level principles of trustworthy AI.

1) Multi-Factor Tolerance Aggregation: At the core of the decision making layer lies the process of *multi-factor tolerance aggregation*, through which the framework computes a final hallucination tolerance level that reflects both user intent and contextual constraints. Unlike traditional approaches that rely on a fixed threshold or a single-source definition of hallucination tolerance, the UV-oriented framework integrates multiple heterogeneous signals into a unified decision profile.

The aggregation process begins with *explicit user preferences*. Users may directly specify tolerance levels through sliders, configuration settings, or natural language instructions. These explicit signals are treated as the primary input, but they are augmented by *implicit inferences* derived from task type, domain context, and historical interaction patterns. For instance, a prompt concerning medical treatment will automatically trigger stricter tolerance constraints, while a brainstorming prompt is interpreted with broader acceptance ranges.

Next, the system incorporates *task-specific risk assessments* generated by the sensing layer. These include difficulty profiling (e.g., reasoning depth, novelty of knowledge), uncertainty forecasting (e.g., semantic entropy scores), and impact profiling (e.g., potential harm in high-stakes domains). Each of these factors is mapped onto a tolerance adjustment axis, ensuring that tolerance is not a static boundary but a context-sensitive measure.

To integrate diverse signals, the framework employs a weighted aggregation model that combines explicit preferences, inferred signals, and task-specific risk scores. Weights can be dynamically adjusted using Bayesian optimization or reinforcement learning approaches, enabling the system to learn optimal aggregation policies over time. This adaptive mechanism ensures that the tolerance calculation evolves with user behavior, domain shifts, and system feedback.

Finally, contextual constraints such as available hardware resources, latency budgets, and cost limitations are factored into the aggregation process. For example, in resource-constrained settings, the framework may adopt more approximate verification strategies while compensating with stricter uncertainty thresholds. Conversely, in critical decision-making environments, it may allocate greater computational resources to ensure maximal factual reliability.

Through this multi-factor aggregation process, the framework derives a final hallucination tolerance profile that balances explicit user intent, task-driven requirements, and operational constraints. This tolerance profile serves as the strategic anchor for all subsequent decisions, guiding method selection, model orchestration, and verification policies within the framework.

2) Method Selection Architecture: Building upon the aggregated tolerance profile, the UV-oriented framework employs a *method selection architecture* that determines which hallucination mitigation strategies should be applied in a given context. Instead of relying on a single universal technique, the framework dynamically orchestrates a repertoire of methods, drawing from a continuously updated database that captures their performance, cost, and suitability across different domains and tasks.

The *hallucination method database* functions as a central knowledge hub that stores structured metadata for each mitigation approach. Entries include quantitative parameters such as computational cost, latency impact, and uncertainty reduction rate, as well as qualitative attributes like potential side effects, domain-specific applicability, and user experience implications. Importantly, this database evolves through continuous feedback: outcomes from prior interactions are logged, analyzed, and integrated to refine the profiles of each method. As a result, the selection process is not static but improves adaptively with time.

The architecture supports a stratified selection protocol aligned with tolerance levels.

- **Zero-Tolerance (Critical Domains):** In safety-critical applications such as medicine, finance, or law, the framework enforces a strict chain of operations: retrieval-augmented generation (RAG) $\rightarrow$ multi-step verification $\rightarrow$ consensus validation (e.g., through multiple models or expert review). This ensures that hallucinations are minimized to the greatest possible extent.
- **Low-Tolerance (Professional/Analytical Domains):** For tasks such as technical Q&A or decision support, the system activates selective verification combined with self-reflection loops and confidence-based filtering. Dynamic sampling adjustments (temperature, top-k, top-p) are also employed to regulate output reliability.
- **High-Tolerance (Creative/Exploratory Domains):** In open-ended or creative tasks, the framework permits controlled hallucinations. Outputs are accompanied by confidence scores or disclaimers, ensuring transparency while allowing the model to generate imaginative or speculative content.

Beyond tolerance-based stratification, the architecture integrates auxiliary mechanisms to enhance robustness. These include *confidence-based filtering*, where outputs below a certain reliability threshold are blocked or flagged; *rejection and abstention protocols*, enabling the system to return "I don't know" responses; and *tool invocation pipelines*, which dynamically call external resources such as search engines, calculators, or plugins when internal knowledge is insufficient.

Overall, the method selection architecture ensures that hallucination mitigation is not approached through rigid prescriptions but through adaptive orchestration. By leveraging a structured method database, stratified tolerance protocols, and auxiliary control mechanisms, the framework provides a flexible yet principled strategy for choosing the right tool at the right time, thereby strengthening both the reliability and trustworthiness of AI outputs.

3) Knowledge & Model Integration: A central element of the decision making layer is the orchestration of knowledge resources and model repositories to ensure that outputs are both contextually aligned and reliable. Rather than treating knowledge and models as independent silos, the UV-oriented framework establishes a cooperative integration pipeline, where each component dynamically supports the other in service of hallucination management.

The knowledge infrastructure is designed as a *hierarchical knowledge base* organized into multiple layers. The *high-confidence layer* contains validated, static knowledge sources such as curated ontologies, peer-reviewed datasets, and trusted databases. The *domain-specialized layer* stores structured knowledge tailored to specific professional areas (e.g., medical guidelines, financial regulations), enabling precision in high-stakes contexts. Finally, the *evolving layer* incorporates continuously updated information streams from retrieval pipelines, APIs, or community inputs, providing adaptability to emerging knowledge while clearly labeling its epistemic status. This layered design ensures both stability and responsiveness, balancing factual reliability with adaptability to new data.

Complementing the knowledge base, the framework maintains a *model repository* composed of heterogeneous models that vary by architecture, size, training data, and domain specialization. These include general-purpose foundation models, lightweight LoRA-adapted variants optimized for specific domains, and fine-tuned expert models trained on specialized datasets. The repository is continuously updated, with performance metrics such as accuracy, latency, robustness, and hallucination frequency logged and stored for reference.

The integration process is governed by an *adaptive selection algorithm*, which matches tasks with the most appropriate knowledge sources and models based on the aggregated tolerance profile. For example, a medical diagnostic query may trigger the selection of a domain-specialized knowledge layer combined with an expert-tuned model, while a creative ideation task may leverage the evolving knowledge layer and a general-purpose model configured with higher generative diversity. The system applies multi-objective optimization, balancing competing goals such as factual accuracy, response time, cost efficiency, and creative flexibility.

In addition, the framework enables *knowledge-model co-optimization*. Knowledge retrieved from external sources is dynamically injected into model prompts or adapters, while model outputs are cross-validated against structured knowledge graphs or retrieval-augmented datasets. Feedback from evaluation loops is also routed to both repositories: hallucination instances inform model fine-tuning, while knowledge gaps detected during generation update the evolving knowledge base. This bidirectional flow ensures that both knowledge and models improve collaboratively over time.

By integrating hierarchical knowledge bases with heterogeneous model repositories, and by orchestrating them through adaptive, feedback-driven selection algorithms, the UV-oriented framework provides a robust mechanism for ensuring trustworthy outputs. This integration not only reduces hallucination risks but also enhances adaptability across domains, thereby reinforcing the system's ability to operate reliably under diverse and dynamic conditions.

D. Action Layer – Execution & Controlled Generation

The Action Layer represents the execution engine of the UV-oriented framework, where strategic decisions derived from the Decision Making Layer are transformed into concrete outputs. This stage operationalizes hallucination management through a combination of generation, verification, adjustment, and safeguard mechanisms. Unlike prior layers that focus on sensing requirements or selecting appropriate strategies, the Action Layer implements these strategies in practice, balancing reliability with flexibility.

Specifically, the Action Layer orchestrates model inference, real-time parameter adjustment, self-reflection and revision protocols, and multi-level uncertainty estimation. It also integrates reliability safeguards, including graduated refusal and abstention mechanisms, multi-stage output filtering, and real-time safety monitoring. Furthermore, this layer incorporates advanced verification protocols—such as multi-source RAG-based checking, consensus-driven multi-model validation, and graph-based knowledge alignment—to ensure factual consistency and reduce the risk of misinformation.

Importantly, the Action Layer does not merely suppress hallucinations but also redefines their role in creative contexts. Through controlled divergence, hallucination can be leveraged as a feature for innovation, where speculative outputs are clearly marked and evaluated on dimensions such as novelty and coherence rather than factual accuracy. This dual orientation—strict control in critical domains and permissive exploration in creative tasks—ensures that the framework remains both trustworthy and versatile across diverse applications.

1) Model Execution & Dynamic Adjustment: The Action Layer begins with orchestrated model execution, where selected strategies from the Decision Making Layer are translated into concrete inference processes. Rather than relying on a single model pathway, the framework employs a multi-model orchestration system that can operate in parallel or sequential pipelines depending on task criticality. For high-stakes applications, multiple heterogeneous models may be executed simultaneously, enabling cross-validation and reducing systematic biases. Each model output is annotated with metadata—such as generation parameters, confidence scores, and potential hallucination indicators—providing transparency for downstream verification and evaluation.

Dynamic adjustment mechanisms are integrated directly into the generation loop. A feedback control system monitors semantic coherence and factual consistency in real time, adjusting sampling parameters such as temperature, top-k, and top-p to minimize hallucination risk. Multi-armed bandit algorithms further optimize these parameters by learning domain- and task-specific configurations that balance diversity with reliability. Repetition penalties are adaptively tuned to prevent both redundant content and the over-suppression of critical factual details.

In addition, the framework incorporates multi-stage self-reflection and revision protocols. Initial outputs are followed by self-critique, where the model identifies unsupported claims or areas of uncertainty. These flagged segments undergo targeted regeneration or retrieval augmentation before a final cross-check is performed against the original query. Through iterative cycles of reflection, correction, and validation, the model progressively enhances output quality. Human feedback at this stage is particularly valuable, as it informs the system about when its self-revisions are effective and when external intervention is necessary.

Together, these mechanisms transform model execution from a static, one-pass generation process into an adaptive, continuously optimized pipeline that can dynamically control hallucination levels while maintaining efficiency and responsiveness.

2) Reliability & Safeguards: To ensure trustworthy outputs, the Action Layer incorporates a comprehensive set of reliability mechanisms and safeguard protocols that operate alongside model execution. These mechanisms are designed to prevent harmful or misleading outputs while maintaining flexibility across different application domains.

First, the framework integrates a multi-layered uncertainty quantification system that captures epistemic and aleatoric uncertainty. Epistemic uncertainty highlights limitations in the model's knowledge, identifying areas where training data is sparse or inconsistent, while aleatoric uncertainty reflects inherent ambiguity in the user query or task context. By combining techniques such as Monte Carlo dropout, deep ensembles, and variational inference, the system provides robust reliability estimates. These are communicated to users through intuitive indicators—ranging from confidence scores to visual uncertainty bars—allowing transparent assessment of output trustworthiness.

Second, the framework implements graduated refusal and abstention mechanisms. Rather than binary acceptance or rejection, the system provides nuanced responses based on confidence levels. Partial answers are provided when some components of the query achieve high confidence, bounded outputs present ranges or intervals instead of exact values, and qualified responses explicitly state underlying assumptions or limitations. In cases of insufficient reliability, the system may invoke explanatory abstention, clarifying why a complete answer cannot be delivered. This graduated strategy builds user trust through transparency and avoids overconfident but misleading outputs.

Third, a multi-stage output management and filtering system ensures that generated content meets safety and quality requirements before reaching the user. Rule-based filters catch obvious hallucinations or policy violations, while machine-learned classifiers identify more subtle issues such as misleading implications or contextual irrelevance. A semantic consistency checker further validates outputs against knowledge bases and prior system responses, detecting contradictions or instability. Multiple output versions can be retained at different filtering levels, providing flexibility for expert users who require access to less-filtered content for research or creative purposes.

Finally, the Action Layer incorporates real-time safety monitoring and intervention. This system tracks semantic drift, pattern matches against known harmful templates, and anomaly detection signals. When risk indicators exceed predefined thresholds, the system can intervene through graduated actions—such as gentle redirection, parameter adjustment, or a full stop with human review. Emergency stop mechanisms allow human supervisors to immediately halt unsafe generation, ensuring that human oversight remains central in high-risk contexts.

Together, these reliability safeguards establish a robust trust boundary for AI generation, mitigating hallucination risks while preserving adaptability for diverse application scenarios.

3) Verification Protocols: Verification constitutes a central safeguard within the Action Layer, ensuring that generated content is systematically validated against trusted knowledge sources and independent reasoning mechanisms before reaching the user. Unlike static filtering, the verification process in the UV-oriented framework is dynamic, multi-sourced, and context-aware, adapting its rigor according to task sensitivity and hallucination tolerance levels.

First, the framework incorporates a multi-source Retrieval-Augmented Generation (RAG) verification system. Generated claims are extracted and classified into verifiable facts, opinions, or creative elements. Factual claims are validated against a hierarchical knowledge structure: (i) primary sources such as peer-reviewed publications or official documents, (ii) secondary sources such as reputable news outlets or encyclopedias, and (iii) tertiary sources such as community-validated repositories. Each source is assigned a reliability score based on domain expertise, provenance, and update recency. The final verification decision is derived through weighted voting, where higher-confidence sources exert greater influence. Discrepancies trigger deeper inspection, distinguishing between hallucinations, outdated information, and legitimate knowledge gaps.

Second, consensus-based multi-model validation strengthens reliability through redundancy and diversity. Multiple independent models—differing in architecture, training data, or reasoning methods—independently generate responses to the same query. Their outputs are compared at a semantic level, and a consensus mechanism inspired by Byzantine fault tolerance aggregates results. Critical claims may require supermajority agreement, while routine outputs can be validated with simple majority. Voting weights are dynamically calibrated based on each model's historical accuracy in the relevant domain. Human arbitration is invoked when disagreement patterns suggest deeper uncertainty, providing ground truth for continuous refinement of consensus mechanisms.

Third, the framework integrates graph-based knowledge validation, enabling logical consistency checking beyond surface-level fact verification. Knowledge is represented as a directed graph, where nodes correspond to entities or concepts and edges encode relationships. The system performs multi-hop reasoning validation, temporal consistency checks, and contradiction detection against this graph structure. For example, historical queries are validated not only against direct facts but also against implied relationships and time-bound constraints, preventing anachronistic hallucinations. Community-contributed relationship rules allow domain experts to refine and extend graph-based validation over time.

Finally, verification results are communicated to end users through a confidence-calibrated reporting mechanism. Outputs include multi-dimensional confidence scores that capture factual accuracy, source reliability, and consensus strength. Intuitive visualization methods—such as uncertainty bars, reliability badges, or color-coded highlights—convey trustworthiness at a glance, while drill-down options allow expert users to inspect supporting evidence, voting patterns, and specific uncertainty points. Natural-language explanations clarify why certain responses are flagged as lower-confidence, enabling transparent and actionable trust assessment.

Through this multi-layered verification architecture, the framework establishes a robust defense against hallucination, ensuring that outputs are not only factually grounded but also contextually aligned and transparently communicated to users.

4) Creative Utilization of Hallucination: While the primary goal of the Action Layer is to minimize harmful hallucinations, the framework also recognizes that hallucination can serve as a valuable feature in creative contexts. Instead of treating all deviations from factual accuracy as failures, this module distinguishes between detrimental misinformation and beneficial imaginative exploration. By reframing hallucination as a controllable design variable, the framework enables AI systems to contribute meaningfully to innovation, ideation, and artistic expression.

First, the system implements a "hallucination-as-a-feature" paradigm. Controlled divergence from training data is encouraged in designated creative modes such as brainstorming, story generation, or hypothesis formation. Outputs generated under these modes are clearly marked with metadata tags indicating their speculative or imaginative status, helping users differentiate between factual content and creative elaboration. Rather than suppressing hallucination, the system actively manages it within a safe sandbox where it can inspire novelty without compromising trust.

Second, the framework offers graduated creative control mechanisms that allow users to tune the degree of imaginative deviation. Adjustable parameters include creativity temperature, novelty thresholds, coherence constraints, and domain boundaries. For example, higher creativity temperatures can increase surprising outputs, while coherence constraints ensure that even highly novel generations remain semantically consistent and narratively coherent. Pre-configured modes are available for common tasks (e.g., fiction writing, design brainstorming, speculative reasoning), but expert users may customize parameters for specialized goals. Real-time previews demonstrate how parameter adjustments impact output characteristics, empowering users to balance novelty with structure.

Third, the framework includes mechanisms for innovation pattern mining and amplification. Pattern recognition algorithms analyze generated outputs to identify promising creative combinations, unexpected connections, or novel structures. These are stored in an "innovation database" that records which patterns yielded valuable insights. Promising patterns can then be amplified through targeted regeneration, enabling the system to build upon creative seeds while retaining their originality. Cross-domain creativity transfer allows insights from one field to inspire innovations in another, supported by abstraction and translation mechanisms.

Finally, the framework supports collaborative human–AI creative processes. Interactive refinement enables users to iteratively guide the system by preserving desirable creative aspects, modifying less relevant ones, or discarding unhelpful outputs. Version branching allows exploration of multiple creative directions in parallel, while session memory preserves context across iterations. Human feedback from creative professionals is especially important, as it teaches the system to distinguish valuable imaginative hallucinations from meaningless randomness. Through this collaboration, the framework positions hallucination not as a flaw but as a catalyst for inspiration and co-creation.

In this way, the Action Layer balances strict control in factual and safety-critical domains with permissive exploration in creative tasks, ensuring that hallucination is managed not only as a risk but also as an opportunity for innovation.

E. Evaluation & Continuous Feedback

The Evaluation Layer closes the loop of the UV-oriented framework by systematically assessing the effectiveness of sensing, decision making, and action components, while continuously feeding insights back into all databases and modules for iterative improvement. Unlike conventional evaluation that treats hallucination detection as a static post-hoc task, this layer implements a dynamic, mitigation-aware assessment process that evolves alongside user interactions and domain requirements.

First, evaluation is multi-dimensional, combining automatic metrics, benchmark datasets, and human judgment. Automatic measures such as BERTScore, FactCC, and QAGS assess factual consistency, while domain-specific benchmarks (e.g., TruthfulQA, FEVER, HaluEval, REALSum, TrueRecall) provide standardized references. Human evaluation complements these automated measures by capturing aspects of subjectivity, domain specificity, and user satisfaction that are not easily quantifiable. Expertise-weighted protocols ensure that feedback from domain experts carries more influence in sensitive contexts, while consensus-based multi-rater assessments reduce subjectivity.

Second, the evaluation system emphasizes continuous calibration. Outputs are not only scored for immediate reliability but also logged as training signals that refine tolerance profiles, update method performance records, and improve knowledge–model alignment. Hallucination instances identified through evaluation are recycled into fine-tuning datasets, while user feedback enriches preference models. Cross-validation and voting mechanisms ensure that assessment outcomes are robust, while online learning dynamically adapts evaluation weights based on real-world performance.

Third, the evaluation process explicitly accounts for mitigation strategies. By measuring outputs under different hallucination control methods, the system develops mitigation-aware evaluation that captures trade-offs between accuracy, efficiency, and creativity. For example, a zero-tolerance RAG+verification pipeline may maximize reliability but introduce latency, while a permissive creative mode may reduce factuality but deliver high novelty and user engagement. Recording these trade-offs allows the framework to optimize for specific application contexts rather than pursuing a one-size-fits-all standard.

Finally, evaluation results are fed back into the framework's core databases—method repository, knowledge base, and model zoo—enabling continuous adaptation. Over time, this feedback loop supports knowledge graph enrichment, model fine-tuning, parameter calibration, and refinement of strategy selection protocols. In doing so, the Evaluation Layer ensures that the framework not only monitors hallucination outcomes but also evolves to anticipate and mitigate them proactively. This continuous improvement cycle establishes a foundation for long-term trustworthiness and resilience in real-world applications.

1) Multi-Level Evaluation: Multi-level evaluation ensures that hallucination management is assessed comprehensively, integrating both quantitative and qualitative perspectives across different layers of the framework. Unlike single-pass evaluation pipelines, this approach evaluates outputs at multiple granularities—token, sentence, document, and interaction levels—while incorporating system-level performance signals.

At the automated level, factual consistency is measured using established metrics such as BERTScore, FactCC, and QAGS, which capture semantic alignment between generated outputs and reference texts. For broader benchmarks, datasets such as TruthfulQA, FEVER, HaluEval, REALSum, and TrueRecall provide standardized environments to test robustness across tasks like factual Q&A, summarization, and reasoning. These automated tools offer scalability and repeatability but require supplementation to capture subtler dimensions of hallucination.

At the human evaluation level, user and expert assessments contribute essential perspectives that cannot be fully automated. Users provide direct judgments on trustworthiness, relevance, and satisfaction, particularly in interactive contexts. Experts contribute weighted feedback in domains where factual accuracy is critical (e.g., medicine, law, scientific reasoning), ensuring that domain knowledge is properly integrated into evaluation. Consensus-based multi-rater protocols are used to reduce subjectivity, while expertise-weighting ensures that specialist opinions carry appropriate influence in sensitive contexts.

At the system evaluation level, outputs are assessed not only as isolated instances but also as part of continuous interaction trajectories. This includes tracking error propagation, long-term consistency, and the cumulative impact of hallucinations across multi-turn dialogues. Additionally, evaluation considers the effectiveness of safeguards, verification protocols, and mitigation strategies deployed in previous layers.

Together, this multi-level approach creates a full-spectrum view of system reliability, balancing automation with human oversight and short-term metrics with long-term behavioral analysis. By capturing token-level precision, task-level robustness, and system-level trustworthiness, the framework establishes a holistic methodology for evaluating and managing hallucination.

2) Feedback & Continuous Learning: Feedback and continuous learning transform evaluation outcomes into actionable improvements for the entire UV-oriented framework. Rather than treating evaluation as a static, end-stage activity, this module establishes a dynamic feedback loop that continuously updates models, knowledge repositories, and strategy selection mechanisms. In doing so, the system evolves toward greater robustness, adaptability, and trustworthiness over time.

First, evaluation results are fed back into the method repository, enabling systematic updates on the effectiveness of hallucination mitigation strategies. Each method is annotated with empirical performance data, including accuracy, efficiency, cost, and side effects across different domains. This allows the system to refine its decision-making policies dynamically, choosing the most context-appropriate strategies for future tasks.

Second, the model zoo benefits from fine-grained feedback through targeted retraining and parameter calibration. Hallucination instances identified during evaluation are stored as curated examples for fine-tuning or reinforcement learning with human feedback (RLHF). Online learning mechanisms allow the system to adapt in near real time, ensuring that models evolve with emerging domains, shifting user preferences, and new knowledge. Domain-specific LoRA adaptations or lightweight fine-tuning can be triggered automatically based on feedback signals, ensuring both efficiency and precision.

Third, the knowledge base is incrementally updated through evaluation signals. When outputs are flagged as outdated, inconsistent, or unverifiable, the system identifies gaps in the knowledge hierarchy and prioritizes updates. Verified corrections are integrated, while hallucination-prone areas are highlighted for human review or external data sourcing. This process supports continuous enrichment of the knowledge graph and alignment of the model's internal representations with evolving ground truth.

Finally, feedback is used to refine user interaction and system governance. User satisfaction scores and expert assessments inform adaptive tolerance calibration, allowing the system to adjust hallucination thresholds dynamically across domains and tasks. Cross-validation and voting protocols ensure that feedback is reliable, while anomaly detection mechanisms prevent bias or adversarial manipulation of evaluation loops. Over time, this collective process creates a self-improving ecosystem in which the framework not only mitigates hallucinations but also anticipates them proactively.

By embedding feedback into every component, the framework establishes continuous learning as a core principle. This ensures that hallucination management is not a one-time intervention but an evolving capability that strengthens with each interaction, aligning the system more closely with the goals of trustworthy AI.

F. Detailed Framework Design for UV Smart Hallucination Control

To complement the high-level feedback-oriented framework introduced earlier, Figure 13 presents a more fine-grained design of the UV Smart Hallucination Control system. This detailed architecture decomposes the framework into modular processing stages, explicit feedback loops, and knowledge management mechanisms that together form a robust closed-loop system.

The design begins with **external inputs and governance constraints**, including design principles, policy and regulation, system objectives, requirements, human preferences, and benchmark information. These signals are first integrated within the **Decision Making** layer, which performs intention understanding and generates strategies tailored to the task context.

The **Action** layer then operationalizes these strategies through a three-stage processing pipeline (pre-generation, in-generation, and post-generation), interacting with the hybrid human–AI system that integrates LLM structure, users, and external knowledge. Outputs from the Action stage are continuously monitored by the **Sensing & Monitoring** module and subsequently assessed within the **Evaluation** layer. Evaluation consolidates sensing data, hallucination detection signals, and user feedback, producing reliability assessments and structured feedback for upstream modules.

A central feature of this architecture is its explicit **multi-loop feedback design** (Loops 1–5). These loops enable continuous communication across layers: from user feedback to decision making, from abstention detection to external knowledge, and from evaluation back to action and knowledge management. The integration of these loops ensures that the system not only controls hallucinations dynamically but also adapts over time by updating objectives, refining knowledge bases, and improving decision strategies.

1) External Inputs and Governance Layer: The external inputs and governance layer establishes the foundational context for the UV Smart Hallucination Control framework. This layer defines the boundary conditions under which the system operates, ensuring that generation strategies are not only technically optimized but also aligned with broader societal, ethical, and regulatory requirements.

First, **design principles, policy, and regulation** provide normative constraints that guide all downstream decisions. These include compliance with legal standards, ethical guidelines, and domain-specific best practices. By embedding such constraints at the entry point of the framework, the system ensures that hallucination control is not handled in isolation but remains consistent with external governance structures.

Second, the system ingests **system objectives and task requirements** that define what constitutes success for a given application. Objectives may range from factual correctness in knowledge-intensive domains to creative divergence in open-ended tasks. Requirements capture operational aspects such as latency, resource constraints, and acceptable risk levels, which in turn influence tolerance thresholds for hallucination.

Third, **human preferences** play a central role in personalizing hallucination management. These preferences, derived from interactive feedback and historical user behavior, shape how the system balances reliability against creativity, detail

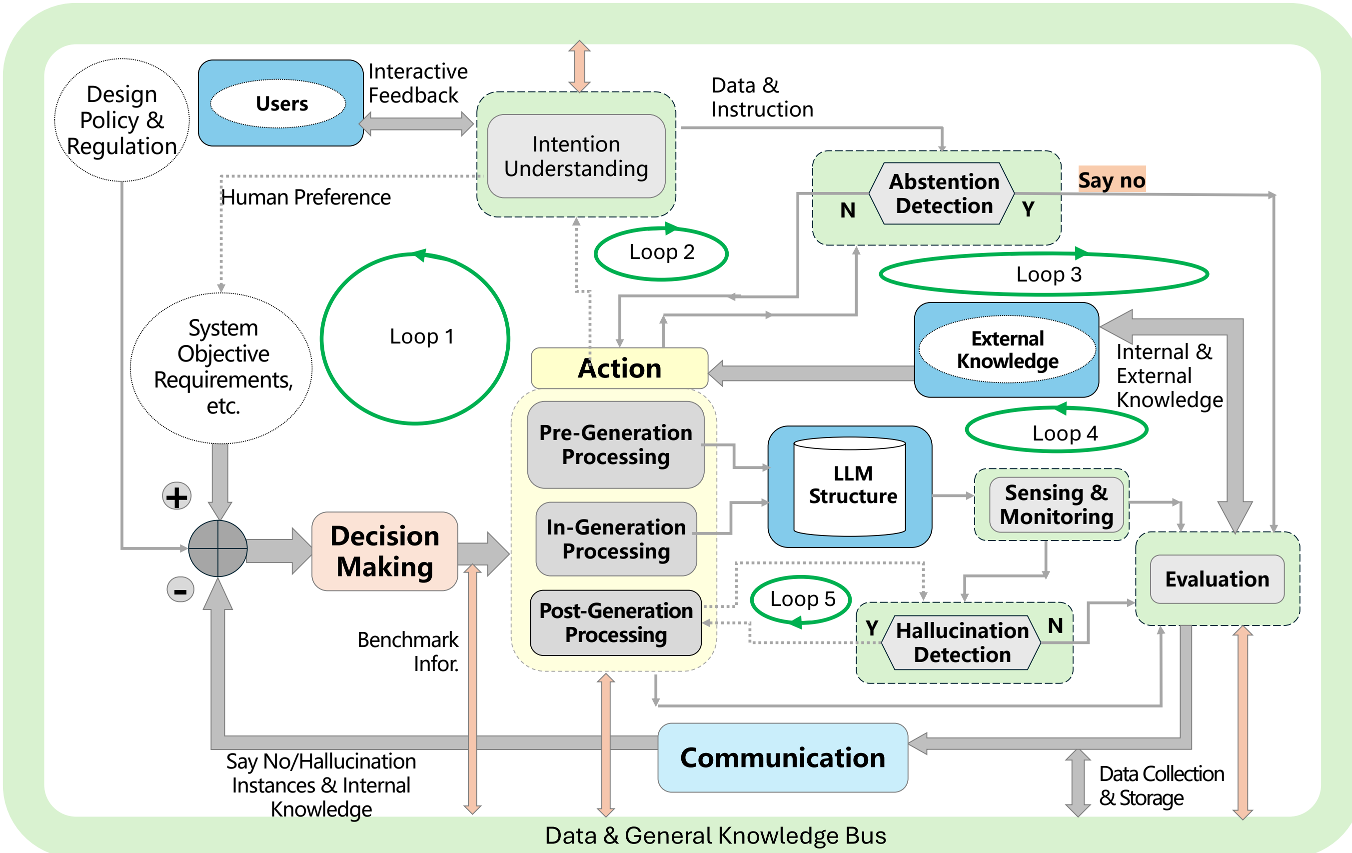

Fig. 13. Detailed Framework Design for UV Smart Hallucination Control

against conciseness, and conservatism against exploration. By capturing such preferences explicitly, the system can dynamically calibrate its behavior to suit individual users and use cases.

Finally, **benchmark information** serves as a calibration reference for system performance. Benchmarks drawn from established datasets and evaluation tasks provide empirical grounding for assessing how well hallucination control mechanisms perform in practice. Integrating benchmarks into the governance layer enables the system to self-assess continuously and remain accountable to transparent, measurable standards.

Together, these external inputs establish the governance scaffolding of the framework. They ensure that hallucination management is simultaneously *externally aligned* with societal and regulatory constraints and *internally adaptive* to user-specific objectives and preferences.

2) Decision Making Layer: The Decision Making layer functions as the central coordination hub of the UV Smart Hallucination Control framework. Its primary role is to interpret external inputs, integrate contextual signals, and generate actionable strategies for hallucination management. By linking governance constraints with operational requirements, this layer ensures that downstream processes operate in alignment with both system objectives and user expectations.

At the core of this layer is **intention understanding**. The system analyzes user queries, contextual requirements, and historical interaction patterns to derive a structured representation of task intent. This process enables the framework to differentiate between high-stakes factual queries, exploratory reasoning tasks, and open-ended creative prompts, each of which requires a different tolerance for hallucination.

The Decision Making layer also integrates **objectives, preferences, and knowledge context**. Inputs from design principles, policies, and regulations are combined with user-specific preferences and benchmark information to form a multidimensional decision space. Within this space, trade-offs between accuracy, efficiency, creativity, and safety are systematically evaluated.

Based on this synthesis, the system generates **strategic outputs** that guide the Action layer. These outputs may include:

- tolerance thresholds for hallucination risk,
- the selection of mitigation strategies (e.g., retrieval augmentation, self-reflection loops, abstention mechanisms),
- routing decisions for whether to rely on internal knowledge, external sources, or hybrid approaches,
- and directives for how uncertainty and verification signals should be handled downstream.

Importantly, the Decision Making layer is not static but adaptive. Feedback from evaluation, sensing, and user interaction continuously flows back into this layer, enabling dynamic recalibration of strategies. Through this feedback-informed adaptation, the Decision Making layer ensures that the framework remains resilient to evolving tasks, shifting user preferences, and emerging regulatory or ethical standards.

3) Processing Pipeline: The Processing Pipeline is the operational core of the Action layer, translating strategic decisions from the Decision Making module into concrete generation processes. To achieve robust hallucination management, the pipeline is structured into three sequential stages: **Pre-Generation Processing**, **In-Generation Processing**, and **Post-Generation Processing**. Each stage contributes distinct safeguards and corrective mechanisms, ensuring that hallucination is addressed throughout the entire lifecycle of content generation.

a) Pre-Generation Processing.: This stage prepares inputs and system configurations before any generation occurs. Tasks include input optimization, ambiguity resolution, and tolerance calibration based on user intent and contextual constraints. Pre-processing may also involve query rewriting, prompt normalization, and alignment with benchmark references to reduce the likelihood of hallucination at its root. By shaping the input space and setting appropriate thresholds in advance, the system minimizes the propagation of errors into later stages.

b) In-Generation Processing.: During the generation process, the system actively monitors and regulates hallucination risk through dynamic control mechanisms. Sampling parameters such as temperature, top-k, and top-p are adjusted in real time based on uncertainty signals, while self-reflection checkpoints allow the model to pause and evaluate its own intermediate outputs. In high-stakes tasks, redundant generation paths or consensus-driven decoding strategies may be invoked, leveraging multiple models or knowledge sources to increase reliability. This stage transforms generation from a static one-pass process into an adaptive, self-regulating pipeline.

c) Post-Generation Processing.: After the initial output is produced, a series of refinement and verification steps are applied. Self-revision loops enable the system to identify unsupported claims, contradictions, or low-confidence statements and regenerate them as necessary. Filtering modules evaluate outputs against knowledge bases, benchmarks, and safety rules to detect and suppress hallucinations. Abstention mechanisms may also be triggered, allowing the system to say "I don't know" when reliability falls below acceptable thresholds. Finally, post-processing prepares the content for delivery to the user, attaching uncertainty estimates and explanatory metadata to promote transparency.

By distributing hallucination control across all three stages, the Processing Pipeline ensures that mitigation is not confined to a single point of intervention. Instead, hallucination is managed as a continuous process, with safeguards embedded before, during, and after content generation.

4) Action Layer and Hybrid Human–AI System: The Action Layer operationalizes the strategies generated by the Decision Making module and connects them to the broader hybrid human–AI ecosystem. This layer serves as the execution hub where selected models, knowledge resources, and interaction protocols are deployed to accomplish the user's task, while ensuring that hallucination risks are actively managed.

At its core, the Action Layer integrates three critical components: the **LLM Structure**, the **Users**, and **External Knowledge**. These components form a hybrid human–AI system characterized by continuous, bidirectional interaction. The LLM Structure provides generative capabilities, the Users contribute task-specific guidance and feedback, and External Knowledge offers grounding and factual verification. Their interplay ensures that the system does not rely solely on internal parametric memory but benefits from external oversight and contextual enrichment.

A central mechanism within this layer is the **abstention and refusal protocol**. When the system detects insufficient knowledge, conflicting evidence, or high uncertainty, it can strategically choose to withhold an answer or respond with an explicit "I do not know." This safeguard prevents the delivery of potentially harmful or misleading hallucinations, while maintaining transparency to the user. Abstention signals are also logged for evaluation, enabling the system to identify recurring knowledge gaps and improve over time.

The Action Layer further enables adaptive interaction protocols. Outputs can be delivered in multiple formats depending on user needs, including direct answers, step-by-step reasoning traces, or evidence-linked summaries. When uncertainty is present, the system communicates not only a result but also its confidence level and relevant knowledge sources. This dual role—delivering actionable content while signaling reliability—empowers users to calibrate their trust in the system appropriately.

In summary, the Action Layer embodies the practical execution of hallucination control, bridging system-level strategies with real-world interactions. By orchestrating LLM computation, user collaboration, and external knowledge access, this layer ensures that the hybrid human–AI system operates as a dynamic, transparent, and trustworthy partner.

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6) Sensing & Monitoring Layer: The Sensing & Monitoring layer acts as the observational backbone of the UV Smart Hallucination Control framework. Its primary role is to continuously track signals from both the Action layer and the hybrid human–AI system, enabling the detection of hallucination events, performance degradation, and contextual drift in real time. By serving as the system's perceptual channel, this layer ensures that deviations from intended behavior are quickly identified and routed to the Evaluation module for structured analysis.

A first responsibility of this layer is the **real-time monitoring of model behavior**. During and after generation, key indicators such as semantic coherence, factual consistency, and alignment with user intent are measured. Low-confidence outputs, logical inconsistencies, or contradictions with external knowledge are flagged as potential hallucination signals. Monitoring is not limited to textual content but also extends to interaction patterns, such as unexpected refusals, erratic response latency, or abrupt shifts in reasoning style, which may indicate instability.

Second, the layer performs **hallucination detection and abstention analysis**. By comparing generated outputs against structured knowledge bases, retrieval results, and historical system responses, the monitoring system can differentiate between acceptable creative divergence and harmful factual errors. When abstention mechanisms are triggered in the Action layer, their frequency, context, and justification are recorded to provide insights into knowledge gaps and decision-making thresholds.

Third, the layer supports **contextual and temporal monitoring**. It tracks external factors such as updates in knowledge sources, user feedback trends, and domain-specific requirements. Temporal monitoring ensures that stale or outdated information is flagged, reducing the risk of hallucination caused by temporal mismatch. Cross-session sensing aggregates user interactions over time, allowing the system to detect evolving patterns in user preference or tolerance.

Finally, the Sensing & Monitoring layer maintains a direct communication channel with the Evaluation module. All sensing signals, detection outcomes, and metadata are transmitted for systematic assessment and storage. This design ensures that monitoring is not a passive process but an active contributor to adaptive feedback loops, enabling the framework to evolve through continuous observation and learning.

In short, this layer provides the perceptual foundation for reliable hallucination management, ensuring that the system remains aware of its own behavior and responsive to the dynamic environment in which it operates.

7) Evaluation Layer: The Evaluation layer serves as the integrative assessment hub of the UV Smart Hallucination Control framework. Its role is to consolidate sensing signals, hallucination detection outcomes, user feedback, and task performance metrics into a structured evaluation process. By closing the loop between monitoring and decision making, this layer ensures that the system continuously learns from its own outputs and adapts its strategies to improve reliability and trustworthiness over time.

First, the Evaluation layer performs **multi-source data consolidation**. Inputs are collected from sensing signals, user interactions, abstention events, and hallucination detection instances. These data streams are aligned with task-specific objectives and benchmark information, creating a comprehensive record of the system's behavior. The collected evidence is systematically organized within a storage module, forming a long-term knowledge base for analyzing hallucination patterns and mitigation effectiveness.

Second, the layer executes **hallucination detection and reliability calibration**. Using both automated metrics (e.g., factual consistency scores such as BERTScore, FactCC, and QAGS) and human-in-the-loop evaluations, the system measures the degree to which generated outputs align with ground-truth knowledge and user expectations. Uncertainty quantification results from the Action layer are cross-validated against actual error rates, enabling continuous calibration of confidence reporting mechanisms. This prevents overconfidence bias and improves the interpretability of uncertainty indicators for end users.

Third, the Evaluation module generates **structured feedback outputs**. These outputs include reliability assessments, error classifications (e.g., factual errors, logical inconsistencies, contextual mismatches), and performance summaries relative to benchmarks. Results are communicated back to users in the form of transparency reports or confidence dashboards, empowering them to make informed judgments about system outputs. At the same time, feedback signals are transmitted upstream to update decision-making strategies, refine user preference models, and adjust knowledge base parameters.

Finally, the Evaluation layer provides the foundation for **adaptive self-improvement**. By identifying recurring hallucination types, correlating them with system states, and recording abstention instances, the framework builds a structured memory that informs future mitigation. Evaluation outcomes are stored in the data collection and storage module, supporting incremental updates to both internal and external knowledge repositories. This creates a continuous learning cycle where the system improves not only on an instance-by- instance basis but also in its long-term handling of hallucination phenomena.

In summary, the Evaluation layer transforms raw sensing and monitoring data into actionable insights, enabling the framework to self-assess, self-correct, and continuously align with both technical requirements and human trust standards.

8) Data & Knowledge Management: The Data & Knowledge Management layer provides the backbone for information flow and long-term learning within the UV Smart Hallucination Control framework. Its function is to connect internal and external knowledge resources, support real-time retrieval and grounding, and maintain structured records of system outputs, hallucination instances, and user feedback. By serving as both a knowledge bus and a memory store, this layer ensures that hallucination control is continuously informed by reliable data and adaptive updates.

At the center of this layer is the **Data & General Knowledge Bus**, which facilitates seamless communication between internal parametric knowledge and external knowledge sources. The bus supports hybrid retrieval strategies, enabling the system to dynamically combine model-encoded representations with structured knowledge bases, benchmark datasets, and domain-specific repositories. This ensures that factual grounding is both comprehensive and context-aware.

A second component is **Data Collection & Storage**, which records outputs, hallucination instances, abstention events, user interactions, and evaluation results. This persistent memory allows the system to track long-term hallucination patterns, measure the effectiveness of different mitigation strategies, and preserve critical cases for future fine-tuning. The storage module also enables longitudinal analysis of user behavior and preference evolution, providing a foundation for adaptive personalization.

Third, the layer manages **internal and external knowledge integration**. Internal knowledge, encoded within model parameters or curated internal databases, is combined with external knowledge retrieved in real time. Mechanisms for conflict resolution, temporal consistency, and source reliability assessment are employed to ensure that integrated knowledge is trustworthy. Updates from evaluation are used to continuously refine both knowledge categories, allowing the system to adapt to emerging information and shifting domains.

Finally, the **communication interface** ensures smooth interaction between knowledge resources and other modules. Decision Making relies on this layer to access contextualized knowledge for strategy generation, the Action layer uses it for real-time retrieval during generation, and Evaluation feeds back new insights to update knowledge representations.

In essence, the Data & Knowledge Management layer transforms the framework into a living system: one that not only consumes and generates knowledge but also curates, organizes, and evolves it over time to strengthen hallucination control and trustworthiness.

9) Feedback Loops: One of the defining characteristics of the UV Smart Hallucination Control framework is its multi-loop feedback design, which establishes a system of interconnected pathways for continuous learning and adaptation. Unlike a linear pipeline, this architecture embeds circular information flows that link objectives, actions, sensing, evaluation, and knowledge. Together, the five loops (Loop 1–5) ensure that hallucination control is addressed not as a one-time filtering step but as a *dynamic, system-wide process*.

a) Loop 1: Strategic Alignment Loop.: This loop connects **System Objectives & Requirements** with **Decision Making**, the **Action** layer, **Intention Understanding**, and **Human Preferences**. Its purpose is to ensure that the system's operational strategies remain aligned with both external governance constraints and personalized user needs. Whenever evaluation reveals gaps between outputs and requirements, this loop feeds corrections back to decision making, updating tolerance thresholds, strategy choices, and preference models. It acts as the top-level control loop, continuously re-aligning the system with evolving goals and values.

b) Loop 2: Abstention Calibration Loop.: Loop 2 links **Intention Understanding**, the **Action** process, **Abstention Detection**, and **Data & Instruction**. This loop governs the conditions under which the system decides to withhold an answer (say "I don't know"). By cross-referencing task intent with data availability and risk signals, the system avoids overconfident responses in areas where knowledge is missing or contradictory. Over time, abstention data collected in this loop becomes a valuable training resource, highlighting knowledge gaps and informing future updates to both internal and external knowledge.

c) Loop 3: Knowledge Validation Loop.: This loop connects the **Action** module with **External Knowledge**, the **Evaluation** layer, and **Abstention Detection**. It ensures that generated outputs are continuously cross-validated against trusted knowledge sources. When inconsistencies arise, evaluation results trigger abstention mechanisms or corrective regeneration. This loop thus acts as a safeguard, preventing hallucinations by grounding responses in verifiable knowledge, while also enriching the system with evidence of where knowledge sources need refinement.

d) Loop 4: Model Monitoring and Adaptation Loop.: Loop 4 integrates the **Action** module, **LLM Structure**, **Sensing & Monitoring**, **Evaluation**, and **External Knowledge**. It establishes a comprehensive oversight cycle where model-level behaviors are tracked in real time, assessed through evaluation, and re-aligned with external knowledge resources. This loop is particularly important for long-term robustness: it not only detects hallucination in the moment but also drives gradual adaptation of the model, ensuring that its internal representations and reasoning processes remain consistent with external truth and system expectations.

e) Loop 5: Post-Generation Detection Loop.: The final loop links **Post-Generation Processing** with **Hallucination Detection**. It is designed to capture errors and hallucinations that slip through earlier layers of control. By systematically logging hallucination instances after generation, this loop provides the framework with a fine-grained dataset of failure cases. These cases feed back into evaluation, data storage, and model training, enabling targeted improvements in specific weak areas. While it operates at the most granular level, Loop 5 is critical for incremental learning, as it transforms each hallucination into a learning opportunity for future iterations.

f) Collective Impact.: Together, these five loops form a multi-layered defense and adaptation architecture. Loop 1 ensures strategic alignment with objectives, Loop 2 governs abstention thresholds, Loop 3 validates against knowledge, Loop 4 monitors and adapts model behavior, and Loop 5 captures errors post-generation. Their interplay transforms hallucination management from a static safeguard into a continuous, adaptive cycle of sensing, evaluation, and correction. This design not only reduces the occurrence of harmful hallucinations but also enables the system to evolve toward greater trustworthiness and resilience over time.

VIII. Impact and Consideration

The phenomenon of AI hallucination carries profound technical, societal, and ethical implications that extend beyond isolated model errors. While early studies of hallucination primarily framed it as a performance limitation of large language models (LLMs) and generative systems, its real-world impact is increasingly evident across healthcare, law, education, finance, creative industries, and media ecosystems. Hallucinations can erode user trust, propagate misinformation, distort cultural narratives, and trigger regulatory and legal challenges when integrated into high-stakes decision-making pipelines [450], [451].

Beyond the immediate risk of factual inaccuracies, hallucinations introduce multi-layered considerations for user behavior, societal stability, and governance. Users may develop overreliance on fluent but unverified outputs, while biases in training data can be amplified through hallucinated narratives, threatening fairness and public confidence [40], [452]. Meanwhile, in multimodal generative AI, hallucinations can manifest as deepfakes or synthetic media, raising concerns about information integrity, election security, and cultural preservation [453].

Consequently, understanding the impact and broader considerations of hallucination requires a multi-dimensional analysis, encompassing domain-specific risks, user trust dynamics, societal and cultural repercussions, and ethical and legal responsibilities. This chapter examines these aspects systematically, highlighting both the immediate harms and the long-term governance challenges associated with deploying hallucination-prone AI systems.

A. Domain-Specific Impact

1) Healthcare and Biomedical Applications: Healthcare and biomedical domains represent some of the highest-stakes scenarios for large language models (LLMs) and generative AI systems. While these models have demonstrated promising capabilities in tasks such as clinical decision support, medical report summarization, biomedical literature review, and patient interaction, the risk of AI hallucination introduces significant safety and ethical concerns.

First, hallucination can directly impact clinical decision-making. Generative models like Med-PaLM or domain-adapted BioGPT can produce factually incorrect medical advice or fabricated treatment recommendations, which could mislead clinicians or patients if unchecked. For instance, Yu et al. observed that hallucinations in healthcare LLMs could propagate incorrect diagnostic or treatment suggestions, potentially resulting in adverse patient outcomes if integrated into decision pipelines without human verification [454].

Second, hallucination in biomedical literature retrieval and summarization undermines scientific reliability. LLMs fine-tuned on biomedical corpora sometimes fabricate citations, misinterpret clinical trial outcomes, or merge unrelated studies into a seemingly coherent summary. This behavior not only reduces trust among medical practitioners but also poses a risk for research reproducibility, as highlighted by Rathkopf, who emphasized that unverified generative outputs can erode the epistemic reliability of scientific knowledge pipelines [450].

Third, hallucination may exacerbate patient misinformation and legal risk. In patient-facing applications such as chatbots for preliminary health advice, a hallucinated diagnosis or incorrect medication recommendation could expose institutions to medical liability. Leiser et al. noted that user interface design and warning mechanisms can partially mitigate this risk, but transparency and human-in-the-loop verification remain critical Leiser [455].

Finally, from a regulatory and ethical perspective, hallucination in healthcare demands strict oversight and compliance with medical standards such as HIPAA in the US or EMA/FDA guidance for AI-assisted tools. Banafa suggests that future clinical deployment of generative AI should incorporate real-time hallucination detection, external knowledge grounding, and clear liability allocation, ensuring that LLMs act as decision-support assistants rather than autonomous prescribers [456].

In summary, while generative AI holds tremendous potential in biomedical informatics, hallucination poses a dual challenge: it threatens patient safety and scientific credibility, necessitating robust detection, human validation, and regulatory safeguards before large-scale clinical deployment.

2) Legal and Policy Applications: Legal and policy domains face unique risks from AI hallucination due to their high reliance on factual accuracy and strict accountability frameworks. LLMs used for legal research, case summarization, or contract drafting can generate hallucinated precedents, misquoted statutes, or fabricated case citations, which may mislead practitioners and lead to serious legal consequences.

A widely reported 2024 incident demonstrated this risk, where lawyers submitted ChatGPT-generated court filings containing nonexistent case citations, resulting in judicial sanctions and public scrutiny [452].

Hallucinations in policy advisory applications present regulatory and reputational risks. Generative models used for policy drafting or legislative summaries may introduce fabricated statistics or misinterpretations of international law, potentially influencing policymaker decisions. Rathkopf emphasizes that the epistemic fragility of LLMs makes them ill-suited for autonomous policy generation without human oversight [450].

Moreover, legal liability and governance are major concerns. Hallucinated legal outputs raise questions of responsibility: whether errors should be attributed to developers, deployers, or users. Coeckelbergh et al. argue that transparent disclosure, rigorous audit, and regulatory alignment (e.g., with EU AI Act provisions for "high-risk" AI systems) are essential to prevent misuse and ensure accountability Coeckelbergh [457].

3) Education and Research: In education and scientific research, AI hallucination poses a dual threat: the spread of misinformation and the erosion of academic trust. LLMs used in automated tutoring, content generation, and research assistance may fabricate historical facts, misinterpret concepts, or invent citations, misleading both students and researchers.

From an educational perspective, reliance on LLMs can lead to misconceptions and overconfidence in AI-generated content. Elsayed reports that student learning outcomes suffer when hallucinated material is accepted uncritically, while overreliance fosters passive learning behaviors [452].

In scientific research, hallucinated outputs threaten knowledge integrity. Rathkopf highlights the risk of fabricated references and misinterpreted literature, which can pollute the academic ecosystem if incorporated into secondary publications. This undermines reproducibility and slows scientific progress, especially when AI-generated summaries circulate widely without verification [450].

Consequently, human-in-the-loop verification and AI-assisted citation validation tools are essential safeguards in education and research applications.

4) Finance and Business: In finance and business, hallucination introduces economic and compliance risks. LLMs are increasingly applied to financial report drafting, market sentiment analysis, risk assessment, and automated consulting, where hallucinated insights can lead to costly decisions.

A single hallucinated claim in financial advisory chatbots or automated report generation can mislead traders or executives, triggering market volatility or regulatory scrutiny. Yu et al. emphasize that AI misreporting in financial contexts can compound risk because users often assume outputs are data-driven and precise [454].

Moreover, compliance risk arises when hallucinated anti-money laundering (AML) reports or audit statements are generated without cross-verification. Leiser et al. argue that confidence calibration and post-generation verification are crucial for business-critical LLM deployment, as hallucinated outputs can trigger regulatory penalties and reputational harm [455].

5) Creative and Media Applications: Creative and media industries experience a paradoxical relationship with hallucination: it is both a source of innovation and a vector for risk.

On the positive side, hallucination can drive creativity in storytelling, game design, advertising, and artistic content generation, producing novel combinations that human creators might not imagine. However, in news media, social platforms, and entertainment, hallucinated outputs can amplify misinformation and public distrust. For instance, generative systems can create photorealistic but fabricated images or videos—blurring the line between imaginative output and deepfake disinformation [456].

Hallucination also intersects with ethical media practices. Bender et al. highlight that stochastic parroting in LLMs can propagate biased or culturally insensitive narratives, raising concerns of social responsibility and cultural erosion [40].

In response, media organizations and creative platforms are exploring content labeling, provenance tracking, and AI watermarking to differentiate creative hallucination from disinformation, balancing innovation with societal safety.

B. User Trust and Behavior

The dynamics of user trust and behavior are central to understanding the real-world impact of AI hallucinations. While LLMs and generative AI produce fluent and coherent outputs, their surface credibility can mislead users into overtrusting content that is factually incorrect or unverifiable.

1) Trust Calibration: Users frequently overestimate the reliability of fluent AI outputs, leading to miscalibrated trust. Hallucination events erode user confidence, especially when errors are discovered in high-stakes tasks such as healthcare or legal document review. Leiser et al. emphasize that unmitigated hallucination reduces long-term trust, and that interface-level interventions—such as confidence visualization or explanation prompts—can improve trust calibration by encouraging skepticism when uncertainty is high [455].

2) Overreliance and Automation Bias: The automation bias phenomenon arises when users default to accepting AI outputs without independent verification. In organizational contexts, this can lead to rubber-stamping of hallucinated content, introducing systemic risk in domains such as finance, research, and legal compliance. Rathkopf notes that overreliance on generative models for decision-making can result in latent risk accumulation, as hallucinated insights may propagate across multi-user or multi-agent workflows [450].

3) Behavioral Adaptation: As users gain experience with generative AI, they often adapt their behavior by cross-checking outputs or using external verification tools, while naive users are more likely to accept hallucinations at face value. Elsayed observed that experienced students and professionals learn to treat AI as a suggestive rather than authoritative source, whereas inexperienced users may propagate errors into downstream tasks, amplifying the societal impact of hallucination [452].

4) Transparency and Interpretability: Providing transparent explanations for why hallucinations occur and how models generate outputs is essential for responsible AI usage. Bender et al. argue that black-box models increase the risk of misplaced trust, whereas interpretability and confidence scoring can promote calibrated skepticism and reduce reliance on unverified outputs [40].

In summary, user trust in AI is fragile and context-dependent. Proper trust calibration, combined with education, interpretability, and interface-level mitigations, is critical to prevent overreliance and minimize the downstream harm of hallucinations in real-world deployments.

C. Societal and Cultural Considerations

The societal and cultural implications of AI hallucination extend beyond technical failures, as hallucinated outputs can influence public opinion, cultural preservation, and the information ecosystem. Unlike errors in isolated computational settings, hallucinations at societal scale may generate misinformation, amplify biases, and distort collective narratives.

1) Misinformation and Disinformation: Generative AI can produce large-scale, coherent, and plausible but false narratives, blurring the boundary between authentic and synthetic content. These hallucinations can be weaponized in disinformation campaigns, threatening election integrity, public health communication, and social stability. For example, Rathkopf warns that unverified AI-generated text can propagate false claims faster than traditional media correction cycles, creating epistemic asymmetry in public discourse [450]. Similarly, Banafa notes that hallucinated narratives in social media ecosystems undermine public trust in both AI and journalism, complicating crisis communication [456].

2) Bias and Fairness: Hallucinations can amplify existing biases in training data, resulting in systematic misrepresentation of marginalized groups. Bender et al. emphasize that stochastic parroting of biased corpora can produce hallucinated outputs that reinforce stereotypes or propagate culturally insensitive misinformation, especially in minority languages or underrepresented cultural contexts [40]. This bias amplification exacerbates inequality in algorithmic media exposure, creating disparate societal impact.

3) Cultural Erosion: Beyond overt misinformation, hallucination may distort or dilute cultural narratives. When generative AI produces hybridized or historically inaccurate representations of folklore, traditions, or linguistic forms, it can contribute to cultural erosion and homogenization. Coeckelbergh argue that hallucinated cultural content without provenance risks undermining local epistemologies and authentic knowledge systems, especially as AI-generated media blends seamlessly with human-authored narratives [457].

4) Media Ecosystem Impact: The integration of AI-generated content into media pipelines presents new challenges for verification and editorial oversight. As hallucinated outputs blend with human journalism, traditional fact-checking mechanisms struggle to keep pace, increasing the risk of synthetic content pollution. Sahoo et al. highlight that visual and multimodal hallucinations—such as AI-generated imagery in breaking news—complicate content provenance tracking, and may erode the baseline of media trust if left unaddressed [458].

In summary, the societal and cultural footprint of hallucination is multifaceted: it propagates misinformation, reinforces structural biases, threatens cultural heritage, and destabilizes media ecosystems. Mitigation demands a cross-disciplinary effort, combining technical solutions, media literacy initiatives, and regulatory frameworks to preserve social trust and cultural integrity in the era of generative AI.

D. Ethical and Legal Implications

The ethical and legal implications of AI hallucination extend beyond technical reliability, encompassing liability, intellectual property (IP), media manipulation, and governance challenges. As LLMs and generative AI systems become deeply embedded in decision-making and public communication, hallucinations present multi-dimensional risks that demand robust ethical frameworks and legal oversight.

1) Responsibility and Liability: AI hallucinations raise complex questions of accountability, particularly in high-stakes domains such as healthcare, law, and finance. A hallucinated medical recommendation or fabricated legal citation can lead to material harm, yet liability allocation between model developers, deployers, and end-users remains ambiguous. Khawaldeh analyzed civil court practices in Jordan, highlighting that hallucinated outputs could trigger legal claims, especially in conversational AI deployed in judicial or contractual workflows [459]. Similarly, Lantyer argues that clear liability frameworks and mandatory disclosure of AI-generated content are needed to combat the legal uncertainty created by hallucination in professional settings [460].

2) Intellectual Property and Academic Integrity: Hallucination often involves fabricated citations, invented authorship, or unintended plagiarism, threatening copyright and scholarly credibility. Williamson emphasize that hallucinated academic outputs, if propagated into research workflows, can undermine the integrity of scientific knowledge and the chain of attribution [451]. This issue becomes especially acute in cross-domain applications, where hallucinated summaries or references may inadvertently pollute institutional repositories or policy recommendations.

3) Deepfakes and Media Manipulation: In multimodal generative AI, hallucinations can manifest as fabricated images, audio, or video, enabling deepfake creation and mass disinformation campaigns. Mylrea frames generative AI as a potential "weapon of mass disinformation", emphasizing that hallucinated and deliberately manipulated content threatens election integrity, public safety, and reputational stability [453]. Such risks demand provenance tracking, watermarking, and cross-platform monitoring to mitigate media ecosystem pollution.

4) AI Governance and Regulatory Frameworks: Mitigating these risks requires integrated governance, combining technical safeguards, regulatory compliance, and ethical design principles. Taeihagh highlights that governance of generative AI must address hallucination explicitly, recommending multi-level regulation, third-party auditing, and standardized hallucination reporting [461].

In summary, ethical and legal considerations of hallucination span liability allocation, IP protection, deepfake prevention, and governance design. Proactive legal adaptation and multi-stakeholder collaboration are essential to ensure safe, accountable, and socially acceptable deployment of generative AI.

IX. Interaction of AI Hallucination

Human–AI interaction constitutes the most direct interface where hallucinations are experienced, interpreted, and managed. Unlike algorithmic or system-level considerations, this dimension focuses on the cognitive, behavioral, and trust-related dynamics that emerge when users engage with AI-generated outputs. Research has shown that users often exhibit *automation bias*, over-trusting fluent but potentially misleading outputs, while at the same time becoming overly skeptical when encountering visible errors [462]. Hallucinations therefore pose a dual challenge: they undermine user confidence in reliable cases, and they encourage uncritical acceptance in risky cases.

A critical issue in this interaction is **trust calibration**. Users require mechanisms that allow them to align their trust with the system's actual reliability. Prior work emphasizes the importance of uncertainty disclosure, confidence estimates, and transparent error reporting as means to support calibrated trust [23], [46]. Without such signals, hallucinations risk propagating unchecked in domains where accuracy is paramount, such as medicine or law.

Moreover, human feedback plays an essential role in shaping system behavior. Approaches such as Reinforcement Learning from Human Feedback (RLHF) and more recent extensions like Reinforcement Learning from AI Feedback (RLAIF) highlight how user interaction can be systematically integrated to refine model outputs [32], [463]. Through such mechanisms, hallucination control is no longer a purely algorithmic challenge but a collaborative process where user preferences, tolerance thresholds, and contextual judgments become part of the adaptive loop.

In summary, Human–AI interaction provides both the *vulnerability point*, where hallucinations directly impact end-users, and the *lever*, through which human oversight and feedback can significantly improve hallucination management. This duality underscores the importance of designing interaction protocols that support transparency, trust calibration, and collaborative adaptation.

A. Human–AI Interaction

1) Dialogue dynamics under hallucination: Hallucination is not just a one-off error; it unfolds through interaction. Once a model states something confidently, conversational norms (politeness, deference to perceived expertise, and momentum in turn-taking) can lead users to accept or build on the claim, which in turn invites the system to elaborate and "rationalize" it in later turns. Real-world incidents show how this dynamic spills into consequences. The Air Canada chatbot provided misleading guidance on bereavement prices, and the company later acknowledged that the language was misleading, illustrating how authoritative dialogue can mask weak grounding [464]. These patterns reinforce the case for human-in-the-loop checks and explicit uncertainty signaling to prevent users from being persuaded of false information by dialogue dynamics.

2) Cognitive biases: Human judgment does not begin from a neutral baseline; it is filtered through cultural frames and cognitive shortcuts. For example, culturally, each community carries its own patterned biases that shape how people interpret information and make decisions [465]. Cognitive biases, mental shortcuts, or heuristics that let us act quickly by leaning on past experience rather than slow deliberative reasoning can streamline decisions but also produce systematic errors and, in extreme cases, human hallucinations, where confident beliefs are weakly grounded [465]. These human tendencies can leak into interactive AI. Microsoft's Tay, designed to learn from Twitter interactions and improve its responses over time, rapidly began to produce racist and Nazi-leaning statements such as 'Hitler was right' after being exposed to toxic inputs; developers later noted that by absorbing interaction data, Tay effectively inherited human prejudices [465]. To reduce such failure modes, systems should be built around human-centered and human-in-the-loop approaches: involve end users early and continuously to surface real needs, gate and monitor what the model learns from public interactions, and implement review and auditing mechanisms that can catch and correct biased behavior before it propagates.

3) User-driven correction and reinforcement: A practical safeguard is to treat users as active correctors whose feedback updates both the answer and the system. In practice, this means visible citations per claim, one-click flags, automatic re-asks for evidence when confidence outruns grounding, and escalation to human review in high-risk contexts. A Vancouver lawyer filed non-existent cases generated by an AI tool, triggering negative publicity and prompting professional guidance on the responsible use of AI [466]. That sequence is precisely why user-driven correction and reinforcement must be built in as first-class features, not afterthoughts. Concretely, legal assistants should enforce 'cite-or-silence' rules (no assertion without a verifiable source), show per-claim provenance (court, docket, reporter, and URL), and run automatic cite-checkers that retrieve the full text to confirm existence and relevance before allowing export. When a user indicates a suspect claim, the system should immediately downgrade confidence, provide alternatives, and record the failure as a hard negative for future training and evaluation, so the same hallucination is less likely to recur. Institutional feedback loops then reinforce these corrections at scale, aligning with a human-centered, human-in-the-loop approach that treats users' interventions as the primary safety mechanism rather than a last line of defense.

4) Human Collaboration for Performance Improvement.: Beyond general feedback mechanisms, recent empirical studies further demonstrate the tangible benefits of human collaboration with LLMs in interactive settings. In particular, ChatGPT's interactive feature enables users to iteratively refine prompts in a multi-turn "prompt engineering" fashion, which has been shown to significantly improve model performance. For example, multi-task evaluations report up to an 8% improvement in ROUGE-1 on summarization tasks and a 2% gain in ChrF++ on machine translation through such collaborative interaction [4]. To facilitate reproducibility, the authors also release code for evaluation set extraction, highlighting the role of transparent benchmarking in supporting systematic human–AI collaboration.

B. Hallucination–Algorithm Interaction

Hallucinations in large language models (LLMs) are not only an artifact of limited data or imperfect user prompts, but also a direct consequence of algorithmic design choices that shape how models learn, adapt, and generalize. The interaction between hallucination and algorithms is therefore inherently *bidirectional*: while some algorithms inadvertently amplify hallucination by overfitting spurious patterns or miscalibrating uncertainty, others offer structured pathways for mitigation by enabling models to learn from human feedback, adapt across domains, or incorporate multiple modalities.

a) Reinforcement Learning Approaches.: Among the most impactful algorithmic interventions is **Reinforcement Learning with Human Feedback (RLHF)**. By integrating human preferences into the optimization loop, RLHF enables models to align more closely with user expectations, thereby reducing hallucination in many high-stakes tasks [32]. Extensions such as Reinforcement Learning from AI Feedback (RLAIF) [463] and iterative preference optimization build on this foundation by incorporating synthetic or model-generated feedback, which allows scaling beyond human annotation limits. These methods illustrate how reinforcement learning transforms hallucination control into an adaptive process: rather than applying static rules, the model continually updates its behavior based on evaluative signals.

b) Adaptive and Meta Learning.: **Adaptive learning** strategies dynamically tune model parameters or decoding processes in response to real-time context. Such adaptability is crucial for managing hallucinations in diverse and evolving domains. Similarly, **meta learning** approaches, such as Model-Agnostic Meta-Learning (MAML) [467], endow models with the ability to rapidly adapt to new tasks with limited supervision. However, this very flexibility can also produce hallucinations when meta-learned priors fail to align with novel task distributions, underscoring the need for careful calibration and domain-aware adaptation.

c) Transfer and Active Learning.: **Transfer learning** [468] enables knowledge reuse across tasks and domains, but domain shift is a known driver of hallucination, as models may incorrectly apply source-domain heuristics to target-domain queries. Mitigation requires explicit domain adaptation, fine-tuning on in-domain data, or uncertainty-aware transfer techniques. Meanwhile, **active learning** [298] provides a complementary mechanism: by prioritizing uncertain or high-risk cases for annotation, active learning can build specialized datasets that capture hallucination-prone scenarios, creating a feedback loop where the model progressively strengthens its weak points.

d) Multimodal and Federated Learning.: Hallucination becomes even more pronounced in **multimodal learning**, where models must integrate heterogeneous signals (e.g., vision and language). Cross-modal misalignment often manifests as object or attribute hallucination, such as describing nonexistent entities in images [469], [470]. Algorithmic advances in alignment, such as contrastive representation learning and attention-based grounding, are crucial for mitigation. On the other hand, **federated learning** introduces a different dimension of complexity: heterogeneous and non-IID client data may amplify hallucinations if local biases propagate into the global model [471]. Robust aggregation, personalized federated learning, and privacy-preserving evaluation are essential to balance decentralization with reliability.

e) Zero-shot and Few-shot Learning.: Hallucinations are especially visible in **zero-shot** and **few-shot** settings, where models must generalize without sufficient supervision. In such cases, overgeneralization and reliance on spurious correlations lead to confident but unsupported outputs. Algorithmic advances such as prompt-based learning, in-context calibration, and retrieval-augmented zero-shot methods help alleviate these risks, yet they remain vulnerable to hallucination in domains with sparse or ambiguous supervision.

f) Algorithmic Trade-offs.: Underlying all these paradigms is a fundamental trade-off: **the pursuit of higher performance often increases hallucination risk**. Techniques that maximize fluency, diversity, or novelty can also amplify unsupported claims. Conversely, overly conservative strategies reduce hallucinations but risk diminishing creativity, adaptability, and user satisfaction. As such, algorithmic choices must navigate competing objectives—accuracy, efficiency, creativity, and safety—depending on the application context.

g) Summary: Taken together, the interaction between hallucination and algorithms highlights that hallucination is not merely a surface-level artifact but a structural phenomenon emerging from learning paradigms themselves. Reinforcement, meta, transfer, active, multimodal, federated, and low-shot learning each impose distinct pressures and opportunities for hallucination control. Understanding these interactions provides a roadmap for designing adaptive, context-aware, and resilient mitigation strategies that can evolve alongside advances in AI learning algorithms.

C. Hallucination and Broader Issues

Hallucinations in large language models (LLMs) are often framed as technical deficiencies, yet their significance extends far beyond system design or algorithmic optimization. They represent a **socio-technical risk factor**, with direct implications for ethics, governance, user trust, safety, and information ecosystems. As AI becomes embedded in high-stakes domains and pervasive in everyday applications, hallucination control must be considered an integral part of broader agendas such as *ethical AI*, *trustworthy AI*, *responsible AI*, *reliable AI*, and *safe AI* [472]–[474].

a) Ethical and Responsible AI.: From an ethical perspective, hallucinations pose risks by generating fabricated content that may appear authoritative yet lack factual grounding. This risk is especially salient in domains such as healthcare, law, finance, and education, where users may act upon incorrect outputs with significant consequences. Studies on responsible AI emphasize that beyond technical fixes, hallucinations raise questions of **accountability, explainability, and fairness** [475], [476]. For example, if a clinical decision support tool fabricates references or misstates treatment guidelines, who bears responsibility—the model developers, the deploying institution, or the end-user? Embedding hallucination control into responsible AI principles therefore requires not only system transparency but also external auditing, clear liability structures, and mechanisms for redress.

b) Trustworthy and Reliable AI.: Trustworthiness and reliability are often highlighted as prerequisites for AI deployment in critical domains. Yet hallucinations undermine these goals by creating a mismatch between model fluency and factual reliability. Research demonstrates that users often struggle with **trust calibration**: they over-trust confident outputs even when incorrect, and conversely, lose trust entirely after encountering a single visible error [462], [477]. This creates a paradox: even models with state-of-the-art benchmark performance can erode trust if hallucination rates remain unpredictable. Addressing this challenge requires systems to disclose uncertainty explicitly, offer interpretable reasoning chains, and provide users with the tools to independently verify content. In this sense, hallucination control is not merely about error reduction, but about designing interaction protocols that **support calibrated trust**.

c) Safety, Adversarial Threats, and Misuse.: Hallucinations are also intertwined with safety risks and adversarial threats. They can be **deliberately induced** through adversarial attacks to produce false or misleading outputs [478], or exploited to generate malicious misinformation such as deepfakes [479]. The ability of generative models to fabricate realistic yet false narratives lowers the barrier for disinformation campaigns, identity fraud, and other forms of malicious use. Addressing these risks requires integrating adversarial robustness, watermarking, and provenance-tracking technologies into hallucination management pipelines. Importantly, hallucination in safety-critical environments (e.g., autonomous driving, medical diagnosis, financial decision-making) may lead not only to reputational harm but to direct physical or economic damage, making robust safeguards imperative.

d) Performance versus Governance.: The relationship between hallucination, performance, and governance is deeply tensional. Pursuing high performance often emphasizes fluency, creativity, and coverage, but these same properties increase the risk of hallucination. On the other hand, applying strict governance and conservative decoding strategies reduces hallucinations but at the cost of innovation, adaptability, and user engagement. This trade-off is reflected in real-world deployment tensions: for instance, highly regulated medical or legal AI systems must prioritize conservatism, while creative applications in art, design, or entertainment benefit from permissive hallucination. This indicates that hallucination management should not aim at universal suppression but rather adopt a **context-sensitive strategy**, calibrated to the stakes of the task and the tolerance of the user.

e) Long-Term and Societal Implications.: Finally, hallucinations have broader implications for information ecosystems and social trust. Widespread exposure to AI-generated but fabricated content can blur the boundary between reliable and unreliable knowledge, leading to an erosion of public trust in digital information. This is especially concerning in domains such as journalism, political discourse, and science communication. At the same time, the normalization of hallucinations in creative applications may shift cultural norms about the acceptability of "imaginative inaccuracies." The long-term challenge is therefore twofold: preventing hallucinations from destabilizing knowledge institutions, while also harnessing their potential in creative, exploratory, and ideation contexts.

f) Summary.: In summary, hallucinations cannot be treated as isolated technical errors; they must be understood as systemic phenomena with ethical, social, and safety implications. Their control requires a multi-layered approach that integrates algorithmic mitigation, governance frameworks, and human-centered design. By embedding hallucination management into the broader discourse on ethical, trustworthy, responsible, reliable, and safe AI, researchers and practitioners can ensure that these systems are both powerful and socially sustainable.

X. Conclusion and Future Direction

A. Conclusion

This survey has provided a comprehensive overview of hallucination phenomena in large language models (LLMs) and multimodal large language models (MLLMs), addressing their manifestations, underlying mechanisms, evaluation approaches, and mitigation strategies. Through a structured taxonomy, we identified and analyzed hallucinations across factuality, faithfulness, logical consistency, and emergent hybrid forms, while also extending the discussion to multimodal-specific risks such as cross-modal inconsistencies and visual overinterpretation. By linking these surface-level manifestations to their generative causes—including data imperfections, training-induced biases, and inference-time vulnerabilities—we demonstrated that hallucinations are not isolated artifacts but systemic outcomes of probabilistic modeling and complex model–environment interactions.

Our review further revealed significant limitations in current evaluation paradigms. Existing benchmarks and metrics often capture only narrow aspects of hallucination, failing to account for mechanism-level explanations or real-world deployment complexities. The analysis showed that without mechanism-aware evaluation protocols, progress risks being fragmented, with mitigation efforts targeting symptoms rather than root causes. Equally important, we highlighted how hallucination control must be understood in relation to application context: while detrimental in high-stakes domains like medicine, finance, or law, controlled hallucinations can be tolerated—or even leveraged—in creative and exploratory settings.

On the mitigation front, we synthesized strategies spanning multiple levels: prompt engineering and decoding control, retrieval-augmented generation and knowledge integration, and system-level solutions such as controllability mechanisms and uncertainty calibration. By comparing their trade-offs in scalability, interpretability, and effectiveness, we emphasized that no single approach is sufficient; rather, a layered and context-sensitive combination is needed. Building on this synthesis, we proposed the UV-oriented framework, which reconceptualizes hallucination as a systemic vulnerability in information and decision-making loops. This framework incorporates dynamic feedback, user-specific tolerance calibration, and consensus-driven verification, offering a more holistic and resilient approach to managing hallucination risks across diverse applications.

In sum, this work contributes to the field in three key ways: (1) offering the most comprehensive taxonomy of hallucination to date, bridging phenomenological descriptions with mechanistic drivers; (2) unifying detection, evaluation, and mitigation research into a coherent lens that clarifies open challenges and trade-offs; and (3) introducing the UV-oriented framework as a novel paradigm for trustworthy AI, emphasizing feedback loops, adaptive strategies, and human-centered design. Together, these contributions provide both a foundation for understanding the complexity of hallucination and a roadmap for building AI systems that are safe, reliable, and aligned with societal needs.

B. Future Directions

While this survey consolidates current progress in understanding, evaluating, and mitigating hallucinations, it also reveals several promising avenues for future research.

First, there is a critical need for unified benchmarks and evaluation protocols that move beyond task-specific or domain-limited assessments. Current resources such as TruthfulQA or HaluEval only partially capture the diversity of hallucination phenomena, especially in multimodal and open-ended contexts. Future work should establish standardized benchmarks that integrate surface-level detection with mechanism-aware evaluations, enabling fairer comparisons and more reliable progress tracking.

Second, advancing toward theoretical and mechanistic models of hallucination is essential. Most existing approaches rely on observational heuristics, which describe when hallucinations occur but not why they arise. Developing causal frameworks—grounded in data distribution properties, optimization dynamics, and inference-time processes—will provide deeper explanatory power and guide targeted mitigation strategies.

Third, the rapid expansion of multimodal systems introduces new alignment and cross-domain challenges. Vision–language and audio–language models often suffer from modality dominance, fabricated object descriptions, and mismatched associations. Future research should develop robust cross-modal grounding methods, better fusion architectures, and transfer-friendly mitigation strategies to ensure reliability in real-world multimodal deployments.

Fourth, researchers must grapple with the trade-off between reliability and creativity. Excessively suppressing hallucinations risks over-constraining models, reducing their generative flexibility and innovative potential. Future work should investigate controllable hallucination mechanisms, enabling users to calibrate acceptable levels of creativity versus factuality depending on application contexts, from scientific reasoning to artistic generation.

Fifth, the human–AI interaction dimension requires more attention. Current safeguards are largely reactive, relying on user detection or post-hoc verification pipelines. Future directions include integrating proactive safeguards, building adaptive feedback loops, and personalizing hallucination tolerance levels. By embedding hallucination awareness into interactive systems, AI can better align with user needs and prevent misinformation propagation.

Finally, building on the UV-oriented framework, future research should explore ecosystem-aware and dynamic feedback architectures. This entails designing AI systems that treat hallucination not as an isolated error but as part of broader information and decision-making loops, with sensing, communication, decision, action, and evaluation tightly coupled. Such architectures, enriched with consensus mechanisms, external verification, and sustainability principles, may provide a resilient foundation for safe, trustworthy, and human-centered AI.

In conclusion, addressing these directions will enable the research community to transition from fragmented solutions toward an integrated vision of hallucination management. This path holds the promise of creating generative AI systems that balance creativity with accountability, ensuring reliability in high-stakes applications while preserving adaptability and value in exploratory domains.

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References

[1] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM computing surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[2] V. Rawte, R. Saha, K. Rudra, and S. Ghosh, "A survey of hallucination in large foundation models," *arXiv preprint arXiv:2309.05922*, 2023.

[3] K. Shuster and et al., "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[4] Y. Bang *et al.*, "A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity," *arXiv preprint arXiv:2302.04023*, 2023.

[5] P. Vaithilingam, T. Zhang, and E. L. Glassman, "Expectation vs. experience: Evaluating the usability of code generation tools powered by large language models," in *CHI Conference on Human Factors in Computing Systems*, 2022, pp. 1–17.

[6] W. X. Zhao and et al., "A survey of hallucination in large language models," *arXiv preprint arXiv:2309.06794*, 2023.

[7] P. Ahadian and Q. Guan, "A survey on hallucination in large language and foundation models," 2025.

[8] Z. Zhu *et al.*, "Can we trust ai doctors? a survey of medical hallucination in large language and large vision-language models," in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025, pp. 6748–6769.

[9] S. M. Abtahi and A. Azim, "Augmenting large language models with static code analysis for automated code quality improvements," in *2025 IEEE/ACM Second International Conference on AI Foundation Models and Software Engineering (Forge)*. IEEE, 2025, pp. 82–92.

[10] D. Gosmar and D. A. Dahl, "Hallucination mitigation using agentic ai natural language-based frameworks," *arXiv preprint arXiv:2501.13946*, 2025.

[11] T. Rahman *et al.*, "Take loads off your developers: Automated user story generation using large language model," in *2024 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. IEEE, 2024, pp. 791–801.

[12] V. Kamra, L. Gupta, D. Arora, and A. K. Yadav, "Evaluating reinforcement learning based models for test time enhancement in rag," in *2025 Second International Conference on Cognitive Robotics and Intelligent Systems (ICC-ROBINS)*. IEEE, 2025, pp. 254–259.

[13] D. Tosi, "Comparing generative ai literature reviews versus human-led systematic literature reviews: A case study on big data research," *IEEE Access*, 2025.

[14] H. D. Menéndez *et al.*, "Testing software and systems," 2025.

[15] S. Cao *et al.*, "Preliminary study on evaluation of smart-cities technologies and proposed uv lifestyles," in *2018 4th International Conference on Universal Village (UV)*. Ieee, 2018, pp. 1–49.

[16] L. Huang *et al.*, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, 2025.

[17] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, "Halueval: A large-scale hallucination evaluation benchmark for large language models," *arXiv preprint arXiv:2305.11747*, 2023.

[18] S. Lin, J. Hilton, and O. Evans, "Truthfulqa: Measuring how models mimic human falsehoods," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 3214–3252.

[19] P. Manakul, A. Liusie, and M. J. Gales, "Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models," *arXiv preprint arXiv:2303.08896*, 2023, verified source: Zero-resource training-free hallucination detection method. [Online]. Available: https://arxiv.org/abs/2303.08896

[20] V. Rawte *et al.*, "Sorry, come again? prompting – enhancing comprehension and reducing hallucination in large language models," *arXiv preprint arXiv:2407.07102*, 2024. [Online]. Available: https://arxiv.org/abs/2407.07102

[21] M. Dahl, V. Magesh, M. Suzgun, and D. E. Ho, "Hallucinating law: Legal mistakes with large language models are pervasive," *arXiv preprint arXiv:2401.16829*, 2024. [Online]. Available: https://arxiv.org/abs/2401.16829

[22] V. Magesh *et al.*, "Hallucination-free? assessing the reliability of leading ai legal research tools," *Journal of Empirical Legal Studies*, vol. 22, no. 2, pp. 216–242, 2025. [Online]. Available: https://arxiv.org/abs/2405.20362

[23] Y. Zhang *et al.*, "Siren's song in the ai ocean: A survey on hallucination in large language models," *arXiv preprint arXiv:2309.01219*, 2023. [Online]. Available: https://arxiv.org/abs/2309.01219

[24] S. Tonmoy *et al.*, "A comprehensive survey of hallucination mitigation techniques in large language models," *arXiv preprint arXiv:2401.01313*, vol. 6, 2024.

[25] Y. Xia *et al.*, "Hallucination diversity-aware active learning for text summarization," *arXiv preprint arXiv:2404.01588*, 2024.

[26] L. Berglund *et al.*, "The reversal curse: Llms trained on "a is b" fail to learn "b is a"," *arXiv preprint arXiv:2309.12288*, 2023.

[27] N. Kandpal, H. Deng, A. Roberts, E. Wallace, and C. Raffel, "Large language models struggle to learn long-tail knowledge," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202, Honolulu, Hawaii, USA, 2023, pp. 15 696–15 707. [Online]. Available: https://proceedings.mlr.press/v202/kandpal23a.html

[28] G. Zheng, X. Oma, J. Jia, Z. Wang, and Y. Song, "Large language models are not strong abstract reasoners," *arXiv preprint arXiv:2305.19555*, 2023.

[29] M. Elaraby *et al.*, "Halo: Estimation and reduction of hallucinations in open-source weak large language models," *arXiv preprint arXiv:2308.11764*, 2023.

[30] F. Shi *et al.*, "Large language models can be easily distracted by irrelevant context," *International Conference on Machine Learning*, pp. 31 210–31 227, 2023.

[31] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[32] L. Ouyang *et al.*, "Training language models to follow instructions with human feedback," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 27 730–27 744.

[33] C. Zhou *et al.*, "Lima: Less is more for alignment," *arXiv preprint arXiv:2305.11206*, 2023.

[34] K. Chen, Q. Chen, J. Zhou, Y. He, and L. He, "Diahalu: A dialogue-level hallucination evaluation benchmark for large language models," in *Findings of the Association for Computational Linguistics: EMNLP 2024*, Y. Al-Onaizan, M. Bansal, and Y. Chen, Eds. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 9057–9079. [Online]. Available: https://aclanthology.org/2024.findings-emnlp.529/

[35] S. Liu *et al.*, "Towards long context hallucination detection," *arXiv preprint arXiv:2504.19457*, 2024.

[36] A. T. Kalai and S. S. Vempala, "Calibrated language models must hallucinate," in *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, May 2024, doi:10.1145/3618260.3649777. [Online]. Available: https://doi.org/10.1145/3618260.3649777

[37] S. Banerjee, A. Agarwal, and S. Singla, "Llms will always hallucinate, and we need to live with this," *CoRR*, vol. abs/2409.05746, Sep. 2024. [Online]. Available: https://arxiv.org/abs/2409.05746

[38] J. White *et al.*, "A prompt pattern catalog to enhance prompt engineering with chatgpt," *arXiv preprint arXiv:2302.11382*, 2023.

[39] B. Wang, G. Li, Y. Li, X. Zhou, and S. Zhang, "Enabling conversational interaction with mobile ui using large language models," *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2023.

[40] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the dangers of stochastic parrots: Can language models be too big?" in *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, 2021, pp. 610–623.

[41] O. P. Mosqueira, M. Geva, J. Berant, and Y. Goldberg, "Measuring and narrowing the compositionality gap in language models," *arXiv preprint arXiv:2210.03350*, 2023.

[42] A. Pal, L. K. Umapathi, and M. Sankarasubbu, "Med-halt: Medical domain hallucination test for large language models," *arXiv preprint arXiv:2307.15343*, 2023.

[43] A. Chang *et al.*, "Medheval: Benchmarking hallucinations and mitigation strategies in medical large vision-language models," *CoRR*, vol. abs/2503.02157, Mar. 2025. [Online]. Available: https://arxiv.org/abs/2503.02157

[44] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, "Building machines that learn and think like people," *Behavioral and Brain Sciences*, vol. 40, 2017.

[45] K. Friston, "The free-energy principle: A unified brain theory?" *Nature Reviews Neuroscience*, vol. 11, no. 2, pp. 127–138, 2010.

[46] S. Kadavath *et al.*, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[47] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 824–24 837, 2022, verified source: Foundational Chain-of-Thought paper with 52

[48] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22 199–22 213, 2022.

[49] B. Peng, S. Narayanan, and C. Papadimitriou, "On limitations of the transformer architecture," *CoRR*, vol. abs/2402.08164, Feb. 2024. [Online]. Available: https://arxiv.org/abs/2402.08164

[50] M. Hahn, "Theoretical limitations of self-attention in neural sequence models," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 156–171, 2020.

[51] Y. Gu *et al.*, "Anah-v2: Scaling analytical hallucination annotation of large language models," *arXiv preprint arXiv:2407.04693*, 2024.

[52] K. Bhattacharya, A. Majumder, and A. Chakrabarti, "A study on effect of reference knowledge choice in generating technical content relevant to sapphire model using large language model," *arXiv preprint arXiv:2407.00396*, 2024. [Online]. Available: https://arxiv.org/abs/2407.00396

[53] D. Hendrycks *et al.*, "Measuring massive multitask language understanding," *arXiv preprint arXiv:2009.03300*, 2020.

[54] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," *International Conference on Machine Learning*, pp. 1321–1330, 2017.

[55] W. Shi *et al.*, "Trusting your evidence: Hallucinate less with context-aware decoding," *arXiv preprint arXiv:2305.14739*, 2023.

[56] R. Aksitov, C.-C. Chang, D. Reitter, S. Shakeri, and Y. Sung, "Characterizing attribution and fluency tradeoffs for retrieval-augmented large language models," *arXiv preprint arXiv:2302.05578*, 2023.

[57] A. Kulkarni *et al.*, "Evaluating evaluation metrics – the mirage of hallucination detection," *arXiv preprint arXiv:2504.18114*, 2024.

[58] Y. Zhang, Q. V. Liao, and R. K. Bellamy, "Effect of confidence and explanation on accuracy and trust calibration in ai-assisted decision making," in *Proceedings of the 2020 conference on fairness, accountability, and transparency*, 2020, pp. 295–305.

[59] H. Yuan *et al.*, "Whispers that shake foundations: Analyzing and mitigating false premise hallucinations in large language models," *arXiv preprint arXiv:2402.19103*, 2024.

[60] Y. Li *et al.*, "Evaluating object hallucination in large vision-language models," *arXiv preprint arXiv:2305.10355*, 2023.

[61] J. Wang *et al.*, "Amber: An llm-free multi-dimensional benchmark for mllms hallucination evaluation," *arXiv preprint arXiv:2311.07397*, 2023, verified external source: AMBER benchmark for multimodal hallucination evaluation with cost and accuracy analysis. [Online]. Available: https://arxiv.org/abs/2311.07397

[62] A. Gunjal, J. Yin, and E. Bas, "Detecting and preventing hallucinations in large vision language models," *arXiv preprint arXiv:2308.06394*, 2024.

[63] Y. Zhou *et al.*, "Analyzing and mitigating object hallucination in large vision-language models," *arXiv preprint arXiv:2310.00754*, 2024. [Online]. Available: https://arxiv.org/abs/2310.00754

[64] Z. Bai *et al.*, "Hallucination of multimodal large language models: A survey," *arXiv preprint arXiv:2404.18930*, 2024. [Online]. Available: https://arxiv.org/abs/2404.18930

[65] H. Liu *et al.*, "A survey on hallucination in large vision-language models," *arXiv preprint arXiv:2402.00253*, 2024. [Online]. Available: https://arxiv.org/abs/2402.00253

[66] T. Liu *et al.*, "A token-level reference-free hallucination detection benchmark for free-form text generation," *arXiv preprint arXiv:2104.08704*, 2021.

[67] M. Zhang, O. Press, W. Merrill, A. Liu, and N. A. Smith, "How language model hallucinations can snowball," *arXiv preprint arXiv:2305.13534*, 2023.

[68] C. Wang and R. Sennrich, "On exposure bias, hallucination and domain shift in neural machine translation," *arXiv preprint arXiv:2005.03642*, 2020.

[69] Z. Wei, S. Wang, X. Rong, X. Liu, and H. Li, "Shadows in the attention: Contextual perturbation and representation drift in the dynamics of hallucination in llms," *arXiv preprint arXiv:2505.16894*, 2025.

[70] L. Yu, M. Cao, J. C. K. Cheung, and Y. Dong, "Mechanistic understanding and mitigation of language model non-factual hallucinations," *arXiv preprint arXiv:2403.18167*, 2024.

[71] Z. Yin, "A review of methods for alleviating hallucination issues in large language models," *Applied and Computational Engineering*, vol. 76, pp. 258–266, 2024.

[72] Z. Xu, S. Jain, and M. Kankanhalli, "Hallucination is inevitable: An innate limitation of large language models," *arXiv preprint arXiv:2401.11817*, 2024.

[73] L. Weidinger *et al.*, "Ethical and social risks of harm from language models," *arXiv preprint arXiv:2112.04359*, 2021.

[74] M. Sharma *et al.*, "Towards understanding sycophancy in language models," *arXiv preprint arXiv:2310.13548*, 2023. [Online]. Available: https://arxiv.org/abs/2310.13548

[75] H. Kim, A. Bibi, P. Torr, and Y. Gal, "Detecting LLM hallucination through layer-wise information deficiency: Analysis of unanswerable questions and ambiguous prompts," *arXiv preprint arXiv:2412.10246*, 2024. [Online]. Available: https://arxiv.org/abs/2412.10246

[76] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, no. 8018, pp. 625–630, 2024.

[77] Y. Qin *et al.*, "Knowledge inheritance for pre-trained language models," *arXiv preprint arXiv:2105.13880*, 2021.

[78] M. Li *et al.*, "Knowledge boundary of large language models: A survey," *arXiv preprint arXiv:2412.12472*, 2024.

[79] C. Fang *et al.*, "Ingest-and-ground: Dispelling hallucinations from continually-pretrained llms with rag," *arXiv preprint arXiv:2410.02825*, 2024.

[80] E. Stringhi, "Hallucinating (or poorly fed) llms? the problem of data accuracy," *i-lex*, vol. 16, no. 2, pp. 54–63, 2023.

[81] C. Jiang *et al.*, "On large language models' hallucination with regard to known facts," *arXiv preprint arXiv:2403.20009*, 2024.

[82] X. Du, C. Xiao, and S. Li, "Haloscope: Harnessing unlabeled llm generations for hallucination detection," *Advances in Neural Information Processing Systems*, vol. 37, pp. 102 948–102 972, 2024.

[83] J. Oh *et al.*, "Erbench: An entity-relationship based automatically verifiable hallucination benchmark for large language models," *Advances in Neural Information Processing Systems*, vol. 37, pp. 53 064–53 101, 2024.

[84] H. Nguyen *et al.*, "Smoothing out hallucinations: Mitigating llm hallucination with smoothed knowledge distillation," 2025. [Online]. Available: https://arxiv.org/abs/2502.11306

[85] A. Ravichander, S. Ghela, D. Wadden, and Y. Choi, "Halogen: Fantastic llm hallucinations and where to find them," *arXiv preprint arXiv:2501.08292*, 2025.

[86] H. Cao *et al.*, "Mitigating knowledge conflicts in language model-driven question answering," *arXiv preprint arXiv:2411.11344*, 2024.

[87] M. Yin *et al.*, "Entropy law: The story behind data compression and llm performance," *arXiv preprint arXiv:2407.06645*, 2024.

[88] J. Kim, Y. J. Kim, and Y. M. Ro, "What if...?: Thinking counterfactual keywords helps to mitigate hallucination in large multi-modal models," *arXiv preprint arXiv:2403.13513*, 2024.

[89] Y. Zhang *et al.*, "Knowledge overshadowing causes amalgamated hallucination in large language models," *arXiv preprint arXiv:2407.08039*, 2024.

[90] N. Varshney *et al.*, "Investigating and addressing hallucinations of llms in tasks involving negation," *arXiv preprint arXiv:2406.05494*, 2024.

[91] H. You *et al.*, "Ferret: Refer and ground anything anywhere at any granularity," *arXiv preprint arXiv:2310.07704*, 2023.

[92] J. Zhang *et al.*, "Eventhallusion: Diagnosing event hallucinations in video llms," *arXiv preprint arXiv:2409.16597*, 2024.

[93] N. McKenna *et al.*, "Sources of hallucination by large language models on inference tasks," *arXiv preprint arXiv:2305.14552*, 2023.

[94] U. Kamath, K. Keenan, G. Somers, and S. Sorenson, "Llm challenges and solutions," in *Large Language Models: A Deep Dive: Bridging Theory and Practice*. Springer, 2024, pp. 219–274.

[95] L. Du *et al.*, "Quantifying and attributing the hallucination of large language models via association analysis," *arXiv preprint arXiv:2309.05217*, 2023.

[96] H. Ye, T. Liu, A. Zhang, W. Hua, and W. Jia, "Cognitive mirage: A review of hallucinations in large language models," *arXiv preprint arXiv:2309.06794*, 2023.

[97] M. Wang, M. Suzuki, H. Sakaji, and K. Izumi, "Interactive dualchecker for mitigating hallucinations in distilling large language models," *arXiv preprint arXiv:2408.12326*, 2024.

[98] Q. Luo *et al.*, "Self-attention and transformers: Driving the evolution of large language models," in *2023 IEEE 6th International conference on electronic information and communication technology (ICEICT)*. IEEE, 2023, pp. 401–405.

[99] S. Serrano and N. A. Smith, "Is attention interpretable?" *arXiv preprint arXiv:1906.03731*, 2019.

[100] X. Gong, T. Ming, X. Wang, and Z. Wei, "Damro: Dive into the attention mechanism of lvlm to reduce object hallucination," *arXiv preprint arXiv:2410.04514*, 2024.

[101] L. Kuhn, Y. Gal, and S. Farquhar, "Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation," *arXiv preprint arXiv:2302.09664*, 2023.

[102] D. Chiang and P. Cholak, "Overcoming a theoretical limitation of self-attention," *arXiv preprint arXiv:2202.12172*, 2022.

[103] V. Rawte *et al.*, "" sorry, come again?" prompting–enhancing comprehension and diminishing hallucination with [pause]-injected optimal paraphrasing," *arXiv preprint arXiv:2403.18976*, 2024.

[104] N. F. Liu *et al.*, "Lost in the middle: How language models use long contexts," *arXiv preprint arXiv:2307.03172*, 2023.

[105] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *International conference on machine learning*. PMLR, 2023, pp. 19 730–19 742.

[106] Z. Fan, Y. Wang, S. Polisetty, and Y. R. Fung, "Unveiling the lack of lvlm robustness to fundamental visual variations: Why and path forward," *arXiv preprint arXiv:2504.16727*, 2025.

[107] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.

[108] Y. Yuan, P. Gao, Q. Dai, J. Qin, and W. Xiang, "Uncertainty guided refinement for fine-grained salient object detection," *IEEE Transactions on Image Processing*, 2025.

[109] J. He *et al.*, "Cracking the code of hallucination in lvlms with vision-aware head divergence," *arXiv preprint arXiv:2412.13949*, 2024.

[110] P.-H. Huang, J.-L. Li, C.-P. Chen, M.-C. Chang, and W.-C. Chen, "Who brings the frisbee: Probing hidden hallucination factors in large vision-language model via causality analysis," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2025, pp. 6125–6135.

[111] A. Rohrbach, L. A. Hendricks, K. Burns, T. Darrell, and K. Saenko, "Object hallucination in image captioning," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Brussels, Belgium: Association for Computational Linguistics, Oct.-Nov. 2018, pp. 4035–4045. [Online]. Available: https://aclanthology.org/D18-1437/

[112] X. Wu *et al.*, "Autohallusion: Automatic generation of hallucination benchmarks for vision-language models," 2024.

[113] J. Li *et al.*, "Fine-grained semantically aligned vision-language pre-training," *Advances in neural information processing systems*, vol. 35, pp. 7290–7303, 2022.

[114] A. Kumar, R. Morabito, S. Umbet, J. Kabbara, and A. Emami, "Confidence under the hood: An investigation into the confidence-probability alignment in large language models," *arXiv preprint arXiv:2405.16282*, 2024.

[115] P. Tupsakhare and N. D. Kulkarni, "Strategies for avoiding gpt hallucinations," *International Research Journal of Modernization in Engineering, Technology and Science*, vol. 6, no. 7, pp. 4131–4136, 2024. [Online]. Available: https://www.researchgate.net/publication/384260920

[116] Y. Sun *et al.*, "Benchmarking hallucination in large language models based on unanswerable math word problem," *arXiv preprint arXiv:2403.03558*, 2024.

[117] X. Chen, L. Wang, W. Wu, Q. Tang, and Y. Liu, "Honest ai: Fine-tuning "small" language models to say "i don't know", and reducing hallucination in rag," *arXiv preprint arXiv:2410.09699*, 2024, this paper demonstrates that smaller language models can be effectively fine-tuned to express uncertainty and reduce hallucinations through honest response generation. The study shows that hybrid RAG and fine-tuning approaches achieved top performance on the CRAG benchmark, demonstrating competitive performance against larger models with reduced computational overhead. [Online]. Available: https://arxiv.org/abs/2410.09699

[118] OpenAI. "Introducing gpt-5." openai.com. https://openai.com/index/introducing-gpt-5/ (accessed Augest 14, 2025).

[119] Y. A. Yadkori *et al.*, "Mitigating llm hallucinations via conformal abstention," *arXiv preprint arXiv:2405.01563*, 2024.

[120] Y. Xu, H. Dong, L. Wang, C. Xiong, and J. Li, "Reward models identify consistency, not causality," *arXiv preprint arXiv:2502.14619*, 2025.

[121] M. Hu *et al.*, "Mitigating large language model hallucination with faithful finetuning," *arXiv preprint arXiv:2406.11267*, 2024.

[122] T. He, J. Zhang, Z. Zhou, and J. Glass, "Exposure bias versus self-recovery: Are distortions really incremental for autoregressive text generation?" *arXiv preprint arXiv:1905.10617*, 2019.

[123] Z. Gekhman *et al.*, "Does fine-tuning llms on new knowledge encourage hallucinations?" *arXiv preprint arXiv:2405.05904*, 2024.

[124] H. Wu *et al.*, "Iter-ahmcl: Alleviate hallucination for large language model via iterative model-level contrastive learning," *arXiv preprint arXiv:2410.12130*, 2024.

[125] W. Ren, X. Li, L. Wang, T. Zhao, and W. Qin, "Analyzing and reducing catastrophic forgetting in parameter efficient tuning," *arXiv preprint arXiv:2402.18865*, 2024.

[126] W. Liu, W. Zeng, K. He, Y. Jiang, and J. He, "What makes good data for alignment? a comprehensive study of automatic data selection in instruction tuning," *arXiv preprint arXiv:2312.15685*, 2023.

[127] H. Rashkin, D. Reitter, G. S. Tomar, and D. Das, "Increasing faithfulness in knowledge-grounded dialogue with controllable features," *arXiv preprint arXiv:2107.06963*, 2021.

[128] W. Shi *et al.*, "Trusting your evidence: Hallucinate less with context-aware decoding," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, 2024, pp. 783–791, verified source: Training-free context-aware decoding method for reducing hallucinations. [Online]. Available: https://aclanthology.org/2024.naacl-short.69/

[129] K. Andriopoulos and J. Pouwelse, "Augmenting llms with knowledge: A survey on hallucination prevention," *arXiv preprint arXiv:2309.16459*, 2023.

[130] B. A. Huberman and S. Mukherjee, "Hallucinations and emergence in large language models," *Available at SSRN 4676180*, 2023.

[131] C. Shi *et al.*, "A thorough examination of decoding methods in the era of llms," *arXiv preprint arXiv:2402.06925*, 2024.

[132] H. Zhang, D. Duckworth, D. Ippolito, and A. Neelakantan, "Trading off diversity and quality in natural language generation," *arXiv preprint arXiv:2004.10450*, 2020.

[133] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, "On faithfulness and factuality in abstractive summarization," *arXiv preprint arXiv:2005.00661*, 2020.

[134] G. Noarov *et al.*, "Foundations of top-k decoding for language models," *arXiv preprint arXiv:2505.19371*, 2025.

[135] L. Yu, M. Cao, J. C. K. Cheung, and Y. Dong, "Mechanisms of non-factual hallucination in language models," *arXiv preprint*

arXiv:2403.18167, 2024. [Online]. Available: https://arxiv.org/abs/2403.18167
[136] A. Gunjal, R. Kumar, S. Patel, and R. Singh, "Mitigating hallucinations in large language models via self-refinement-enhanced knowledge retrieval," *arXiv preprint arXiv:2405.06545*, 2024. [Online]. Available: https://arxiv.org/abs/2405.06545
[137] Z. Wei *et al.*, "Detecting and mitigating the ungrounded hallucinations in text generation by llms," in *Proceedings of the 2023 International Conference on Artificial Intelligence, Systems and Network Security*, 2023, pp. 77–81.
[138] S. Wang *et al.*, "Developing a reliable, fast, general-purpose hallucination detection and mitigation service," *arXiv preprint arXiv:2407.15441*, 2024.
[139] A. Voznyuk, G. Gritsai, and A. Grabovoy, "Advacheck at semeval-2025 task 3: Combining ner and rag to spot hallucinations in llm answers," in *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*, 2025, pp. 1204–1210.
[140] V. Rawte *et al.*, "Factoid: Factual entailment for hallucination detection," *arXiv preprint arXiv:2403.19113*, 2024.
[141] Á. Kovács and G. Recski, "Lettucedetect: A hallucination detection framework for rag applications," *arXiv preprint arXiv:2502.17125*, 2025.
[142] E. Fadeeva *et al.*, "Faithfulness-aware uncertainty quantification for fact-checking the output of retrieval augmented generation," *arXiv preprint arXiv:2505.21072*, 2025.
[143] C. George and A. Stuhlmüller, "Factored verification: detecting and reducing hallucination in summaries of academic papers," *arXiv preprint arXiv:2310.10627*, 2023.
[144] A. Agrawal, M. Suzgun, L. Mackey, and A. T. Kalai, "Do language models know when they're hallucinating references?" *arXiv preprint arXiv:2305.18248*, 2023.
[145] T. Brown *et al.*, "Language models are few-shot learners," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
[146] Q. Dong *et al.*, "A survey on in-context learning," *arXiv preprint arXiv:2301.00234*, 2022.
[147] Z. Ji *et al.*, "Towards mitigating hallucination in large language models via self-reflection," *arXiv preprint arXiv:2310.06271*, 2023.
[148] B. P. Allen, F. Polat, and P. Groth, "Shroom-indelab at semeval-2024 task 6: Zero-and few-shot llm-based classification for hallucination detection," *arXiv preprint arXiv:2404.03732*, 2024.
[149] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate before use: Improving few-shot performance of language models," in *International conference on machine learning*. PMLR, 2021, pp. 12 697–12 706.
[150] X. Shi, Z. Zhu, Z. Zhang, and C. Li, "Hallucination mitigation in natural language generation from large-scale open-domain knowledge graphs," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 12 506–12 521.
[151] X. Chen, S. Jia, and Y. Xiang, "A review: Knowledge reasoning over knowledge graph," *Expert systems with applications*, vol. 141, p. 112948, 2020.
[152] H. Sansford, N. Richardson, H. P. Maretic, and J. N. Saada, "Grapheval: A knowledge-graph based llm hallucination evaluation framework," 2024. [Online]. Available: https://arxiv.org/abs/2407.10793
[153] R. Hasegawa and R. Ichise, "Cokglm: Detecting hallucinations generated by large language models via knowledge graph verification," in *International Knowledge Graph and Semantic Web Conference*. Springer, 2024, pp. 212–224.
[154] J. Zhang *et al.*, "Knowhalu: Hallucination detection via multi-form knowledge based factual checking," *arXiv preprint arXiv:2404.02935*, 2024.
[155] S. Gilbert, J. N. Kather, and A. Hogan, "Augmented non-hallucinating large language models as medical information curators," *NPJ digital medicine*, vol. 7, no. 1, p. 100, 2024.
[156] F. Petroni *et al.*, "Language models as knowledge bases?" in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds. Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 2463–2473. [Online]. Available: https://aclanthology.org/D19-1250/
[157] S. Borgeaud *et al.*, "Improving language models by retrieving from trillions of tokens," in *International conference on machine learning*. PMLR, 2022, pp. 2206–2240.
[158] Y. Gao *et al.*, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2312.10997*, 2023.
[159] X. Guan *et al.*, "Mitigating large language model hallucinations via autonomous knowledge graph-based retrofitting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, 2024, pp. 18 126–18 134.
[160] N. Varshney, W. Yao, H. Zhang, J. Chen, and D. Yu, "A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation," *arXiv preprint arXiv:2307.03987*, 2023.
[161] X. Zhang *et al.*, "Self-alignment for factuality: Mitigating hallucinations in llms via self-evaluation," *arXiv preprint arXiv:2402.09267*, 2024.
[162] W. Sun *et al.*, "Contrastive learning reduces hallucination in conversations," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 11, 2023, pp. 13 618–13 626.
[163] X. Wang, J. Pan, L. Ding, and C. Biemann, "Mitigating hallucinations in large vision-language models with instruction contrastive decoding," *arXiv preprint arXiv:2403.18715*, 2024.
[164] S. Leng *et al.*, "Mitigating object hallucinations in large vision-language models through visual contrastive decoding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 13 872–13 882, verified source: Training-free visual contrastive decoding method for reducing object hallucinations. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Leng_Mitigating_Object_Hallucinations_in_Large_Vision-Language_Models_through_Visual_Contrastive_CVPR_2024_paper.html
[165] R. Sennrich, J. Vamvas, and A. Mohammadshahi, "Mitigating hallucinations and off-target machine translation with source-contrastive and language-contrastive decoding," *arXiv preprint arXiv:2309.07098*, 2023.
[166] D. Chen *et al.*, "Lower layer matters: Alleviating hallucination via multi-layer fusion contrastive decoding with truthfulness refocused," *arXiv preprint arXiv:2408.08769*, 2024.
[167] F. Li, P. Zhang *et al.*, "Mitigating hallucination issues in small-parameter llms through inter-layer contrastive decoding," in *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024, pp. 1–8.
[168] A. Manevich and R. Tsarfaty, "Mitigating hallucinations in large vision-language models (lvlms) via language-contrastive decoding (lcd)," *arXiv preprint arXiv:2408.04664*, 2024.
[169] Z. Chen *et al.*, "Halc: Object hallucination reduction via adaptive focal-contrast decoding," *arXiv preprint arXiv:2403.00425*, 2024.
[170] D. Chen, S. Wang, Z. Fan, X. Hu, and C. Li, "Freeze-cd: Alleviating hallucination of large language models via contrastive decoding with local freezing training," in *2024 IEEE International Conference on Smart Internet of Things (SmartIoT)*. IEEE, 2024, pp. 325–329.
[171] L. Yoffe, A. Amayuelas, and W. Y. Wang, "Debunc: mitigating hallucinations in large language model agent communication with uncertainty estimations," *arXiv preprint arXiv:2407.06426*, 2024.
[172] E. Quevedo, J. Yero, R. Koerner, P. Rivas, and T. Cerny, "Detecting hallucinations in large language model generation: A token probability approach," *arXiv preprint arXiv:2405.19648*, 2024.
[173] J. Kossen *et al.*, "Semantic entropy probes: Robust and cheap hallucination detection in llms," *arXiv preprint arXiv:2406.15927*, 2024.
[174] I. Halperin, "Prompt-response semantic divergence metrics for faithfulness hallucination and misalignment detection in large language models," *arXiv preprint arXiv:2508.10192*, 2025.
[175] G. Zhou *et al.*, "Mitigating modality prior-induced hallucinations in multimodal large language models via deciphering attention causality," 2025. [Online]. Available: https://arxiv.org/abs/2410.04780
[176] J. Yu *et al.*, "A cause-effect look at alleviating hallucination of knowledge-grounded dialogue generation," 2024. [Online]. Available: https://arxiv.org/abs/2404.03491
[177] W. Xiao *et al.*, "Detecting and mitigating hallucination in large vision language models via fine-grained ai feedback," 2025. [Online]. Available: https://arxiv.org/abs/2404.14233
[178] C. Tomani, K. Chaudhuri, I. Evtimov, D. Cremers, and M. Ibrahim, "Uncertainty-based abstention in llms improves safety and reduces hallucinations," *arXiv preprint arXiv:2404.10960*, 2024.
[179] K. Kang, E. Wallace, C. Tomlin, A. Kumar, and S. Levine, "Unfamiliar finetuning examples control how language models hallucinate," *arXiv preprint arXiv:2403.05612*, 2024.

[180] S. Schulhoff *et al.*, "The prompt report: A systematic survey of prompt engineering techniques," 2025. [Online]. Available: https://arxiv.org/abs/2406.06608

[181] D. Shah. "What is in-context learning, and how does it work: The beginner's guide." lakera.ai. https://www.lakera.ai/blog/what-is-in-context-learning (accessed Aug 18, 2025).

[182] L. Barkley and B. van der Merwe, "Investigating the role of prompting and external tools in hallucination rates of large language models," *arXiv preprint arXiv:2410.19385*, 2024.

[183] J. Varughese. "What is in-context learning?" ibm.com. https://www.ibm.com/think/topics/in-context-learning (accessed Aug 18, 2025).

[184] B. Wang *et al.*, "Towards understanding chain-of-thought prompting: An empirical study of what matters," *arXiv preprint arXiv:2212.10001*, 2022.

[185] V. Gadesha. "What is prompt engineering?" ibm.com. https://www.ibm.com/think/topics/prompt-engineering (accessed Aug 18, 2025).

[186] M. Liang *et al.*, "Thames: An end-to-end tool for hallucination mitigation and evaluation in large language models," *arXiv preprint arXiv:2409.11353*, 2024.

[187] Z. Duan and J. Wang, "Prompt-based monte carlo tree search for mitigating hallucinations in large models," *arXiv preprint arXiv:2501.13942*, 2025.

[188] J. Kim, Y. J. Kim, and Y. M. Ro, "What if...?: Thinking counterfactual keywords helps to mitigate hallucination in large multi-modal models," 2024. [Online]. Available: https://arxiv.org/abs/2403.13513

[189] J. Chen, X. Huang, and Y. Li, "Dynamic supplementation of federated search results for reducing hallucinations in llms," 2024.

[190] P. Feldman, J. Foulds, and S. Pan, "Trapping llm hallucinations using tagged context prompts," *Preprint posted online on June*, vol. 9, 2023.

[191] W. Li *et al.*, "Visual evidence prompting mitigates hallucinations in multimodal large language models," 2024.

[192] Y. Mu, P. Bai, K. Bontcheva, and X. Song, "Addressing topic granularity and hallucination in large language models for topic modelling," 2024. [Online]. Available: https://arxiv.org/abs/2405.00611

[193] S. Cui *et al.*, "Prompt2gaussia: Uncertain prompt-learning for script event prediction," 2023. [Online]. Available: https://arxiv.org/abs/2308.02103

[194] P.-F. Guo, Y.-D. Tsai, and S.-D. Lin, "Benchmarking large language model uncertainty for prompt optimization," 2024. [Online]. Available: https://arxiv.org/abs/2409.10044

[195] S. Diao *et al.*, "Active prompting with chain-of-thought for large language models," 2024. [Online]. Available: https://arxiv.org/abs/2302.12246

[196] X. Zou *et al.*, "Look twice before you answer: Memory-space visual retracing for hallucination mitigation in multimodal large language models," 2025. [Online]. Available: https://arxiv.org/abs/2410.03577

[197] H. Yuan *et al.*, "Whispers that shake foundations: Analyzing and mitigating false premise hallucinations in large language models," 2024. [Online]. Available: https://arxiv.org/abs/2402.19103

[198] S. Li *et al.*, "Mitigating object hallucinations in mllms via multi-frequency perturbations," *arXiv preprint arXiv:2503.14895*, 2025.

[199] J. Li, Y. Hou, M. Sachan, and R. Cotterell, "What do language models learn in context? the structured task hypothesis," *arXiv preprint arXiv:2406.04216*, 2024.

[200] E. K. Vrunda Gadesha, Vanna Winland. "What is chain of thought (cot) prompting?" ibm.com. https://www.ibm.com/think/topics/chain-of-thoughts (accessed Aug 18, 2025).

[201] Y. Chang, L. Jing, X. Zhang, and Y. Zhang, "A unified hallucination mitigation framework for large vision-language models," 2024. [Online]. Available: https://arxiv.org/abs/2409.16494

[202] H. Ding, L. Pang, Z. Wei, H. Shen, and X. Cheng, "Retrieve only when it needs: Adaptive retrieval augmentation for hallucination mitigation in large language models," *arXiv preprint arXiv:2402.10612*, 2024.

[203] C. Wei, Z. Chen, S. Fang, J. He, and M. Gao, "Opdai at semeval-2024 task 6: Small llms can accelerate hallucination detection with weakly supervised data," *arXiv preprint arXiv:2402.12913*, 2024.

[204] V. G. Meredith Syed. "What is zero-shot prompting?" ibm.com. https://www.ibm.com/think/topics/zero-shot-prompting (accessed Aug 18, 2025).

[205] S. Jha *et al.*, "Dehallucinating large language models using formal methods guided iterative prompting," in *2023 IEEE International Conference on Assured Autonomy (ICAA)*. IEEE, 2023, pp. 149–152.

[206] H. Lee *et al.*, "Context-informed grounding supervision," *arXiv preprint arXiv:2506.15480*, 2025.

[207] A. S. Silberstein, R. Braynard, C. Ellis, K. Munagala, and J. Yang, "A sampling-based approach to optimizing top-k queries in sensor networks," in *22nd International Conference on Data Engineering (ICDE'06)*. IEEE, 2006, pp. 68–68.

[208] S. Liu, C. Ling, and D. Stehlé, "Decoding by sampling: A randomized lattice algorithm for bounded distance decoding," *IEEE transactions on information theory*, vol. 57, no. 9, pp. 5933–5945, 2011.

[209] T. Li, Z. Li, and Y. Zhang, "Improving faithfulness of large language models in summarization via sliding generation and self-consistency," 2024. [Online]. Available: https://arxiv.org/abs/2407.21443

[210] M. Geva, R. Schuster, J. Berant, and O. Levy, "Transformer feed-forward layers are key-value memories," *arXiv preprint arXiv:2012.14913*, 2020.

[211] Y.-S. Chuang *et al.*, "Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps," 2024. [Online]. Available: https://arxiv.org/abs/2407.07071

[212] R. Busa-Fekete, B. Szorenyi, W. Cheng, P. Weng, and E. Hüllermeier, "Top-k selection based on adaptive sampling of noisy preferences," in *International Conference on Machine Learning*. PMLR, 2013, pp. 1094–1102.

[213] F. Harrington, E. Rosenthal, and M. Swinburne, "Mitigating hallucinations in large language models with sliding generation and self-checks," *Authorea Preprints*, 2024.

[214] S. Yao *et al.*, "Tree of thoughts: Deliberate problem solving with large language models," *Advances in neural information processing systems*, vol. 36, pp. 11 809–11 822, 2023.

[215] Y. Xiao and W. Y. Wang, "On hallucination and predictive uncertainty in conditional language generation," *arXiv preprint arXiv:2103.15025*, 2021.

[216] S. Zhang, T. Yu, and Y. Feng, "Truthx: Alleviating hallucinations by editing large language models in truthful space," *arXiv preprint arXiv:2402.17811*, 2024.

[217] D. Chen *et al.*, "Expanding before inferring: Enhancing factuality in large language models through premature layers interpolation," *arXiv preprint arXiv:2506.02973*, 2025.

[218] M. Tremblay, S. Gervais, and D. Maisonneuve, "Unveiling the role of feed-forward blocks in contextualization: An analysis using attention maps of large language models," 2024.

[219] E. Ainsworth, J. Wycliffe, and F. Winslow, "Reducing contextual hallucinations in large language models through attention map optimization," *Authorea Preprints*, 2024.

[220] Y. Xing, Y. Li, I. Laptev, and S. Lu, "Mitigating object hallucination via concentric causal attention," *Advances in Neural Information Processing Systems*, vol. 37, pp. 92 012–92 035, 2024.

[221] Y.-J. Huang *et al.*, "What if we recaption? addressing multimodal hallucination via data correction," *arXiv preprint arXiv:2403.04031*, 2024. [Online]. Available: https://arxiv.org/abs/2403.04031

[222] C. Jiang *et al.*, "Hallucination augmented contrastive learning for multimodal large language model," 2024. [Online]. Available: https://arxiv.org/abs/2312.06968

[223] X. L. Li *et al.*, "Contrastive decoding: Open-ended text generation as optimization," *arXiv preprint arXiv:2210.15097*, 2022.

[224] Y. Zhang, L. Cui, W. Bi, and S. Shi, "Alleviating hallucinations of large language models through induced hallucinations," 2024. [Online]. Available: https://arxiv.org/abs/2312.15710

[225] H. Zhang, H. Chen, M. Chen, and T. Zhang, "Active layer-contrastive decoding reduces hallucination in large language model generation," *arXiv preprint arXiv:2505.23657*, 2025.

[226] J. Chen, R. Thirukovalluru, J. Wang, K. Luo, and B. Dhingra, "Atomic consistency preference optimization for long-form question answering," *arXiv preprint arXiv:2505.09039*, 2025.

[227] G. Bebis and M. Georgiopoulos, "Feed-forward neural networks," *IEEE Potentials*, vol. 13, no. 4, pp. 27–31, 1994.

[228] H. Li *et al.*, "Redundancylens: Revealing and exploiting visual token processing redundancy for efficient decoder-only mllms," 2025. [Online]. Available: https://arxiv.org/abs/2501.19036

[229] I. Gerber, "Attention is not all you need: The importance of feedforward networks in transformer models," 2025. [Online]. Available: https://arxiv.org/abs/2505.06633

[230] H. Duan, Y. Yang, and K. Y. Tam, "Do llms know about hallucination? an empirical investigation of llm's hidden states (2024)," *URL https://arxiv. org/abs/2402.09733*, 2024.

[231] F. Liu *et al.*, "Mitigating hallucination in large vision-language models via robust instruction tuning," *International Conference on Learning Representations*, 2024.
[232] D. Park, Z. Qian, G. Han, and S.-N. Lim, "Mitigating dialogue hallucination for large vision language models via adversarial instruction tuning," 2024. [Online]. Available: https://arxiv.org/abs/2403.10492
[233] K. Li, O. Patel, F. Viégas, H. Pfister, and M. Wattenberg, "Inference-time intervention: Eliciting truthful answers from a language model," *Advances in Neural Information Processing Systems*, vol. 36, pp. 41 451–41 530, 2023.
[234] Y. Xia *et al.*, "Hallucination diversity-aware active learning for text summarization," 2024. [Online]. Available: https://arxiv.org/abs/2404.01588
[235] X. Qu, M. Song, W. Wei, J. Dong, and Y. Cheng, "Mitigating multilingual hallucination in large vision-language models," *arXiv preprint arXiv:2408.00550*, 2024.
[236] J. Zhang, T. Wang, H. Zhang, P. Lu, and F. Zheng, "Reflective instruction tuning: Mitigating hallucinations in large vision-language models," in *European Conference on Computer Vision*. Springer, 2024, pp. 196–213.
[237] X. Lyu, B. Chen, L. Gao, H. Shen, and J. Song, "Alleviating hallucinations in large vision-language models through hallucination-induced optimization," *Advances in Neural Information Processing Systems*, vol. 37, pp. 122 811–122 832, 2024.
[238] Z. Sun *et al.*, "Aligning large multimodal models with factually augmented rlhf," *arXiv preprint arXiv:2309.14525*, 2023.
[239] T. Yu *et al.*, "Rlhf-v: Towards trustworthy mllms via behavior alignment from fine-grained correctional human feedback," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 13 807–13 816.
[240] R. Rafailov *et al.*, "Direct preference optimization: Your language model is secretly a reward model," *Advances in neural information processing systems*, vol. 36, pp. 53 728–53 741, 2023.
[241] H. Lee *et al.*, "Rlaif vs. rlhf: Scaling reinforcement learning from human feedback with ai feedback," *arXiv preprint arXiv:2309.00267*, 2023.
[242] J. Wang, "Hallucination reduction and optimization for large language model-based autonomous driving," *Symmetry*, vol. 16, no. 9, p. 1196, 2024.
[243] W. Xiao *et al.*, "A comprehensive survey of direct preference optimization: Datasets, theories, variants, and applications," 2025. [Online]. Available: https://arxiv.org/abs/2410.15595
[244] S. Mangrulkar *et al.*, "Peft: State-of-the-art parameter-efficient fine-tuning methods," 2022.
[245] J. Pfeiffer *et al.*, "mmt5: Modular multilingual pre-training solves source language hallucinations," 2023. [Online]. Available: https://arxiv.org/abs/2305.14224
[246] A. Chakraborty, M. Alam, V. Dey, A. Chattopadhyay, and D. Mukhopadhyay, "Adversarial attacks and defences: A survey," *arXiv preprint arXiv:1810.00069*, 2018.
[247] Z. Zhang, E. Strubell, and E. Hovy, "A survey of active learning for natural language processing," *arXiv preprint arXiv:2210.10109*, 2022.
[248] A. Tharwat and W. Schenck, "A survey on active learning: State-of-the-art, practical challenges and research directions," *Mathematics*, vol. 11, no. 4, 2023. [Online]. Available: https://www.mdpi.com/2227-7390/11/4/820
[249] C.-H. Chiang and H. yi Lee, "Re-examining human annotations for interpretable nlp," 2022. [Online]. Available: https://arxiv.org/abs/2204.04580
[250] A. M. Palmer, "Semi-automated annotation and active learning for language documentation," 2009.
[251] N. Pangakis, S. Wolken, and N. Fasching, "Automated annotation with generative ai requires validation," *arXiv preprint arXiv:2306.00176*, 2023.
[252] J. Li, Y. Yuan, and Z. Zhang, "Enhancing llm factual accuracy with rag to counter hallucinations: A case study on domain-specific queries in private knowledge-bases," *arXiv preprint arXiv:2403.10446*, 2024.
[253] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, "Fine-tuning or retrieval? comparing knowledge injection in llms," 2024. [Online]. Available: https://arxiv.org/abs/2312.05934
[254] S. Dhuliawala *et al.*, "Chain-of-verification reduces hallucination in large language models," *arXiv preprint arXiv:2309.11495*, 2023.
[255] E. Perez, P. Lewis, W. tau Yih, K. Cho, and D. Kiela, "Unsupervised question decomposition for question answering," 2020. [Online]. Available: https://arxiv.org/abs/2002.09758
[256] A. J. Oche, A. G. Folashade, T. Ghosal, and A. Biswas, "A systematic review of key retrieval-augmented generation (rag) systems: Progress, gaps, and future directions," 2025. [Online]. Available: https://arxiv.org/abs/2507.18910
[257] J. Kirchenbauer and C. Barns, "Hallucination reduction in large language models with retrieval-augmented generation using wikipedia knowledge," 2024.
[258] S. Barnett, S. Kurniawan, S. Thudumu, Z. Brannelly, and M. Abdelrazek, "Seven failure points when engineering a retrieval augmented generation system," in *Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering-Software Engineering for AI*, 2024, pp. 194–199.
[259] C. Amiraz, F. Cuconasu, S. Filice, and Z. Karnin, "The distracting effect: Understanding irrelevant passages in rag," *arXiv preprint arXiv:2505.06914*, 2025.
[260] G. Sng, Y. Zhang, and K. Mueller, "A novel approach to eliminating hallucinations in large language model-assisted causal discovery," 2024. [Online]. Available: https://arxiv.org/abs/2411.12759
[261] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity," 2024. [Online]. Available: https://arxiv.org/abs/2403.14403
[262] M. Niu, H. Li, J. Shi, H. Haddadi, and F. Mo, "Mitigating hallucinations in large language models via self-refinement-enhanced knowledge retrieval," *arXiv preprint arXiv:2405.06545*, 2024.
[263] Q. Lv *et al.*, "Coarse-to-fine highlighting: Reducing knowledge hallucination in large language models," *arXiv preprint arXiv:2410.15116*, 2024.
[264] J. Herzig, P. K. Nowak, T. Müller, F. Piccinno, and J. Eisenschlos, "Tapas: Weakly supervised table parsing via pre-training," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, doi:10.18653/v1/2020.acl-main.398. [Online]. Available: http://dx.doi.org/10.18653/v1/2020.acl-main.398
[265] S. Pan *et al.*, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 7, pp. 3580–3599, 2024.
[266] E. Dinan *et al.*, "Wizard of wikipedia: Knowledge-powered conversational agents," *arXiv preprint arXiv:1811.01241*, 2018.
[267] A. Martino, M. Iannelli, and C. Truong, "Knowledge injection to counter large language model (llm) hallucination," in *European Semantic Web Conference*. Springer, 2023, pp. 182–185.
[268] X. Xu *et al.*, "A survey on knowledge distillation of large language models," *arXiv preprint arXiv:2402.13116*, 2024.
[269] A. Sharma *et al.*, "Mitigating hallucination with zerog: An advanced knowledge management engine," 2024. [Online]. Available: https://arxiv.org/abs/2411.05936
[270] D. McDonald, R. Papadopoulos, and L. Benningfield, "Reducing llm hallucination using knowledge distillation: A case study with mistral large and mmlu benchmark," *Authorea Preprints*, 2024.
[271] A. Sharma, L. Lara, C. J. Pal, and A. Zouaq, "Reducing hallucinations in language model-based sparql query generation using post-generation memory retrieval," *arXiv preprint arXiv:2502.13369*, 2025.
[272] R. Capellini, F. Atienza, and M. Sconfield, "Knowledge accuracy and reducing hallucinations in llms via dynamic domain knowledge injection," 2024.
[273] F. Wan *et al.*, "Knowledge verification to nip hallucination in the bud," *arXiv preprint arXiv:2401.10768*, 2024.
[274] H. Zhang *et al.*, "Visual question decomposition on multimodal large language models," *arXiv preprint arXiv:2409.19339*, 2024.
[275] P. Sahu, K. Sikka, and A. Divakaran, "Pelican: Correcting hallucination in vision-llms via claim decomposition and program of thought verification," *arXiv preprint arXiv:2407.02352*, 2024.
[276] H. Zheng *et al.*, "Thinking before looking: Improving multimodal llm reasoning via mitigating visual hallucination," *arXiv preprint arXiv:2411.12591*, 2024.
[277] S. Fairburn and J. Ainsworth, "Mitigate large language model hallucinations with probabilistic inference in graph neural networks," 2024.
[278] Q. Yu *et al.*, "Hallucidoctor: Mitigating hallucinatory toxicity in visual instruction data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 12 944–12 953.

[279] X. Amatriain, “Measuring and mitigating hallucinations in large language models: amultifaceted approach,” 2024.

[280] J. Luo, C. Xiao, and F. Ma, “Zero-resource hallucination prevention for large language models,” *arXiv preprint arXiv:2309.02654*, 2023.

[281] S. Lee, S. H. Park, Y. Jo, and M. Seo, “Volcano: Mitigating multimodal hallucination through self-feedback guided revision,” 2024. [Online]. Available: https://arxiv.org/abs/2311.07362

[282] A. NAUG, A. GUILLEN, and S. SARKAR, “Enhancing large language models with ensemble of critics for mitigating toxicity and hallucination.”

[283] D. Lei *et al.*, “Chain of natural language inference for reducing large language model ungrounded hallucinations,” *arXiv preprint arXiv:2310.03951*, 2023.

[284] D. Gosmar, D. A. Dahl, and E. Coin, “Conversational ai multi-agent interoperability, universal open apis for agentic natural language multimodal communications,” *arXiv preprint arXiv:2407.19438*, 2024.

[285] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, “Improving factuality and reasoning in language models through multiagent debate,” in *International Conference on Machine Learning*, 2023, pp. 8595–8611.

[286] A. Chen, P. Pasupat, S. Singh, H. Lee, and K. Guu, “Purr: Efficiently editing language model hallucinations by denoising language model corruptions,” *arXiv preprint arXiv:2305.14908*, 2023.

[287] J. Li, J. Zhang, Z. Jie, L. Ma, and G. Li, “Mitigating hallucination for large vision language model by inter-modality correlation calibration decoding,” *arXiv preprint arXiv:2501.01926*, 2025.

[288] E. Sharkey and P. Treleaven, “Optimising large language models: Taxonomy and techniques,” *Available at SSRN*, 2025.

[289] X. Han, “Heterogeneous model collaborations in language models,” Ph.D. dissertation, University of Washington, 2025.

[290] A. Favero *et al.*, “Multi-modal hallucination control by visual information grounding,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 14 303–14 312.

[291] A. Addlesee, “Grounding LLMs to in-prompt instructions: Reducing hallucinations caused by static pre-training knowledge,” in *Proceedings of Safety4ConvAI: The Third Workshop on Safety for Conversational AI @ LREC-COLING 2024*, T. Dinkar *et al.*, Eds. Torino, Italia: ELRA and ICCL, May 2024, pp. 1–7. [Online]. Available: https://aclanthology.org/2024.safety4convai-1.1/

[292] X. Li, “Teach large language models the concept of meta-cognition to reduce hallucination text generation.”

[293] S. Xu *et al.*, “Fine-tuned llms know more, hallucinate less with few-shot sequence-to-sequence semantic parsing over wikidata,” 2023. [Online]. Available: https://arxiv.org/abs/2305.14202

[294] B. Galitsky, “Truth-o-meter: Collaborating with llm in fighting its hallucinations,” in *Interdependent Human-Machine Teams*. Elsevier, 2025, pp. 175–210.

[295] D. Lee, E. Park, H. Lee, and H.-S. Lim, “Ask, assess, and refine: Rectifying factual consistency and hallucination in llms with metric-guided feedback learning,” in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 2422–2433.

[296] Z. Xu, S. Jain, and M. Kankanhalli, “Hallucination is inevitable: An innate limitation of large language models,” 2025. [Online]. Available: https://arxiv.org/abs/2401.11817

[297] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the national academy of sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[298] B. Settles, “Active learning literature survey,” *University of Wisconsin-Madison Department of Computer Sciences*, 2009.

[299] Y. Gal and Z. Ghahramani, “Dropout as a bayesian approximation: Representing model uncertainty in deep learning,” in *International Conference on Machine Learning*, 2016, pp. 1050–1059.

[300] A. Malinin and M. Gales, “Predictive uncertainty estimation via prior networks,” *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[301] X. L. Li *et al.*, “Contrastive decoding: Open-ended text generation as optimization,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023, pp. 12 286–12 312.

[302] A. K. Vijayakumar *et al.*, “Diverse beam search: Decoding diverse solutions from neural sequence models,” *arXiv preprint arXiv:1610.02424*, 2016.

[303] A. Vaswani *et al.*, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[304] A. Rogers, O. Kovaleva, and A. Rumshisky, “A primer in bertology: What we know about how bert works,” *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 842–866, 2020.

[305] N. F. Liu *et al.*, “Lost in the middle: How language models use long contexts,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[306] S. Min *et al.*, “Factscore: Fine-grained atomic evaluation of factual precision in long form text generation,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 12 076–12 100.

[307] O. Honovich *et al.*, “True: Re-evaluating factual consistency evaluation,” *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 3905–3920, 2022.

[308] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning, “A large annotated corpus for learning natural language inference,” in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, 2015, pp. 632–642.

[309] A. Williams, N. Nangia, and S. Bowman, “A broad-coverage challenge corpus for sentence understanding through inference,” in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2018, pp. 1112–1122.

[310] S. Lin, J. Hilton, and O. Evans, “Truthfulqa: Measuring how models mimic human falsehoods,” *arXiv preprint arXiv:2109.07958*, 2021.

[311] W. Kryściński, B. McCann, C. Xiong, and R. Socher, “Evaluating the factual consistency of abstractive text summarization,” *arXiv preprint arXiv:1910.12840*, 2019.

[312] A. R. Fabbri *et al.*, “Summeval: Re-evaluating summarization evaluation,” *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 391–409, 2021.

[313] W. Luo *et al.*, “Halludial: A large-scale benchmark for automatic dialogue-level hallucination evaluation,” *arXiv preprint arXiv:2406.07070*, 2024.

[314] A. Simhi, J. Herzig, I. Szpektor, and Y. Belinkov, “Constructing benchmarks and interventions for combating hallucinations in llms,” *arXiv preprint arXiv:2404.09971*, 2024.

[315] D. Jin *et al.*, “What disease does this patient have? a large-scale open domain question answering dataset from medical exams,” *Applied Sciences*, vol. 11, no. 14, p. 6421, 2021.

[316] B. Yan, J. Zhang, Z. Yuan, S. Shan, and X. Chen, “Evaluating the quality of hallucination benchmarks for large vision-language models,” *arXiv preprint arXiv:2406.17115*, 2024.

[317] Z. Yang *et al.*, “Hotpotqa: A dataset for diverse, explainable multi-hop question answering,” *arXiv preprint arXiv:1809.09600*, 2018.

[318] D. Zhu *et al.*, “Pollmgraph: Unraveling hallucinations in large language models via state transition dynamics,” *arXiv preprint arXiv:2404.04722*, 2024.

[319] S. Qi, L. Gui, Y. He, and Z. Yuan, “A survey of automatic hallucination evaluation on natural language generation,” *arXiv preprint arXiv:2404.12041*, 2024.

[320] X. Yu, H. Cheng, X. Liu, D. Roth, and J. Gao, “Reeval: Automatic hallucination evaluation for retrieval-augmented large language models via transferable adversarial attacks,” *arXiv preprint arXiv:2310.12516*, 2023.

[321] J. Duan *et al.*, “Truthprint: Mitigating lvlm object hallucination via latent truthful-guided pre-intervention,” 2025.

[322] P. Zablocki and Z. Gajewska, “Assessing hallucination risks in large language models through internal state analysis,” *Authorea Preprints*, 2024.

[323] M. Amirizaniani *et al.*, “Llmauditor: A framework for auditing large language models using human-in-the-loop,” *arXiv preprint arXiv:2402.09346*, 2024, verified source: LLMAuditor framework for human-in-the-loop LLM auditing using probe generation and TruthfulQA evaluation. [Online]. Available: https://arxiv.org/abs/2402.09346

[324] M. Giuffrè, K. You, and D. L. Shung, “Evaluating chatgpt in medical contexts: the imperative to guard against hallucinations and partial accuracies,” *Clinical Gastroenterology and Hepatology*, vol. 22, no. 5, pp. 1145–1146, 2024.

[325] S. Petryk *et al.*, “Aloha: A new measure for hallucination in captioning models,” *arXiv preprint arXiv:2404.02904*, 2024.

[326] K. Kuang *et al.*, “Causal inference,” *Engineering*, vol. 6, no. 3, pp. 253–263, 2020.

[327] A. Basit and W. Burgard, "Causalbench benchmark: Testing the causal inference abilities of large language models," *arXiv preprint arXiv:2404.06349*, 2024.
[328] R. Kaushik, "The axiom of choice and its influence on llm hallucinations: An exploration," *Available at SSRN 4722440*, 2023.
[329] C. Cui *et al.*, "Holistic analysis of hallucination in gpt-4v (ision): Bias and interference challenges," *arXiv preprint arXiv:2311.03287*, 2023.
[330] Z. Chen *et al.*, "How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites," *Science China Information Sciences*, vol. 67, no. 12, p. 220101, 2024.
[331] M. Arslan, H. Ghanem, S. Munawar, and C. Cruz, "A survey on rag with llms," *Procedia computer science*, vol. 246, pp. 3781–3790, 2024.
[332] Z. Zhu, Y. Yang, and Z. Sun, "Halueval-wild: Evaluating hallucinations of language models in the wild," *arXiv preprint arXiv:2403.04307*, 2024.
[333] X. Liang *et al.*, "Uhgeval: Benchmarking the hallucination of chinese large language models via unconstrained generation," *arXiv preprint arXiv:2311.15296*, 2023.
[334] M. Baqar and R. Khanda, "Hallucinations and truth: A comprehensive accuracy evaluation of rag, lora and dora," 2025.
[335] V. Rawte *et al.*, "Exploring the relationship between llm hallucinations and prompt linguistic nuances: Readability, formality, and concreteness," *arXiv preprint arXiv:2309.11064*, 2023.
[336] N. Mündler, J. He, S. Jenko, and M. Vechev, "Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation," *arXiv preprint arXiv:2305.15852*, 2023.
[337] R. Azamfirei, S. R. Kudchadkar, and J. Fackler, "Large language models and the perils of their hallucinations," *Critical Care*, vol. 27, no. 1, p. 120, 2023.
[338] T. R. McIntosh *et al.*, "A culturally sensitive test to evaluate nuanced gpt hallucination," *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 6, pp. 2739–2751, 2023.
[339] A. Jesson *et al.*, "Estimating the hallucination rate of generative ai," *Advances in Neural Information Processing Systems*, vol. 37, pp. 31 154–31 201, 2024.
[340] R. Hu, J. Zhong, M. Ding, Z. Ma, and M. Chen, "Evaluation of hallucination and robustness for large language models," in *2023 IEEE 23rd International Conference on Software Quality, Reliability, and Security Companion (QRS-C)*. IEEE, 2023, pp. 374–382.
[341] Z. Wang, "Causalbench: A comprehensive benchmark for evaluating causal reasoning capabilities of large language models," in *Proceedings of the 10th SIGHAN Workshop on Chinese Language Processing (SIGHAN-10)*, 2024, pp. 143–151.
[342] M. V. Carro, F. G. Selasco, D. A. Mester, and M. A. Leiva, "Are ufos driving innovation? the illusion of causality in large language models," *arXiv preprint arXiv:2410.11684*, 2024.
[343] C. Jiang *et al.*, "Hal-eval: A universal and fine-grained hallucination evaluation framework for large vision language models," in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024, pp. 525–534.
[344] M. Wu *et al.*, "Evaluating and analyzing relationship hallucinations in large vision-language models," *arXiv preprint arXiv:2406.16449*, 2024.
[345] A. Ferrando *et al.*, "A mechanistic analysis of a transformer trained on a symbolic multi-step reasoning task," *arXiv preprint arXiv:2402.11917*, 2024.
[346] C. Sun, Y. Li, D. Wu, and B. Boulet, "Onioneval: An unified evaluation of fact-conflicting hallucination for small-large language models," *arXiv preprint arXiv:2501.12975*, 2025.
[347] P. Kaul *et al.*, "Throne: An object-based hallucination benchmark for the free-form generations of large vision-language models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 228–27 238.
[348] Z. Li, X. Wu, H. Du, H. Nghiem, and G. Shi, "Benchmark evaluations, applications, and challenges of large vision language models: A survey," *arXiv preprint arXiv:2501.02189*, vol. 1, 2025.
[349] Q. Li *et al.*, "How llms react to industrial spatio-temporal data? assessing hallucination with a novel traffic incident benchmark dataset," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track)*, 2025. [Online]. Available: aclanthology.org/2025.naacl-industry.4/
[350] P. Xu *et al.*, "Lvlm-ehub: A comprehensive evaluation benchmark for large vision-language models," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
[351] F. Liu *et al.*, "Exploring and evaluating hallucinations in llm-powered code generation," *arXiv preprint arXiv:2404.00971*, 2024.
[352] D. Wan *et al.*, "Acueval: Fine-grained hallucination evaluation and correction for abstractive summarization," in *Findings of the Association for Computational Linguistics ACL 2024*, 2024, pp. 10 036–10 056.
[353] M. Feng, Y. Tang, Z. Zhang, and C. Xu, "Do more details always introduce more hallucinations in lvlm-based image captioning?" *arXiv preprint arXiv:2406.12663*, 2024.
[354] W. Zhao, Y. Tao, R. Liu, L. Kong, and Y. Chen, "Wildhallu: Evaluating long-form factuality in llms with real-world entities," *arXiv preprint arXiv:2407.09043*, 2024.
[355] Q. Cheng *et al.*, "Evaluating hallucinations in chinese large language models," *arXiv preprint arXiv:2310.03368*, 2023.
[356] C. Ravuri and S. Amarasinghe, "Eliminating hallucination-induced errors in llm code generation with functional clustering," *arXiv preprint arXiv:2506.11021*, 2025.
[357] G. Bai *et al.*, "Mt-bench-101: A fine-grained benchmark for evaluating large language models in multi-turn dialogues," *arXiv preprint arXiv:2402.14762*, 2024.
[358] Y. Sun *et al.*, "Why and how llms hallucinate: Connecting the dots with subsequence associations," *arXiv preprint arXiv:2504.12691*, 2025.
[359] C. Niu *et al.*, "Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models," *arXiv preprint arXiv:2401.00396*, 2023.
[360] F. S. Bao *et al.*, "Faithbench: A diverse hallucination benchmark for summarization by modern llms," *arXiv preprint arXiv:2410.13210*, 2024.
[361] M. Cossio, "A comprehensive taxonomy of hallucinations in large language models," *arXiv preprint arXiv:2508.01781*, 2025.
[362] T. Liu *et al.*, "A token-level reference-free hallucination detection benchmark for free-form text generation," *arXiv preprint arXiv:2104.08704*, 2021.
[363] N. Jiang, Q. Li, L. Tan, and T. Zhang, "Collu-bench: A benchmark for predicting language model hallucinations in code," *arXiv preprint arXiv:2410.09997*, 2024.
[364] Y. Tian *et al.*, "Codehalu: Investigating code hallucinations in llms via execution-based verification," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 24, 2025, pp. 25 300–25 308.
[365] J. Wang and Y. Chen, "A review on code generation with llms: Application and evaluation," in *2023 IEEE International Conference on Medical Artificial Intelligence (MedAI)*. IEEE, 2023, pp. 284–289.
[366] Y. Qin *et al.*, "Toolllm: Facilitating large language models to master 16000+ real-world apis," *arXiv preprint arXiv:2307.16789*, 2023.
[367] M. Z. Afzal *et al.*, "Next generation xr systems-large language models meet augmented and virtual reality," *IEEE computer graphics and applications*, 2025.
[368] Y. Sun, D. Sheng, Z. Zhou, and Y. Wu, "Ai hallucination: towards a comprehensive classification of distorted information in artificial intelligence-generated content," *Humanities and Social Sciences Communications*, vol. 11, no. 1, pp. 1–14, 2024.
[369] N. Chakraborty, M. Ornik, and K. Driggs-Campbell, "Hallucination detection in foundation models for decision-making: A flexible definition and review of the state of the art," *ACM Computing Surveys*, vol. 57, no. 7, pp. 1–35, 2025.
[370] K. Zuo and Y. Jiang, "Medhallbench: A new benchmark for assessing hallucination in medical large language models," *arXiv preprint arXiv:2412.18947*, 2024.
[371] D. Li *et al.*, "Overview of the scihal25 shared task on hallucination detection for scientific content," in *Proceedings of the Fifth Workshop on Scholarly Document Processing (SDP 2025)*, 2025, pp. 307–315.
[372] H. Kang and X.-Y. Liu, "Deficiency of large language models in finance: An empirical examination of hallucination," *arXiv preprint arXiv:2311.15548*, 2023.
[373] R. Chen *et al.*, "Rethinking domain-specific llm benchmark construction: A comprehensiveness-compactness approach," *arXiv preprint arXiv:2508.07353*, 2025.
[374] G. Hong *et al.*, "The hallucinations leaderboard: An open effort to measure hallucinations in large language models," *arXiv preprint arXiv:2404.05904*, 2024.
[375] J. Novikova, C. Anderson, B. Blili-Hamelin, D. Rosati, and S. Majumdar, "Consistency in language models: Current landscape, challenges, and future directions," *arXiv preprint arXiv:2505.00268*, 2025.

[376] E. Asgari *et al.*, "A framework to assess clinical safety and hallucination rates of llms for medical text summarisation," *npj Digital Medicine*, vol. 8, no. 1, p. 274, 2025.
[377] W. Y. Choong, Y. Guo, and M. Kankanhalli, "Vidhal: Benchmarking temporal hallucinations in vision llms," *arXiv preprint arXiv:2411.16771*, 2024, verified source: Benchmark for temporal hallucinations in video understanding with multimodal models. [Online]. Available: https://arxiv.org/abs/2411.16771
[378] L. Sun, L. Wang, J. Sun, and T. Okatani, "Temporal insight enhancement: Mitigating temporal hallucination in video understanding by multimodal large language models," in *International Conference on Pattern Recognition*. Springer, 2024, pp. 455–473.
[379] C. Li, E. W. Im, and P. Fazli, "Vidhalluc: Evaluating temporal hallucinations in multimodal large language models for video understanding," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 13 723–13 733.
[380] N. Pham and M. Schott, "H-pope: Hierarchical polling-based probing evaluation of hallucinations in large vision-language models," *arXiv preprint arXiv:2411.04077*, 2024.
[381] W. H. Walters and E. I. Wilder, "Fabrication and errors in the bibliographic citations generated by chatgpt," *Scientific Reports*, vol. 13, no. 1, p. 14045, 2023.
[382] N. F. Liu, T. Zhang, and P. Liang, "Evaluating verifiability in generative search engines," *arXiv preprint arXiv:2304.09848*, 2023.
[383] Y. A. Yadkori, I. Kuzborskij, A. György, and C. Szepesvári, "To believe or not to believe your llm," *arXiv preprint arXiv:2406.02543*, 2024.
[384] N. Barrowman, "Correlation, causation, and confusion," *The New Atlantis*, pp. 23–44, 2014.
[385] G. Carloni, A. Berti, and S. Colantonio, "The role of causality in explainable artificial intelligence," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 15, no. 2, p. e70015, 2025.
[386] C. Fierro and A. Søgaard, "Factual consistency of multilingual pretrained language models," *arXiv preprint arXiv:2203.11552*, 2022.
[387] Z. Yang, Z. Meng, X. Zheng, and R. Wattenhofer, "Assessing adversarial robustness of large language models: An empirical study," *arXiv preprint arXiv:2405.02764*, 2024.
[388] R. Cantini, A. Orsino, M. Ruggiero, and D. Talia, "Benchmarking adversarial robustness to bias elicitation in large language models: Scalable automated assessment with llm-as-a-judge," *arXiv preprint arXiv:2504.07887*, 2025.
[389] D. Muhlgay *et al.*, "Generating benchmarks for factuality evaluation of language models," *arXiv preprint arXiv:2307.06908*, 2023.
[390] X. Chen *et al.*, "Unified hallucination detection for multimodal large language models," *arXiv preprint arXiv:2402.03190*, 2024, this paper presents the UNIHD framework for unified hallucination detection across multiple modalities. Evaluated on MHaluBench, it achieves 81.91% accuracy at claim level and 84.60% at segment level, demonstrating effective multi-granularity evaluation at both sentence-level and entity-level for comprehensive factual assessment. [Online]. Available: https://arxiv.org/abs/2402.03190
[391] S. Tong *et al.*, "Eyes wide shut? exploring the visual shortcomings of multimodal llms," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9568–9578, verified source: Study on visual shortcomings and language priors in multimodal LLMs. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Tong_Eyes_Wide_Shut_Exploring_the_Visual_Shortcomings_of_Multimodal_LLMs_CVPR_2024_paper.html
[392] OpenAI, "How much does gpt-4 cost?" https://openai.com/api/pricing/, 2023, official OpenAI pricing documentation - $0.03 per 1,000 prompt tokens and $0.06 per 1,000 completion tokens for GPT-4.
[393] J. Liu *et al.*, "Phd: A prompted visual hallucination evaluation dataset," *CoRR*, 2024.
[394] N. Elhage *et al.*, "A mathematical framework for transformer circuits," *Transformer Circuits Thread*, 2021, verified source: Foundational work on transformer circuits and mechanistic interpretability. [Online]. Available: https://transformer-circuits.pub/2021/framework/index.html
[395] A. Zou *et al.*, "Representation engineering: A top-down approach to ai transparency," *arXiv preprint arXiv:2310.01405*, 2023, verified source: Representation engineering for understanding internal model representations. [Online]. Available: https://arxiv.org/abs/2310.01405
[396] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, "Locating and editing factual associations in gpt," *Advances in Neural Information Processing Systems*, vol. 35, pp. 17 359–17 372, 2022, verified source: Causal tracing and editing factual associations in language models. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html
[397] Y. Li *et al.*, "Mitigating hallucinations in large language models via causal reasoning," *arXiv preprint arXiv:2508.12495*, 2025.
[398] S. Raghava, "Classification of hallucinations in large language models using a novel weighted metric," *UC Merced Undergraduate Research Journal*, vol. 17, no. 1, 2024, risk-weighted and Severity-aware Metrics.
[399] Y. Gao, H. Wang, Y. Liu, K. Huang, and Y. Chen, "Toolbehonest: A multi-level hallucination diagnostic benchmark for tool-augmented large language models," *arXiv preprint arXiv:2406.20015*, 2024.
[400] N. Dziri *et al.*, "Faithdial: A faithful benchmark for information-seeking dialogue," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 1473–1490, 2022.
[401] L. Jiao *et al.*, "Causal inference meets deep learning: A comprehensive survey," *Research*, vol. 7, p. 0467, 2024.
[402] P. Islam *et al.*, "Financebench: A new benchmark for financial question answering," *arXiv preprint arXiv:2311.11944*, 2024.
[403] R. Shan, Q. Ming, G. Hong, and H. Wu, "Benchmarking the hallucination tendency of google gemini and moonshot kimi," *OSFPreprints*, 2024.
[404] Z. Zhang *et al.*, "Multimodal chain-of-thought reasoning in language models," *arXiv preprint arXiv:2302.00923*, 2023, verified source: Multimodal Chain-of-Thought reasoning for vision-language models. [Online]. Available: https://arxiv.org/abs/2302.00923
[405] Y. Feng, "Counterfactual probing for hallucination detection and mitigation in large language models," *arXiv preprint arXiv:2508.01862*, 2025.
[406] M. Canby, A. Davies, C. Rastogi, and J. Hockenmaier, "How reliable are causal probing interventions?" *arXiv preprint arXiv:2408.15510*, 2024.
[407] P.-H. Huang, J.-L. Li, C.-P. Chen, M.-C. Chang, and W.-C. Chen, "Who brings the frisbee: Probing hidden hallucination factors in large vision-language model via causality analysis," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2025, pp. 6125–6135.
[408] R. Zhang, H. Zhang, and Z. Zheng, "Vl-uncertainty: Detecting hallucination in large vision-language model via uncertainty estimation," 2024.
[409] P. Liang *et al.*, "Holistic evaluation of language models," *arXiv preprint arXiv:2211.09110*, 2022.
[410] P. Laban, T. Schnabel, P. N. Bennett, and M. A. Hearst, "Summac: Re-visiting nli-based models for inconsistency detection in summarization," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 163–177, 2022.
[411] A. Wang, K. Cho, and M. Lewis, "Asking and answering questions to evaluate the factual consistency of summaries," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, July 2020, pp. 5008–5020.
[412] L. Xu, H. Xie, S.-Z. J. Qin, X. Tao, and F. L. Wang, "Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment," *arXiv preprint arXiv:2312.12148*, 2023.
[413] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in neural information processing systems*, vol. 33, pp. 9459–9474, 2020.
[414] RAGAS, "Ragas: Retrieval augmented generation assessment – metrics documentation," docs.ragas.io, https://docs.ragas.io/en/stable/concepts/metrics/, 2025, accessed 2025-08-18.
[415] N. F. Liu *et al.*, "Lost in the middle: How language models use long contexts," *arXiv preprint arXiv:2307.03172*, 2023.
[416] S. Barnett, S. Kurniawan, S. Thudumu, Z. Brannelly, and M. Abdelrazek, "Seven failure points when engineering a retrieval augmented generation system," *arXiv preprint arXiv:2401.05856*, 2024.
[417] R. Rafailov *et al.*, "Direct preference optimization: Your language model is secretly a reward model," *arXiv preprint arXiv:2305.18290*, 2023, verified source: Direct Preference Optimization foundational paper for preference-based learning. [Online]. Available: https://arxiv.org/abs/2305.18290
[418] Z. Zhao *et al.*, "Beyond hallucinations: Enhancing lvlms through hallucination-aware direct preference optimization," *arXiv preprint arXiv:2311.16839*, 2023.
[419] E. J. Hu *et al.*, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.

[420] S.-Y. Liu *et al.*, "Dora: Weight-decomposed low-rank adaptation," *arXiv preprint arXiv:2402.09353*, 2024.
[421] X. Zhou *et al.*, "An empirical study on parameter-efficient fine-tuning for multimodal large language models," *arXiv preprint arXiv:2406.05130*, 2024.
[422] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.
[423] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," *arXiv preprint arXiv:2310.11511*, 2024.
[424] Y. Fu, R. Xie, X. Sun, Z. Kang, and X. Li, "Mitigating hallucination in multimodal large language model via hallucination-targeted direct preference optimization," *arXiv preprint arXiv:2411.10436*, 2024.
[425] X. Leng *et al.*, "V-dpo: Mitigating hallucination in large vision language models via vision-guided direct preference optimization," *arXiv preprint arXiv:2411.02712*, 2024, verified source: Vision-guided DPO with specific results on CHAIR and RefCOCO benchmarks. [Online]. Available: https://arxiv.org/abs/2411.02712
[426] X. Zhang *et al.*, "Chain of preference optimization: Improving chain-of-thought reasoning in llms," *arXiv preprint arXiv:2406.09136*, 2024, verified source: Chain-of-Preference Optimization for improving reasoning and reducing cascading errors. [Online]. Available: https://arxiv.org/abs/2406.09136
[427] B. Peng *et al.*, "Check your facts and try again: Improving large language models with external knowledge and automated feedback," *arXiv preprint arXiv:2302.12813*, 2023.
[428] Q. Huang *et al.*, "Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 13 418–13 427.
[429] Y. Liu *et al.*, "G-eval: Nlg evaluation using gpt-4 with better human alignment," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 2511–2522.
[430] B. Wang, S. Chern, E. Chern, and P. Liu, "Halu-j: Critique-based hallucination judge," *arXiv preprint arXiv:2407.12943*, 2024.
[431] D. Deutsch, R. Dror, and D. Roth, "On the limitations of reference-free evaluations of generated text," *arXiv preprint arXiv:2210.12563*, 2022.
[432] H. Hu, J. Zhang, M. Zhao, and Z. Sun, "Ciem: Contrastive instruction evaluation method for better instruction tuning," *arXiv preprint arXiv:2309.02301*, 2023.
[433] L. Tang *et al.*, "Understanding factual errors in summarization: Errors, summarizers, datasets, error detectors," *arXiv preprint arXiv:2205.12854*, 2022.
[434] W. Mo, Y. Peng, and Y. Liu, "Viscon: Identifying and benchmarking vision hallucination for large vision-language model," *arXiv preprint arXiv:2506.19513*, 2025. [Online]. Available: https://openreview.net/forum?id=cjpTu0Op5t
[435] S. Qi, Y. He, and Z. Yuan, "Can we catch the elephant? a survey of the evolvement of hallucination evaluation on natural language generation," *arXiv preprint arXiv:2404.12041*, 2024.
[436] H. Ghanem and C. Cruz, "Enhancing knowledge graph construction: Evaluating with emphasis on hallucination, omission, and graph similarity metrics," in *International Knowledge Graph and Semantic Web Conference*. Springer, 2024, pp. 32–46.
[437] N. Guha *et al.*, "Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models," *arXiv preprint arXiv:2308.11462*, 2023.
[438] C. Qiu, X. Huang, J. Liu, and D. Tang, "Detecting hallucinated content in conditional neural sequence generation," *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 1393–1408, 2023.
[439] A. Baez, M. Wong, A. Douglas, and N. Guntvedt, "Hallucibert: Efficiently predicting likelihood of hallucation pre-inference," *arXiv preprint arXiv:2409.02976*, 2024.
[440] S. S. Ravi, B. Mielczarek, A. Kannappan, D. Kiela, and R. Qian, "Lynx: An open source hallucination evaluation model," *arXiv preprint arXiv:2407.08488*, 2024.
[441] F. Gao *et al.*, "Large language models on wikipedia-style survey generation: an evaluation in nlp concepts," *arXiv preprint arXiv:2308.10410*, 2023.
[442] S. Yao *et al.*, "React: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR)*, 2023.
[443] GitHub, "Outlines: Structured outputs," dottxt-ai.github.io, https://dottxt-ai.github.io/outlines/, 2025, accessed 2025-08-18.
[444] GitHub, "Llguidance: Super-fast structured outputs," github.com, https://github.com/guidance-ai/llguidance, 2025, accessed 2025-08-18.
[445] T. Stening. "vllm docs: Structured outputs." docs.vllm.ai. https://docs.vllm.ai/en/v0.9.2/features/structured_outputs.html (accessed Aug.18, 2025).
[446] A. R. Fabbri, C.-S. Wu, W. Liu, and C. Xiong, "Qafacteval: Improved qa-based factual consistency evaluation for summarization," *arXiv preprint arXiv:2112.08542*, 2021.
[447] C. Chen *et al.*, "Inside: Llms' internal states retain the power of hallucination detection," *arXiv preprint arXiv:2402.03744*, 2024.
[448] OpenTelemetry, "Opentelemetry specification – traces overview," opentelemetry.io, https://opentelemetry.io/docs/specs/otel/overview/, 2025, accessed 2025-08-18.
[449] Arize, "What is openinference?" arize.com, https://arize.com/docs/ax/learn/tracing-concepts/what-is-openinference, 2025, accessed 2025-08-18.
[450] C. Rathkopf, "Hallucination, reliability, and the role of generative ai in science," *arXiv preprint arXiv:2504.08526*, 2025.
[451] S. M. Williamson and V. Prybutok, "The era of artificial intelligence deception: unraveling the complexities of false realities and emerging threats of misinformation," *Information*, vol. 15, no. 6, p. 299, 2024.
[452] H. Elsayed, "The impact of hallucinated information in large language models on student learning outcomes: A critical examination of misinformation risks in ai-assisted education," *Northern Reviews on Algorithmic Research, Theoretical Computation, and Complexity*, vol. 9, no. 8, pp. 11–23, 2024.
[453] M. Mylrea, "The generative ai weapon of mass destruction: Evolving disinformation threats, vulnerabilities, and mitigation frameworks," in *Interdependent Human-Machine Teams*. Elsevier, 2025, pp. 315–347.
[454] P. Yu, H. Xu, X. Hu, and C. Deng, "Leveraging generative ai and large language models: a comprehensive roadmap for healthcare integration," in *Healthcare*, vol. 11, no. 20. MDPI, 2023, p. 2776.
[455] F. Leiser *et al.*, "From chatgpt to factgpt: A participatory design study to mitigate the effects of large language model hallucinations on users," in *Proceedings of Mensch und Computer 2023*, 2023, pp. 81–90.
[456] A. Banafa, *Artificial Intelligence in Action: Real-World Applications and Innovations*. CRC Press, 2025.
[457] M. Coeckelbergh and D. J. Gunkel, *Communicative AI: A critical introduction to large Language models*. John Wiley & Sons, 2025.
[458] P. Sahoo *et al.*, "A comprehensive survey of hallucination in large language, image, video and audio foundation models," *arXiv preprint arXiv:2405.09589*, 2024.
[459] A. M. Khawaldeh, "Generative ai hallucinations and legal liability in jordanian civil courts: Promoting the responsible use of conversational chat bots," *International Journal for the Semiotics of Law-Revue internationale de Sémiotique juridique*, vol. 38, no. 2, pp. 381–401, 2025.
[460] V. Habib Lantyer, "The phantom menace: Generative ai hallucinations and their legal implications," *The Phantom Menace: Generative AI Hallucinations and Their Legal Implications*, 2025.
[461] A. Taeihagh, "Governance of generative ai," pp. 1–22, 2025.
[462] G. Bansal *et al.*, "Does the most accurate model always win? the impact of accuracy on human trust in ai," in *Proceedings of the 26th ACM Conference on Intelligent User Interfaces*. ACM, 2021, pp. 450–460.
[463] Y. Bai *et al.*, "Training a helpful and harmless assistant with reinforcement learning from human feedback," *arXiv preprint arXiv:2204.05862*, 2022. [Online]. Available: https://arxiv.org/abs/2204.05862
[464] L. Cecco, "Air canada ordered to pay customer who was misled by airline's chatbot," *The Guardian*, online article; case involves a chatbot misinforming a customer about eligibility for a bereavement fare. [Online]. Available: https://www.theguardian.com/world/2024/feb/16/air-canada-chatbot-lawsuit
[465] A. Tlili and D. Burgos, "Ai hallucinations? what about human hallucination?!: Addressing human imperfection is needed for an ethical ai," *IJIMAI*, vol. 9, no. 2, pp. 68–71, 2025.
[466] L. Cecco, "Canada lawyer under fire for submitting fake cases created by ai chatbot," *The Guardian*, accessed via online edition of The Guardian. [Online]. Available: https://www.theguardian.com/world/2024/feb/29/canada-lawyer-chatgpt-fake-cases-ai
[467] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proceedings of the 34th*

International Conference on Machine Learning (ICML), 2017, pp. 1126–1135. [Online]. Available: https://arxiv.org/abs/1703.03400

[468] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on knowledge and data engineering*, vol. 22, no. 10, pp. 1345–1359, 2009.

[469] C.-Y. Lee, P. Zhang *et al.*, "Hallucination of multimodal large language models: Taxonomy, challenges, and opportunities," *arXiv preprint arXiv:2309.04020*, 2023. [Online]. Available: https://arxiv.org/abs/2309.04020

[470] Y. Zhou, Z. Zhou, M. Xu *et al.*, "A survey of multimodal hallucination in large vision-language models," *arXiv preprint arXiv:2310.01419*, 2023. [Online]. Available: https://arxiv.org/abs/2310.01419

[471] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282. [Online]. Available: https://arxiv.org/abs/1602.05629

[472] L. Floridi and J. Cowls, "Ai4people—an ethical framework for a good ai society: Opportunities, risks, principles, and recommendations," *Minds and Machines*, vol. 28, no. 4, pp. 689–707, 2018.

[473] A. Jobin, M. Ienca, and E. Vayena, "The global landscape of ai ethics guidelines," *Nature Machine Intelligence*, vol. 1, no. 9, pp. 389–399, 2019.

[474] L. Weidinger *et al.*, "Taxonomy of risks posed by language models," *arXiv preprint arXiv:2112.04359*, 2022. [Online]. Available: https://arxiv.org/abs/2112.04359

[475] M. Mitchell *et al.*, "Model cards for model reporting," *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pp. 220–229, 2019.

[476] J. Morley *et al.*, "The ethics of ai in health care: A mapping review," *Social Science & Medicine*, vol. 260, p. 113172, 2020.

[477] P. Liang, R. Bommasani, A. Zeng *et al.*, "Holistic evaluation of language models," in *NeurIPS 2022*, 2022. [Online]. Available: https://arxiv.org/abs/2211.09110

[478] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *International Conference on Learning Representations (ICLR)*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6572

[479] Y. Mirsky and W. Lee, "The creation and detection of deepfakes: A survey," *ACM Computing Surveys (CSUR)*, vol. 54, no. 1, pp. 1–41, 2021.